Toward Neuroprotective Cooling of the Eye: A Scleral-Contact Thermoelectric Device

BY

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DISSERTATION

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To my parents and brothers, whose love and help were indispensable for this work: thank you for helping me push through.
To my friends, far and close, but particularly to those in Chicago: you were the support I needed, my second family.

Accessibility Statement

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Contribution of Authors

Chapter 1 is an original introduction by the author.

Chapter 2 includes work published as Y. Nakamura, L. Mecacci, and J. R. Hetling, "Therapeutic Levels of Hypothermia Achieved in Isolated Porcine Eyes Using a Scleral Contact Interface," *Annals of Biomedical Engineering*, vol. 54, 2026 (published online November 2025) [**1**]. Y. Nakamura and the author contributed equally to that article, as stated in its author-contributions declaration; Y. Nakamura worked with the author on the preserved- and fresh-porcine experiments during the author's final year of master's study. J. R. Hetling supervised the work. The article is published open access under a Creative Commons Attribution 4.0 license. The fresh-porcine recordings from that study were reprocessed by the author for Chapter 2 together with the human-donor data.

Chapter 2 also draws on two manuscripts in preparation of which the author is first author: an ex-vivo methods paper with Y. Nakamura, K. Fong, N. Shirdade, and J. R. Hetling, and a human–porcine comparison with J. R. Hetling. K. Fong and N. Shirdade carried out earlier work on the eye-cooling system on which the methods paper builds. As an undergraduate research assistant in 2018–2019, K. Fong designed, machined, and prototyped an earlier ocular cooling device and the instrumentation for measuring its temperature. In 2018, N. Shirdade carried out the thermal simulation of the cooling of the porcine eye.

Chapter 3 is the original work of the author.

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Chapter 5 is an original discussion by the author.

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List of Abbreviations

3D	Three-dimensional
ACC	Animal Care Committee
AI	Artificial intelligence
AP	Anteroposterior
ARVO	Association for Research in Vision and Ophthalmology
ASA	American Society of Anesthesiologists
ATP	Adenosine triphosphate
AVMA	American Veterinary Medical Association
BRL	Biologic Resources Laboratory
CAD	Computer-aided design
CI	Confidence interval
CIRBP	Cold-inducible RNA-binding protein
CLEAr	Contact lens electrode array
CRAO	Central retinal artery occlusion
CV	Coefficient of variation
DAQ	Data acquisition
DC	Direct current
DSC	Differential scanning calorimetry
ERC	European Resuscitation Council
ERG	Electroretinogram
ESICM	European Society of Intensive Care Medicine
FDA	U.S. Food and Drug Administration
HACA	Hypothermia After Cardiac Arrest (trial)
LED	Light-emitting diode
Micro-CT	Micro-computed tomography
MOSFET	Metal-oxide-semiconductor field-effect transistor
MRI	Magnetic resonance imaging
NEVL	Neural Engineering Vision Laboratory
NTC	Negative temperature coefficient
ONH	Optic nerve head
OP	Oscillatory potential
PWM	Pulse-width modulation
RBM3	RNA-binding motif protein 3
RM	RetMap electrode (corneal ERG electrode)
RPE	Retinal pigment epithelium
SCEC	Scleral-contact eye cooler
SCR	Scleral contact ring
SD	Standard deviation
TEC	Thermoelectric cooler
TTM	Targeted temperature management
UIC	University of Illinois Chicago
USDA	U.S. Department of Agriculture

Summary

Ischemia of the retina and optic nerve, as in retinal artery occlusion, can cause permanent blindness within hours, and no established treatment protects the tissue. Mild cooling protects neural tissue from ischemic injury in laboratory studies, but cooling the whole body carries systemic risks. This dissertation concerns the scleral-contact eye cooler (SCEC), a thermoelectric device that holds a ring on the anterior sclera at 4 °C. It asks how much of that cooling reaches the back of the eye, whether the device can be made smaller, and whether the cooling can be detected, and tolerated, in a living eye.

In the first aim, temperature was recorded at five sites, from front to back, in isolated porcine and human donor eyes held in a warm bath. At the proximal optic nerve, the deepest site, human eyes cooled by nearly 4 °C within 20 min and porcine eyes slightly less. No difference between the species was detected there, and a difference above about 2 °C is unlikely.

In the second aim, a reduced ring of about half the original volume and a ceramic ring each cooled the proximal optic nerve of donor eyes nearly as well as the original ring. In a model of the perfused human eye, body and ring temperature set the posterior temperature far more than ring design did. An integrated control board was built and tested on the bench.

In the third aim, the device with the reduced ring was applied to one eye of anesthetized pigs for 30 to 120 min, with the fellow eye as control. Retinal cooling was sought through the electroretinogram. In the six animals the ring stayed near 9.5 °C, after correction for a calibration error, in the earlier sessions, or near 4 °C in the later ones, throughout every session. In all 14 analyzed sessions of the primary animals the b-wave of the cooled eye became slower than before cooling, at both primary flash strengths, by a mean of 5.1 to 5.8 ms, and in all but one wider, while that of the control eye did not slow; its oscillatory potentials were reduced. Both changes largely reversed after switch-off. The intraocular pressure rose no more in the cooled eye than in the control eye, redness and staining were worse only occasionally, and no adverse event was attributed to the device. One animal died of a complication of intubation. In the histology the cooled eye showed no damage absent from the control eye.

The device therefore cools the back of an isolated eye by a few degrees without surgery and, in the living eye, changes retinal function as a cooler retina would. The temperature it reaches in a perfused eye, the safety of longer exposures, and protection against ischemia remain to be tested.

CHAPTER 1

Introduction

1.1. Clinical Motivation

Acute retinal ischemia is a cause of sudden and usually permanent monocular vision loss. Its prototype, central retinal artery occlusion (CRAO), is a form of acute ischemic stroke [**2**] with a population incidence of about 2 per 100,000 person-years [**3, 4**]. In data that also count branch occlusions, retinal artery occlusion as a whole is more than three times as common [**5**]. Few affected eyes recover useful vision. Among eyes with permanent non-arteritic CRAO without cilioretinal sparing presenting at counting fingers or worse, acuity improved at all in only 22% of those seen within a week of onset [**6**], and pooled natural-history data give a 17.7% rate of recovery to 20/100 or better [**7**].

No treatment has proven effective, and the 2026 guideline of the American Heart Association and American Stroke Association for acute ischemic stroke rates the usefulness of intravenous thrombolysis within 4.5 h for disabling non-arteritic CRAO as uncertain [**8**]. In pooled observational data, conservative measures such as ocular massage and anterior-chamber paracentesis were associated with worse recovery than natural history [**7**]. Intra-arterial fibrinolysis matched conservative care at a higher complication rate, and its randomized trial was stopped early [**9**]. The first two randomized trials of intravenous thrombolysis within 4.5 h, THEIA and TenCRAOS, found no significant visual benefit over aspirin [**10, 11**]; one fatal intracranial hemorrhage occurred with tenecteplase [**11**]. A third, placebo-controlled trial, REVISION, was stopped for futility at a prespecified interim analysis after 127 participants had been enrolled, and its results have not been published [**12, 13**]. The TenCRAOS investigators concluded that effective therapy remains elusive and that

alternative approaches warrant investigation [**11**], and the THEIA investigators that the low recovery rates raise questions about how long the retina tolerates ischemia [**10**]. Time is a further barrier. Vision loss is painless, and patients present late. Median symptom-to-door times were about 13 h in a Swiss national stroke registry and a US comprehensive stroke center [**14, 15**], and 17 h to first medical contact in an Australian audit [**16**]. In the registry and stroke-center cohorts, only about a quarter of patients arrived within 4–4.5 h.

Retinal and optic-nerve ischemia also complicate nonocular surgery. Perioperative visual loss follows about 8.6 per 10,000 cardiac operations, where retinal vascular occlusion accounts for most cases, and 3.1 per 10,000 spinal fusions [**17**]. Retinal artery occlusion alone complicates 7.8 per 10,000 cardiac procedures [**18**], and ischemic optic neuropathy accounts for most cases reported to the American Society of Anesthesiologists (ASA) Postoperative Visual Loss Registry after prone spine surgery [**19, 20**]. No treatment has been shown to be effective, and guidance for visual loss after spine surgery is directed at prevention [**20, 21**]. Unlike spontaneous occlusion, the period of surgical risk is known in advance, and a protective intervention could be applied before and during it. External pressure on the globe is itself a recognized cause of perioperative retinal artery occlusion [**19, 21**], so a device that rests on the eye must not raise the intraocular pressure, a safety endpoint of Aim 3 (Section 1.8).

These settings share a need for an intervention that extends the retina's tolerance of ischemia, in addition to reperfusion. Preclinical evidence motivates localized cooling for that purpose (Section 1.3). Delivering the cooling to the posterior eye through an accessible surface is the engineering problem addressed in this dissertation.

1.2. The Eye

A device that cools the retina from outside the eye works through the layered wall of the globe and against its two circulations. This section describes these structures and the temperature field they sustain.

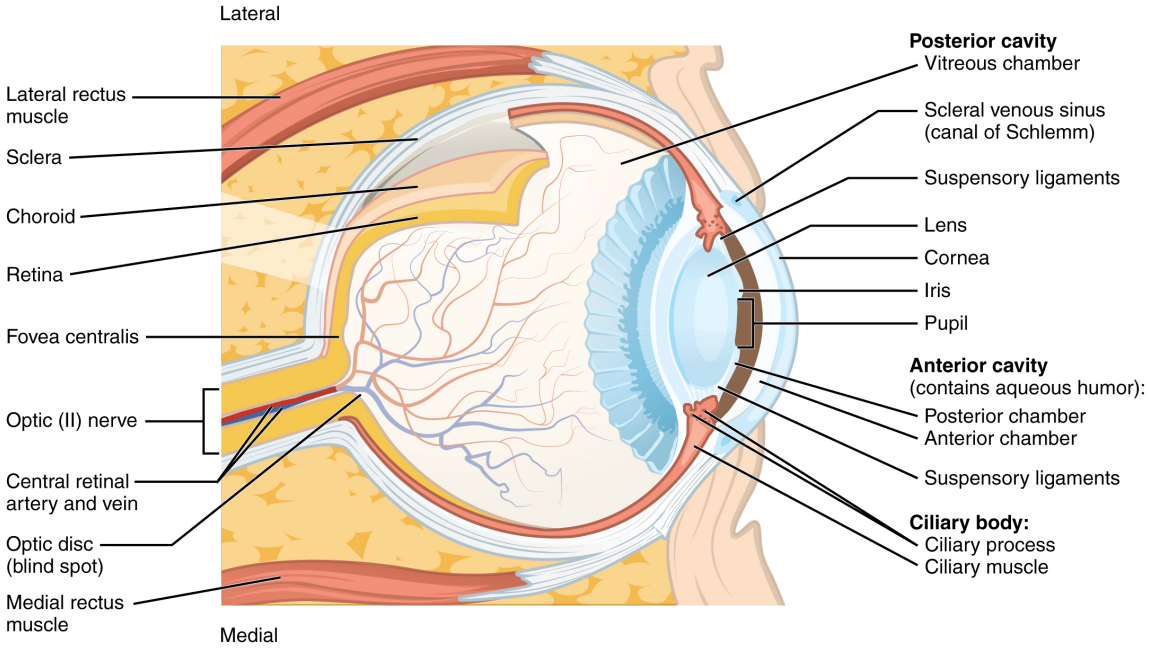


FIGURE 1.1. Cross-sectional anatomy of the human eye: the optical media, the anterior structures, the sclera, choroid, and retina around the vitreous cavity, and the extraocular muscles. From OpenStax College, *Anatomy & Physiology* (Houston: OpenStax, 2013), Fig. 14.13, "Structure of the Eye," via Wikimedia Commons, CC BY 3.0; unaltered.

1.2.1. Gross Anatomy. The human eye is a roughly spherical globe about 24 mm across [**23, 24**], with three concentric coats (Figure 1.1). The outer *fibrous* coat is the transparent cornea, continuous with the opaque sclera. The middle *vascular* coat, or uvea, comprises the iris, ciliary body, and choroid; it carries the greater part of ocular blood flow, although the inner retina is served by a separate retinal circulation (Section 1.2.2). The inner *neural* coat is the retina, the tissue that this work aims to protect. The cooler acts from the outer surface of the fibrous coat, so the anatomy that matters here is the layered wall between the point of contact and the retina, and the optical path. The cornea, aqueous humor, lens, and vitreous, a transparent gel filling the posterior cavity, form that path [**25**] and also carry heat between an anterior cooling interface and the posterior eye.

At the posterior pole the eye wall comprises, from the vitreous outward, the retina, the retinal pigment epithelium (RPE), Bruch's membrane, the choroid, and

the sclera [**25**]. Within the retina the photoreceptors lie outermost, apposed to the RPE, and the ganglion-cell axons that form the optic nerve lie toward the vitreous. The retina is among the most metabolically active tissues in the body [**26**], and its outer, photoreceptor layers are avascular, depending on the underlying choroid across Bruch's membrane and the RPE (Figure 1.2).

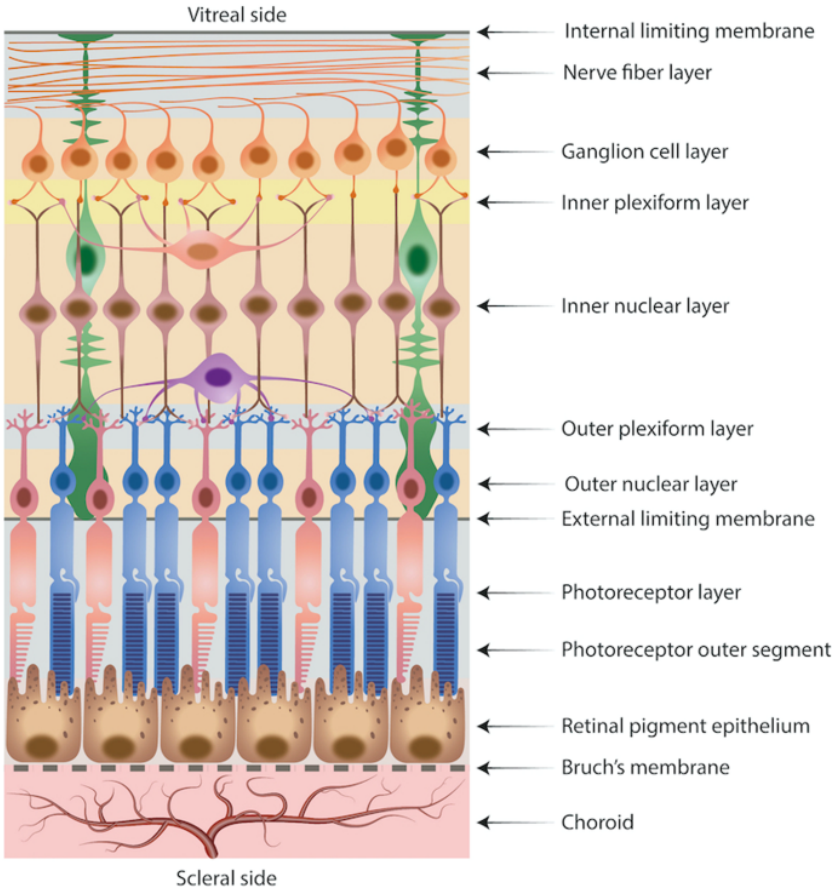


FIGURE 1.2. Retinal layers and their relation to the retinal pigment epithelium, Bruch's membrane, and choroid. Unmodified from Ferrara et al., Fig. 3 [**27**], CC BY 4.0.

The sclera forms five-sixths of the outer coat and is the contact surface of the cooler. Its collagenous stroma is normally avascular and is nourished by the adjacent episcleral and choroidal networks [**28**]. Its thickness is markedly non-uniform: 1.0–1.35 mm at the posterior pole, 0.4–0.6 mm at the equator, about 0.3 mm under the rectus muscles, and roughly 0.8 mm at the limbus [**28**]. Histological measurements give lower values, 0.50 ± 0.11 mm at the limbus (Table 1.3).

1.2.2. Blood Supply. Blood flow matters here as a source of heat. The ocular circulation delivers blood at core temperature, and a cooling device must remove that heat as well as the heat stored in the tissue. The eye's high metabolic demand is met by two largely separate vascular beds, the retinal and the uveal, fed from a common source and drained into a common sink. Arterial inflow arises almost entirely from the ophthalmic artery, usually the first major branch of the internal carotid artery; the anterior ciliary arteries arise not from it directly but from the muscular arteries of the rectus muscles. Venous outflow returns through the superior and inferior ophthalmic veins, mainly to the cavernous sinus [**25**].

Figure 1.3 shows the major ocular arterial pathways and venous drainage around the globe, and Figure 1.4 traces the path of the blood from the internal carotid artery to the venous outflow of the orbit.

The retinal circulation is end-arterial. The central retinal artery enters the globe through the optic nerve head (ONH) and supplies the inner retinal layers with little collateral supply, which helps explain the vulnerability of the retina to arterial occlusion [**45, 46**]; the outer, photoreceptor layers depend on the choroid beneath them. Total retinal blood flow in healthy human eyes is roughly 33–80 $\mu\mathrm{L\,min^{-1}}$ by laser Doppler and Doppler optical coherence tomography [**29, 30, 31, 32, 33, 34**]. The uveal circulation, dominated by the choroid, carries the large majority of ocular blood flow. It is supplied by the posterior ciliary arteries, whose short branches also form the circle of Zinn–Haller around the optic nerve, the supply of the ONH (Figure 1.5), and its venous return converges almost entirely into the vortex veins [**25**].

Per gram of tissue the choroid has the highest blood flow of any tissue in the body [**35, 36**]: about 2000 $\mathrm{mL\,min^{-1}}$ per 100 g, a value the same text attributes to the choriocapillaris, against 60 $\mathrm{mL\,min^{-1}}$ per 100 g for the retina [**25**], and roughly four times the flow of the renal cortex [**37**]. The commonly quoted choroidal share of 85% of ocular blood flow [**37, 38**] rests on monkey measurements in which the choroid received 607 of the 738 $\mathrm{mg\,min^{-1}}$ reaching the retina and uvea, or 82% [**39**].

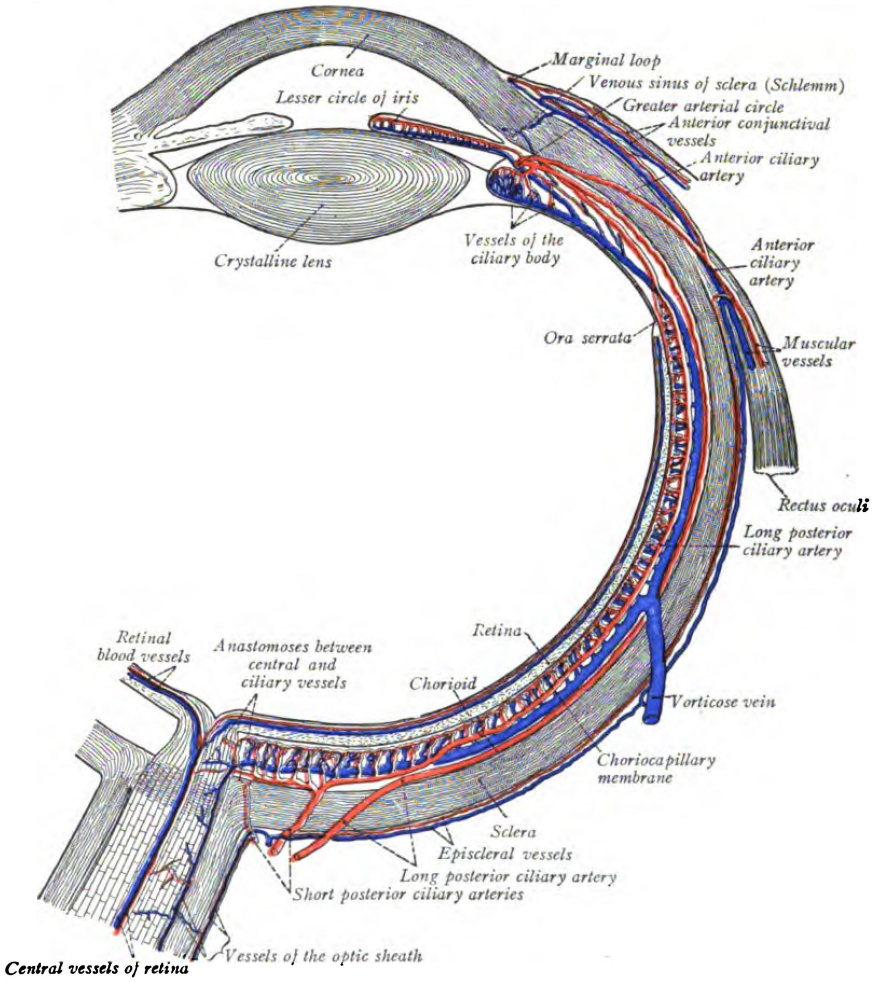


FIGURE 1.3. Blood vessels of the eyeball, diagrammatic section; arteries red, veins blue. From J. Sobotta and J. P. McMurrich, *Atlas and Text-book of Human Anatomy*, vol. 3 (Philadelphia: W. B. Saunders, 1914), Fig. 743, after Th. Leber; artist K. Hajek. Public domain; scan from AnatomyTOOL, with the printed caption cropped and the drawing otherwise unaltered.

These figures come from microsphere studies in monkeys and cats, and the model of Chapter 3 likewise takes the perfusion of its choroid from microsphere measurements in monkeys [**40**] (Appendix D). In humans, magnetic-resonance perfusion imaging of the posterior pole gives 149 ± 48 mL per 100 mL of tissue per minute for retina and choroid together, in four healthy volunteers [**41**].

The two beds are regulated very differently, which matters for how each clears heat. Retinal blood flow is strongly autoregulated, held nearly constant across changes in perfusion pressure and matched to metabolic demand [**42**], a control also central to

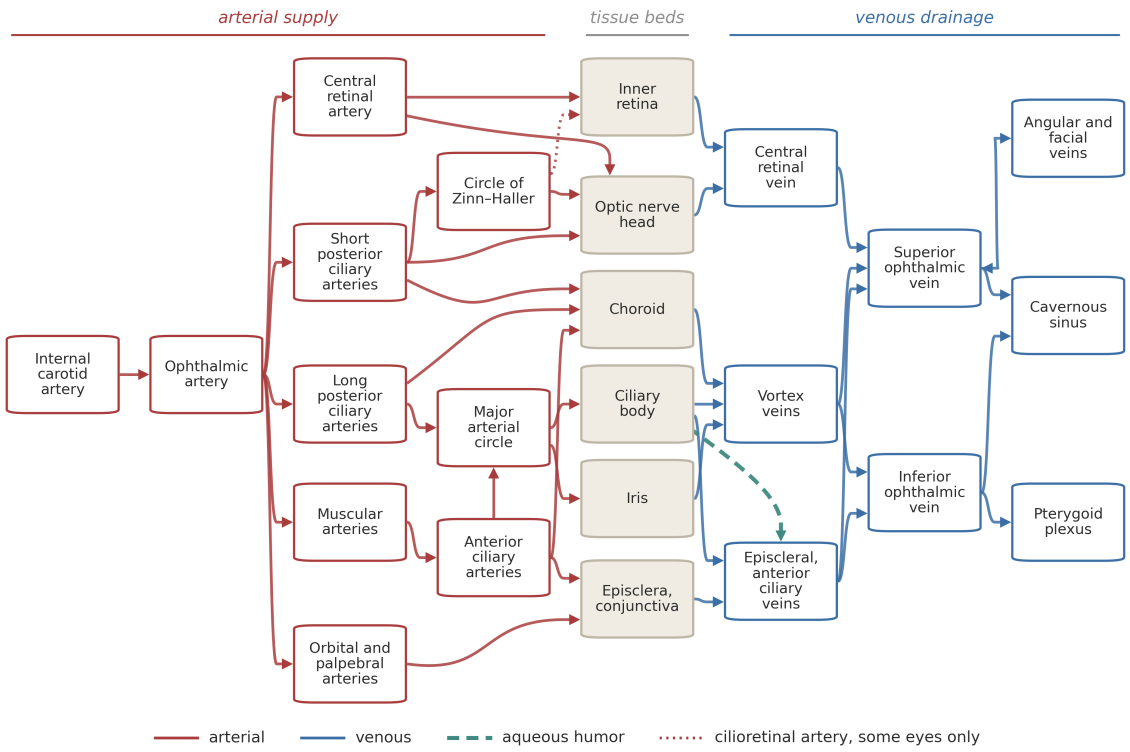


FIGURE 1.4. Flow path of the ocular circulation, from the internal carotid artery to the venous outflow of the orbit; arrows give the direction of flow, and tissue beds are shaded. The two-headed arrow is the connection with the angular and facial veins, in which flow can run either way because the orbital veins have few or no valves [**25, 35**]; the arrow from the central retinal artery to the optic nerve head is its supply to the surface nerve-fiber layer. Not drawn: the central retinal vein may drain directly into the cavernous sinus, and the prelaminar optic nerve head drains in part to the peripapillary choroidal veins [**25, 43**].

optic-nerve-head perfusion [**43**]. The choroid lacks this classical pressure autoregulation; its flow is governed largely by autonomic tone, sympathetic stimulation producing vasoconstriction [**25, 36**]. Some regulation is present: in healthy subjects performing isometric exercise, a 67% rise in mean ocular perfusion pressure raised subfoveal choroidal blood flow by only 12%, and the compensation broke down at larger rises [**44**]. Thermally, choroidal flow is very high, only partially responsive to perfusion pressure, and essentially uncoupled from local retinal metabolic demand [**36**], and it exchanges heat with the posterior tissues (Section 1.2.3).

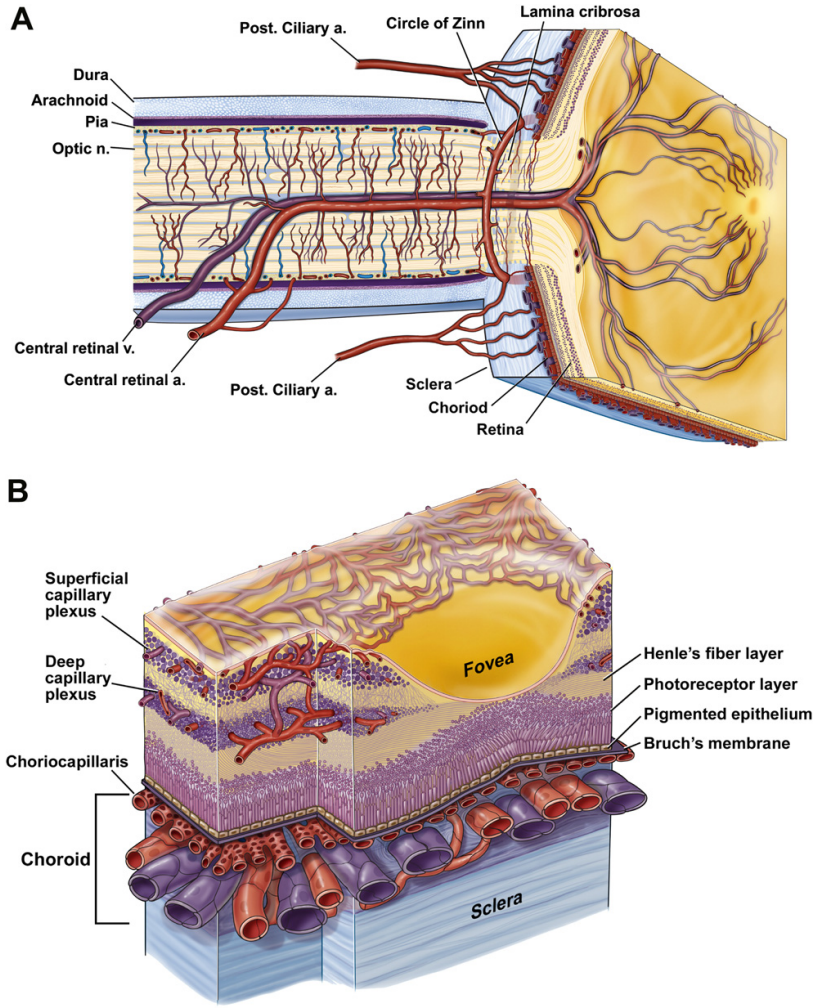


FIGURE 1.5. Retinal and choroidal circulation. (A) Vascular supply at the optic nerve and posterior globe. (B) Retinal capillary plexuses and underlying choroidal vessels. Reprinted from *Progress in Retinal and Eye Research*, vol. 31, J. Kur, E. A. Newman, and T. Chan-Ling, "Cellular and physiological mechanisms underlying blood flow regulation in the retina and choroid in health and disease," pp. 377–406, Fig. 1, Crown Copyright (2012), with permission from Elsevier [**26**]. The drawings, by Dave Schumick, are reprinted from *Encyclopedia of the Eye*, B. Anand-Apte and J. G. Hollyfield, "Developmental anatomy of the retinal and choroidal vasculature," pp. 9–15, Copyright (2010), with permission from Elsevier.

1.2.3. Ocular Thermal Environment. In patients, temperature rises from the front of the eye to the back (Table 1.1). Shinoda and colleagues measured it at four sites with a needle thermoprobe at the start of vitrectomy, before the infusion was opened, in an operating room set at 25 °C. They attribute the gradient partly to the cooler surroundings and limit their values to these conditions [**47**]. Two later intraoperative series give different absolute values [**48, 49**]. In the second, the vitreous was warmer in front of the optic disc than behind the lens, a rise the authors attribute to the nearby choroidal circulation [**49**].

TABLE 1.1. Intraocular temperatures measured in patients during vitreoretinal surgery, before vitrectomy. The posterior values are vitreous, not retinal, temperatures. —: not measured.

Study	**Anterior chamber**	**Behind the lens**	**Mid-vitreous**	**In front of the optic disc**
Shinoda et al. [**47**]: 30 eyes with intact vitreous	30.0 ± 1.7 °C	32.8 ± 1.2 °C	34.0 ± 0.9 °C	35.0 ± 0.9 °C
Kobylianskyi et al. [**48**]: 23 eyes with proliferative diabetic retinopathy	—	—	35.6 ± 0.7 °C	—
Ríos-Nequis et al. [**49**]: 12 eyes, combined cataract and vitreous surgery[a]	28.7 ± 2.8 °C	29.6 ± 0.7 °C	—	31.9 ± 1.1 °C

[a]Anterior chamber at the start of surgery; vitreous before vitrectomy, after the cataract step.

At the anterior surface, convection, radiation, and tear evaporation remove heat and help explain the cooler cornea and aqueous [**50, 51**]. Their contributions depend on ambient and surface conditions: convection and radiation can supply heat in sufficiently warm surroundings, while evaporation removes it. Within the globe, heat is conducted through the ocular wall and contents, including the vitreous, and exchanged with the circulation.

The choroid can supply or remove heat according to the blood–tissue temperature difference. In anesthetized monkeys under dim illumination, reducing choroidal flow lowered macular retinal–choroidal temperature. Under a focused microscope lamp, tissue temperature exceeded blood temperature and reducing flow raised it further

[**37, 38**]. The same circulation that removes absorbed radiant heat can thus oppose externally imposed cooling, and it does so continuously, because choroidal flow is high and only partly pressure-regulated [**44**]. The ex-vivo experiments stand in for this load with a 37 °C circulating bath (Chapter 2), and the device was also tested in vivo (Chapter 4).

Thermal models differ mainly in how they represent heat delivery: a fixed-temperature boundary, a heat-transfer coefficient, a Pennes perfusion term, discrete vessels, or multiphase blood [**52, 53, 54**]. In the model of Geiser and colleagues, choroidal and ciliary perfusion dominated retinal flow as a heat source, and ambient temperature from −20 to +40 °C changed the predicted retinal temperature by about 1.9 °C [**55**]. Flyckt and colleagues found the bioheat approach adequate for first-order predictions and discrete vasculature necessary for finer spatial detail [**53**].

The measurements of this dissertation give temperatures inside the eye during contact cooling, for comparison with related models from this research program that directly address that cooling configuration. Gongal and colleagues modeled a scleral ring as a 4 °C boundary on the anterior sclera; with Pennes perfusion the optic nerve head settled at 34.0 °C [**56**], whereas a multiphase formulation, in which cooled blood returning from the front of the eye carries the cold backward, predicted 29.6 °C [**54**]. These models motivated the device.

Temperatures inside the eye were measured directly, ex vivo, before the device was tested in the living eye, because the temperature applied at the front does not set the temperature at the back. Holding a steady temperature requires heat extraction that balances the remaining net heat inputs, so an anterior contact temperature defines an applied boundary, while the posterior temperature depends on transport through the eye and on physiological heat delivery.

1.3. Therapeutic Hypothermia and Neuroprotection

Therapeutic hypothermia is the deliberate lowering of tissue temperature to slow metabolism and limit ischemic and hypoxic injury. The clinical evidence concerns mild-to-moderate hypothermia, 32–35 °C, which Polderman describes as the first treatment with proven efficacy for postischemic neurological injury [**57**], a judgement resting principally on the post-cardiac-arrest literature of that time and not confirmed by later trials (Section 1.3.2).

1.3.1. Mechanisms. Hypothermia acts on many targets at once; in Yenari and Han's review of the experimental literature it affects nearly every investigated cell-death pathway [**61**]. In ischemia, the failure of oxidative metabolism depletes adenosine triphosphate (ATP), disrupts transmembrane ion gradients, and promotes glutamate accumulation and calcium overload, followed on reperfusion by free-radical production and inflammation [**57, 61**]. The basic effect of cooling is metabolic: brain oxygen consumption and glucose metabolism fall by roughly 5% for every degree Celsius, and ATP stores are preserved, which may limit calcium overload and excitotoxic glutamate [**61**]. Cooling also curbs apoptosis, the post-ischemic inflammatory response, free-radical production, and edema [**57, 61**].

Metabolic suppression is the mechanism most likely to matter in the retina, among the most metabolically active tissues in the body (Section 1.2.1). The rate of rewarming also affects the protection conferred by cooling in experimental models, and the systemic hypothermia literature treats controlled rewarming as part of the protocol [**57**].

1.3.2. Evidence in the Central Nervous System. Clinical outcomes depend on the indication and protocol [**61**]. Therapeutic cooling has an established role in neonatal encephalopathy. After cardiac arrest, early trials reported benefits [**65, 66**], but the TTM and TTM2 trials found no improvement with 33 °C over a controlled

36 °C or over targeted normothermia [**58, 59**]. Current post-arrest guidelines prescribe temperature control at a constant target between 32 and 37.5 °C with active prevention of fever, rather than mandatory hypothermia [**60**], and the 2025 ERC–ESICM guideline emphasizes fever prevention [**67**]. In acute ischemic stroke, the class to which CRAO belongs, the 2026 guideline recommends against induced hypothermia in normothermic patients and lists the benefit of selective intra-arterial brain cooling as unknown [**8**].

These systemic trials do not establish the efficacy of localized retinal cooling, but they document a reason to investigate local delivery. Systemic cooling brings coagulopathy and an infection risk that rises with treatment duration [**57**], and in TTM2 more arrhythmia with hemodynamic compromise (24% against 17%) [**59**]. It is also slow: in eleven randomized trials after cardiac arrest the median time to 32–34 °C ranged from 38 to 360 min [**68**]. During stroke thrombectomy, systemic intravascular cooling below 34 °C was followed by more serious adverse events than standard care, a difference the trial was not powered to test [**69**]. Selective cooling of the brain remains investigational [**60**]. Limiting the cooled region is an engineering strategy to limit this exposure.

1.3.3. Evidence in the Retina. How long the retina survives complete arterial occlusion is uncertain, and depends on the model and the perfusion conditions. In Hayreh's experimental model, transient central retinal artery occlusion in elderly atherosclerotic rhesus monkeys produced no detectable damage at 97 min, with progressively more extensive irreversible damage at longer occlusion times up to 240 min [**45**]. Tobalem and colleagues, in a critical review, argue that this figure is an overestimate, attributing it to barbiturate anesthesia, uncontrolled hypothermia, and residual collateral perfusion in the clamp preparation, and propose that infarction of retinal ganglion cells is likely after only 12–15 min of complete occlusion [**46**]. Loss of function and irreversible injury can also diverge. Ex vivo, light responses of ON-bipolar cells recovered after overnight incubation in oxygenated medium at room temperature

in human donor retinas collected 50 to 60 min after circulatory arrest, and in two of five pairs of eyes enucleated 3 to 4 h post mortem. Mouse retinas recovered fully after 1 h of hypoxia but only in part after 3 h. These preparations lack the vascular and inflammatory components of ischemia in vivo [**70**].

In each of these accounts the window is short relative to the delays with which patients present (Section 1.1) [**14, 15**]. The uncertainty in the deadline argues for an intervention that can be applied promptly and can extend tolerance (Section 1.8).

Hypothermia has protected the retina in cell-culture, ex-vivo, and animal models at temperatures from about 34 °C down to 30 °C and below, and protection grew as cooling deepened. In isolated rabbit retina after 1 h of simulated ischemia, recovery of light-evoked compound action potentials was 15% at 36 °C, 36% at 33 °C, and 53% at 30 °C [**62**]. Cooling to 30 °C protected porcine retinal explants during a three-hour hydrogen-peroxide exposure [**63**], cooling to 30 and 20 °C protected retinal ganglion cells in hypoxic organ culture [**71**], and cooling to 34 °C reduced the secretion of vascular endothelial growth factor by cultured RPE cells by 38% [**72**]. Cooling for 20 min at a retro-ocular temperature of about 32 °C, 24 h before the insult, protected the rat retina from ischemia–reperfusion damage [**64**].

In those injury studies, electrophysiological recovery is evidence of preserved function; in the present work the electroretinogram (ERG) is used instead to detect temperature-sensitive retinal responses and their recovery after rewarming. Cooling the eye in vivo reduces ERG amplitudes and prolongs their timing [**73, 74**]. Temperature-dependent photoresponse features have been used to estimate temperature in isolated mouse retinas [**75**], temperature-dependent photopic ERG kinetics have been shown in isolated porcine retinas and anesthetized pigs [**76**], and the same group has since used focal ERG kinetics to determine retinal temperature during laser heating of the living pig retina, with a standard error of 0.6 °C [**77**]. That calibration applies to focal, light-adapted responses heated above body temperature, so for the full-field flash ERG recorded during cooling in this work, ERG changes are indirect

evidence of cooling. Tools developed in this laboratory that could show where, and how widely, the retina cools are discussed in Section 5.5.

1.4. Ocular Cooling: Prior Approaches and Their Limitations

Attempts to cool the eye span six decades, from research techniques never intended for clinical use to devices now in early clinical trials (Table 1.2; representative devices in Figure 1.6). Grouped by invasiveness, each falls short of the goal of this dissertation, sustained, controlled hypothermia at the posterior pole.

Invasive routes reached the retina or the optic nerve. Intraocular perfusion cooled the cat retina to 22 °C [**73**]. Anterior-chamber perfusion with an 11 °C solution, held by a feedback-controlled thermoelectric heat pump, cooled the rabbit retinal surface by 2 to 4 °C [**78**]. Cold irrigation during vitrectomy protected the rabbit retina [**79**], and trans-nasal cooling protected the goat optic nerve only with a cold-protectant [**80**]. Cyclocryotherapy shows that cold can cross the sclera, but it destroys tissue by design [**81**]. All of these need intraocular or surgical access.

Research preparations in living animals showed that cold applied at the surface can reach the back of the eye [**73, 74, 82, 64**]. The experiments were short and used exposed eyes or rats. Temperature was measured in few animals or not at all, and none was under feedback control.

Apart from those probe readings [**73, 64**], no non-invasive cooling method has reported a temperature behind the lens. Compresses, a water-cooled contact lens and handheld cryoanesthesia applicators act on the surface for minutes or seconds [**83, 84, 87, 88, 89**]. The eyelid dressing of Anatychuk and colleagues shows that thermoelectric cooling can be packaged for ocular contact, with feedback control for up to 48 h, but its posterior effect has only been modeled [**85, 86**]. Scleral-contact cooling interfaces are clinically tractable, as devices now in early clinical testing for chronic ocular surface pain confirm independently of the present work [**91, 90, 92, 93**]. They

target surface pain through the long ciliary nerves. None of these devices was designed for, or has measured, sustained hypothermia at the retina or the ONH.

TABLE 1.2. Prior approaches to ocular cooling, grouped by invasiveness. Temperatures are of the cooling medium or application site unless a tissue is named. ERG: electroretinogram.

Approach	**Deepest site and temperature reached**	**Why it falls short**
Invasive routes		
Intraocular perfusion, cat [**73**]	Retina, 22 °C; ERG reduced and slowed	Intraocular access; physiology, not rescue
Anterior-chamber perfusion, rabbit [**78**]	Retinal surface 2–4 °C cooler (11 °C, 10 min)	Intraocular access; rewarms within minutes
Cold irrigation, rabbit [**79**]	Retina: ERG and histology better at 8 than 38 °C	Operating room only (vitrectomy)
Trans-nasal nerve cooling, goat [**80**]	Optic nerve, 4 °C; protected only with a cold-protectant	Surgical access; retina not cooled; vision not restored
Cyclocryotherapy [**81**]	Ciliary body ablated, −80 °C probe, 60 s	Destructive by design
Research preparations in living animals		
Cold-saline cap, cat [**73**]	Retina not below 32 °C (probe)	Exposed eye; body and brain cooled too
Scleral cooling coil, rabbit [**74**]	Retina, estimated 15–18 °C (one animal); ERG changed	Eyelids and extraocular tissues removed
Ice pack, rat [**82**]	Inner retina preserved in ischemia	No temperature measured or controlled
Cold gel, rat [**64**]	Behind the globe, about 32 °C (pilot, 4 rats)	Not controlled; rat eye
Non-invasive and clinical devices		
Ice or cold compress [**83**]	Surface: less injection pain	No temperature measured or controlled
Water-cooled contact lens, rabbit [**84**]	Cornea: less laser damage	Anterior only; posterior not tested
Peltier eyelid dressing [**85, 86**]	Retina and vitreous 1–2 °C cooler, modeled only	No measured or clinical data
Cryoanesthesia applicators [**87, 88, 89**]	Conjunctiva, −15 °C within 10 s	Seconds of cooling; surface anesthesia only
ETX-4143 scleral cooling [**91, 90, 92, 93**]	Surface; nothing reported inside the eye	Targets surface pain, not the retina

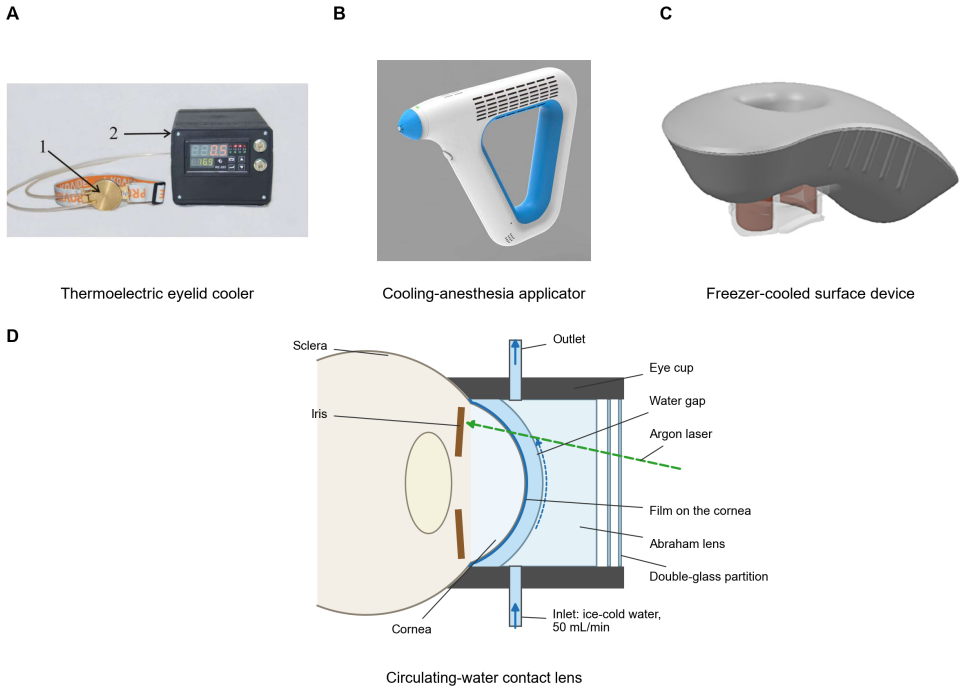


FIGURE 1.6. Representative ocular cooling devices. (A) Thermoelectric eyelid dressing and control unit (1: cooling plate; 2: control and power unit), from Anatychuk et al., Fig. 1 [**85**] (CC BY 4.0). (B) Handheld thermoelectric applicator for injection anesthesia, made by RecensMedical, the sponsor of OcuCool [**89**]; from Chao et al., Fig. 2A, cropped to the device [**88**] (CC BY-NC 3.0). (C) Freezer-cooled ocular-surface device, from Galor et al., Fig. 1 [**91**] (CC BY-NC-ND 4.0); the rabbit-study version used a single conductor. (D) Circulating-water contact lens used during argon laser iridotomy in rabbits: ice-cold water ($50\ \mathrm{mL\,min^{-1}}$) flows between an Abraham lens in an eye cup and a film on the cornea; drawn from the description by Kataoka et al. [**84**].

1.5. Thermoelectric Cooling

Thermoelectric cooling is how the device of this dissertation moves heat, by the Peltier effect. When current is driven across a junction between dissimilar conductors, heat is absorbed at that junction and released at the other junction of the circuit, and the pair works as a heat pump rather than as a power source [**94**]. A conventional thermoelectric cooler (TEC) module connects many n-type and p-type semiconductor legs electrically in series and thermally in parallel between two thin ceramic plates, which insulate electrically while conducting heat (Figure 1.7) [**95**]. Current pumps heat from one face, the cold side, to the other; reversing the polarity

reverses which face is cold, so one module can cool or heat. The module as a whole is not thermodynamically reversible: Joule heating (I^2R) and conduction back along the temperature gradient ($K\,\Delta T$) [**94**] are losses in both modes, where I is the drive current, R the electrical resistance of the module, K its thermal conductance, and ΔT the temperature difference between the hot and cold faces.

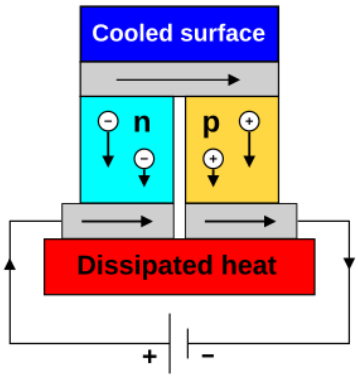


FIGURE 1.7. Operating principle of a thermoelectric couple: an applied current drives heat from the cold side to the hot side through n-type and p-type semiconductor legs. Illustration by Ken Brazier, Wikimedia Commons, CC BY-SA 4.0.

The first consequence for design is that cooling capacity and temperature difference trade off, so a module's headline ratings are not available together. The rated maximum heat-pumping capacity applies at zero temperature difference, and the maximum temperature difference is reached only at zero cooling load [**96**]. Because the Peltier term is proportional to current and Joule heating to its square, the coldest attainable temperature is reached at an optimal current, not at maximum current [**94**]. A module must therefore be selected against the cooling load at the actual temperature difference, hot-side temperature, and electrical operating point, not against no-load ratings; this check is made for the device in Chapter 3 (Section 3.4).

The second consequence is that all the heat delivered to the hot side must be rejected to the ambient. This heat is $Q_h = Q_c + P_{\text{in}}$, where Q_c is the heat pumped from the cold side, P_{in} the electrical power supplied to the module, and Q_h the heat released at the hot side. Hot-side heat rejection is a principal constraint on realizing

rated performance [**96, 97**], and the thermal resistance of the whole path, from the tissue through the module to the ambient heat sink, governs how much of a module's capability reaches the point of contact. A perfused eye adds a load that is continuously renewed.

The module is solid-state, without moving parts or a working fluid, and makes no vibration or acoustic noise [**97, 98, 96**], although heat rejection still needs a system around it: the cooler developed here (Section 1.7), like the eyelid device of Section 1.4 [**85**], uses a liquid loop. The annular module adopted here is 26.0 mm in outside diameter with a 14.0 mm central hole and a thickness of 4.7 mm [**96**], which leaves an aperture for the cornea and the optical path. Temperature is set by the magnitude and direction of the drive current and held by closed-loop feedback, and the modules are compatible with battery operation, as in portable thermoelectric cooling for personal thermal management [**99**].

1.6. The Porcine Model

Among nonprimate species, the pig is a widely used large-animal model of the human eye, and its use in retinal research continues to grow [**100**]. Its globe approaches human scale, 23.9 ± 0.08 mm in anteroposterior length in enucleated eyes [**101**] against roughly 24 mm in humans, which supports the testing of device-interface tools intended for human use [**100**].

Porcine scleral geometry and water content support its use for interface development, but differences remain; Table 1.3 gives the values with their measurement conditions [**102, 103**]. Two matter for a scleral-contact device. Porcine sclera is thinnest only a few millimeters behind the limbus, whereas the human sclera, away from the muscle insertions, is thinnest near the equator, so a device 5 to 6 mm from the limbus in the pig is analogous to one near the equator in a human eye [**102**]. And the porcine orbit is small and its extraocular musculature unusually strong, so

extraocular devices are difficult to place behind the muscle insertions [**102**], which limits where a ring can sit in this species.

Porcine measurements show a cooler aqueous compartment and warmer posterior vitreous, qualitatively resembling the human gradient [**47, 104**]. Thermal similarity must be assessed experimentally, as in Aim 1.

The retina itself is comparably organized. Porcine retinal circulation is holangiotic, as in humans, but the pig has no central retinal artery, the retina being supplied by several cilioretinal arteries, and its orbital supply arises from the external carotid [**105, 106**]. The retinal layers are distributed comparably [**101, 107, 108**]. The pig lacks a true fovea and has an elongated visual streak of high cone density (Table 1.3) [**100, 109**]; foveal function is not an endpoint here, since the work targets cooling of the posterior eye, measured at the proximal optic nerve.

The living pig is expected to be a conservative test of posterior cooling, because perfusion is the principal heat load that a cooling device must overcome in vivo (Section 1.2.3) and porcine choroidal flow is high. It exceeds the per-eye totals measured in monkeys by the same microsphere method and the per-area values measured in humans by magnetic resonance imaging (MRI) (Table 1.3) [**110, 40, 111**], and human choroidal flow declines further with age [**112**]. The methods differ, however.

Porcine models of ocular hypertension and retinal ischemia are available for future efficacy studies [**114, 115**], but the present work induces neither. The same species serves both phases: eyes from a commercial supplier for ex-vivo thermal characterization, and the live animal for functional and safety assessment with perfusion.

TABLE 1.3. Human and porcine ocular parameters from separate studies. Human scleral thicknesses are, unless another source is given, histological means from non-axially elongated eyes enucleated for melanoma or end-stage glaucoma [**113**]; porcine values are means from formalin-fixed globes grouped by body weight [**102**]. Blood-flow methods are given in each cell.

Parameter	Human	Porcine	Source
Globe axial length	≈24 mm (postmortem AP mean 24.44 ± 1.03 mm)	23.9 ± 0.08 mm (5 eyes, 6–8 months)	[**23, 24, 101**]
Scleral thickness, limbus	0.50 ± 0.11 mm	0.83, 0.91, 1.12 mm (small, medium, large)	[**102, 113**]
Scleral thickness, equator	0.42 ± 0.15 mm	0.56, 0.73, 0.86 mm (small, medium, large)	[**102, 113**]
Thinnest point away from the muscle insertions, and its location	∼0.4 mm near the equator, ≈13 mm behind the limbus	0.31, 0.35, 0.43 mm at only 5–6 mm behind the limbus	[**102**]
Scleral thickness, under the rectus muscles	∼0.3 mm	Not reported by the cited sources	[**28**]
Scleral thickness, posterior pole	0.94 ± 0.18 mm measured; 1.0–1.35 mm quoted in review	Thickness plateaus from ≈10–12 mm behind the limbus; posterior-pole value not tabulated	[**28, 102, 113**]
Total scleral surface area	≈16–17 cm^2	7.78, 9.66, 11.92 cm^2 (small, medium, large)	[**102**]
Scleral water content (wet mass)	$71.6 \pm 0.63\%$	$69.5 \pm 1.18\%$	[**103**]
Choroidal blood flow	≈1.1 $\mu L\,min^{-1}\,mm^{-2}$ (MRI, retina + choroid, posterior pole); monkey 0.6–1.3 $mL\,min^{-1}$ per eye (microspheres)	3.2–3.7 $\mu L\,min^{-1}\,mm^{-2}$, ≈1.6–1.7 $mL\,min^{-1}$ per eye (microspheres, 19–27 kg, pentobarbital)	[**110, 40, 111**]
Choroidal response to perfusion pressure	Partial compensation for moderate rises	Falls in proportion to perfusion pressure	[**44, 110**]
Retinal vascular pattern	Holangiotic	Holangiotic; no central retinal artery; orbital supply arises from the external carotid	[**101, 105**]
Fovea	True fovea present	Absent; elongated visual streak (20,000–35,000 cones/mm^2, higher local peaks)	[**109**]
Experimentally induced disease models	—	Ocular hypertension (episcleral vein cauterization); retinal ischemia (endovascular occlusion)	[**114, 115**]

1.7. The Scleral-Contact Eye Cooler

The scleral-contact eye cooler (SCEC) applies a controlled thermal boundary at the accessible anterior sclera to remove heat from the globe. It has been developed in the Neural Engineering Vision Laboratory over several student projects. The first prototype (Gen 1) was built by S. J. Dreyer as a portable cooler for traumatic eye injury [**117**]: a machined aluminum ring on the anterior sclera, with three thermoelectric modules on its faces and finned heat sinks cooled by forced air, driven at a fixed current from a bench supply without closed-loop control. It was tested on thawed porcine eyes with three scleral thermistors in a 37 °C canola-oil bath, chosen because its conductivity is close to that of orbital fat, and the same bath is used in Chapter 2. The posterior sclera cooled by about 6.6 °C on average (four of five eyes), some tissue in contact with the ring froze, and the removal of heated air limited how cold the ring could become; liquid cooling was judged more efficient but set aside for a field device. The second prototype (Gen 2), designed and built in 2018–2019 by K. Fong, kept the Gen 1 ring and replaced the air-cooled heat sinks with water cooling, and in 2018 N. Shirdade carried out thermal simulations of the cooling of the porcine eye. The ex-vivo methods manuscript in preparation builds on both projects (Contribution of Authors).

The third generation (Gen 3), the device of this dissertation, was completed by the author during his M.S. study [**118, 1**] (Figure 1.8). A scleral contact ring (SCR), shaped to conform to the anterior sclera, forms the cooled boundary against the eye while leaving a central aperture over the cornea and pupil. The device therefore does not touch the cornea, although the cornea can still cool indirectly, and the visual axis stays unobstructed, so assessments such as the pupillary light reflex remain accessible. A single annular thermoelectric cooler (Section 1.5) pumps heat from the SCR to a liquid-cooled heat exchanger, or water block, and the coolant carries it to a copper coil in ice water. Control electronics provide power, thermoelectric drive, and thermal sensing. A temperature sensor embedded in the SCR is the feedback signal by which

the ring is held at its set point, 4 °C, the lower limit of hypothermic organ storage (Section 2.2.1); the drive was adjusted at first by the operator and later switched automatically by a comparator. The controlled variable is the ring temperature, and tissue temperatures elsewhere in the globe are responses to the imposed boundary. The tolerance of living ocular tissue to prolonged cooling at this boundary is assessed in Aim 3.

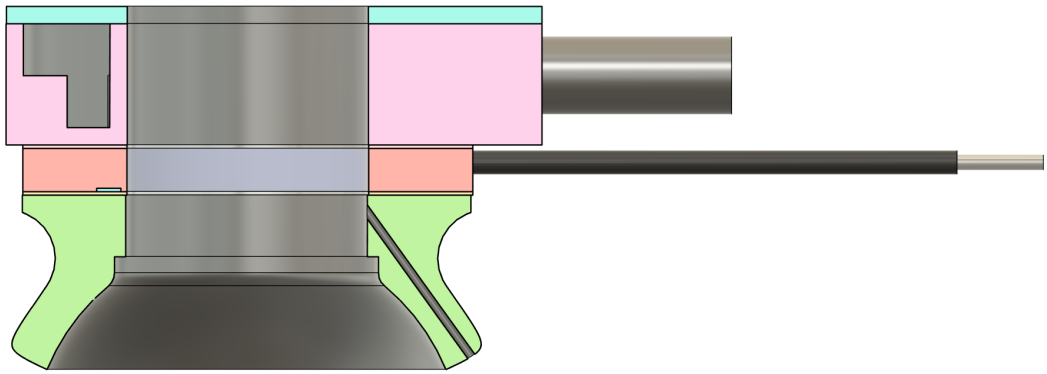


FIGURE 1.8. Cross-sectional computer-aided design (CAD) view of the scleral-contact eye cooler (SCEC): contoured scleral contact ring (green), annular thermoelectric module (orange), and liquid-cooled heat exchanger, or water block (pink and blue). The central aperture gives optical access; the coolant connection and electrical lead extend laterally, and an oblique bore through the ring holds the thermistor.

In the M.S. work, fresh porcine eyes with thermistors were cooled in the oil bath and imaged to document the sensor positions. The next steps set out there, a ceramic ring, automatic control of the module, a smaller system, and tests in perfused eyes and human donor eyes, are the subject of this dissertation. Results from the porcine eyes, obtained with Y. Nakamura, were presented at two conferences [**119, 120**] and published in 2025 [**1**], and the human donor experiments were presented in 2026 [**121**]. Within the same research program, Gongal and colleagues published two finite-element models of the human eye cooled by a scleral ring [**56, 54**] (Section 1.2.3).

Two version names are used for the remainder of the dissertation, because the device evolves across the three aims. *SCEC Gen 3* denotes the ex-vivo prototype validated in Aim 1 (Chapter 2). *SCEC Gen 4* denotes the enhanced device developed in

Aim 2 (Chapter 3), which targets improved efficiency, portability, safety, and patient comfort, including the consolidation of drive and sensing onto an integrated board. The in-vivo study of Aim 3 (Chapter 4) used a build that carried over some, but not all, of the Gen 4 enhancements; its configuration is reconciled against the Gen 4 target in Section 4.2.4, so that "the device" refers to a definite object in every chapter. Heating and analgesic applications are discussed as future work in Section 5.5.

1.8. Knowledge Gap and Specific Aims

Acute retinal ischemia lacks a proven treatment, and most patients present beyond the window of time-limited reperfusion (Section 1.1). Preclinical evidence supports cooling as a means of extending the retina's tolerance of ischemia, but a practical delivery strategy must provide localized, active, and controlled cooling through an accessible interface, sustain it, produce a characterized response at the posterior eye, and be safe in the living eye. To the author's knowledge, no existing approach provides all of these (Table 1.2).

The obstacle is measurement as much as hardware: the temperature of a contact device says nothing about the temperature at the retina or optic nerve (Section 1.2.3). Before this work, no scleral-contact cooling device had been characterized with spatially verified temperatures throughout the globe, none had been characterized thermally in human eyes, and the porcine eye had not been shown to predict the human thermal response. The scleral-contact devices already applied in patients target surface pain and did not measure the temperatures they produced inside the eye, and in living animals the posterior eye had been cooled from outside only in short research preparations, none under feedback control (Section 1.4). No scleral-contact device had been used for sustained, controlled cooling of the posterior segment of an intact living eye of human size.

The central hypothesis of this dissertation is that a scleral-contact thermoelectric device can produce reproducible posterior ocular cooling under controlled ex-vivo conditions, retain cooling performance through selected design refinements, and produce functional responses consistent with retinal cooling in vivo while maintaining ocular and systemic safety under the tested conditions. It is addressed through three Specific Aims, which assess thermal delivery and preclinical tolerability. Neuroprotective efficacy after ischemia and thermal analgesia motivate the work, but establishing either requires outcome studies with an injury model or a pain measure, which are left to future work (Section 5.5).

Specific Aim 1: Validate SCEC Gen 3 and characterize its thermal performance in ex-vivo eyes. The objective is to characterize, across ex-vivo models using formalin-preserved porcine, fresh porcine, and fresh human donor eyes, the cooling the scleral-contact interface produces at defined measurement sites, and how much it varies between eyes. Temperature is recorded at five ocular sampling locations (anterior, equatorial, and posterior sclera; vitreous; and the proximal optic nerve), whose positions are verified by micro-computed tomography (micro-CT), together with the SCR feedback channel. As a secondary comparison, preserved porcine eyes are compared with fresh porcine eyes to determine whether preserved tissue can serve as a benchtop surrogate for device development.

Two hypotheses are tested. First, SCEC Gen 3 produces a reproducible reduction in temperature at the proximal optic nerve within approximately 15 min (90% of the 20-min excursion), with its size interpreted against a 3 °C benchmark. Second, the cooling excursion at this site differs between fresh human and fresh porcine eyes by an amount that is small against the excursion itself. The absence of a difference cannot be shown directly. This comparison is therefore reported as the difference between the species with its confidence interval, whose limits give the largest difference that the data allow. Cooling is measured as the decrement from each preparation's own baseline, because baseline temperature differs between preparations. The 3 °C

benchmark is the step at which functional recovery improved in isolated retina under simulated ischemia [**62**]. It is a benchmark and not an established threshold (Section 2.4), so absolute temperatures are reported alongside the decrement and read against two reference intervals: the 32–35 °C interval of systemic therapeutic hypothermia (Section 1.3) and the roughly 30–34 °C range of preclinical retinal protection (Section 1.3.3). Performance is characterized by temperature–time profiles, baseline temperature, cooling excursion, time to 90% of the excursion (t_{90}, a measure of kinetics), and variability across preparations; the variability between eyes within each preparation and the thermal difference between species are evaluated separately (Chapter 2).

Specific Aim 2: Enhance the device (SCEC Gen 4) for efficiency, portability, safety, and patient comfort. The objective is to develop and evaluate SCEC Gen 4 refinements through computational modeling, experimental assessment of selected scleral contact ring designs, and development of an integrated printed circuit board for device control. Refinements include reduced-profile ring geometries with and without an integral corneal lens, alternative ring materials, an insulating outer layer to focus heat extraction, water-block and cooling-circuit miniaturization, TEC potting (present in every device), and consolidation of control electronics. Reduced form factor and bulk are intended to improve portability, handling, and patient comfort, while material selection and TEC potting address safety considerations at the device–tissue interface and within the thermoelectric assembly.

Computational modeling provides the primary basis for the comparisons, and experiments on selected fabricated rings provide complementary evidence. The hypothesis is that selected geometric and material refinements can reduce device bulk while preserving cooling performance at the posterior measurement sites. Patient comfort itself is not directly measured; improvements relevant to it are assessed through reductions in device dimensions and bulk.

Specific Aim 3: Validation and Safety Assessment in an In-Vivo Porcine Model. The objective is to evaluate evidence of retinal cooling and assess ocular and systemic safety in six live pigs under a protocol approved by the University of Illinois Chicago Animal Care Committee (ACC), the university's institutional animal care and use committee (Section 4.2.1). Each animal undergoes three cooling sessions, lasting 30, 60, and 120 min, followed by a terminal session for final assessments and tissue collection. Testing in the perfused living eye evaluates device performance in the presence of physiological heat delivery, including the choroidal circulation (Section 1.2.3), and allows the systemic response to its application to be assessed.

Changes in the ERG are used as indirect functional evidence of retinal cooling, which reduces ERG amplitudes and prolongs their timing [**73, 74, 122, 76**]. The b-wave implicit time and amplitude are used together, a longer implicit time and a smaller amplitude being the expected response. Amplitude is not interpreted alone, because it also varies with electrode contact and position. Changes in the cooled eye are compared with those in the fellow eye, which removes systemic and anesthetic effects shared by both eyes. These measurements indicate cooling but do not measure retinal temperature or its spatial extent.

Safety endpoints are retinal functional recovery after rewarming, corneal integrity assessed by fluorescein staining and surface analysis, intraocular pressure measured by tonometry, tissue-level histology, and systemic stability. ERG responses during cooling and after rewarming are assessed separately to distinguish reversible functional changes from persistent abnormalities within the observation period.

The hypothesis is that device application produces ERG changes consistent with retinal cooling, that the ERG returns to its baseline values after rewarming, and that the other measured safety endpoints show no increase beyond those of the fellow control eye across the tested cooling sessions.

CHAPTER 2

Specific Aim 1: Validate SCEC Gen 3 and Characterize Its Thermal Performance in Ex-Vivo Eyes

Portions of this chapter have been published or presented [**118, 119, 120, 1, 121**]; the Contribution of Authors gives the details.

The fresh-porcine recordings, first reported in the author's M.S. thesis [**118**] and published in [**1**], were reprocessed for this chapter together with the human data, through the common quality-control pipeline described in Section 2.2.6. All fresh-porcine values reported here are the reprocessed values. They differ slightly from the published ones, and none of the published conclusions changes. The two sets are compared in Appendix B.

2.1. Background and Rationale

This chapter measures how far cooling applied at the surface reaches into the eye. An ex-vivo preparation permits temperature sensors to be placed at multiple sites throughout the globe, their positions to be verified by imaging, and the resulting temperature fields to be recorded without the variability introduced by blood flow, systemic physiology, anesthesia, and thermoregulation. The absence of perfusion is also the preparation's principal limitation. It was addressed in part by immersing the posterior globe in a continuously mixed 37 °C bath (Section 2.2.2). Like the choroidal circulation, the bath resupplies heat to the globe during cooling. It does so from outside the sclera, however, whereas the choroid delivers heat directly beneath the retina, and the heat the bath delivered was not measured (Section 2.4). The preparation thus establishes the tissue-level response to scleral-contact cooling against a physiological warm boundary.

Three preparations were used because each answers a different question. Formalin-preserved porcine eyes, which can be stored and used without the timing constraints of fresh tissue, were used to develop the platform and to test whether it resolved region-specific cooling reproducibly. Fresh porcine eyes established proof of concept [**1**] and provided the reference against which preserved tissue was assessed as a practical benchtop surrogate. Fresh human donor eyes were required because agreement between the two species cannot be assumed. Porcine and human globes have similar axial lengths, but in separate histological series the porcine sclera is roughly 1.3 to 2.2 times as thick as the human sclera at the limbus and equator, and it is thinnest close to where the ring sits (Section 1.6, Table 1.3). The sclera is the first conductive path between the contact ring and the posterior eye. These differences could alter both the magnitude and the spatial distribution of cooling. The porcine eye can inform device development (Chapter 3) and interpretation of the in-vivo study (Chapter 4) only to the extent that its thermal response matches the human response at the site of interest.

The hypotheses are stated in Section 1.8. The cooling at the proximal optic nerve was judged against a 3 °C benchmark derived from retinal evidence [**62**], and preserved porcine eyes were compared with fresh porcine eyes as a secondary comparison (Section 2.2.6).

2.2. Methods

2.2.1. Device Description: SCEC Gen 3. SCEC Gen 3 (Figure 2.1) consisted of three thermally coupled components: an aluminum scleral contact ring (SCR), an annular thermoelectric cooler (TEC), and a liquid-cooled aluminum heat exchanger (water block). The SCR formed the interface with the anterior sclera and was machined with a curved ocular-contact surface having an intended contact area of approximately 390 mm^2. A 14-mm-diameter central aperture left the cornea exposed

and preserved access to the optical axis. A negative-temperature-coefficient thermistor embedded in the SCR provided continuous measurement of its temperature.

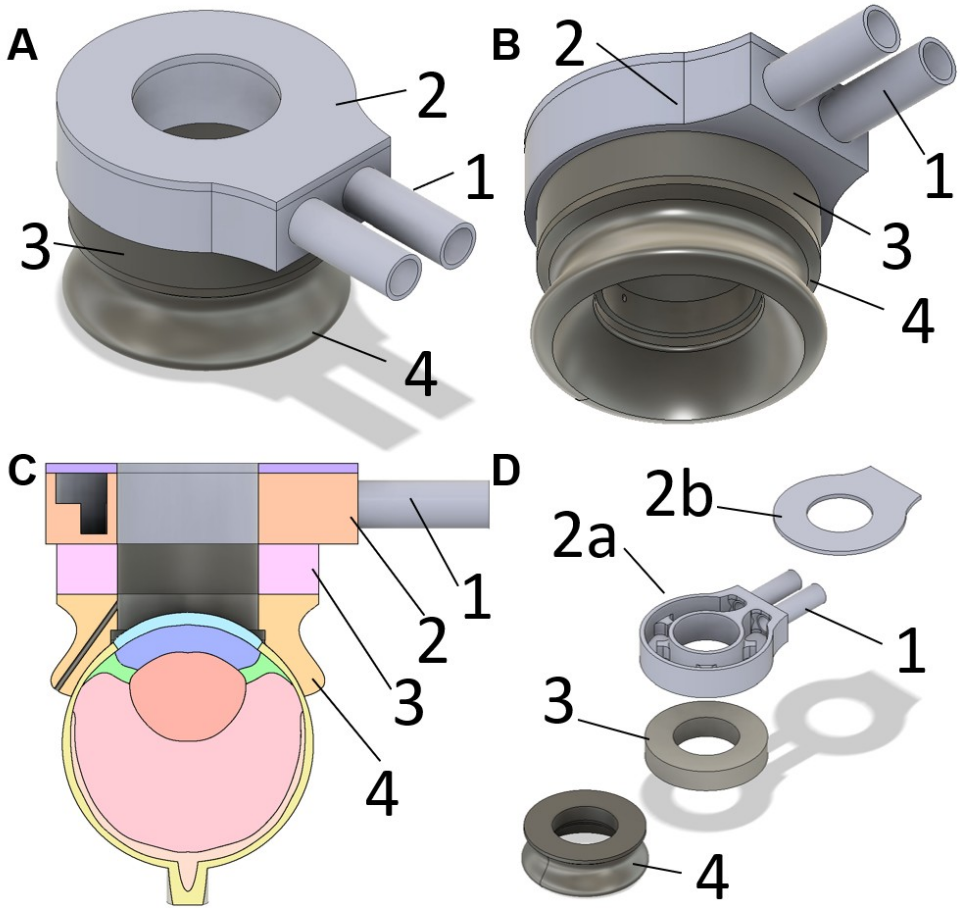


FIGURE 2.1. Components of SCEC Gen 3. (A) Solid model seen from above and (B) from below. (C) Cross-section of the device in position on a porcine eye. (D) Exploded view. 1, coolant inlet and outlet; 2, heat exchanger (2a, body; 2b, lid); 3, thermoelectric cooler (TEC); 4, scleral contact ring (SCR). Panel C is adapted from Nakamura et al., Fig. 2A [**1**], CC BY 4.0.

Heat extracted from the eye entered the aluminum SCR and was pumped across an annular TEC (RH14-14-10-L1-W4.5, Tark Thermal Solutions, Durham, NC, USA). The hot side of the TEC was coupled to the custom-machined aluminum heat exchanger, through which coolant was continuously circulated. The interfaces between the SCR, TEC, and heat exchanger were bonded with thermally conductive epoxy

(OB-101-2, Omega Engineering, USA). The assembled original prototype was approximately 23 mm in height.

The SCR temperature was the controlled variable throughout the experiments, and the cold set point was 4.0 °C in all cooling experiments described in this chapter. The TEC was driven in current-controlled mode at up to 3.9 A, the module's maximum current, about 8 W of electrical power at the supply (3.9 A at 2.1 V). In the porcine experiments, with preserved and with fresh eyes, the operator adjusted the supply to the TEC manually while monitoring the SCR thermistor, holding the SCR at the set point. For the human donor eyes, the platform had been upgraded with automatic feedback control. A comparator with hysteresis (Schmitt trigger) switched the TEC drive in response to the SCR thermistor (Figure A.2 in Appendix A). Once at the set point, the ring temperature of the human runs cycled over a band of about 0.2 °C with the original aluminum ring, 0.1 °C with the reduced-bulk ring, and 0.5 °C with the ceramic ring. Because the control method differed between the two cohorts, device performance was evaluated from the measured SCR temperature together with the simultaneously recorded tissue temperatures.

The set point was chosen at the lower limit of hypothermic organ storage. Organ preservation aims to hold tissue between 4 and 8 °C, because temperatures below about 2 °C increase the risk of cold injury [**116**], and donor corneal endothelium is preserved for up to two weeks at 4 °C [**123**]. Freezing injury requires ice formation, with lethal temperatures from about −2 to −70 °C depending on the tissue [**81, 124**], so a 4 °C boundary cannot produce it. Clinical ocular devices briefly expose the conjunctiva and sclera to much colder surfaces [**89, 92, 91**]. Tissue in contact with the ring remains warmer than the set point: in fresh porcine eyes the anterior sclera equilibrated near 11 °C (Section 2.3.2).

2.2.2. Ex-Vivo Apparatus. The ex-vivo apparatus was built for this work from custom subsystems: a positioning system for the eye and the cooling device, an

intraocular-pressure system, a warm oil bath, and a coolant circuit for the heat exchanger (Figures 2.2 and 2.3). It accepts excised porcine or human eyes instrumented with up to five temperature sensors, and it holds the induced tissue temperatures for as long as cooling is applied.

Eye and cooling-ring positioning system. An aluminum frame carried two independent vertical translation stages, one for the 3D-printed eye holder, on which the eye rested cornea-up on a perforated support ring, and one for a 3D-printed holder for the SCEC, so that the device could be lowered onto the eye.

Intraocular pressure. Intraocular pressure was maintained near the physiological range using a gravity-fed reservoir of room-temperature phosphate-buffered saline connected to the anterior chamber through a 27-Ga needle inserted through the cornea. In all three cohorts, the reservoir was set about 21 cm above the eye. A column of 20.7 cm of water corresponds to the mean intraocular pressure of normal porcine eyes, 15.2 mmHg [**114**]. In all fresh-eye experiments, fluid movement into the eye was minimal.

Warm oil bath. The posterior globe was partially submerged in a canola-oil bath maintained at approximately 37 °C. The oil was mixed continuously, at first by air bubbled through the oil and later, to reduce oil splatter, by a submersible pump. The bath stood in for the orbital tissues that surround the globe and, as a warm boundary, resupplied heat to the tissue in place of the absent perfusion. The oil level was approximately 4 mm posterior to the limbus, immediately below the scleral-contact device.

Coolant circuit. Heat rejected by the TEC was removed through the heat exchanger by coolant that a 12-V pump circulated continuously at approximately 150 $\mathrm{mL\,min^{-1}}$ through a copper coil immersed in a beaker of ice and water. The pump and its setting were the same as in the porcine runs of [**1**]. The coolant was a 50:50 mixture of propylene glycol and distilled water in the porcine experiments and distilled water in the human-donor experiments.

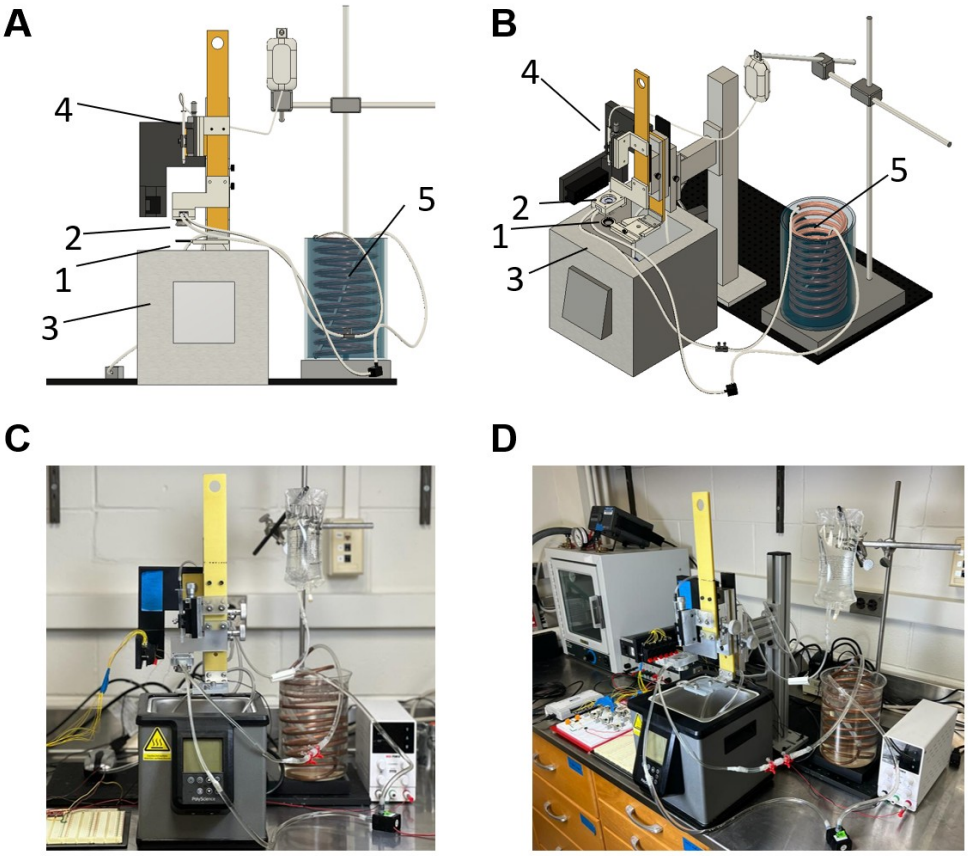


FIGURE 2.2. Ex-vivo apparatus. (A) Frontal and (B) panoramic views of the CAD model, and (C, D) the corresponding photographs of the assembled apparatus. 1, eye holder; 2, cooling-device holder; 3, oil bath; 4, intraocular-pressure syringe (fed from the elevated saline reservoir); 5, ice bath.

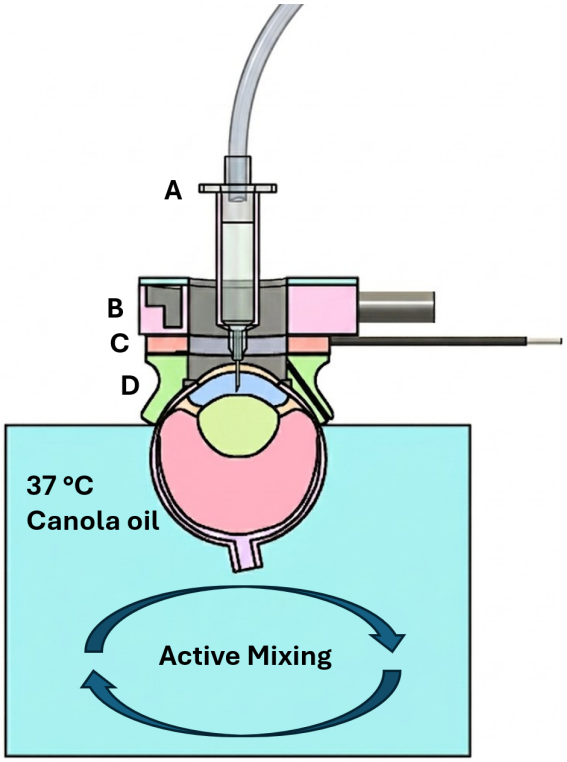


FIGURE 2.3. Schematic cross-section of the ex-vivo cooling setup. The posterior globe is partly submerged in the stirred 37 °C canola-oil bath, and an infusion needle (A) in the anterior chamber connects the eye to the intraocular-pressure reservoir. SCEC Gen 3 rests on the anterior sclera: heat exchanger (B), TEC (C), and SCR (D).

2.2.3. Ex-Vivo Eye Preparation. Three ex-vivo tissue cohorts were used, in the order in which they were studied: formalin-preserved porcine eyes, fresh porcine eyes, and fresh human donor eyes.

Formalin-preserved porcine eyes (Pig Eyes, Plain Preserved) were obtained from VWR International LLC (Pittsburgh, PA, USA). The eyes are sold as formalin-preserved specimens, without color staining, in a holding fluid that contains less than 0.1% formaldehyde by weight according to the supplier's safety data sheet. The preserved-eye experiments were used to assess the reproducibility and utility of the ex-vivo platform. Five preserved eyes were included in the thermal analysis.

Fresh porcine eyes were obtained through a commercial supplier (Animal Biotech Industries, Doylestown, PA, USA). Globes were harvested with about 13 mm (half an inch) of proximal optic nerve attached and shipped overnight in an insulated box with cold packs; they arrived the day after harvest and were tested on the day of arrival or the day after. Five fresh porcine eyes were included in the thermal analysis.

Fresh human donor eyes were obtained from VisionFirst Indiana Lions Eye Bank (Carmel, IN, USA). Eyes were eligible if they were received within 72 h of enucleation, were grossly intact, and retained at least 10 mm of proximal optic nerve. Eyes with penetrating trauma or gross ocular pathology were excluded. Nine eyes from five donors were included. Donors were male, 43–77 years of age, and the cohort supplied for this study was exclusively Caucasian. Enucleation took place 8.5 to 19.4 h after death (median 12.4 h). The tissue was de-identified and supplied with authorization for research use.

In all three cohorts, excess extraocular tissue, including rectus muscle and orbital fat, was removed to expose the scleral surface, and each eye was then mounted cornea-up on the eye holder (Section 2.2.2) with a small quantity of cyanoacrylate adhesive.

2.2.4. Temperature Measurement and Spatial Verification. Temperature was measured with negative-temperature-coefficient thermistors (GA10K3MCD1, TE Connectivity Ltd., China) [**125**]. The sensing tip of each sensor was at most 3.3 mm

long and 0.5 mm in diameter, with a nominal response time of 200 ms in liquid. Five sensors were implanted at anatomically defined locations: anterior sclera, equatorial sclera, posterior sclera, vitreous, and the proximal optic nerve, adjacent to the optic nerve head. The three scleral thermistors were introduced through small needle incisions and aligned approximately parallel to the scleral surface. The vitreous sensor was advanced toward the center of the globe. The proximal optic nerve sensor was inserted along the axis of the nerve in the porcine eyes, and parallel to the posterior scleral surface, at the junction of the nerve with the globe, in the human eyes (Figure 2.4). Cyanoacrylate adhesive was used to stabilize the leads at the insertion sites.

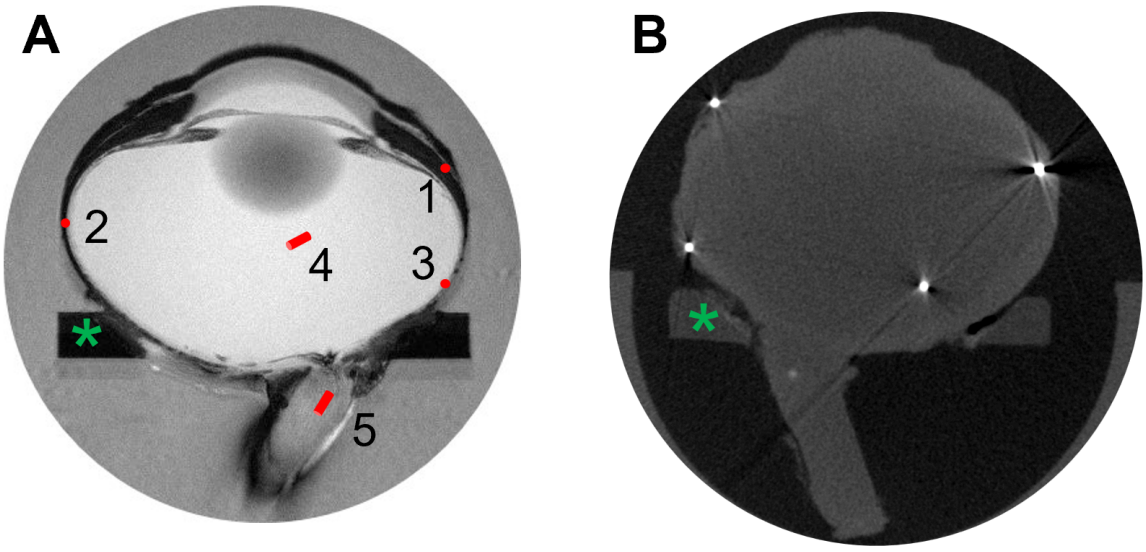


FIGURE 2.4. Target and measured positions of the implanted sensors in porcine eyes [**1**]. Left (A): MRI slice near the midline of the globe, with the target position and orientation of each sensor as a red cylinder: 1, anterior sclera; 2, equatorial sclera; 3, posterior sclera; 4, vitreous; 5, proximal optic nerve. The dark band between the front of the eye and the bath is the displacement of the oil bath relative to the tissue in the image, not anatomy (Section C.4). Right: micro-CT slice of an instrumented eye; the sensors are the bright spots, and the green asterisk marks the acrylic eye support. Adapted from Nakamura et al., Fig. 1 [**1**] (scale bars omitted), CC BY 4.0.

A sixth thermistor embedded in the SCR measured the imposed boundary temperature and provided the feedback signal used for device control. Each thermistor formed one arm of a voltage divider whose output was amplified by a non-inverting amplifier, giving approximately −8.0 V at 37 °C and +8.0 V at 4 °C (about 0.48 V °C^{-1}). A Hall-effect current sensor (ACS723 breakout, SparkFun Electronics, Niwot, CO, USA) in series with the TEC supply recorded the drive current. All six temperature channels were acquired simultaneously at 10 Hz using a DI-2108 data acquisition unit (DataQ Instruments, Akron, OH, USA) and recorded with WinDaq software (Figure A.1 in Appendix A).

Porcine eyes were imaged by 9.4-T magnetic resonance imaging before implantation to document gross ocular anatomy, and micro-computed tomography after the cooling experiment localized the implanted thermistors in the porcine and human eyes (settings in Section C.4). Reconstructed volumes were imported into 3D Slicer, and sensor coordinates were used to assess spatial consistency and identify sensors that did not sample the intended anatomical compartment.

The calibration procedure evolved during the experimental program. In the porcine experiments, the channels were calibrated before each session in a Fluke 9102S dry-block calibrator [**126**] fitted with a custom insert. With the thermistors held at 20.5 and 37.0 °C, the two potentiometers of each channel's amplifier, which set the intercept and slope, were trimmed until the output read 0 V and −8.00 V, respectively. These are the values of the circuit's linear voltage–temperature transfer function at those temperatures, and this single fixed function then converted voltage to temperature for all channels. For the later human-donor experiments, the channels were trimmed in the same way and three calibration plateaus, at 4.0, 20.5, and 37.0 °C, were then recorded in each session. Each plateau was read as the median of its last 3 min before the ramp to the next temperature, the last 10 s excluded; where those 3 min still drifted by more than 5 mV per minute, it was read as the median of the longest stretch whose 10-s medians stayed within ±0.01 V. The 37.0 °C

plateau had settled for more than 7 min before each recording started, and where less than 3 min of it was recorded its value is the median of the whole recorded plateau. The same rule reads the in-vivo calibrations (Section 4.2.6). The voltage of each channel was converted to temperature with the least-squares straight line through its three plateaus. The conditioning circuit is linearized. On a bench recording of eleven dry-block temperatures between 2 and 39 °C, a straight line reproduced the block temperature within 0.34 °C.

The expanded uncertainty of an individual absolute temperature measurement was ±1.8 °C ($k = 2$). It is dominated by sensor self-heating, which can only make a reading too low, by at most 1.4 °C, and no correction was applied for it. Because the primary comparative endpoint was a within-run cooling excursion, static common-mode contributions largely canceled. In the human eyes, the corresponding differential uncertainty was ±0.40 °C ($k = 2$) for a small excursion, such as that of the proximal optic nerve, and ±0.85 °C for a large one, such as that of the SCR. In the porcine eyes, whose calibration was trimmed at two points, it was ±0.33 and ±1.1 °C. The uncertainty budget is given in Appendix A.

2.2.5. Experimental Protocol. For each experiment, the instrumented eye was positioned cornea-up in the 37 °C bath and the SCR was lowered onto the anterior sclera using a linear translation stage. Circumferential contact was established by directing light through the central aperture and lowering the device until light was no longer visible between the ring and scleral surface. The sequence of steps is summarized in Figure 2.5.

The porcine experiments used an active thermal-equilibration step before cooling. The SCEC was operated in reverse polarity to bring the interface toward approximately 34.5 °C, the temperature of the exposed anterior ocular surface, and the preparation was allowed to equilibrate for approximately 30 min. The temperature at each sensor at the first sample of the record defined its initial value.

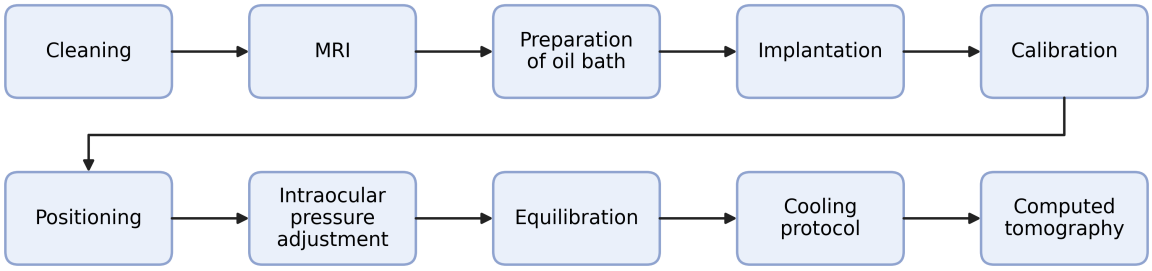


FIGURE 2.5. Experimental protocol for the porcine eyes, from receipt of the eye to post-experiment imaging. Human donor eyes followed the same sequence without MRI and equilibrated passively in the 37 °C bath.

At time zero, the device was switched to cooling polarity and the SCR set point was changed to 4.0 °C. Temperature at the SCR and all five implanted locations was recorded continuously for 20 min. The duration was sufficient for all measured compartments to approach a stable cooled state.

Human donor eyes were tested using the same 4.0 °C, 20-min cooling protocol, but were allowed to establish their baseline temperatures by passive equilibration in the 37 °C bath. Each human eye was cooled on the same day with three rings, the original ring reported here and the reduced-bulk and ceramic rings compared in Chapter 3, on a single set of implanted sensors and in an order that varied between eyes. Before each run the eye returned passively in the bath to about its resting state. The porcine and human experiments therefore began from different absolute baselines. For this reason, the cross-species comparison was based on the cooling excursion and not on the final absolute temperature.

2.2.6. Data Analysis. All recordings were processed with a single analysis pipeline, implemented as a Python package that runs the complete analysis from the raw files (Appendix C).

Denoising. After conversion to temperature (Section 2.2.4), each trace was median filtered (51 samples, 5.1 s) to suppress the spikes produced by the switching of the TEC relay, and then low-pass filtered with a zero-phase, second-order Butterworth filter (0.01 Hz cutoff). Before filtering, the start of each trace was extended by point

reflection about its first sample, and the median filter padded the trace with its edge values to avoid underestimating the initial temperature.

Quality control. Three objective gates and a final visual review determined which traces entered the analysis. The gates do not use the size of the cooling excursion. The visual review, by its nature, was made with the traces in view. The gates are listed in order of precedence, and a trace failing more than one gate was excluded once, under the first reason in this order. First, sensor placement was verified from the co-registered micro-CT coordinates. Within each cohort, the positions of the four sensors fixed to tissue (anterior, equatorial, and posterior sclera, and proximal optic nerve) of every eye were rigidly aligned, by rotation and translation without scaling, to the mean set of the cohort (generalized Procrustes alignment, Equation C.8 in Section C.2). A tissue-fixed sensor was flagged when its aligned offset from the mean position of its site exceeded one thermistor length, 3.3 mm, the maximum length of the sensing tip given by the manufacturer. A shift of the whole eye in the scanner therefore does not flag a sensor, and the fence is the same for every cohort (Figure 2.6). The vitreous sensor is not fixed to tissue, so it did not enter the alignment and had no offset fence. It was flagged when its micro-CT position lay more than 75% of the distance from the center of the segmented globe to its wall in the same direction (Section C.4).

Second, channels whose raw voltage was flat-lined, or lay at the input limit of the acquisition unit (± 10 V) for more than 75% of the record, indicating a disconnected or saturated sensor, were excluded. One further human trace was excluded as a hardware fault although it passed this test. The anterior-sclera sensor of eye 9 was failing. It had reached the input limit in the recording made before the run, and it gave no signal in the two later runs on that eye. Third, a single-exponential decay was fitted to each remaining filtered trace, and traces with $R^2 < 0.85$ were excluded as noisy or non-monotonic. The fit served only this screen. Finally, every trace that passed the three gates was inspected against the other traces of its cohort and site.

Traces showing an obvious transient artifact, a sustained oscillation or re-warming while the SCR held its set point, were excluded. Three traces were removed in this way. Every trace excluded by the hardware gate, the fit gate, or visual review is shown in Figure 2.7. Because the visual review was not blind to the traces, every comparison was repeated with the visually excluded traces retained, as a sensitivity analysis.

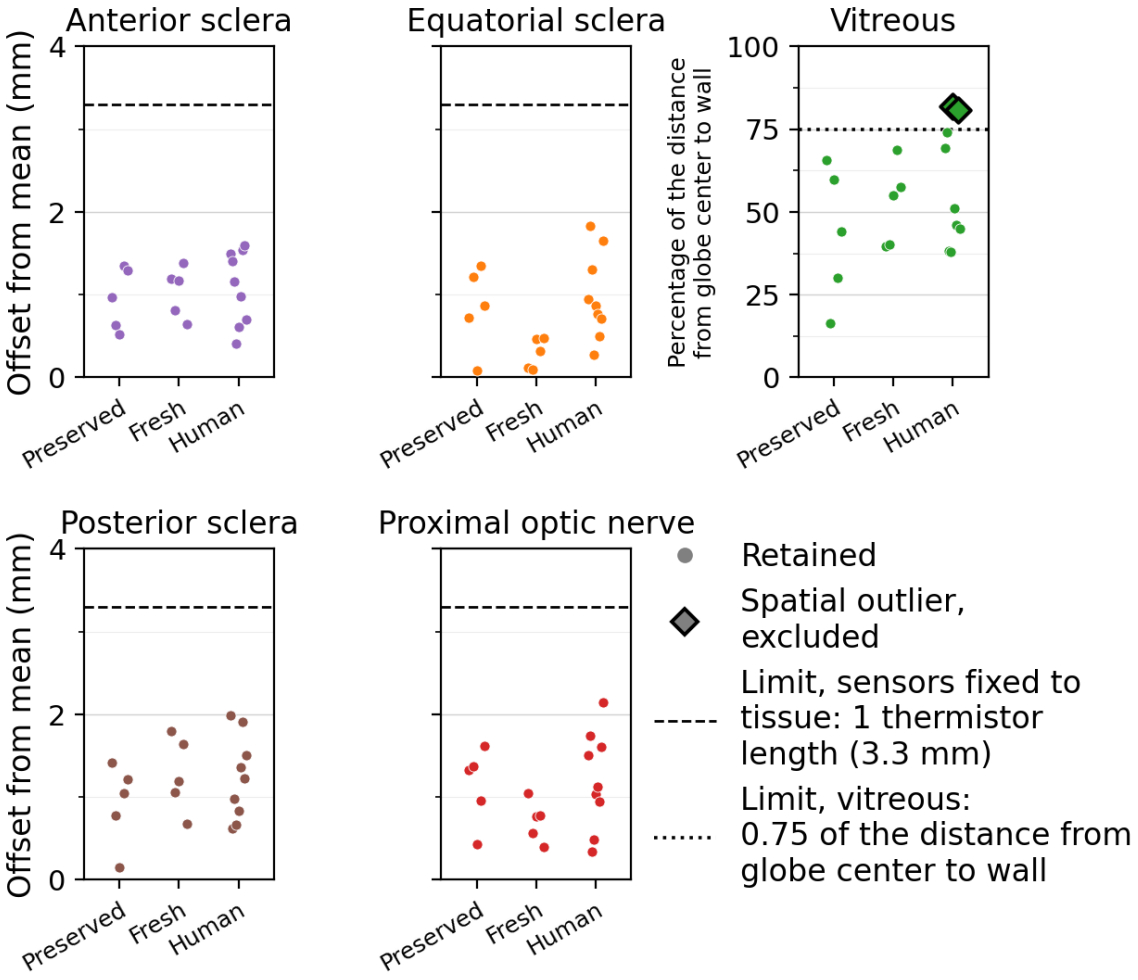


FIGURE 2.6. Spatial screen of sensor placement in preserved porcine, fresh porcine, and human eyes. Scleral and proximal optic nerve panels: offset $r_{e,s}$ of each sensor from the mean position of its site, after the four tissue-fixed sensors of each eye were rigidly aligned to the cohort mean (Equation C.8); dashed lines, the exclusion fence of one thermistor length (3.3 mm). Vitreous panel: position of each vitreous sensor on micro-CT as a percentage of the distance from the center of the globe to its wall in that direction; dotted line, the 75% limit; diamonds, the two excluded sensors (micro-CT sections in Figures C.2 and C.3).

Endpoints. For each included trace, the primary endpoint was the cooling excursion,

$$\Delta T = T_{\text{initial}} - T_{\text{final}}, \tag{2.1}$$

where T_{initial} is the temperature at the first sample of the record ($t = 0$), at device activation, and T_{final} the mean temperature over the last 4 min of the 20-min record (minutes 16–20). In the porcine runs the SCR had already been running for a few seconds at $t = 0$. Cooling kinetics were summarized by t_{90}, the first time at which the trace had fallen by 90% of ΔT, and every trace had reached t_{90} before the final window began. The stability of the cooled plateau was summarized by the standard deviation (SD) of the median-filtered trace over the same 4-min window. Final temperature was reported descriptively.

Group comparisons. Two comparisons addressed the aim: preserved against fresh porcine eyes, to assess preserved tissue as a benchtop surrogate, and fresh human against fresh porcine eyes, both cooled with SCEC Gen 3, to test porcine–human thermal concordance. Both used the same procedure. Per-site contrasts used two-sided Welch t-tests with Holm correction across the five tissue sites. The SCR, a device endpoint, was analyzed separately and kept outside the Holm family. Effect sizes were reported as small-sample-corrected Hedges' g (the difference over the pooled SD), as a point estimate; the uncertainty of each comparison is given by the confidence interval of the difference in °C. Tests that treat the two eyes of a subject as independent can inflate the type I error, whereas a test on the per-subject average holds it at the nominal level [**127**]. The nine human eyes came from five donors, and each test was repeated with fellow eyes averaged to one value per donor, to verify that treating fellow eyes as independent did not drive the result. Each porcine eye was treated as its own donor, since the animal of origin was not recorded and the five fresh eyes came from one harvest. Because some sites retained only a few traces, failure to detect a difference was not interpreted as evidence that there is none.

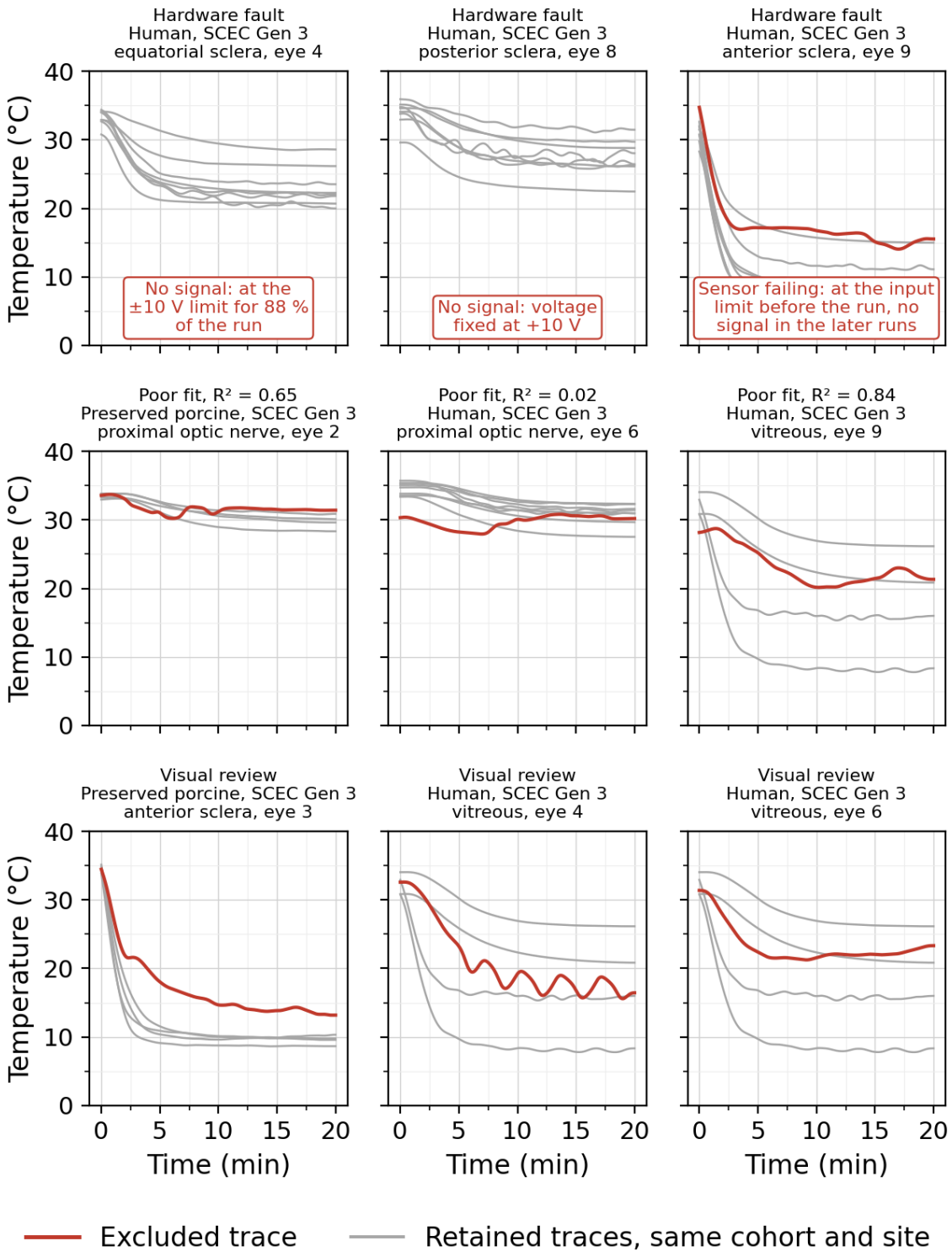


FIGURE 2.7. Traces excluded by the hardware gate, the fit gate, and on visual review (red), against the retained traces of the same cohort and site (grey). A channel with a hardware fault has no temperature signal, so its panel states the fault in the raw voltage; the failing sensor of human eye 9 had a signal in this run and is drawn with its note. R^2 is that of the single-exponential fit. The two sensors excluded by the spatial screen are in Figure 2.6.

Largest difference compatible with the data. Each comparison is also reported with the 90% confidence interval of the difference in ΔT. A difference beyond the interval limit farther from zero is called unlikely, which corresponds to a one-sided test at the 5% level. That limit is judged against the excursion at the site and the 3 °C benchmark [**62**]. Differences between groups are judged by these intervals, not by the measurement uncertainty of a single excursion (Section 2.2.4). Statistical significance was assessed at $\alpha = 0.05$.

2.3. Results

Three cohorts were cooled with SCEC Gen 3 under the original ring: five preserved porcine eyes, five fresh porcine eyes, and nine fresh human eyes. Of the 114 channel traces recorded, 103 passed quality control (Table 2.1). No scleral or proximal optic nerve sensor, in any cohort, lay beyond the placement fence. The largest offset was 2.1 mm. Two vitreous sensors, both in human eyes, lay more than 75% of the distance from the center of the globe to its wall (82 and 81%) and were excluded by the spatial screen. Three human channels had a hardware fault, three traces failed the fit gate, and three were removed on visual review. The vitreous, whose sensor is the least mechanically constrained, accounted for five of the eleven exclusions.

Table 2.1. Quality-control outcome by cohort; each eye contributed six traces. A trace that failed more than one gate is counted once, under the first reason listed (the reasons are in order of precedence).

Cohort	Traces	Spatial	Hardware	Fit	Visual	Included
Preserved porcine (5 eyes)	30	0	0	1	1	28
Fresh porcine (5 eyes)	30	0	0	0	0	30
Fresh human (9 eyes)	54	2	3	2	2	45
Total	114	2	3	3	3	103

2.3.1. Preserved Porcine Eyes. Cooling of the formalin-preserved porcine eyes was rapid and spatially ordered (Figure 2.8). All five preserved eyes were analyzed, and 28 of their 30 traces passed quality control. No sensor lay beyond the

spatial fence. One proximal optic nerve trace failed the fit criterion, and one anterior-sclera trace was removed on visual review. The SCR held the commanded 4 °C set point, with a mean final temperature of 4.1 °C (Table 2.2).

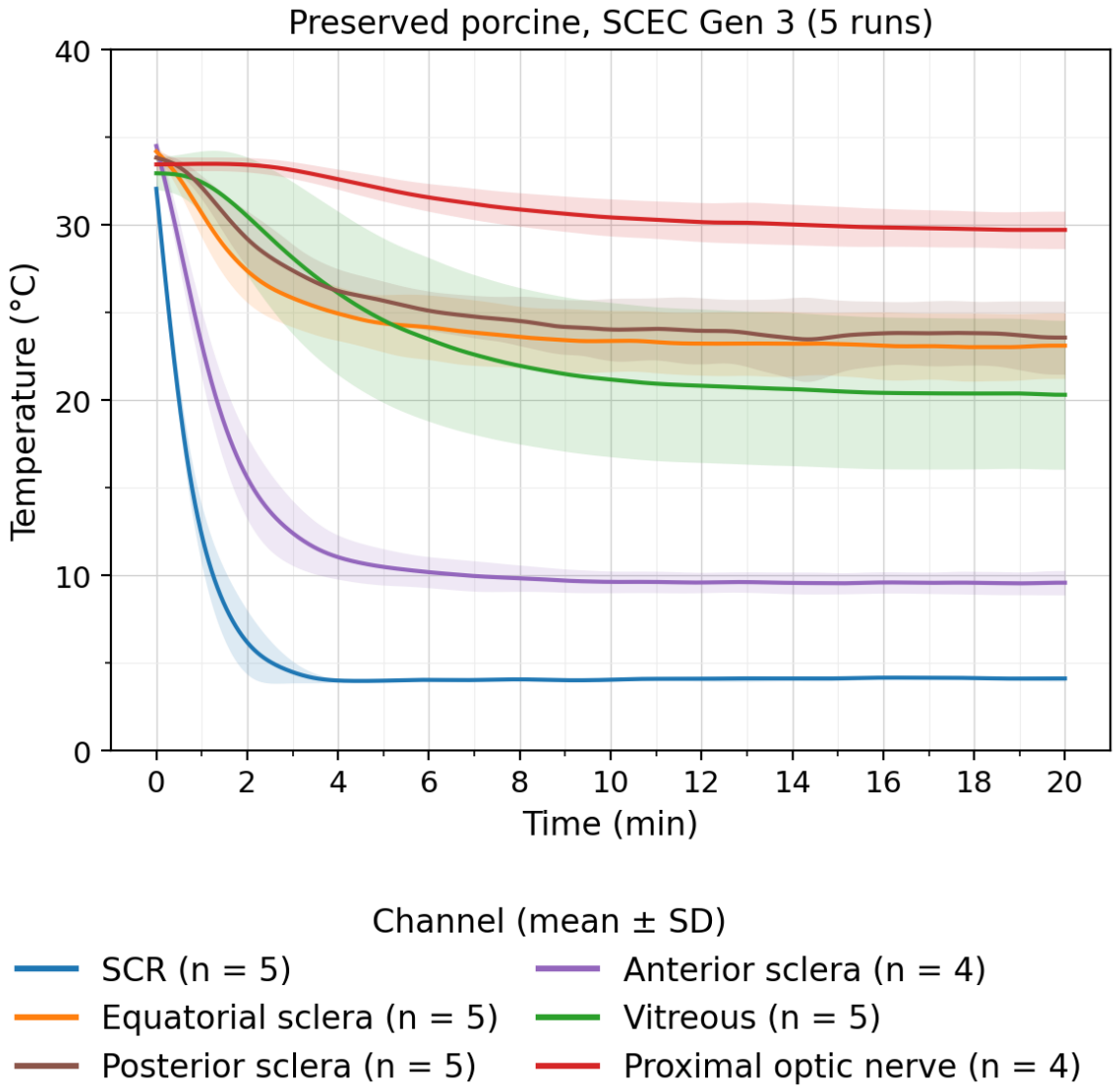


FIGURE 2.8. Temperature at each site during SCEC Gen 3 cooling of five preserved porcine eyes: cohort means and ±1 SD (bands) of the included traces; the legend gives the number of traces per site.

Cooling varied by site (Table 2.2): the anterior sclera showed the largest excursion, 24.9 °C, and the proximal optic nerve the smallest and slowest, 3.7 ± 0.8 °C with t_{90} of 783 s (13.1 min). Over the final 4 min the deeper sites had approached but not

reached a steady state. The largest dispersion between specimens was in the vitreous (final temperature SD 4.3 °C).

TABLE 2.2. Thermal response of preserved ex-vivo porcine eyes during SCEC Gen 3 cooling: mean (SD) over the included traces (n). Initial temperature, final temperature, and ΔT were averaged separately, so ΔT need not equal the difference of the rounded means. Plateau SD is the within-trace SD over the final 4 min.

Location	n	**Initial**	**Final**	ΔT	t_{90}	**Plateau SD**
Units	–	(°C)	(°C)	(°C)	(s)	(°C)
SCR	5	32.1 (0.5)	4.1 (0.1)	27.9 (0.6)	111 (26)	0.06
Anterior sclera	4	34.5 (0.6)	9.6 (0.6)	24.9 (0.9)	188 (46)	0.06
Equatorial sclera	5	34.2 (0.3)	23.1 (1.9)	11.1 (2.0)	324 (123)	0.07
Vitreous	5	33.0 (0.9)	20.4 (4.3)	12.6 (3.4)	515 (72)	0.06
Posterior sclera	5	33.8 (0.4)	23.8 (1.9)	10.1 (1.8)	417 (83)	0.16
Proximal optic nerve	4	33.5 (0.4)	29.8 (1.0)	3.7 (0.8)	783 (95)	0.06

2.3.2. Fresh Porcine Eyes: Proof of Concept. SCEC Gen 3 produced a rapid and spatially ordered decrease in temperature throughout the isolated porcine eye (Figure 2.9). The SCR reached 90% of its excursion in 95 s and held a mean final temperature of 4.1 °C.

TABLE 2.3. Thermal response of fresh ex-vivo porcine eyes during SCEC Gen 3 cooling, as in Table 2.2.

Location	n	**Initial**	**Final**	ΔT	t_{90}	**Plateau SD**
Units	–	(°C)	(°C)	(°C)	(s)	(°C)
SCR	5	32.1 (0.4)	4.1 (0.1)	27.9 (0.4)	95 (12)	0.03
Anterior sclera	5	35.1 (0.5)	10.8 (3.8)	24.3 (3.9)	200 (67)	0.06
Equatorial sclera	5	34.5 (0.2)	25.1 (2.9)	9.4 (3.1)	340 (92)	0.05
Vitreous	5	34.0 (0.3)	22.4 (3.6)	11.6 (3.8)	566 (63)	0.04
Posterior sclera	5	34.3 (0.3)	24.5 (2.1)	9.8 (2.1)	519 (64)	0.08
Proximal optic nerve	5	33.8 (0.3)	30.3 (1.5)	3.5 (1.5)	676 (110)	0.06

The anterior sclera again cooled most, by 24.3 °C, and the proximal optic nerve least, by 3.5 ± 1.5 °C, above the 3 °C benchmark, with t_{90} of 11.3 min (Table 2.3). Its final temperature lay below the 32–35 °C interval of systemic hypothermia [**57**], within which the baseline already fell, and inside the roughly 30–34 °C range of preclinical

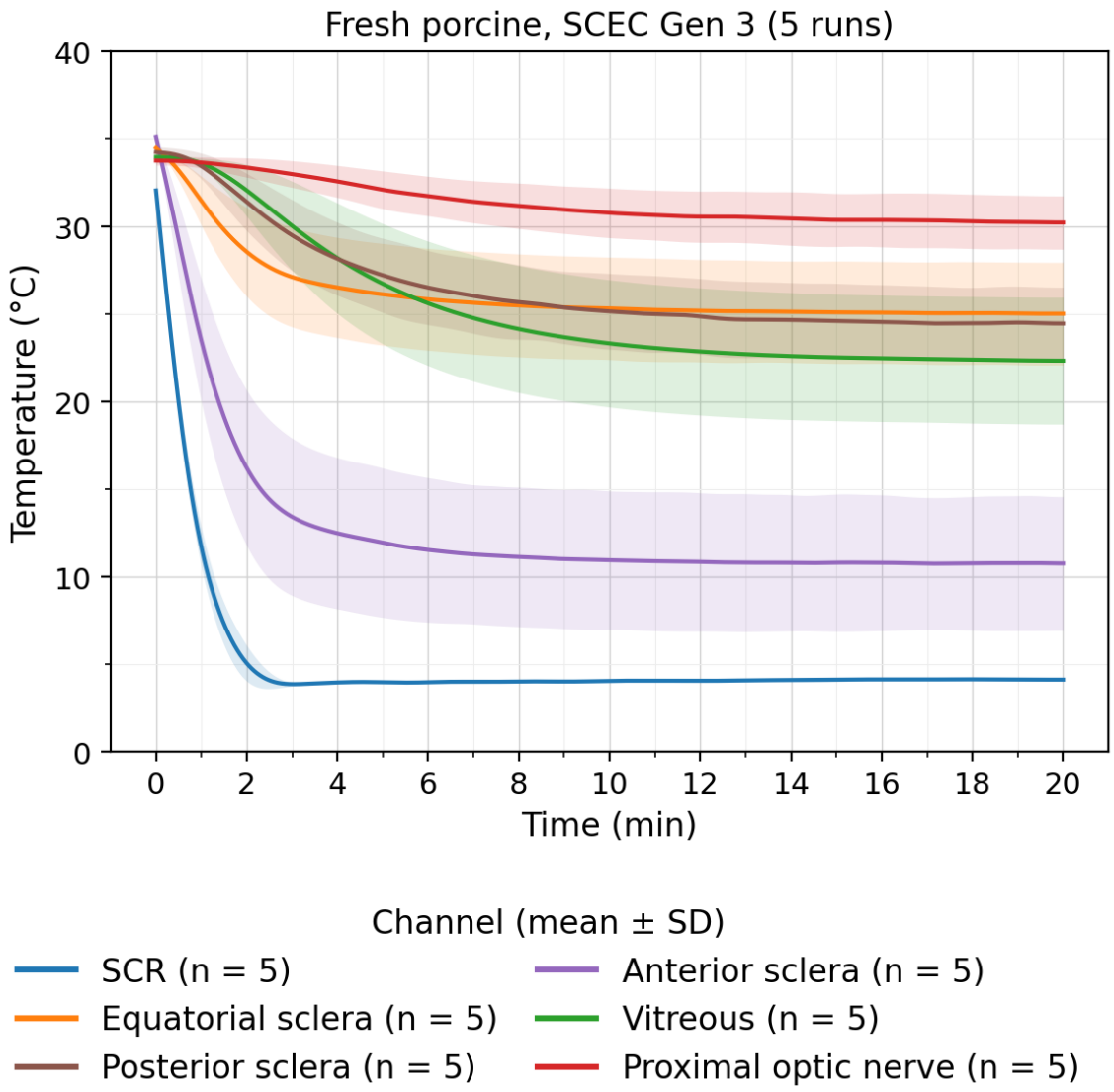


FIGURE 2.9. Temperature at each site during SCEC Gen 3 cooling of five fresh porcine eyes, as in Figure 2.8.

retinal protection (Section 1.3.3). All 30 traces passed quality control. Relative to its small excursion the proximal optic nerve varied most (coefficient of variation 43%).

2.3.3. Fresh Human Donor Eyes. The same device was then evaluated in fresh human donor eyes (Figure 2.10). Nine eyes from five donors were tested; 45 of 54 traces passed quality control, leaving four to nine traces per site (Table 2.4).

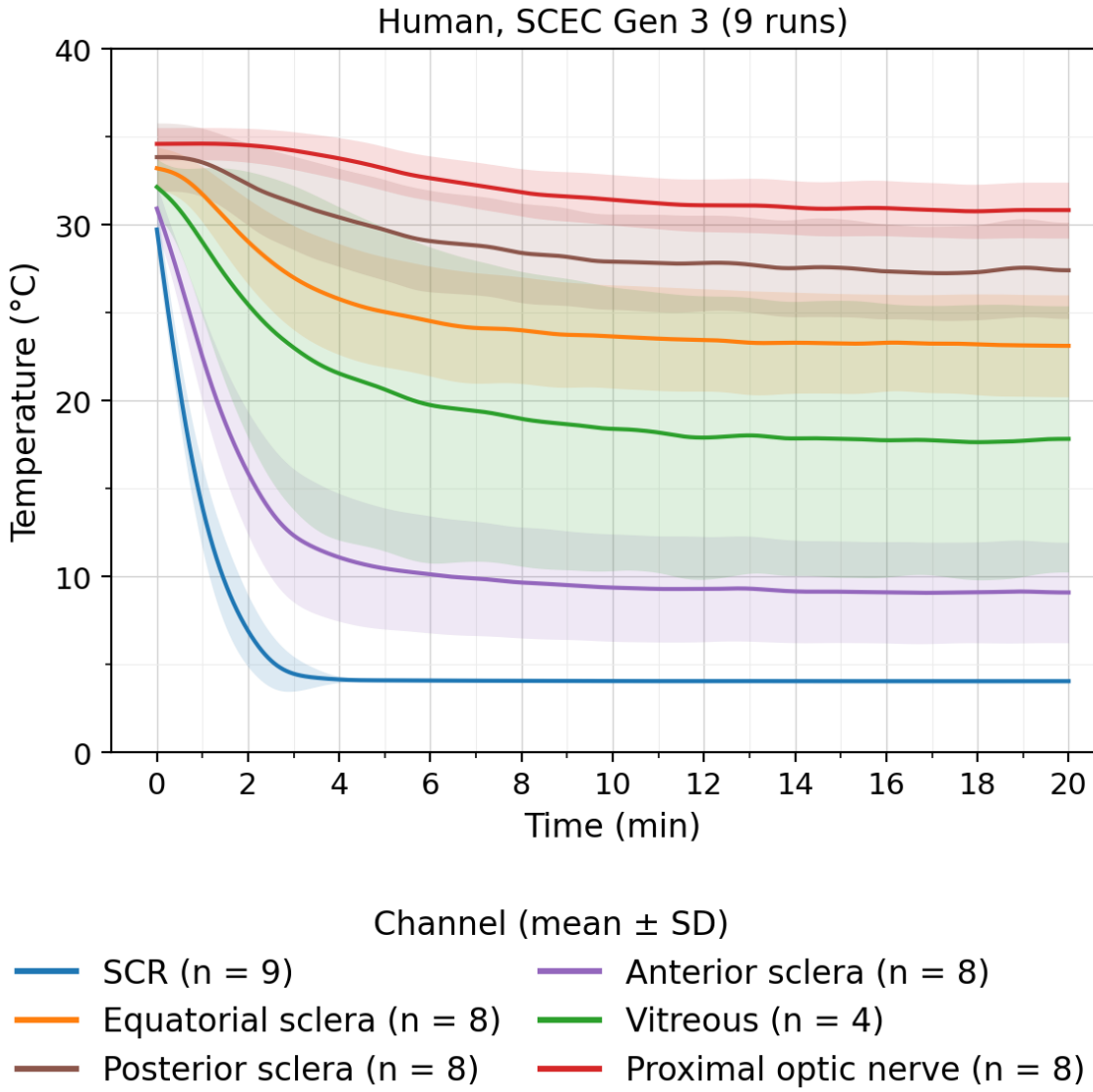


FIGURE 2.10. Temperature at each site during SCEC Gen 3 cooling of nine fresh human donor eyes, as in Figure 2.8.

The spatial pattern seen in porcine eyes was reproduced (Table 2.4). The anterior sclera cooled most, by 21.8 °C, and the proximal optic nerve least, by 3.8 ± 1.1 °C, to 30.8 °C, with t_{90} of 11.1 min. Plateaus were stable (mean within-plateau SD at most 0.24 °C). Human starting temperatures differed from the porcine ones because the cohorts were equilibrated differently (Section 2.2.5). Before cooling, the human

eyes showed a gradient of about 4 °C from the anterior sclera to the proximal optic nerve, similar to that measured in patients [**47**].

TABLE 2.4. Thermal response of fresh human donor eyes during SCEC Gen 3 cooling, as in Table 2.2.

Location	n	**Initial**	**Final**	ΔT	t_{90}	**Plateau SD**
Units	–	(°C)	(°C)	(°C)	(s)	(°C)
SCR	9	29.7 (1.4)	4.1 (0.1)	25.7 (1.4)	124 (27)	0.07
Anterior sclera	8	30.9 (1.4)	9.1 (2.9)	21.8 (2.8)	233 (82)	0.10
Equatorial sclera	8	33.2 (1.2)	23.2 (2.8)	10.0 (2.6)	420 (131)	0.14
Vitreous	4	32.2 (1.6)	17.7 (7.7)	14.4 (6.8)	437 (233)	0.15
Posterior sclera	8	33.9 (1.9)	27.4 (2.7)	6.5 (1.3)	531 (122)	0.24
Proximal optic nerve	8	34.6 (0.9)	30.8 (1.6)	3.8 (1.1)	667 (79)	0.09

2.3.4. Repeatability of Cooling at the Optic Nerve. Cooling at the posterior target was present in every eye but varied in magnitude between eyes (Table 2.5). All 17 included proximal optic nerve traces cooled, by between 2.3 and 5.9 °C. All 17 reached 90% of their excursion within 15 min, the slowest at 14.3 min. Twelve of the 17 traces reached the 3 °C benchmark, one of them at 2.999 °C. The between-eye coefficient of variation of the proximal optic nerve excursion was 22–43%, against 1–6% at the SCR, whose temperature was controlled. The boundary was thus reproduced closely from eye to eye, and the variation at the posterior target arose within the tissue.

TABLE 2.5. Repeatability of cooling at the proximal optic nerve across eyes. Counts are of included proximal optic nerve traces. CV is the between-eye coefficient of variation of ΔT; [a]at the proximal optic nerve; [b]one trace at 2.999 °C.

Cohort	n	**Traces reaching**		ΔT **range**	**CV of** ΔT **(%)**	
(eyes)	(traces)	$\Delta T \geq 3$ **°C**	$t_{90} \leq 15$ **min**	**(°C)**	**Nerve**[a]	**SCR**
Preserved porcine	4	3	4	2.95–4.56	22	2
Fresh porcine	5	3	5	2.26–5.92	43	1
Fresh human	8	6[b]	8	2.51–5.79	29	6
Total	17	12[b]	17	2.26–5.92		

2.3.5. Fresh Versus Preserved Porcine Eyes. In preserved eyes, cooling closely followed that of fresh eyes (Figure 2.11). The SCR held 4.1 °C in both preparations, reaching 90% of its excursion in 111 s in preserved eyes and 95 s in fresh eyes. The spatial pattern of cooling survived chemical fixation. In both preparations the anterior sclera cooled most and fastest, and the proximal optic nerve least and slowest, with t_{90} of 783 s (13.1 min) in preserved and 676 s (11.3 min) in fresh eyes.

No difference in cooling excursion was detected between preserved and fresh tissue at any of the five implanted sites (Table 2.6; all Holm-adjusted $p = 1.00$). The largest nominal difference was at the equatorial sclera, where preserved eyes cooled 1.7 °C more. At the proximal optic nerve the difference was −0.2 °C (fresh minus preserved). The 90% confidence intervals (CI) of the difference at the SCR and the proximal optic nerve (Table 2.6) make a difference larger than 0.6 °C at the SCR, or 1.7 °C at the proximal optic nerve, unlikely. At the remaining sites the intervals were wider and allow differences of 2.6 to 5.2 °C. Retaining the visually excluded anterior-sclera trace changed the anterior difference from −0.6 to +0.2 °C and left every other value, and every conclusion, unchanged.

TABLE 2.6. Cooling excursion ΔT in preserved and fresh porcine eyes, in °C: mean (SD), with the number of included traces in brackets. Difference, fresh minus preserved, with its 90% confidence interval (Welch), whose limit farther from zero is the largest difference the data allow. p_{Holm} is corrected across the five tissue sites; the SCR is tested separately, and its p (*) is unadjusted. g, Hedges' g (unitless). Anterior, equatorial, and posterior are the scleral sites.

Location	**Preserved**	**Fresh**	**Diff.**	**90% CI**	p_{Holm}	g
SCR	27.9 (0.6) [5]	27.9 (0.4) [5]	0.0	−0.6, 0.6	0.96*	0.03
Anterior	24.9 (0.9) [4]	24.3 (3.9) [5]	−0.6	−4.4, 3.2	1.00	−0.18
Equatorial	11.1 (2.0) [5]	9.4 (3.1) [5]	−1.7	−4.8, 1.5	1.00	−0.58
Vitreous	12.6 (3.4) [5]	11.6 (3.8) [5]	−1.0	−5.2, 3.3	1.00	−0.25
Posterior	10.1 (1.8) [5]	9.8 (2.1) [5]	−0.3	−2.6, 2.0	1.00	−0.14
Proximal optic nerve	3.7 (0.8) [4]	3.5 (1.5) [5]	−0.2	−1.7, 1.3	1.00	−0.15

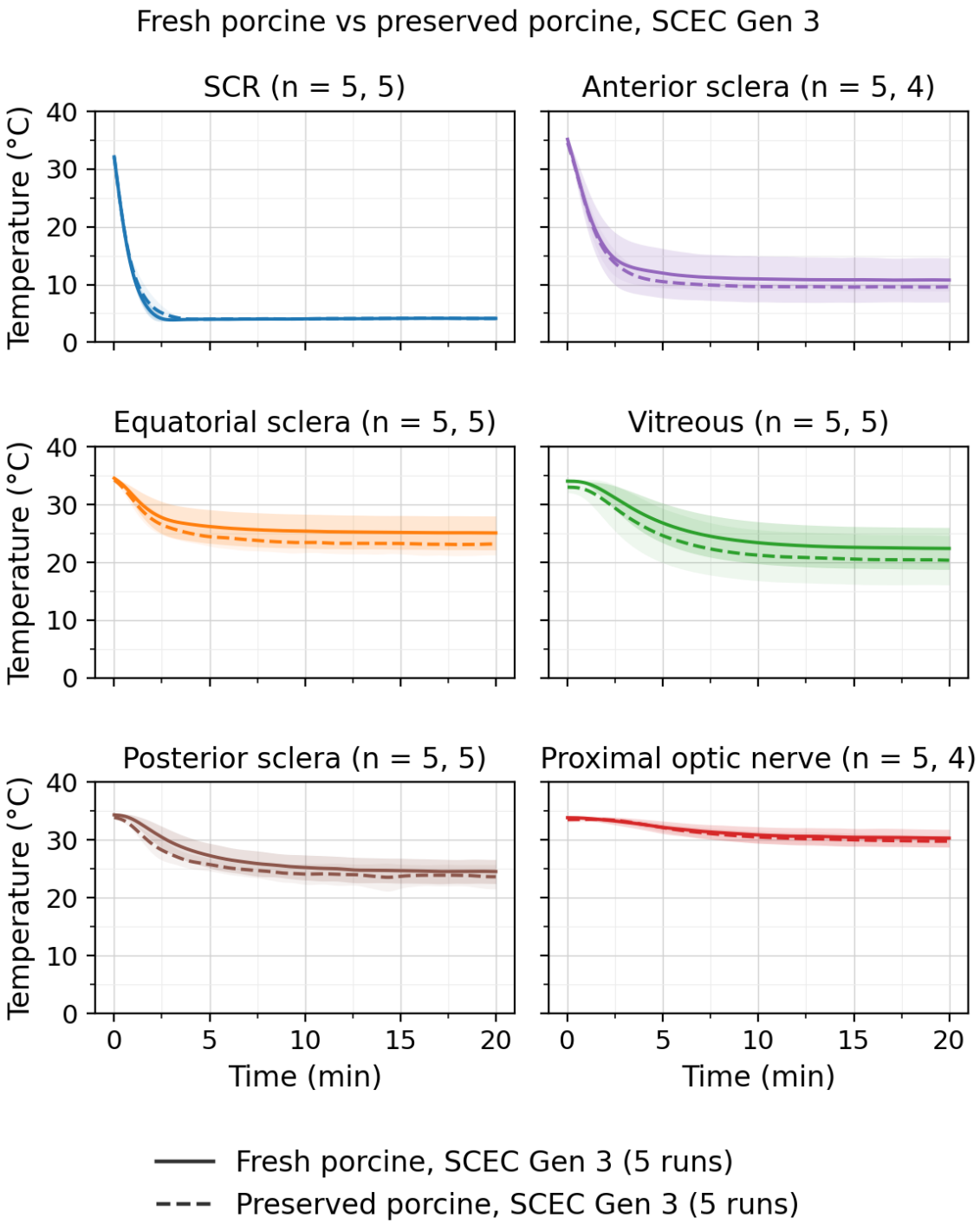


FIGURE 2.11. Temperature at each site during SCEC Gen 3 cooling of fresh (solid) and preserved (dashed) porcine eyes: cohort means and ±1 SD (bands) of the included traces; panel titles give the number of traces (fresh, preserved).

2.3.6. Human–Porcine Thermal Concordance. For the cross-species analysis, human and fresh porcine eyes were compared under the same ring, so that the species effect was not confounded by device geometry (Figure 2.12, Table 2.7).

At the proximal optic nerve, the two species cooled by similar amounts. The mean excursion was 3.8 ± 1.1 °C in human eyes and 3.5 ± 1.5 °C in fresh porcine eyes. The human-minus-porcine difference was 0.30 °C (Table 2.7). No difference was detected, and one larger than 1.8 °C is unlikely, since it lies outside the 90% confidence interval and is excluded by a one-sided test at the 5% level (Section 2.2.6). With fellow human eyes averaged to one value per donor, the difference was much the same, and the result did not depend on treating fellow eyes as independent. Leaving out either of two of the five porcine eyes raised the upper limit of the interval, and pooling the fresh and preserved porcine traces narrowed it, without changing the conclusion.

TABLE 2.7. Cooling excursion ΔT in fresh human and fresh porcine eyes under SCEC Gen 3; difference, human minus porcine; otherwise as in Table 2.6.

Location	**Human**	**Porcine**	**Diff.**	**90% CI**	p_{Holm}	g
SCR	25.7 (1.4) [9]	27.9 (0.4) [5]	−2.3	−3.2, −1.3	0.001*	−1.79
Anterior	21.8 (2.8) [8]	24.3 (3.9) [5]	−2.5	−6.4, 1.4	1.00	−0.71
Equatorial	10.0 (2.6) [8]	9.4 (3.1) [5]	0.6	−2.6, 3.7	1.00	0.19
Vitreous	14.4 (6.8) [4]	11.6 (3.8) [5]	2.8	−5.1, 10.7	1.00	0.47
Posterior	6.5 (1.3) [8]	9.8 (2.1) [5]	−3.3	−5.3, −1.2	0.10	−1.88
Proximal optic nerve	3.8 (1.1) [8]	3.5 (1.5) [5]	0.3	−1.2, 1.8	1.00	0.22

Of the five implanted tissue sites, only at the proximal optic nerve was the species difference bounded below 2 °C. Away from the proximal optic nerve, the species differences varied by site. Human eyes cooled slightly more at the equatorial sclera, whereas porcine eyes cooled more at the anterior and posterior sclera. The largest consistent site-specific contrast was at the posterior sclera, where porcine tissue cooled 3.3 °C more than human tissue. Its effect-size interval was the only one among the tissue sites to exclude zero ($g = -1.88$). The contrast did not remain significant after correction for multiple comparisons (nominal $p = 0.02$, Holm-adjusted $p = 0.10$). The

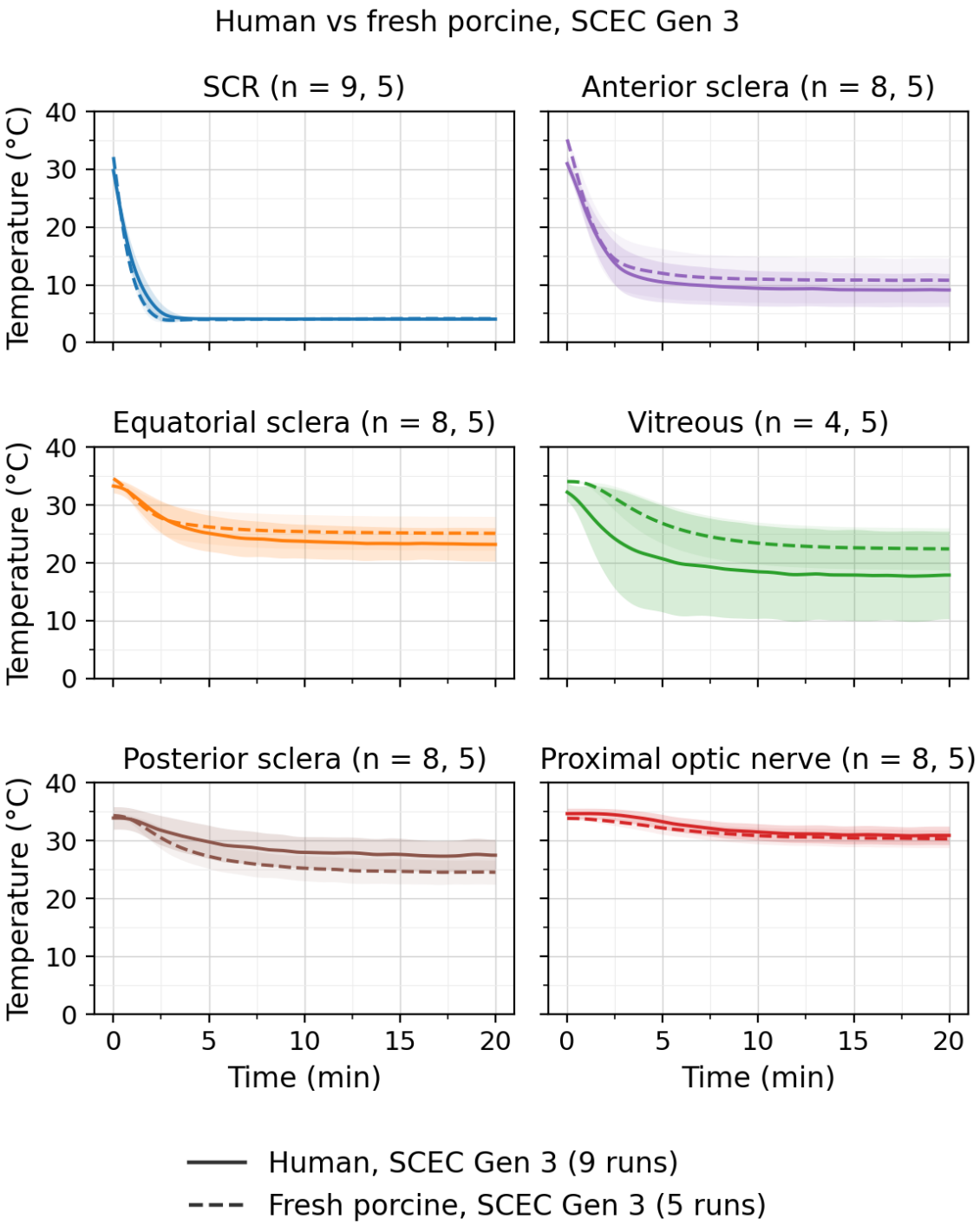


FIGURE 2.12. Temperature at each site during SCEC Gen 3 cooling of fresh human (solid) and fresh porcine (dashed) eyes, as in Figure 2.11; panel titles give the number of traces (human, porcine).

differences at the equatorial sclera and in the vitreous were small against their wide intervals (Table 2.7). With the two visually excluded human vitreous traces retained, the per-site vitreous difference fell from 2.8 to 2.1 °C. No proximal optic nerve trace was excluded visually, so the proximal optic nerve results, including the confidence interval of the species difference, were identical in both analyses. The SCR excursion was 2.3 °C smaller in human eyes ($p = 0.001$). Both cohorts ended at 4.1 °C, and the difference reflects the lower human starting temperature.

2.4. Discussion

Specific Aim 1 established that SCEC Gen 3 delivers controlled trans-scleral cooling to the deep posterior eye. The first hypothesis was supported in occurrence and timing but not in magnitude: every eye cooled at the proximal optic nerve and all 17 reached 90% of the excursion within 15 min, but only 12 of 17 reached the 3 °C benchmark (Section 2.3.4). The test of the second hypothesis found no detectable difference between the species at that site and a bound of 1.8 °C. Preserved and fresh porcine eyes did not differ detectably at the SCR or the proximal optic nerve.

Magnitude of cooling at the posterior target. The mean proximal optic nerve excursion exceeded the 3 °C benchmark in every cohort. The benchmark is the smallest step tested in isolated retina subjected to simulated ischemia [**62**]. That step ran from 36 to 33 °C, whereas the proximal optic nerve started ex vivo at a lower temperature (Tables 2.3 and 2.4). A 3 °C decrement here is therefore comparable in size to that step but does not reproduce it. The final proximal optic nerve temperatures in fresh porcine and human eyes lie within the roughly 30–34 °C range over which preclinical retinal studies report protection (Section 1.3.3). Both are also close to the 30 °C at which retinal ganglion cells were protected in hypoxic organ culture [**71**]. These absolute values depend on each preparation's starting temperature and carry the absolute measurement uncertainty of Section 2.2.4, most of it a possible underreading from sensor self-heating.

The decrement reached in vivo is expected to be smaller than ex vivo, because perfusion resupplies heat that the bath did not (see *Absence of perfusion*, below). Delivery at this magnitude for 20 min from a boundary held at 4 °C, well above freezing, is the engineering premise on which the remaining aims depend.

The proximal optic nerve as a conservative index. Most of the retina was likely cooled further than the proximal optic nerve. The proximal optic nerve was chosen as the deep posterior target because it is the site farthest from the ring; its final temperature, 30.3 to 30.8 °C, lies at the lower end of the preclinical retinal range. It does not, however, represent the retina as a whole. The retina lines the globe from just behind the ring to the optic disc, and the cooling imposed at the ring reaches it along a gradient. The retina itself was not instrumented. It lies, however, between the vitreous and the posterior sclera, which in every cohort ended far colder than the proximal optic nerve (Tables 2.2, 2.3, and 2.4), and the peripheral retina lies closer still to the ring. The computational model of Chapter 3 resolves this gradient along the retina.

Preserved tissue as a benchtop surrogate. Preserved and fresh porcine eyes behaved alike at the two sites where the comparison was decisive. The SCR followed the same trajectory in both, and the proximal optic nerve excursions differed by 0.2 °C, with a difference larger than 1.7 °C unlikely. The proximal optic nerve responded more slowly in preserved eyes (Section 2.3.5). At the scleral and vitreous sites, with four or five traces per group, the confidence intervals were too wide to show either a difference or its absence. Preserved eyes are therefore suitable for what the platform is most often asked to do, comparing design variants under identical conditions, where a change in region-specific cooling is the quantity of interest and tissue availability is otherwise limiting. Absolute performance, and any conclusion about sites other than the posterior target, should rest on fresh tissue.

Porcine tissue as a surrogate for the human eye. The cross-species comparison is compatible with using the porcine eye, and a porcine in-vivo model, as a surrogate for

the human eye at the posterior target, and only there. At the proximal optic nerve the species differed by 0.3 °C, and by much the same with fellow eyes averaged by donor. The other tissue sites gave a mixed picture. Porcine eyes cooled 3.3 °C more at the posterior sclera, the largest consistent site-specific contrast, although it did not survive correction for multiple comparisons. At the equatorial sclera and in the vitreous the intervals were too wide to interpret. The posterior-sclera contrast is not explained by the starting temperature, which was nearly the same in the two species at this site, and it is consistent with the anatomy. The porcine sclera is thinnest only 5 to 6 mm behind the limbus, close to the ring, whereas the human sclera is thickest at the posterior pole (Table 1.3).

Differences in globe size, scleral thickness, and the depth of each site below the warm bath may contribute to these contrasts. Sensor placement also contributes. The vitreous sensor has no rigid landmark, cannot be seen during insertion, and showed the largest variance in position in both species, and in final temperature in the human eyes.

The state of the vitreous may also differ, as the human vitreous liquefies progressively with age [**128**], and the two tissues differed in post-mortem age. Neither was measured, and both remain untested explanations.

Comparability of the cohorts. The porcine and human experiments differed in three ways that bear on the comparison at the proximal optic nerve, and two of them would both tend to enlarge the human excursion relative to the porcine one, in the direction of the observed difference. First, the control method was manual in the porcine and automatic in the human experiments. The SCR, however, reached the same 4.1 °C plateau in every cohort, so the imposed boundary was comparable. Second, the starting temperature differed. After equilibration with the unpowered ring resting on the eye, the human SCR and anterior sclera began 2.3 and 4.2 °C cooler than in the porcine experiments, and the human proximal optic nerve about 0.8 °C warmer. An excursion toward a fixed boundary grows with the distance from

that boundary. Third, the proximal optic nerve sensor was oriented differently, along the nerve in the porcine eyes and parallel to the posterior sclera at the junction of the nerve with the globe in the human eyes. The human sensor thus sampled tissue closer to the cooled globe. The size of these second and third effects could not be estimated from these data.

Absence of perfusion. The ex-vivo measurements likely overestimate the cooling the device produces at depth in a perfused eye, most at the sites farthest from the ring. All experiments were performed in non-perfused tissue. The continuously mixed 37 °C bath restored, in part, the warm boundary that surrounds the globe in vivo, but it did not reproduce the choroidal circulation, which delivers heat directly beneath the retina. The perfused model of Chapter 3 cooled the deep sites less than the human eyes did ex vivo (Section 3.3.6). Blood could in principle carry cooling from the ring toward the vortex veins [**1**], but the model indicates that perfusion opposes the cooling at depth. The main indication for the device is ischemia, in which ocular blood flow is by definition reduced, so a non-perfused preparation corresponds to part of the clinical scenario. The present measurements are a tissue-level thermal baseline and do not predict in-vivo temperatures.

Measurement limitations. Thermistor implantation was necessarily invasive, and sensor position varied between eyes, most of all in the vitreous. Micro-CT localization and the spatial screen removed sensors that lay far from the intended position of their site but could not remove smaller placement differences. Small air bubbles introduced during implantation could insulate a sensor locally, and variation in the contact pressure of the SCR could change the thermal resistance of the ring–sclera interface. Both would add to the variation between eyes. Sample sizes were modest, particularly for the vitreous after quality-control exclusions, which limited the power of every site-level comparison. The species comparison at the proximal optic nerve is limited in the same way. Its bound of 1.8 °C is about half of the excursion itself, of about 3.5–3.8 °C, and rests on five porcine eyes. The cohorts also differed in starting

temperature and sensor orientation. The data therefore show that the two species differ by less than about 2 °C at this site, not that they cool alike. The human cohort was demographically narrow, five male Caucasian donors aged 43–77, and cannot show whether the ocular thermal response varies with sex, age, ancestry, or other donor characteristics.

Scope. Aim 1 evaluated thermal delivery only. An ex-vivo preparation cannot show whether the local exposure, about 10 °C at the anterior sclera, is tolerated by living tissue, or whether cooling preserves retinal function during an ischemic insult. The first is assessed in Aim 3 and the second remains future work. The remaining limitations of SCEC Gen 3, its bulk, heat-rejection hardware, control architecture, and the temperatures it imposes at the tissue interface, motivate Aim 2.

CHAPTER 3

Specific Aim 2: Enhance the Device (SCEC Gen 4) for Efficiency, Portability, Safety, and Patient Comfort

3.1. Background and Rationale

The design choices of SCEC Gen 3 were acceptable for establishing feasibility but limit any use beyond it. Specific Aim 1 showed that SCEC Gen 3 can lower the temperature of the deep posterior eye by holding a ring on the anterior sclera at 4 °C (Chapter 2). Its use outside a laboratory remained to be shown. The ring was machined from bulk aluminum, 2336 mm^3 of metal resting on the conjunctiva and under the eyelids, and it placed an electrical conductor in direct contact with the eye. Heat was carried away by a heat exchanger of similar volume and an ice-bath circuit, and the controller was assembled from separate boards.

The bulk of the ring also bears on where its cooling goes. A ring that fills the fornices contacts the eyelids over a larger area than the sclera. In the computational model of Gen 3 developed in this chapter, about 66% of the heat extracted over a 35-minute session came from the eyelids and only 33% from the eye. Heat drawn from the eyelids does not cool the target tissue, yet the module, the heat-rejection hardware, and the power supply are all sized by the total heat they must move.

Efficiency, the first of the four enhancement targets, is the share of the extracted heat drawn from the eye. Portability is the size of the ring, the heat-rejection hardware, and the control electronics. Safety covers the electrical isolation of the tissue from the powered module, surface temperatures below 43 °C, the limit for applied parts [**129**] that a clinical device will have to meet, and the behavior of the controller when a sensor or the module fails. Comfort is judged only by the dimensions and volume of the parts that contact the eye and eyelids.

No change may cost the cooling that Aim 1 established. Tissue is limited, each eye carries a single set of implanted sensors, and a new ring takes weeks to fabricate. For these reasons the refinements were explored first in a computational model, which holds every other condition fixed and evaluates each variant over a range of operating conditions. The designs that the model favored and that could be built were then tested in human donor eyes on the ex-vivo platform of Aim 1, paired within the same eyes. The control electronics were consolidated into a single board and verified on the bench.

The hypothesis is stated in Section 1.8. Whether cooling is preserved is judged at the proximal optic nerve, the one site at which the hypothesis is tested, from the difference from Gen 3 and the largest difference that its confidence interval allows, set against the excursion itself. Changes in efficiency and in the other targets are reported as measured differences from Gen 3.

3.2. Methods

3.2.1. Design Requirements and Refinements. Each refinement addresses one or more of the four targets and is judged against the Gen 3 device of Chapter 2. Table 3.1 records how far each was taken and where its evidence is reported.

Ring material. Aluminum conducts heat well but is an electrical conductor in contact with the eye. Candidate replacements were screened for thermal conductivity, electrical insulation, optical transparency, hardness, and use in medical devices, and two were carried forward. Aluminum oxide is a ceramic that combines moderate thermal conductivity with electrical insulation and can be made translucent or transparent, depending on how it is processed [**130, 131**]. Fused silica glass is transparent and insulating but conducts heat poorly. Fused silica was also retained because of its manufacturing route, which seemed of interest at first: parts of fused silica glass

TABLE 3.1. Refinements of SCEC Gen 4, the target of each, and how far each was taken; modeled refinements were evaluated only with the computational model of Section 3.2.2.

Refinement	**Target**	**Status**	**Evidence**
Ceramic and glass rings	Safety	Ceramic ring built and tested; glass modeled	Model (three materials); paired tests in human donor eyes (aluminum against ceramic)
Reduced-bulk ring (v2)	Portability, comfort	Built and tested	Model; paired tests in human donor eyes
Integral corneal lens (v3)	Efficiency, safety, comfort	Modeled	Model
Insulating outer layer	Efficiency	Modeled	Model (0.5 and 1.0 mm silicone)
Smaller heat exchanger	Portability	Modeled	Water-block model (three designs)
Integrated control board	Portability, safety	Built and bench-tested	Bench verification
Potting of the TEC	Safety	Built	Silicone injected with a curved-tip syringe to fill the annular module between its two ceramic plates and waterproof it, in every device used in this work
Coolant circuit	Portability	Built	Replacement pump and push-to-connect tubing fittings
Higher post density and printing on the ring	Efficiency, portability	Concept only	None; future work
Contact-pressure sensor, ERG electrode, injection port	Safety	Concept only	None; future work
Patient comfort	Comfort	Not measured; implied	Dimensions and volume only

can be molded or printed from a silica composite and then sintered [**132, 133**] (Glassomer GmbH, Freiburg, Germany), which would combine transparency with ease of manufacture.

Ring geometry. The Gen 3 ring (v1) is a bulk aluminum part that conforms to the anterior sclera and leaves the cornea exposed, with a volume of 2336 mm^3 and a scleral contact area of approximately 390 mm^2. The reduced-bulk ring (v2) keeps a similar contact area, 393 mm^2, with a thinner profile that nearly halves the volume, to 1191 mm^3. The third design (v3) adds an integral lens of 105 mm^3 over the cornea to the v2 profile (1296 mm^3, 45% less than v1). The lens extends the contact area to 619 mm^2, protects the cornea, and is intended to limit corneal drying.

Outer layer, heat rejection, and control. A silicone layer, 0.5 or 1.0 mm thick, on the outer walls of the ring was evaluated as a way to direct the extracted heat toward the eye and away from the eyelids. Three heat exchangers of decreasing volume were evaluated (Section 3.2.2.3), cooled with distilled water (Appendix E). The controller must hold the ring at its set point, nominally 4 °C, and stop the module when a sensor or the module fails. A clinical device will also have to keep every patient-contacting surface, in practice the hot side of the module and the heat exchanger, below 43 °C. Here that limit was used only in the heat-exchanger model, as the safety limit for the hot side.

3.2.2. Computational Model. Two finite-element models were first built and solved in COMSOL Multiphysics 6.4 (COMSOL AB, Stockholm, Sweden): one of the eye, orbit, and eyelids with the ring in place, to compare ring materials, geometries, and the insulating layer, and one of the heat exchanger on the hot side of the module. A Python program then replaced COMSOL, and the results of this chapter are those of the Python runs. The analysis is described in Section C.6.

3.2.2.1. *From COMSOL to Python.* Both COMSOL models were exploratory, built to compare design variants under identical conditions rather than to predict the temperatures of a particular eye. The eye model coupled conduction to flow in

the humors and the air in front of the eye, represented the module as a layer on the ring, and began cooling with the ring held at ambient temperature. Its results were computed on the finest of three meshes per design, about one million elements (Table 3.2 and Figure D.3; settings in Section C.6).

Errors in how the boundary conditions were applied were then found, among them the starting state of the ring and the emissivity of the ceramic and glass rings. By then the COMSOL license had ended, so the models could not be corrected in COMSOL. A finite-element program was therefore written in Python with Claude Code (Appendix J; folder `code/ch3-heat-replica` of the code repository, Section C.7), reading the meshes, materials, and boundary conditions from the COMSOL model files and solving heat transfer only. It was validated first against exact solutions and then against the stored COMSOL solutions, with the same set-up (Section 3.3.1).

The errors were then fixed. The eye now starts at rest, with the device in place and switched off, so that the anterior sclera begins near 36 °C instead of 27 °C. The air domain and the module layer are removed, each ring takes the emissivity of its material, and the tissues take the values of the property workbook (Appendix D). The heat is computed from the solved temperature field, across each surface of the ring and from the sources of the model. The same 27 operating points, three materials, six insulating layers, and rewarming cases were solved in this set-up. COMSOL has no solution of it, so these runs were validated internally, by their energy balance, time step, and mesh. Runs beyond the COMSOL cases are called extension runs. For the heat exchanger, the Python runs give each design the same heat.

3.2.2.2. *Scleral Contact Ring and Eye.*

Geometry. The model represents a human eye in the orbit, with the ring on the anterior sclera and the eyelids closed over its outer surface (Figure 3.1). The globe is divided into cornea, tear film, aqueous humor, lens, ciliary body, vitreous humor with Cloquet's canal, retina, choroid, and sclera, and is surrounded by orbital fat, the extraocular muscles, and the eyelids. The eye has the dimensions of an adult

TABLE 3.2. Meshes on which the results were computed: the finest mesh of the eye model for each ring, without the air and the module layer, and the intermediate mesh of each heat exchanger. Element quality is the equiangular skewness of the dihedral angles of each tetrahedron (1 for a regular tetrahedron).

	Eye model			Heat exchanger		
Statistic	**v1**	**v2**	**v3**	**v1**	**v2**	**v3**
Mesh vertices	165 472	163 912	171 043	35 977	54 051	35 449
Tetrahedra	946 374	940 447	979 821	207 341	309 912	192 068
Boundary triangles	123 762	121 606	130 148	20 643	34 437	27 349
Minimum element quality	0.100	0.030	0.008	0.066	0.159	0.030
Average element quality	0.669	0.669	0.666	0.675	0.669	0.673
Mesh volume (cm^3)	19.88	19.53	19.59	4.52	2.65	0.94

human globe (Appendix D), and the orbit is a cylinder of radius 17.5 mm. The three rings were placed on the same eye. The model is symmetric about the vertical plane through the optical axis and only half was solved; heat flows are reported for the whole device.

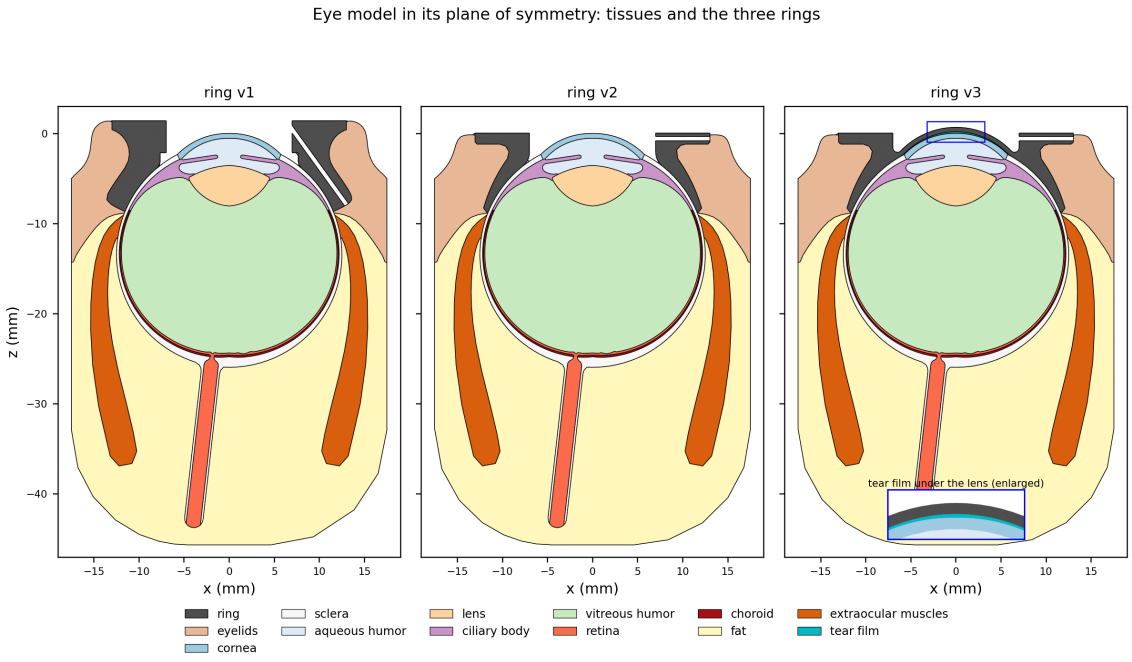


FIGURE 3.1. The eye model with the three scleral contact rings: v1, the original ring of SCEC Gen 3; v2, the reduced-bulk ring; v3, the reduced-bulk ring with an integral corneal lens. The plane shown is the plane of symmetry, the cut of the half model that was solved; colors mark the domains, and the inset enlarges the tear film beneath the lens of v3.

Heat transfer and boundaries. Heat conduction was solved in the tissues and the ring, with the humors at rest. In the perfused tissues (choroid, retina, ciliary body, extraocular muscles, eyelids, and orbital fat), the heat equation carries the Pennes terms for perfusion and metabolic heat [**134**],

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \omega \, \rho_b c_b \, (T_b - T) + Q_{\mathrm{met}}, \tag{3.2}$$

where T is the temperature of the tissue at time t, ρ, c_p, and k are the density, specific heat, and thermal conductivity of the tissue, ω its perfusion rate, ρ_b and c_b those of blood, T_b the body temperature, and Q_{met} the metabolic heat (Table 3.3; sources in Appendix D). The choroid, with the highest perfusion, is the dominant perfused heat source opposing cooling. Its perfusion comes from microsphere measurements in monkey eyes [**40**].

The outer shell of the orbit is held at T_b. The exposed surfaces lose heat to the room, at the ambient temperature T_0, through a coefficient of 10 $\mathrm{W\,m^{-2}\,K^{-1}}$ and by radiation, and the cornea also by tear evaporation. In v3 the lens covers the ocular surface, so neither tear evaporation nor exposure applies to the eye (Figure D.1). The face of the ring that met the module is free in the starting state, the eye at rest with the device switched off, and is then brought from its starting temperature to the set point T_c along a step of 90 s. The transient was computed for 35 min. Further details are given in Appendix D.

Conditions. At the baseline condition (ambient 25 °C, body 37 °C, set point 4 °C) the ring material was varied among aluminum, aluminum oxide, and fused silica, and the three geometries were compared directly. Each geometry was also solved for every combination of ambient temperature (10, 25, and 35 °C), body temperature (35, 37, and 39 °C), and set point (4, 8, and 12 °C), 27 conditions per ring, to test whether the comparison depends on the operating point. A silicone layer of 0.5 or 1.0 mm was added to the outer walls, and rewarming after switch-off was simulated for a subset of conditions.

TABLE 3.3. Thermal properties, perfusion rates, and metabolic heat of the domains of the eye model. The properties of the tear film and the ring depend on temperature and are given at 20 °C (aluminum oxide, for example, 37.0 at 4 °C and 32.1 $\mathrm{W\,m^{-1}\,K^{-1}}$ at 37 °C).

Domain	**Conductivity**	**Density**	**Specific heat**	**Perfusion**	**Metabolic heat**
Symbol (units)	k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	ρ ($\mathrm{kg\,m^{-3}}$)	c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	ω ($\mathrm{s^{-1}}$)	Q_{met} ($\mathrm{W\,m^{-3}}$)
Cornea	0.56	1070	3707	–	–
Aqueous humor	0.593	1005	4159	–	–
Lens	0.43	1126	3255	–	–
Vitreous humor	0.59	1010	4134	–	–
Ciliary body	0.49	1088	3556	0.0064	685
Retina	0.55	1014	3958	0.0039	3443
Choroid	0.51	1074	3682	0.0927	624
Sclera	0.545	1102	3431	–	–
Extraocular muscles	0.49	1090	3421	0.0045	686
Eyelids (skin)	0.34	1070	3660	0.0011	644
Orbital fat	0.2245	920	2500	0.0004	199
Tear film	0.60	998	4180	–	–
Ring, aluminum	236	2700	903	–	–
Ring, alumina	34.3	3990	764	–	–
Ring, fused quartz	1.37	2220	705	–	–
Silicone layer	0.26	1280	1781	–	–

3.2.2.3. *Heat Exchanger.* Each heat exchanger is an aluminum annulus around the module with fins on alternating walls that force the coolant along a zig-zag path (Figure 3.2 and Table 3.4). The v3 block has no lid of its own and is closed by the hot-side ceramic plate of the module, which the coolant touches directly. Each was modeled whole. The heat from the module entered as a uniform flux over its footprint, an annulus of inner radius 7 mm and outer radius 13 mm, the same in the three designs, and each received the same heat: 10 W, the operating load, and 15.5 W, a conservative upper value. In v3 most of that annulus is the coolant itself, because the model has no plate closing the block. The heat left with the coolant; the outer walls and the module plane outside the footprint lost a small amount of heat to the ambient air.

The water flow and the temperature were solved with the Python program for every combination of flow rate (0.10, 0.25, and 0.50 $\mathrm{L\,min^{-1}}$) and inlet temperature (4, 10, and 20 °C), and the temperature was advanced in time with the flow where

the flow was unsteady. The surfaces on which these conditions act are shown in Figure D.2.

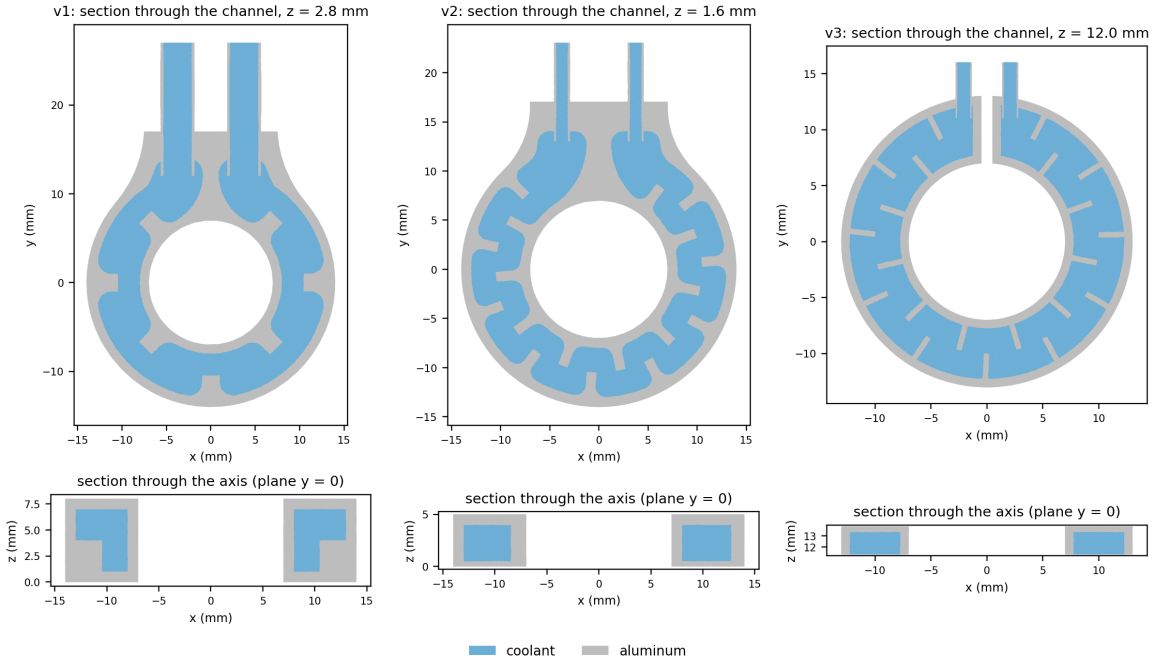


FIGURE 3.2. The three heat-exchanger designs as modeled: v1, that of SCEC Gen 3, with seven fins filling half the channel height; v2, a lower channel and 18 thinner fins; v3, the lowest channel and 21 fins, closed by the hot-side ceramic plate of the module instead of a lid. Coolant (blue) and aluminum (gray) in a horizontal section at 30% of the channel height, where the fins of v1 are (top), and in the vertical section through the axis (bottom). Dimensions in Table 3.4.

TABLE 3.4. Geometry of the three heat-exchanger designs. Volumes measured on the CAD models.

Feature	**v1 (Gen 3)**	**v2**	**v3**
Volume (mm^3)	2337	1619	435
Annulus, inner/outer radius (mm)	7.00/14.00	7.00/14.00	7/13
Channel height (mm)	6.0	3.5	2.0
Fins: number, thickness (mm)	7, 2.0	18, 1.0	21, 0.5
Fin height	half channel	full channel	full channel
Inlet and outlet diameter (mm)	4.00	2.25	1.41
Lid	separate	separate	module plate

3.2.2.4. *Outcome Measures.* Temperatures were read at the sites measured in Chapter 2 (Figure 3.3): the anterior, equatorial, and posterior sclera, the vitreous,

and the proximal optic nerve, at its junction with the globe. A distal optic-nerve probe, 1 cm behind the globe, recorded the effect of cooling behind the eye, and domain probes the average, minimum, and maximum temperature of the retina and the whole eye. The steady temperature at each site is a weighted mean of the body and ring temperatures. The principal outcome was thus the cooling effectiveness

$$\varepsilon = \frac{T_b - T_{\text{site}}}{T_b - T_{\text{ring}}} \times 100\%, \tag{3.3}$$

where T_b is the body temperature, T_{site} the steady temperature at the site, and T_{ring} the final temperature of the ring. It is 0% for a site left at body temperature and 100% for a site cooled to the temperature of the ring. It does not depend on the operating point: a linear fit of the steady temperatures over the 27 operating points of each ring reproduced every site to within 0.003 °C, and less than 1% of a change in ambient temperature reached any site. It also gives directly the ring temperature needed to reach a target. Kinetics were summarized by t_{90}, rewarming by the time each site stayed below 35 °C after switch-off, and efficiency by the share of the heat drawn from the eye rather than the eyelids and air. For the heat exchanger, the outcomes were the rise of the average temperature of the heated face above the coolant and the peak temperature of that face against the 43 °C limit.

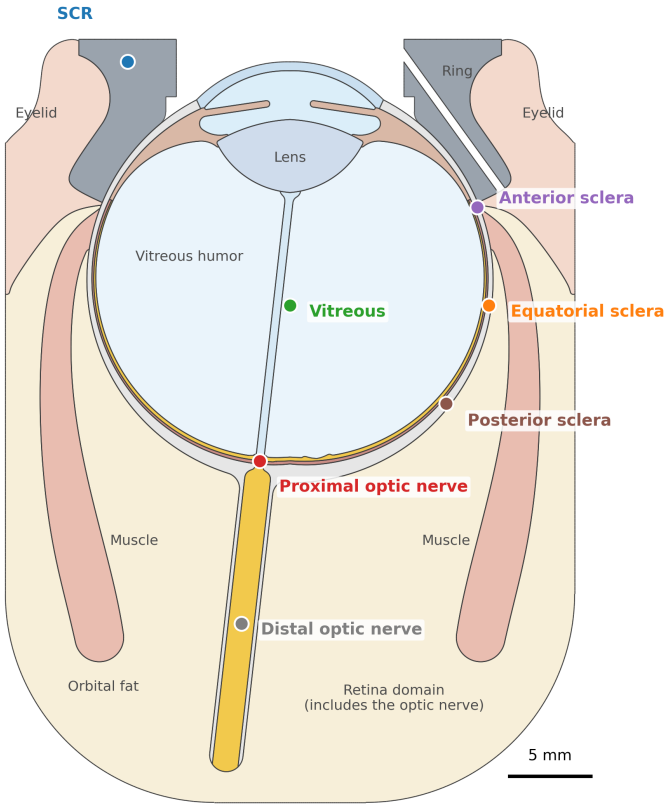


FIGURE 3.3. Probe locations on the section of the model in its plane of symmetry (v1 ring): the anterior, equatorial, and posterior sclera, the vitreous, the proximal optic nerve, the distal optic nerve (1 cm behind the globe), and the SCR. The SCR probe, on the rim of the ring 10 mm from the axis and out of this plane, is drawn turned into it. Yellow, the retina domain, including the optic nerve, over which the average retina is taken.

3.2.3. Experimental Ring Comparisons. A ceramic ring of the original geometry and the reduced-bulk ring were each compared with the original aluminum ring in human donor eyes, to test whether an electrically insulating material, or halving the volume, costs cooling.

Rings. The ceramic ring reproduced the original geometry in alumina, fired from the same design file without further machining (Ceramco, Inc., Center Conway, NH, USA). The reduced-bulk ring (v2) was machined from aluminum with the contact surface of the original ring and a thinner profile, which reduced the height of the assembled device from 23 to 20 mm. Each ring was a complete device built as SCEC Gen 3 (Figure 3.4).

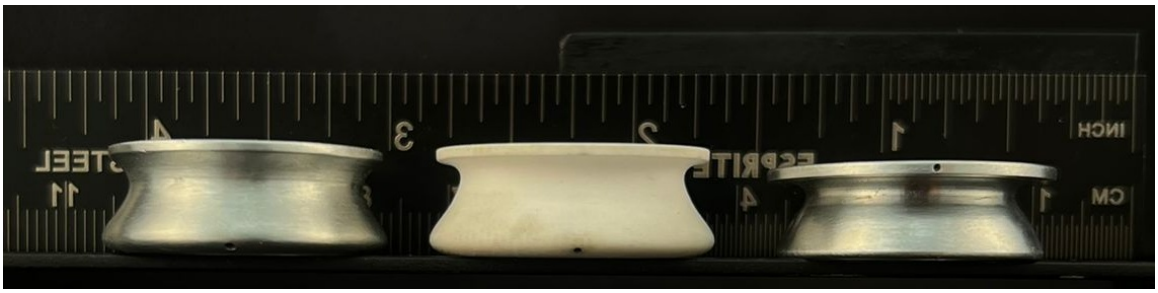


FIGURE 3.4. The three scleral contact rings tested in human donor eyes, side view on a ruler (inches above, centimeters below): (left) the original aluminum ring of SCEC Gen 3 (v1), (center) the alumina ring of the original geometry, and (right) the reduced-bulk aluminum ring (v2). The photograph is mirrored, so the ruler reads in reverse.

Protocol. The nine human donor eyes of Chapter 2 were each cooled with all three rings on the same day, with the same implanted sensors, under the protocol of Section 2.2.5, re-equilibrating in the bath between rings, in an order that varied between eyes. The comparisons are therefore paired within each eye, and differences in sensor placement, donor, and anatomy cancel.

Analysis. The recordings were processed with the pipeline and endpoints of Section 2.2.6. A sensor excluded by the spatial screen was excluded under all three rings, and one further trace, the vitreous sensor of eye 4 under the reduced-bulk ring, was removed on visual review for the same oscillation it showed under the original ring. In all, 23 of the 162 traces recorded with the three rings were excluded (11 for a hardware fault, 3 for a poor fit, 6 by the spatial screen, and 3 on visual review). This left 45, 47, and 47 traces for the original, ceramic, and reduced-bulk rings. One of the excluded traces is the proximal optic nerve of eye 6 under the original ring, and the comparisons with that ring at the site of the hypothesis rest on eight eyes. Each ring was compared with the original by the procedure of the species comparison in Chapter 2. Paired t tests were run at each site with Holm correction across the five tissue sites. Each difference in °C is reported with its 90% confidence interval. The paired standardized difference g_z (the mean paired difference over the SD of the differences, with the same small-sample correction) is reported with its 95% confidence

interval. The limit of that interval farther from zero is the largest difference that the data allow (Section 2.2.6). The hypothesis is tested at the proximal optic nerve.

Comparison with the model. As a supplementary step, the measurements were set beside the model through the same quantity for both, the fraction of each site's own starting difference from the ring that the device removed,

$$\varepsilon_{\Delta} = \frac{T_{\text{initial}} - T_{\text{final}}}{T_{\text{initial}} - T_{\text{ring}}} \times 100\%, \tag{3.4}$$

where T_{initial} and T_{final} are the temperatures of the site when the ring was switched on and at the end of cooling, and T_{ring} the final temperature of the ring. Model and measurements were also compared through t_{90}. Both start from an eye at rest with the unpowered ring in place: in the model the eye has settled with the device off, and ex vivo the ring equilibrated with the eye in the 37 °C bath. The model is read after 35 min and the experiments over minutes 16 to 20. Each ring was matched to its model design (original ring to v1, reduced-bulk ring to v2, ceramic ring to the alumina v1 run), at a body temperature of 37 °C and a set point of 4 °C. Because the model includes perfusion and the eyelids and the experiments do not, the comparison is not a validation of the model. It was read for the order of the sites, by Spearman's rank correlation (not tested, with five sites), and for the direction of the differences, by the ratio of measured to modeled effectiveness at each site.

3.2.4. Integrated Control Board. The separate boards of the Gen 3 controller, a thermistor conditioning circuit, a comparator that switched the module through a relay, and a laboratory supply (Section 2.2.1), were consolidated for SCEC Gen 4 onto a single printed circuit board designed for this work (Figure 3.5). Its firmware, schematics, and drawings are given in Appendix F.

The microcontroller runs a digital proportional–integral–derivative loop on the ring thermistor at 5 Hz and sets the drive of a thermoelectric-module controller. Hard limits on the module current and voltage are set by trimmers, independently of the firmware, and the module is off by default. A safety routine running every 50 ms

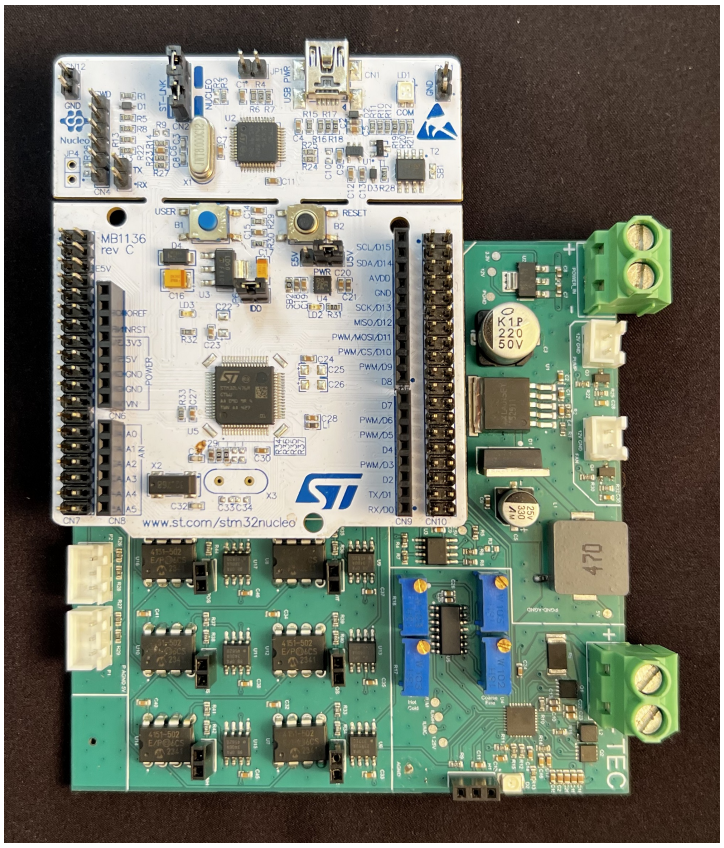


FIGURE 3.5. The integrated control board (revision 1.0): the microcontroller board on its headers (upper left), the six thermistor channels (lower left), and the power and drive stage (right), with the screw terminals for the 12 V supply (upper right) and the thermoelectric module (lower right) and the pump and fan connectors between them.

stops the module and latches the fault if the ring thermistor reads open or out of range, the 2.5 V reference leaves its band, or the module current is abnormal, and a hardware watchdog resets the microcontroller if its program stops (Section F.1).

Bench verification. The board was verified on the bench in stages, from unpowered checks and operation into dummy loads to closed-loop regulation of a ring on a preserved porcine eye in the bath, with an 18-min pull-down and hold at 4 °C (Section F.2). Three safety cutouts were triggered on purpose, by disconnecting the ring thermistor and the module and by stalling the program. The current monitor was compared with a multimeter in series with the module. Regulation was summarized

by the time to reach the set point and the largest deviation while held, the current monitor by its largest deviation from the multimeter.

3.3. Results

3.3.1. Validation of the Python Program. The Python program agreed with the solutions stored by COMSOL within 0.06 °C at every probe and stored time of the 104 cooling cases, and within 0.008 °C at the end of cooling. In the 22 rewarming cases it agreed within 0.054 °C at the end, and within 0.075 °C at every tissue probe during the run with short steps after the release; the largest difference, 0.11 °C, was at the v3 ring 0.3 min after the release in the coldest case. All nine material cases agree within 0.03 °C.

These checks show that the program solves the COMSOL models correctly and that it is consistent with the experiment. The corrected runs use the same solver and meshes. They do not validate the model against a perfused eye. The comparisons with the eyes of patients and with the published models are in Section 3.4.

The eye model converged at the deep sites: from the intermediate to the finest mesh the final temperature changed by at most 0.04 °C at the optic nerve and the posterior sclera and 0.11 °C at the equatorial sclera, the vitreous, and the average retina, and differences between designs smaller than this are not interpreted. The anterior sclera, next to the ring, did not converge, and refinement around the ring put it up to about 1.3 °C higher, so temperatures there are lower bounds (Section D.6).

In the COMSOL heat-exchanger model the heat carried away by the coolant matched the heat supplied to within 1.2% on average and about 5% at every operating point, and the Python runs closed within 0.2%. The COMSOL solution changed by 0.03 °C or less between the two finest meshes, but the Python program agreed with it only within 0.6 °C at the module face and 0.04 °C at the outlet. The temperature of the module face is thus known to about 1 °C, and the values on these meshes are low (Section D.6). The COMSOL sweep gave v3 about a fifth less heat than v1 and

v2, so the results below come from the Python runs with the same heat on the three designs, on the intermediate mesh. The flow was steady only in v1, at 0.10 $\mathrm{L\,min^{-1}}$ and at 0.25 $\mathrm{L\,min^{-1}}$ with the 4 °C inlet, and elsewhere fluctuated by 0.6 to 7.3% of the pressure drop.

3.3.2. Ring Material. In the model, an aluminum-oxide ring cooled the eye almost as well as an aluminum ring of the same geometry, and a fused-silica ring did not (Figure 3.6A). After 35 min at the baseline condition, the aluminum-oxide ring kept 89 to 95% of the cooling effectiveness of aluminum at every site and with every geometry. With the original geometry it kept 94% at the proximal optic nerve, and its final temperatures were 0.2 °C warmer there and 2.2 °C warmer at the anterior sclera, the site next to the ring. Fused silica kept only 29 to 46% of the effectiveness of aluminum. With the original geometry its proximal optic nerve ended at 35.7 °C against 33.8 °C with aluminum, and its anterior sclera at 28.4 °C against 9.3 °C. The ring itself, held at the same cold face, ended at 7.7 °C instead of 4.1 °C. The glass conducted too little heat to cool the eye in a useful way, and on this result fused silica was dropped without a ring being built.

In the human donor eyes, the ceramic ring and the aluminum ring of the original geometry cooled the proximal optic nerve equally, by 3.7 and 3.8 °C on average. The paired difference, aluminum minus ceramic, was −0.20 °C (90% confidence interval −0.82 to +0.43 °C, eight eyes), and no difference was detected. A difference larger than 1.0 °C is unlikely with the eyes of each donor averaged, and one larger than 0.8 °C with each eye counted separately (Figure 3.7A; Section 3.3.3). The same held for the ring itself (+0.16 °C, −0.41 to +0.73), which ended at 4.1 °C with either material. The ceramic ring took longer, however, to reach 90% of its excursion, 174 s against 124 s on average. At the other four tissue sites the ceramic ring cooled more on average, by 0.9 to 1.7 °C (Figure 3.7A). No difference was significant after Holm correction ($p \geq 0.38$). With intervals that allow differences of 2.4 to 4.3 °C, these sites say little either way.

The model and the experiment agree at the proximal optic nerve, where both find no meaningful difference between the two materials. The model predicts 0.23 °C more cooling with aluminum there, inside the experimental confidence interval. At the other tissue sites they differ in direction. The model predicts a small advantage for aluminum, at most 2.3 °C, whereas the experiment found a trend, not significant, toward more cooling with ceramic, and the model's difference lies outside the experimental interval at the anterior and posterior sclera. An electrically insulating alumina ring thus preserved cooling at the proximal optic nerve, and the glass alternative was excluded by the model alone.

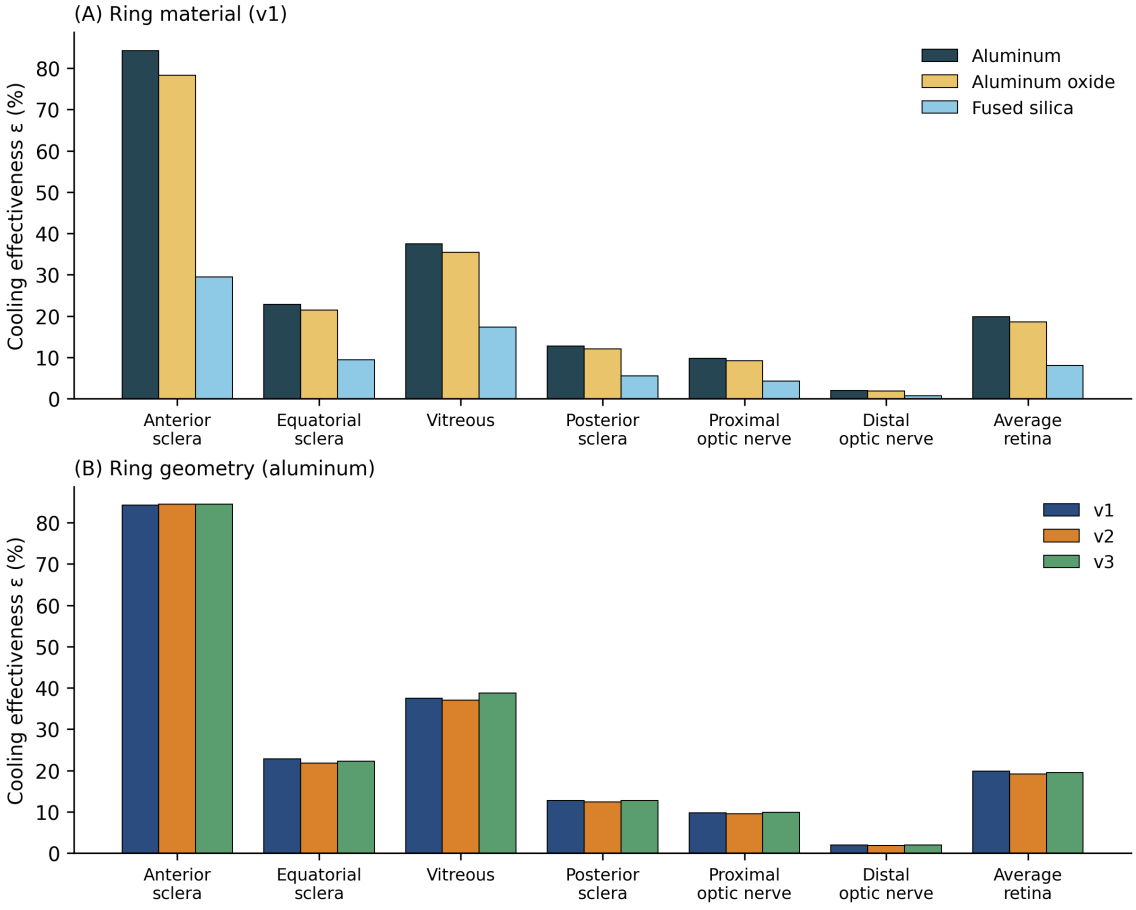


FIGURE 3.6. Cooling effectiveness ε at each site in the model after 35 min of cooling at the baseline condition: (A) v1 rings of aluminum, aluminum oxide, and fused silica; (B) aluminum rings of the three geometries.

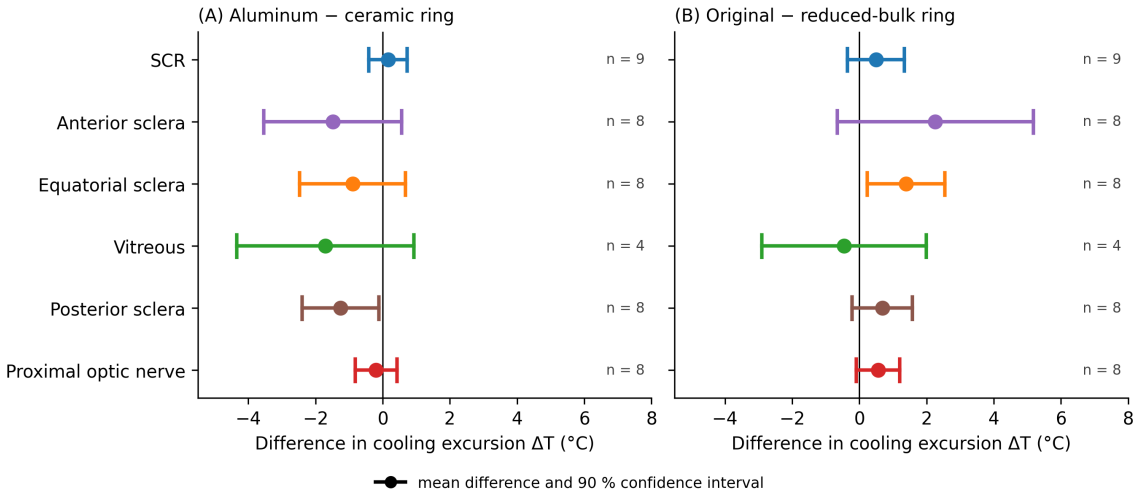


FIGURE 3.7. Paired differences in cooling excursion at each site in human donor eyes between the original ring and (A) the ceramic ring, (B) the reduced-bulk ring: mean with its 90% confidence interval, whose limit farther from zero is the largest difference the data allow. n, eyes with a valid pair.

3.3.3. Ring Geometry. In the model, the three geometries cooled the eye almost identically (Figure 3.8). The cooling effectiveness of the three rings differed by at most 2 percentage points at any site, and by 3 for the average over the whole eye. At the proximal optic nerve it was 9.8, 9.6, and 9.9% for v1, v2, and v3 (Figure 3.6B).

Across all 27 operating points the cooling excursion with each ring was a nearly fixed fraction of that with the original ring at every site (with the reduced-bulk ring, 97.8 to 99.5% of it at the proximal optic nerve). Thus the comparison does not depend on the body, ambient, or set-point temperature. At the baseline condition the reduced-bulk ring cooled the deep sites slightly less than the original: by 0.05 °C at the proximal optic nerve, 0.09 °C at the posterior sclera, 0.1 °C at the vitreous, and 0.3 °C at the equatorial sclera. These differences exceed the change of the results between the two finest meshes (Table D.2) but are small, 0.6 to 3.9% of the excursion of each of these sites (at the proximal optic nerve, 1.7% of an excursion of 3.1 °C). Adding the corneal lens (v3) restored the original ring's final temperatures at the proximal optic nerve and posterior sclera, to within 0.04 °C. It also cooled the vitreous more than either open ring, by 0.7 °C relative to v1 and 0.8 °C relative to v2, the

largest difference between any two designs. The time to 90% of the final cooling was the same for all three rings to within 1 min at every site (Section 3.3.5).

The reduced bulk did change how much heat the ring drew (Figure 3.9). Over 35 min of cooling the reduced-bulk ring drew 14% less heat than the original, and almost all of the saving came from the eyelids. The heat drawn from the eye was nearly unchanged, and its share, the measure of efficiency, rose from 33% to 37%. The ring with the corneal lens (v3) drew about as little heat as the reduced-bulk ring, and 38% of it from the eye, because the lens adds contact with the cornea. Followed back to its sources, the 7.7 kJ drawn by the original ring came 48% from the wall of the orbit, 33% from the blood, 14% from heat stored in the tissues, and 6% from the room.

In the human donor eyes, the reduced-bulk ring cooled the proximal optic nerve by 3.1 °C on average against 3.8 °C for the original ring. The paired difference, original minus reduced-bulk, was +0.56 °C (90% confidence interval −0.09 to +1.20 °C, eight eyes). The reduced-bulk ring thus cooled the proximal optic nerve 0.6 °C less on average, not significantly, and by no more than 1.2 °C (Figure 3.7B). At the posterior sclera the difference was +0.68 °C (−0.21 to +1.58). At the other three sites the intervals were wider, 2.3 to 5.8 °C across (Figure 3.7B). No difference was significant after Holm correction ($p \geq 0.29$). The ring itself held its set point equally well with either geometry, ending at 4.0 to 4.1 °C, and the reduced-bulk ring reached 90% of its excursion sooner, in 82 s against 124 s on average. The comparisons were repeated with the eyes of each donor averaged (five donors) and with the vitreous traces removed on visual review put back. Neither repeat changed a conclusion.

Model and experiment agree at the proximal optic nerve, the site of the hypothesis. The model predicts that the original ring cools it 0.05 °C more than the reduced-bulk ring, inside the experimental interval, and both find that the two rings cool it almost equally. Both also find the original ring cooling the posterior and equatorial sclera slightly more, by 0.7 and 1.4 °C in the experiment and by 0.09 and 0.3 °C in the model.

Compared, as for the materials, on the cooling excursion, the model's differences lie inside the experimental confidence intervals at all five sites. At the vitreous and the anterior sclera the two disagree in sign. Halving the volume of the ring kept the cooling at the proximal optic nerve, the site at which the hypothesis of this aim is tested, at a cost of about 0.6 °C, and of no more than 1.2 °C, of the 3.8 °C excursion.

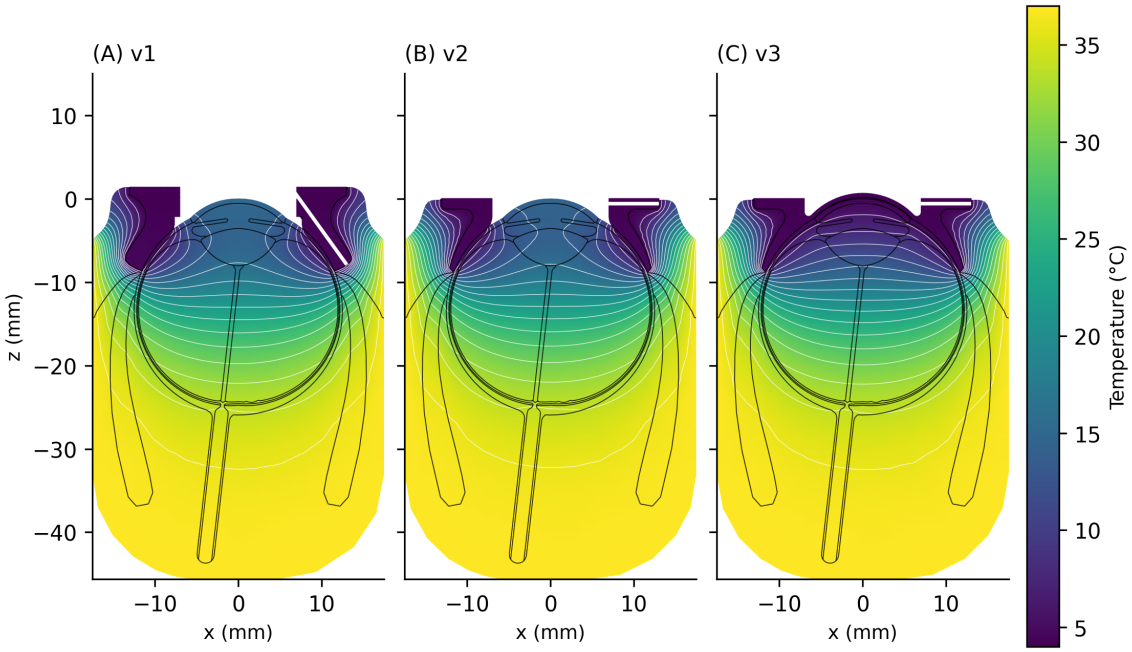


FIGURE 3.8. Temperature in the plane of symmetry after 35 min of cooling with (A) v1, (B) v2, and (C) v3, at the baseline condition, on a common color scale. White lines are isotherms every 2 °C; black lines outline the ring and the tissues.

3.3.4. Insulating Outer Layer. The insulating layer did not improve cooling. It was evaluated only in the model. With a layer of 0.5 or 1.0 mm on the outer walls of any of the three rings, the sites ended within −0.05 to +0.33 °C of the ring without a layer after 35 min at the baseline condition. The change was largest at the anterior sclera, 0.3 °C with the 1.0-mm layer on the three rings, the size of the change between meshes at that site. It was smallest at the deep sites: at most 0.03 °C at the proximal optic nerve, comparable to the change between the two finest meshes there, and at

most 0.01 °C at the distal optic nerve. The cooling effectiveness fell by no more than 1 percentage point at any site.

The layer did change where the ring drew its heat (Figure 3.9). For the original ring the total heat drawn over 35 min fell by 5 and 7% with layers of 0.5 and 1.0 mm and that drawn from the eyelids by 8 and 11%, so the share drawn from the eye rose from 33% to 35 and 36%, and the other rings behaved alike.

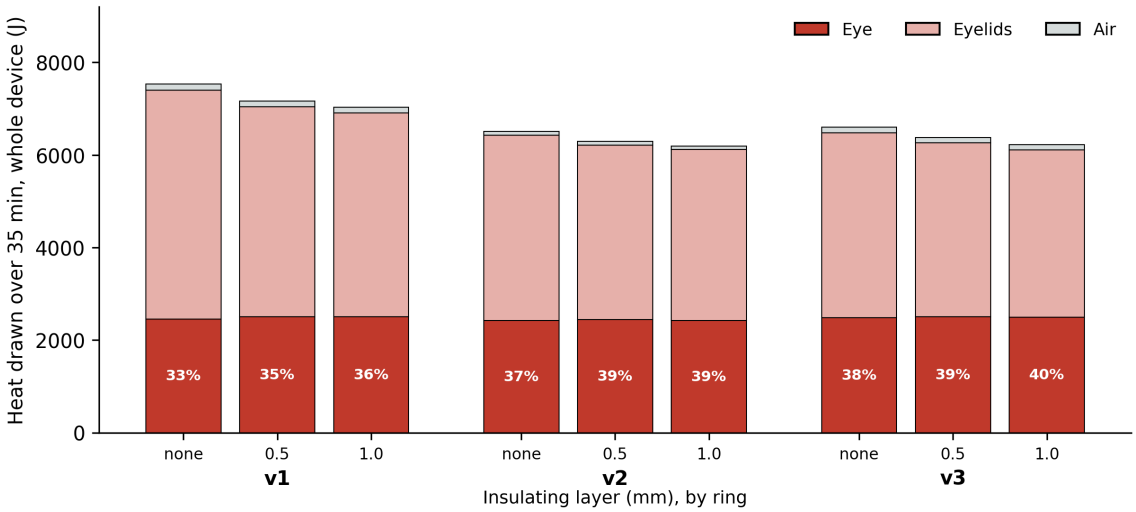


FIGURE 3.9. Heat drawn by the whole device over 35 min from the eye, the eyelids, and the air, for each ring without and with a 0.5 or 1.0 mm insulating layer; the share from the eye is printed in the eye part of each bar. Heat reaching the lens of v3 through the tear film counts with the eye, and heat through a layer with the eyelids.

3.3.5. Cooling Targets and Kinetics. The steady temperature at each site depends linearly on the body and ring temperatures. Each site is thus a weighted mean of the body and ring temperatures, $T_{\text{site}} = (1 - \varepsilon)\, T_b + \varepsilon\, T_{\text{ring}}$, with ε as a fraction and the symbols as in Equation 3.3, which gives the ring temperature that brings a site to a target (Figure 3.10). The retina itself is cooled along a gradient, from its anterior edges next to the ring to the posterior pole (Figure 3.12). At a body temperature of 37 °C, the proximal optic nerve reaches 35 °C with the ring at 16.6 °C or colder, and with the ring at the 4 °C floor of the device it settles at 33.8 °C. Bringing it to 32 °C would need a ring at −14.1 °C, which the device cannot

provide. The shallower sites need less: the average retina reaches 32 °C with the ring at 11.7 °C (Figure 3.10). The distal optic nerve, 1 cm behind the globe, stayed within 1 °C of body temperature at every operating point and cannot be brought to either target.

The body temperature shifts every requirement: at 35 °C the proximal optic nerve reaches 32 °C with the ring at 4.3 °C, while at 39 °C it needs −1.9 °C to reach even 35 °C. The three ring geometries require ring temperatures within about 2.1 °C of one another (2.9 °C at a body temperature of 39 °C) for every site other than the distal optic nerve and for both targets.

Cooling was fastest next to the ring and slowest at depth (Figure 3.11). Over the operating points, the median time to 90% of the final cooling rose with depth, from the anterior sclera to the distal optic nerve (Table 3.5). These times varied by at most about 0.5 min with the operating point and the ring geometry.

After the device was switched off, the eye rewarmed over minutes (Figure 3.11). At a body temperature of 37 °C, the proximal optic nerve and the posterior sclera rose above 35 °C first and the vitreous and the anterior sclera last, the time at each site depending on the set point and the geometry (Table 3.5). At 39 °C the proximal optic nerve never fell below 35 °C.

TABLE 3.5. Times in the model: median time to 90% of the final cooling over the operating points; and, at a body temperature of 37 °C over the set points and ring geometries, the time from switch-on until each site fell below 35 °C and the time it stayed below 35 °C after switch-off. –, the distal optic nerve did not fall below 35 °C.

Site	**Time to 90% (min)**	**Time to below 35 °C after switch-on (min)**	**Time below 35 °C after switch-off (min)**
Anterior sclera	1.6	0.3	13.5 to 15.9
Equatorial sclera	8.8	2.0 to 2.5	7.9 to 9.8
Vitreous	10.4	2.5 to 2.8	13.1 to 15.1
Average retina	10.0	1.6 to 2.4	7.4 to 9.5
Posterior sclera	12.4	5.5 to 7.2	6.7 to 9.0
Proximal optic nerve	14.3	8.4 to 12.1	5.8 to 8.6
Distal optic nerve	20.0	–	–

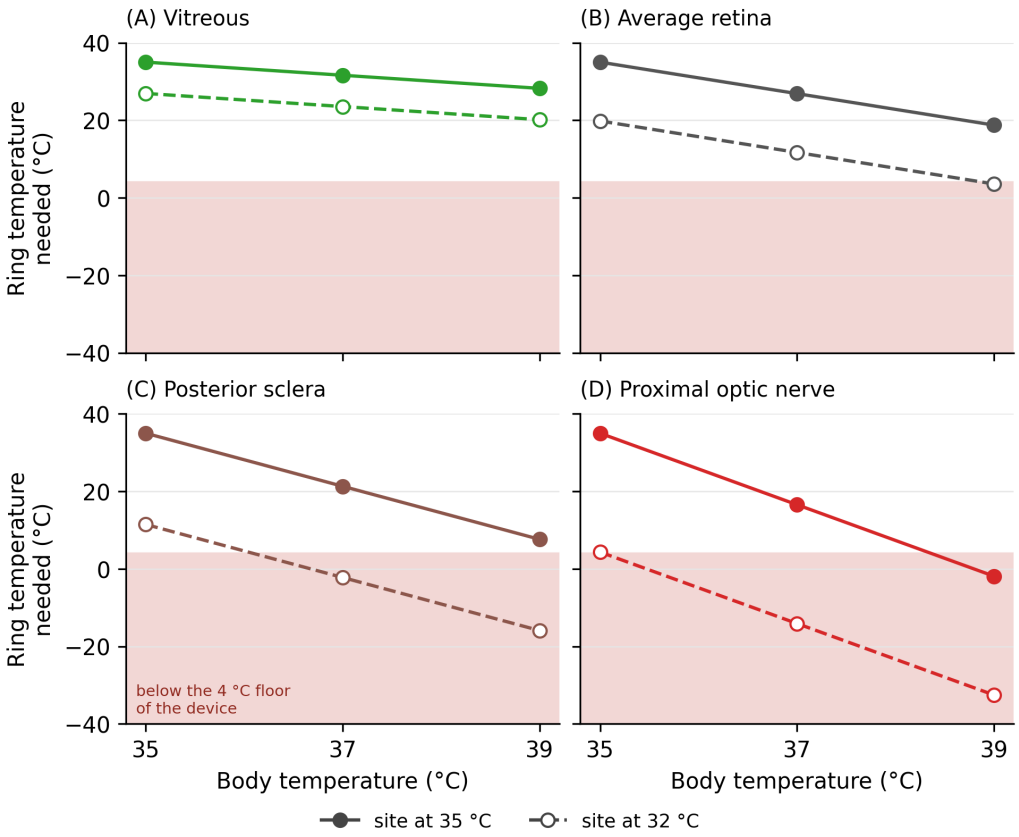


FIGURE 3.10. Ring temperature needed, from the effectiveness of v1, to bring (A) the vitreous, (B) the average retina, (C) the posterior sclera, and (D) the proximal optic nerve to 35 °C (solid, filled markers) and 32 °C (dashed, open markers) at steady state, against body temperature. Shaded, below the 4 °C floor of the device.

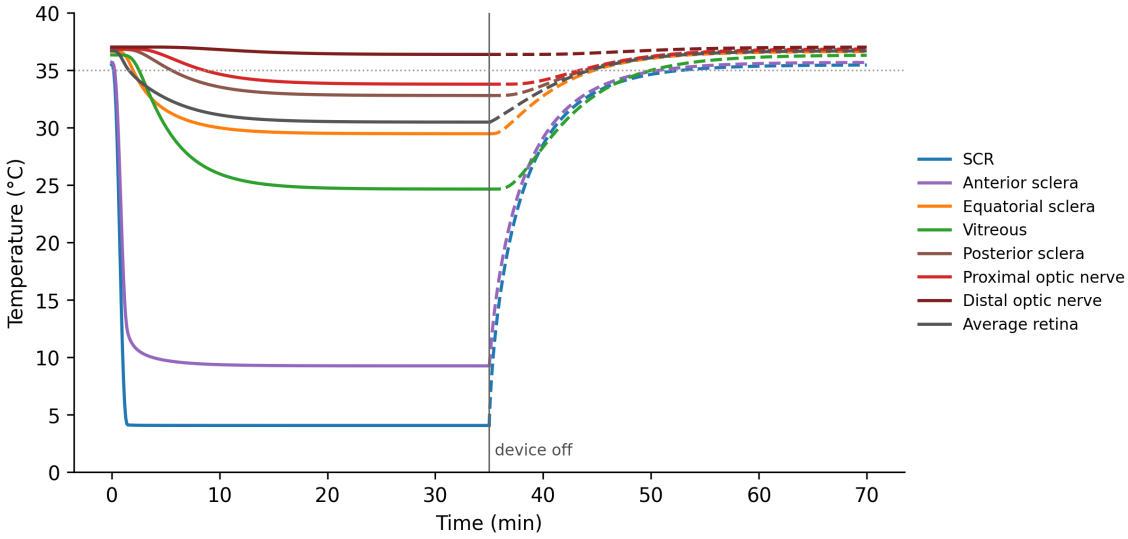


FIGURE 3.11. Temperature at each site of the model during 35 min of cooling with the original ring (v1, solid) and after switch-off (dashed), at the baseline condition (body 37 °C, ambient 25 °C, set point 4 °C); dotted line, 35 °C.

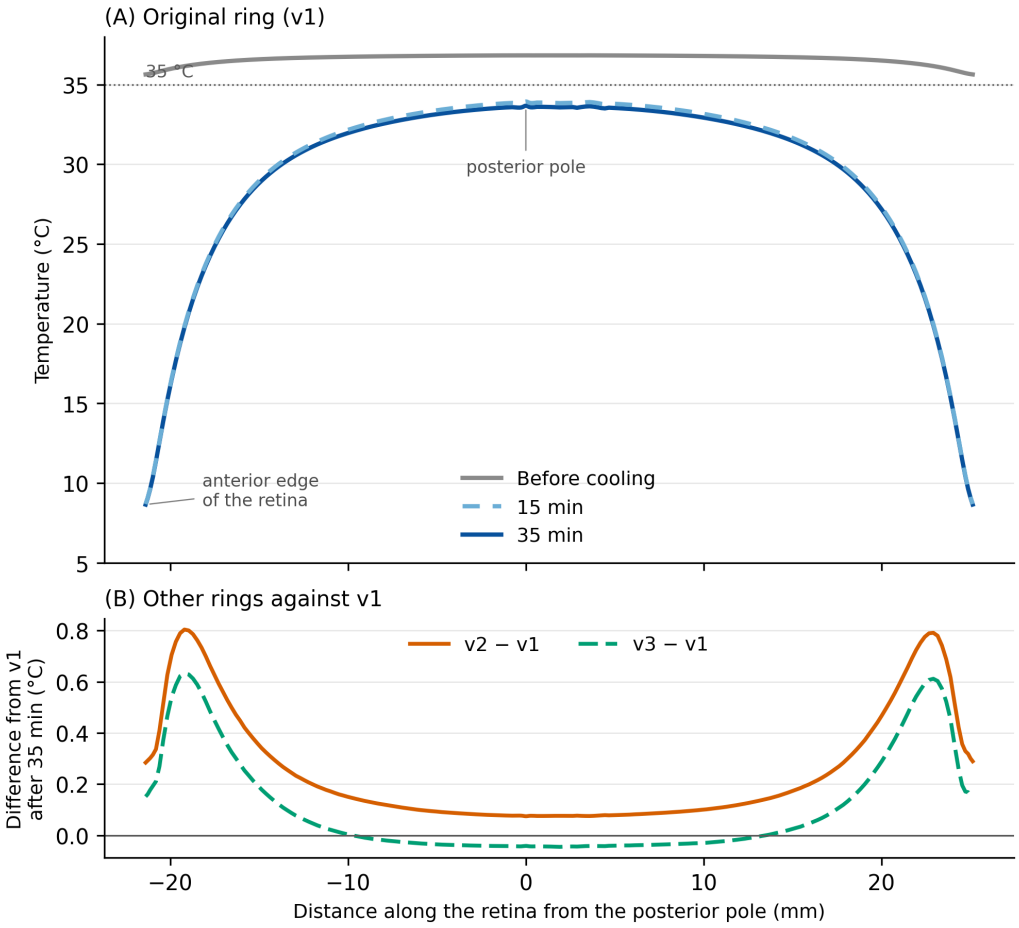


FIGURE 3.12. Temperature along the whole arc of the retina in the plane of symmetry of the model, from one anterior edge through the posterior pole to the other, at the baseline condition. (A) Original ring (v1), before cooling and after 15 and 35 min (these two nearly coincide); dotted line, 35 °C. (B) Difference of v2 and v3 from v1 after 35 min.

3.3.6. Model versus Ex-Vivo Measurements. With each of the three rings, the model and the donor eyes ranked the five sites in the same order (Spearman $\rho = 1$): both cooled the anterior sclera the most and the proximal optic nerve the least. The amounts matched only at the anterior sclera, next to the ring, where the measured ε_Δ was 0.87 to 1.12 times the modeled value (Figure 3.13). At the deep sites the donor eyes cooled more than the model, 1.1 to 2.2 times as much, with the smallest gap at the proximal optic nerve (1.1 to 1.4 times). These factors are approximate, because the sensors at depth did not sit exactly at the points of the model. The sensors named posterior sclera lay at 66% of the axial length from the anterior pole on average, and the posterior point of the model at 81%.

Most of this difference comes from the difference between the two conditions: modeled as in the experiment, enucleated, in oil at 37 °C and without perfusion, the model cooled the equatorial and posterior sclera and the vitreous about as much as the donor eyes or more, depending on the heat-transfer coefficient of the oil, which was not measured. The donor eyes also cooled faster at depth, reaching 90% of their cooling in 11.0 to 11.4 min at the proximal optic nerve against 14.3 to 14.6 min in the model, 8.8 to 9.1 against 12.4 to 12.7 min at the posterior sclera, and 6.9 to 7.6 against 10.4 to 10.8 min at the vitreous, and more slowly at the anterior sclera, 3.6 to 3.9 against 1.5 to 2.2 min.

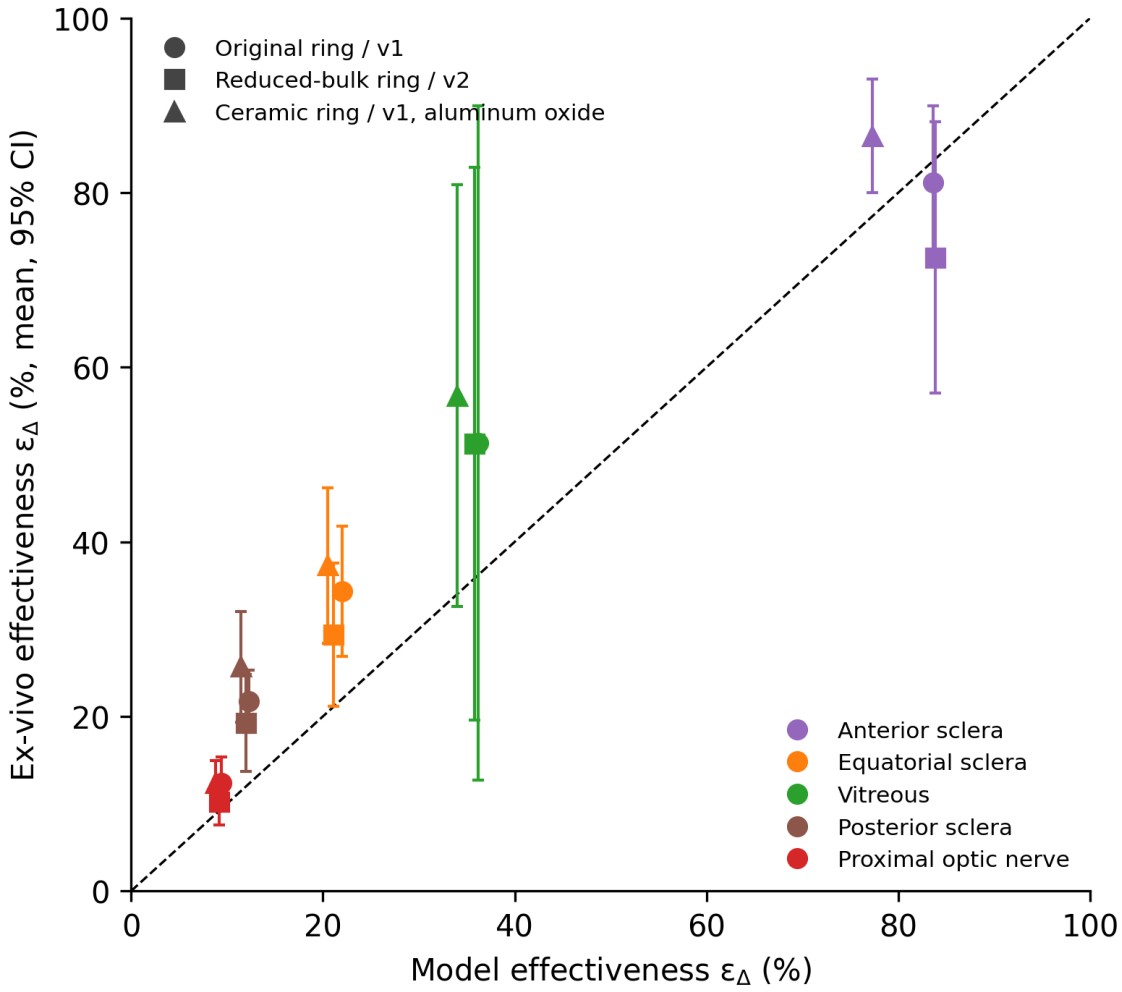


FIGURE 3.13. Cooling effectiveness ε_Δ (Equation 3.4) computed in each human donor eye (mean over the eyes, with its 95% confidence interval from the t distribution with $n-1$ degrees of freedom) against the model, for each site and ring; dashed, the line of identity. Points above it cooled more ex vivo than in the model. n, 4 or 5 eyes at the vitreous and 8 or 9 elsewhere. The model values are lower than ε in Figure 3.6 because ε_Δ is referred to the starting temperature of each site, not to body temperature (proximal optic nerve, v1: 9 against 10%).

3.3.7. Heat Exchanger and Heat Rejection. With the flow fixed the temperature problem is linear, so each solved run gives the temperature of the heated face for any heat load and ambient temperature,

$$T_f - T_{in} = R\,P + c\,(T_a - T_{in}), \tag{3.5}$$

where T_f is the average temperature of the heated face in °C, T_{in} the coolant inlet temperature in °C, R the rise per watt in °C W^{-1}, P the heat in W, c the dimensionless coupling to the air, and T_a the ambient temperature in °C. Checked against direct calculations, this relation held within 0.013 °C. With R and c of the nine runs of each design, the two heat loads and ambient temperatures of 10 and 35 °C give 36 conditions per design.

With the same heat, the smaller heat exchangers kept the hot side closer to the coolant than the original did at the same flow (Figure 3.14). Over the 36 operating points the heated face stayed 0.7 to 3.9 °C above the coolant inlet in v1, 0.4 to 2.3 °C in v2, and 0.3 to 1.9 °C in v3. At the same flow and heat, v2 rose about 60% as much as v1 and v3 about 45%. These are lower bounds, because the meshes do not resolve the wall layers (Section D.5).

In all three the temperature of the hot side was set almost entirely by the temperature of the coolant. The inlet temperature explained 97.6 to 99.3% of the variance of the heated-face temperature, and the flow rate less than 2%. The 43 °C limit applies to the hottest point, so it was checked against the peak temperature of the heated face, from runs on the mean flow. At the lowest flow, the warmest inlet, and the higher heat load the peak was 24.5 °C in v1, 23.7 °C in v2, and 28.4 °C in v3, at least 14.6 °C below the limit (Figure 3.14). The peak of v3 reached 10 °C above the inlet at 0.10 L min^{-1} and 1.9 °C at 0.50 L min^{-1}. It is probably too high, because in the model the heat enters the coolant directly, with no ceramic plate to spread it. The pressure drop rose steeply as the channels narrowed (Table 3.6). The smaller channels of v2 and v3 need about 40 and 200 times the pressure of the original, so

at the same pumping power the order reverses: with water at 4 °C and 10 mW the rise was 0.067, 0.105, and 0.143 °C per watt in v1, v2, and v3. The 12-V diaphragm pump of the coolant loop (CrocSee CRS-7A016) is listed at 1.3 L min^{-1} and a head of at least 5 m, about 49 kPa. It covers v1 at every flow of the table. The 42 kPa of v2 at 0.50 L min^{-1} is close to that head, and v3 at 0.25 L min^{-1} and above exceeds it, before the tubing is counted, so these blocks would run at lower flow on this pump or need a stronger one.

TABLE 3.6. Pressure drop of the three heat exchangers at each coolant flow rate, computed with the Python program from the solved flow, and hydraulic pumping power at the highest flow rate (pressure drop times volume flow rate; pump efficiency not included).

Flow rate	**v1 (Gen 3)**	**v2**	**v3**
Pressure drop (kPa)			
0.10 L min^{-1}	0.05	1.9	9
0.25 L min^{-1}	0.25	11	52
0.50 L min^{-1}	0.9	42	206
Pumping power (W)			
0.50 L min^{-1}	0.007	0.35	1.7

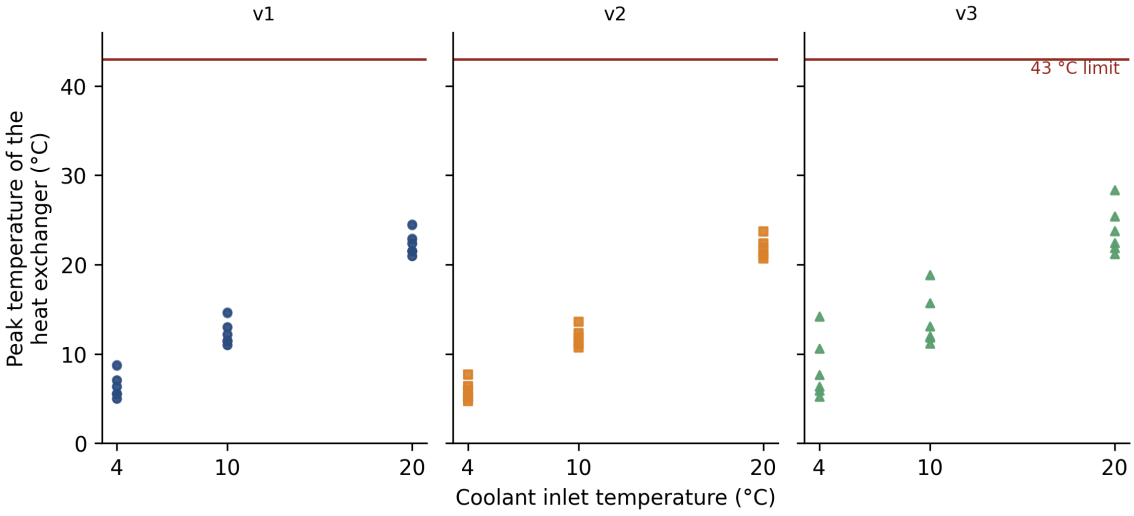


FIGURE 3.14. Peak temperature of each heat exchanger, the hottest point of the aluminum and the heated face, at its 36 operating points, with 10 and 15.5 W on every design, against the coolant inlet temperature; line, the 43 °C limit. The hottest point always lay on the heated face.

3.3.8. Integrated Control Board. The planned bench stages were completed, except that the reference did not stay in regulation under drive (below). The hard limits clamped the module at 3.93 A in cooling, 0.94 A in heating, and ±1.69 V. In closed loop the ring reached a cooling set point in about 30 s and held it within ±0.3 °C for 8 min. With the device on a preserved porcine eye in the bath, as in the ex-vivo experiments, the ring was pulled down from 31 to 4 °C in about 5 min and undershot to 2.0 °C. It then held 4.0 °C (3.8 to 4.3 °C) for the last 7 min of a run of 18 min (Figure 3.15A). The current monitor agreed with a multimeter in series within ±0.03 A from 0.48 to 3.93 A in cooling (Figure 3.15B). The safety cutouts that were tested stopped the module when triggered on purpose (Figure 3.15C).

The other functions of the board worked: the flow, pressure, and ambient sensors read, and the board switched the pump and the fan (Section F.2).

One fault remains to be fixed. While the module is driven, the 2.5 V reference of the board at times loses regulation, at the enable step and also during steady drive. The hardware current limits are set from that reference and cannot be relied on while it is disturbed. The module current is then limited only in duration, by the firmware cutouts, which stopped the module in every logged fault. At the start of the soak in Figure 3.15A the reference dipped and the current reversed for a few seconds without a cutout, and the run continued. The cause was not found, and the board was therefore not used in the in-vivo study (Section 4.2.4).

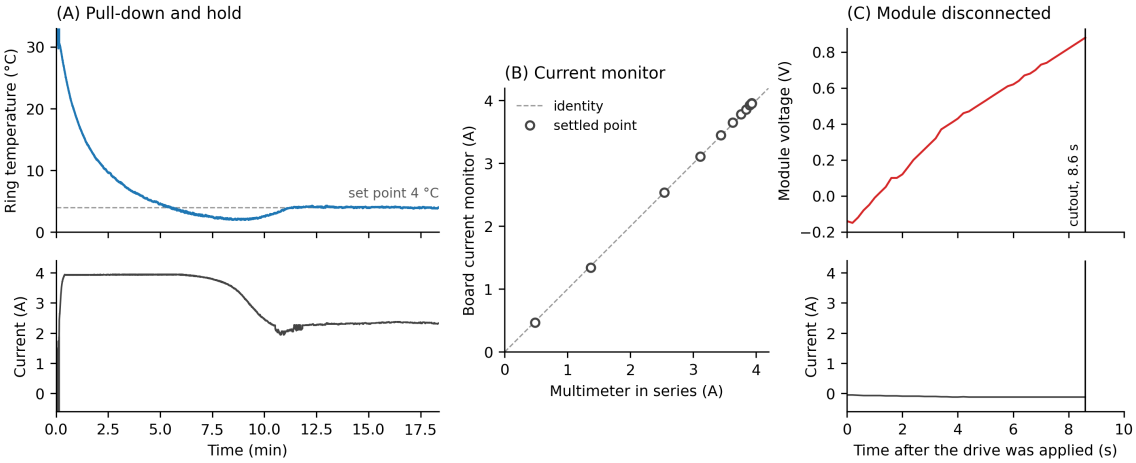


FIGURE 3.15. Bench performance of the control board. (A) Closed-loop pull-down of the ring from 31 °C to the 4 °C set point (dashed) and hold, with the module current, over 18.4 min; the current axis is clipped at −0.6 A, and in the first 9 s the current reversed to −1.5 A. (B) Current reported by the board against a multimeter in series, at thirteen settled drive levels in cooling into a resistor; dashed line, identity. (C) Open-circuit cutout with the module disconnected on purpose: the voltage applied by the control loop rose while almost no current flowed, and the cutout stopped the drive 8.6 s after it was applied.

3.4. Discussion

Both refinements that were built kept the cooling at the proximal optic nerve, the site of the hypothesis: the alumina ring within 1.0 °C of aluminum in paired human donor eyes, and the reduced-bulk ring about 0.6 °C less on average, by no more than 1.2 °C (0.05 °C in the model). The smallest heat exchanger was 81% smaller than that of Gen 3, and the controller was consolidated onto one board. None of them cooled the proximal optic nerve more than Gen 3 did.

Alumina kept 89 to 95% of the effectiveness of aluminum, whereas fused silica conducted too little heat to cool the eye as well and was excluded without being built. At the other tissue sites the donor eyes showed a trend, not significant after correction, opposite to the model.

That no refinement cooled the proximal optic nerve more than Gen 3 is consistent with the model. At every site the three ring geometries differed in cooling effectiveness ε by no more than 2 percentage points (9.6 to 9.9% at the proximal optic nerve), and

the time to 90% of cooling did not change with the geometry. The temperature reached at depth is set mainly by the body temperature. At the proximal optic nerve a change of 1 °C in body temperature moves the final temperature by 0.90 °C and a change of 1 °C in ring temperature by 0.10 °C. At steady state in the model, with its nominal perfusion, the posterior eye sits between a cold source on the anterior sclera and a warm, perfused choroid and orbit, and its temperature is a fixed weighting of the two over the operating points tested, with an effectiveness of about 10% at the proximal optic nerve. The geometry of the ring therefore has a small effect on the cooling at depth, and deeper cooling would have to come from a colder ring or a different route to the posterior eye.

The insulating layer shows the difference between efficiency and effectiveness: on the original ring it raised the share of heat drawn from the eye from 33 to 36% without improving the temperatures reached. It is worth having for the module and the heat exchanger, which carry less load, and possibly for the patient, since the eyelids are cooled less.

Around the optic nerve head the retina is cooled no more than the proximal optic nerve, and the distal optic nerve, 1 cm behind the globe, is barely cooled at all. At a body temperature of 39 °C the proximal optic nerve would not reach 35 °C (35.6 °C with the ring at 4 °C), although the drop below body temperature, 3.4 °C, is no smaller than at 37 °C. The average retina reaches 32 °C with the ring at 12 °C, inside the 30 to 34 °C range of preclinical retinal protection, but the average is carried by the periphery. With the ring at 4 °C and the body at 37 °C the retina ranges from about 9 °C at its anterior edge to 33.7 °C at the optic nerve head (Figure 3.12).

In the model of a perfused eye, at its nominal perfusion rates (Table 3.3) and a body temperature of 37 °C, a ring at the 4 °C floor of the device brings the proximal optic nerve to 33.8 °C, inside the 32 to 35 °C range of systemic therapeutic hypothermia, but 32 °C there would need a ring well below freezing. This value depends more on blood flow than on anything else in the model. In extension runs of the Python

program under the same conditions (ring v1 at 4 °C, body at 37 °C, 35 min of cooling), with every perfusion rate halved the proximal optic nerve reached 32.1 °C, and with every rate doubled 35.2 °C. The Pennes term assumes that blood leaves the tissue at tissue temperature. A lumped model of retinal heat balance that uses nearly the same choroidal perfusion rate (9.63×10^{-2} s^{-1}) assumes instead an equilibration factor of 0.5 between blood and choroid, which halves the heat exchanged, as in the halved case [**135**]. The wall of the orbit, held at body temperature in the model, is the second influence: held there, it keeps supplying heat to the orbital tissues, which can then cool only as far as their perfusion and metabolic heat allow. When the wall instead lets heat out (an assumed 30 W m^{-2} K^{-1}), the orbital tissues cool as well, and under the same conditions the proximal optic nerve reached 33.0 °C. The statement that 32 °C cannot be reached at the optic nerve thus holds at the nominal perfusion rates with either wall, and at half the nominal rates only with the wall held at body temperature. With half perfusion and heat leaving through the wall, the optic nerve reaches 30.5 °C.

An earlier Pennes model of a scleral ring from this research program reached a similar value at the optic nerve head, 34.0 °C against 33.7 to 33.8 °C here, with different boundary conditions and perfusion rates [**56**]. Its multiphase successor, in which cooled blood carries the cold backward, predicts deeper cooling, 29.6 °C [**54**].

Before cooling the model is warmer than the eyes of patients measured at the start of surgery [**47**], 31.7 against 30.0 °C in the anterior chamber and 35.9 against 35.0 °C in front of the optic disc, so the cooling it predicts for a given target temperature could be conservative.

Model and donor eyes rank the sites identically and agree next to the ring, so the model reproduces the path of heat from the anterior sclera into the globe. At depth the donor eyes cooled more, mostly because of the difference between the two conditions (Section 3.3.6). The cooling measured ex vivo at depth is therefore likely to overstate that of a perfused eye, by a factor that remains to be measured.

Heat rejection did not constrain the design. The hot side ran within a few degrees of the coolant in every design, well under the 43 °C limit, which was examined in the model only because the hot side is not instrumented. With the coolant touching the module directly, the smallest heat exchanger kept the hot side closer to the coolant than the largest at the same heat and flow, so most of the volume of the Gen 3 heat exchanger could be removed without thermal penalty. The price is pressure: the narrower channels of v2 and v3 need about 40 and 200 times the pressure of v1 (Table 3.6), and at the same pumping power v1 had the smallest rise. At 0.10 L min^{-1} v3 needs about 9 kPa, within the reach of a small pump.

The module itself has less margin than the heat exchanger. After 35 min the ring of the model drew 3.42 W through the module face held at 4 °C (v1; 2.93 and 2.95 W for v2 and v3). In the heat-exchanger model the hot face of the module ran 0.3 to 20 °C above that, from the coldest to the warmest coolant. At its maximum current of 3.9 A the module pumps about 3.5, 3.0, and 2.5 W with 0, 10, and 20 °C across it, read from the manufacturer's curves [**96**]. The margin is therefore small or negative: at full current the module covers the load of the original ring only with coolant at 4 °C, and falls short by up to 0.4 W with coolant at 10 °C and 0.9 W at 20 °C; v2 and v3 fall short by up to 0.3 W at 20 °C. A module of the same series with shorter elements, RH14-14-06-L1-W4.5, is listed at 5.4 W with no temperature difference and is 3.8 mm thick against 4.7 mm [**136**]. Scaled in the same way it would cover the load of every ring at all three coolant temperatures, but it needs 6.0 A, above the 3.9 A to which the present board is limited.

The integrated control board met its requirements on the bench, with one exception that prevented its use: the reference that sets the hardware current limits and the temperature measurements can lose regulation under drive. The firmware stopped the module each time, but a safety argument cannot rest on the firmware alone, and the next revision will set the limits from an independent reference.

The model was not validated against a perfused eye, and its tissue properties and perfusion rates come from the literature (Appendix D). Choroidal perfusion was varied only in extension runs. The points sampled in the model are not in the same places as the implanted sensors at depth, and the model and the measurements are therefore compared only approximately there. They agree quantitatively only at the anterior sclera. The lens of v3, the insulating layer, and the smaller heat exchangers were evaluated only in the model, and the ceramic ring only in the original geometry. The ex-vivo comparisons rest on nine eyes from five donors, with four to eight valid pairs per site. One ring of each kind was tested, so a difference between rings cannot be separated from the individual ring and its seating on the eye. The integrated control board was tested on the bench only. Patient comfort remains to be measured. The refinements carried into the in-vivo study were those that had been built and verified: the reduced-bulk aluminum ring on the Gen 3 heat exchanger, regulated by the Gen 3 controller (Section 4.2.4).

CHAPTER 4

Specific Aim 3: Validation and Safety Assessment in an In-Vivo Porcine Model

4.1. Background and Rationale

Neither the isolated eyes nor the computational model of Chapters 2 and 3 can answer the two questions on which any clinical use depends: whether cooling of this kind is tolerated by living ocular tissue, and whether it reaches the retina against an intact circulation. The isolated eye has no perfusion, and the bath that stood in for it resupplied less heat than a perfused eye receives, so the ex-vivo excursions are likely an upper estimate of the cooling reached at depth (Section 2.4). Only in vivo is the choroidal heat load present (Section 1.2.3), and only in vivo can retinal function be measured while the eye is cooled (Section 1.3.3).

The tolerance of living ocular tissue to non-freezing cooling lasting tens of minutes to hours has not, to the author's knowledge, been established. The storage evidence that motivated the 4 °C set point (Section 2.2.1) comes from metabolically arrested, non-perfused tissue held in preservation solution, and prolonged non-freezing cold can injure perfused tissue, with vascular and sensory stages [**137**]. Cooling also has reversible effects specific to the eye: donor corneas swell when stored at 4 °C and thin again over the first hours of rewarming, as the endothelial pump resumes [**138**], and pressure applied to the globe, as by a contact device, can raise intraocular pressure and can itself occlude the retinal circulation [**6, 21**] (Section 1.1). These effects define the safety endpoints of this aim, listed in Section 1.8.

The objective of this aim was to evaluate, in a live porcine model, whether the device produces changes in the electroretinogram (ERG) consistent with retinal cooling and whether repeated cooling sessions of increasing duration are tolerated by the ocular surface, the globe, and the animal. The hypothesis is stated in Section 1.8.

4.2. Methods

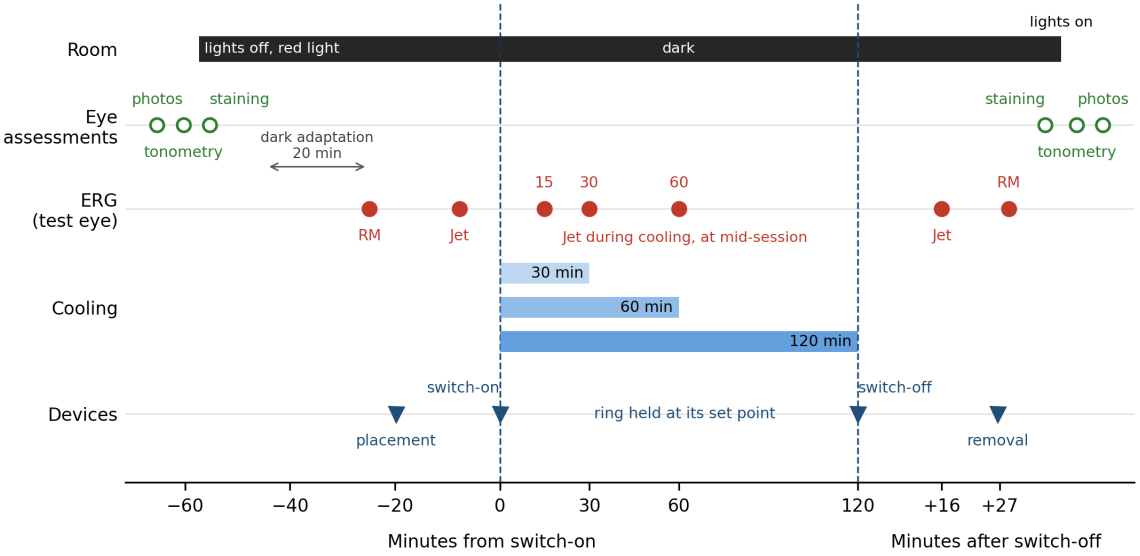


FIGURE 4.1. Sequence of a cooling session: baseline assessments, dark adaptation and baseline ERG, placement of both devices, switch-on, cooling with ERG recordings, switch-off, removal, and post-cooling assessments. Device and ERG times are the medians of the fifteen sessions of animals 2 to 6; in animals 1 and 2 the number and times of the recordings during cooling varied. Black bar, room lights off and red light, from the corneal staining before cooling to that after.

4.2.1. Study Design and Approval. All procedures were approved by the University of Illinois Chicago Animal Care Committee under protocol `IPROTO202600000055` (approved April 30, 2026), with amendments `IAMEND202600000339`, `IAMEND202600000409`, `IAMEND202600000587`, and `IAMEND202600000642`. Each animal underwent three cooling sessions of increasing length, 30, 60, and 120 min, and a terminal session, with daily observation on the intervening days. Figure 4.1 shows the sequence of a cooling session. The sessions were planned for days 1, 4, 7, and 10; the days performed and the weight of

each animal are given in Table 4.1. In the terminal session, except in animal 1, both devices were placed and a Jet ERG recorded, without cooling, before euthanasia, enucleation, and histology. Baseline assessments were repeated before every cooling session, so that each session was referenced to the state of the eye on that day. Each session was performed under general anesthesia. In every session the right eye was the test eye, fitted with the active device, and the left eye was the control eye, fitted with an identical device that was not activated. Each animal thus served as its own control for the mechanical effects of the device, for anesthesia, and for systemic changes shared by both eyes.

FIGURE 4.2. Animals 3 and 4 in a pen of the Biologic Resources Laboratory the day after arrival, with wood-shaving bedding and toys for enrichment. During the study each animal had its own pen.

4.2.2. Animals. The protocol provides for six domestic pigs (*Sus scrofa*; farm strain, not genetically modified), obtained from an approved vendor (Oakhill Genetics) and housed individually in the Biologic Resources Laboratory (BRL) of the University of Illinois Chicago (Figure 4.2). All six were male. Five completed the study and one died during it (Section 4.3.1); their weights are given in Table 4.1.

They were less than a year old. The species was chosen for the similarity of its eye to the human eye in size, structure, and retinal vascular pattern (Section 1.6). The study was classified in U.S. Department of Agriculture (USDA) pain category D.

The number of animals was set for a paired comparison. Assuming a 30% change in the primary ERG measure, a standard deviation of 20%, a two-sided α of 0.05, and 80% power, the normal approximation gave 3.5, rounded to four animals, and two further animals were approved to allow for attrition, including equipment failure. The quantity of interest is the change of each eye from its own recording before cooling: a change in the test eye and none in the control eye over the same interval. The power obtained with the measured variability is given with the results (Section 4.3.3).

TABLE 4.1. Sex of each animal, day of each session, and weight (kg) from the BRL record of the session; ∼, written as approximate on the record. Animal 3 died after session 3.

Animal	Sex	Days	Session 1	Session 2	Session 3	Session 4
1	Male	1, 4, 8, 11	26.3	26.2	28.0	28.6
2	Male	1, 4, 8, 12	32.5	32.5	33.8	36.7
3	Male	1, 4, 8, –	33.4	∼33	34.8	–
4	Male	1, 4, 8, 11	32.8	∼33	35.1	36.0
5	Male	1, 5, 9	28.9	30	32.8	–
6	Male	1, 7, 10	30.4	32.7	∼33	–

4.2.3. Anesthesia and Physiological Monitoring. Anesthesia was induced and maintained by BRL staff under the approved protocol; the drugs and doses of each session are given in Appendix G. Each pig was sedated, intubated, ventilated, and maintained on isoflurane (1–2%).

From animal 2 onward, after the protocol had been amended, rocuronium was given intravenously once adequate anesthetic depth had been verified, to abolish eye movement; no neuromuscular block was used in animal 1. Animal 2 moved once, during session 2, and the active device came briefly out of the eye and was reinserted at once.

Flunixin was given for analgesia. Propofol, hydromorphone, and an intravenous infusion of lidocaine and ketamine were given at the discretion of the veterinarian or veterinary technician. The infusion was added after animal 2 moved, as an adjunct to isoflurane.

Electrocardiogram, pulse oximetry, capnography, indirect blood pressure, and rectal temperature were monitored continuously and recorded every 10 to 20 min. Anesthetic depth was assessed from jaw tone and the palpebral reflex together with heart rate, respiratory rate, and blood pressure. Room temperature and relative humidity were logged at the start and end of each session. Forced warm air was used when needed and was withdrawn when core temperature rose.

If physiological parameters left the safety range during cooling, the protocol suspended cooling for 15–20 min. A session was ended if core temperature fell below 35 °C and did not stabilize with warming, if oxygen saturation remained below 90%, or if heart rate fell below the limits set by the veterinary staff. Heart rate was watched at device placement and at the start of cooling, because of the oculocardiac reflex.

4.2.4. Device Configuration. The device used in vivo was SCEC Gen 3 fitted with the reduced-bulk ring, the geometry modeled as v2 in Chapter 3, made in aluminum and mounted on the Gen 3 heat exchanger. In the donor eyes this ring gave the proximal optic nerve 0.6 °C less cooling than the original ring on average, a difference unlikely to exceed 1.2 °C. The Gen 3 controller was set to hold the ring at a nominal 4 °C (Section 2.2.1), with the interruptions and the calibration error of Sections 4.3.2 and 4.2.6. In the first two animals the switching element was a relay, as in Aim 1. After the relay had failed in session 2 of animal 1 and in session 3 of animal 2, it was replaced, from session 1 of animal 3 onward, by a trigger switch module with two metal-oxide-semiconductor field-effect transistors (MOSFETs; ANMBEST, rated by the seller at DC 5–36 V and 15 A), which switched reliably and silently. The heat exchanger was supplied by a pump with distilled water cooled in an ice bath

(Appendix E). The control eye was fitted with an identical, inactive replica of the device, so that the two eyes received the same mechanical contact.

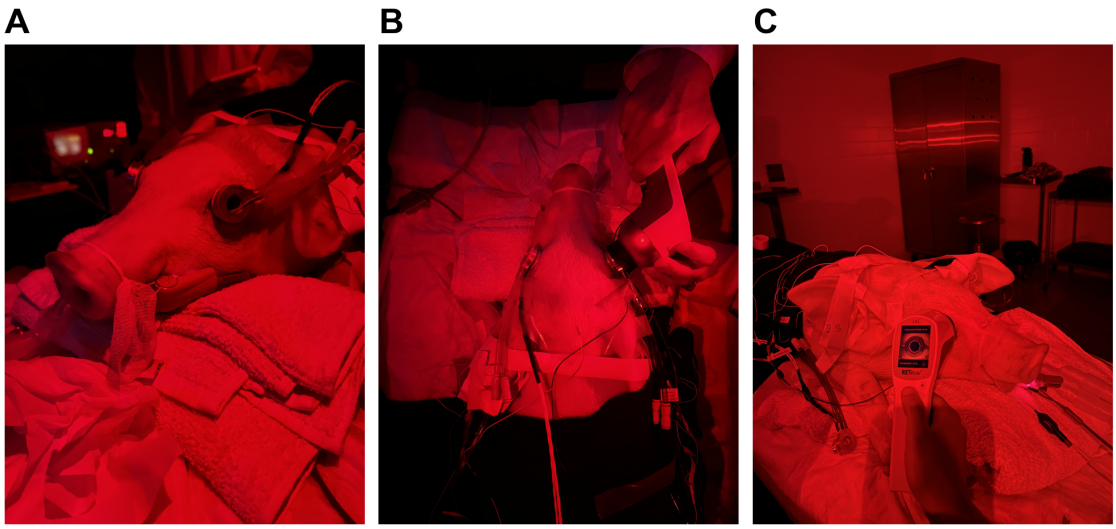


FIGURE 4.3. The cooling session in the animal laboratory, under red light: (A) the device on the eye of an anesthetized pig and (B) an ERG recorded with the Jet electrode through its central opening (animal 1, session 1); (C) the screen of the ERG recorder during a recording (animal 2, session 3).

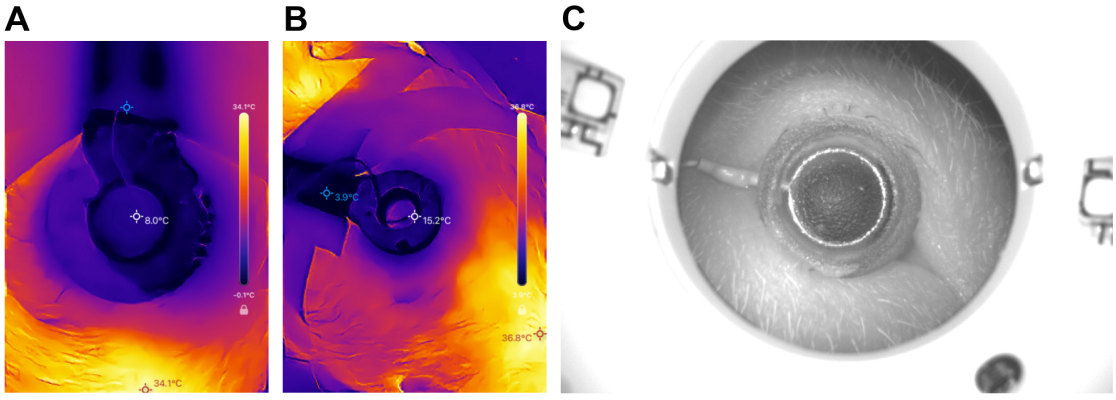


FIGURE 4.4. The device on the eye in thermal images, (A) animal 6, session 1, and (B) animal 4, session 3, and (C) the RM electrode on the eye seen by the pupil camera of the ERG recorder (animal 5, session 1). The thermal images are from a phone camera without emissivity correction; the reflective metal parts do not read true temperature, so the images show only the pattern of cooling.

4.2.5. Device Application. From animal 2 the room lights were turned off for corneal staining, and the room was kept dark (Figure 4.3), with every light source covered by a red filter screen, from the start of dark adaptation until after the last corneal staining of the post-cooling assessments. After the baseline assessments (Section 4.2.8), an ophthalmic lubricant ointment (Puralube Vet Ointment, white petrolatum and mineral oil, Dechra) was applied to the eye-contacting side of the ring of each device, and the ring was placed on the eye by sliding its skirt under the eyelids with the help of tissue tweezers. The active device was placed on the right eye first and the control device on the left eye 2 to 6 min later, and both were secured with tape (Figure 4.4). The coolant circuit, temperature sensor, and power of the active device were connected with the coolant pump and the thermoelectric drive off; both were started together at switch-on. A corneal ERG electrode (Jet) was placed through the central opening of each device onto the cornea.

The ring of the active device read about 16 to 21 °C before it was placed. Each device stayed on for the duration of the session and both devices were then removed, when the ring had rewarmed to between 20 and 35 °C, followed by the post-cooling assessments.

4.2.6. Temperature Monitoring. The temperature of the test ring was measured by the thermistor embedded in it, through the signal-conditioning chain and data-acquisition unit of Aim 1 (Section 2.2.4; DI-2108 unit, WinDaq software), at 10 Hz. The drive current was recorded on a second channel.

Before each session the temperature channel was calibrated in the dry-block calibrator at 37.0, 28.7, 20.5, 12.3, and 4.0 °C. This channel is not the linearized channel of Aim 1, so the readings were converted to temperature with a model of the measuring circuit fitted to the plateaus of that session (Section A.1). A fit to the three original plateaus reproduced the other two within 0.06 to 0.23 °C, and within 0.03 °C with the corrected calibration described below; this is taken as the accuracy of the ring temperature.

Ring temperatures of the sessions before the well thermistor was held against the aluminum are reported with 5.5 °C added, and the set point that read 4 °C in those sessions was about 9.5 °C. Until session 1 of animal 5 the well thermistor used for this calibration did not touch the aluminum of the well, so it could not shed its self-heating as the ring thermistor does in the ring. From session 1 of animal 6 and session 2 of animal 5 a custom piece held it in contact with the aluminum, and this calibration, the corrected calibration, is taken as correct. Before session 2 of animal 6 the channel was calibrated both ways on the same morning: against the corrected calibration, the earlier arrangement read the ring 5.5 to 5.6 °C low from 4 to 25 °C and 5.2 °C low at 35 °C. Animal 5 shows the same shift between its first two sessions, 5.3 to 6.0 °C before placement and on the eye after switch-on.

No sensors were implanted in vivo. Core temperature was recorded rectally by the anesthesia team.

4.2.7. Assessment of Retinal Function. Full-field ERGs were recorded under dark adaptation. The pupils were dilated with tropicamide (1%), phenylephrine hydrochloride (2.5%), and proparacaine hydrochloride (0.5%), one to two drops each, and the eyes were dark adapted in the darkened room (Section 4.2.5). Dark adaptation was planned at 15 min in animal 1 and at 20 min from animal 2 onward, and the first recording was made at its end. In the 120-min sessions the drops were applied again during the session.

An ophthalmic lubricant gel (Celluvisc, carboxymethylcellulose sodium 1%, Allergan) was applied to the cornea as the coupling medium of the electrode. Sterile subcutaneous needle electrodes served as the ground, on top of the neck, and as the references, one for each eye between the eye and the ear, and stayed in place for the whole session. The active electrode was an RM (RetMap) corneal electrode before the device was installed and after it was removed, and a Jet electrode, through the central opening of the device, with the device in place. Responses were recorded with a handheld full-field ERG system with an integrated Ganzfeld stimulator (RETeval,

LKC Technologies, Gaithersburg, MD, USA), using its scotopic protocol for rabbit, mouse, and pig, with white flashes of three strengths: 0.06, 8, and 25 cd·s/m^2, presented at 0.5, 0.1, and 0.05 Hz. The two eyes were recorded in turn, the test eye first.

In animals 1 and 2 the number and the times of the recordings differed between sessions. In three of these sessions (session 2 of animal 1, sessions 1 and 2 of animal 2) the Jet recording after switch-off was made within 2 min of switch-off, with the ring still below 13.5 °C, and so records the end of cooling rather than recovery. Session 3 of animal 1 had no Jet recording before cooling or after switch-off. From animal 3 the times were fixed: one recording at about the middle of cooling, and the Jet recording after switch-off at about 15 min, once the ring temperature had stabilized. The last recording of each session was made with the RM electrode after the devices had been removed. The ring temperature at each recording was read from the cooling record (mean of the 2 min around the recording); it is not available for session 3 of animal 3, whose cooling was not recorded because the recording software stopped with an error shortly after it was started, which was not noticed during the session.

The analysis used the raw record of each flash step, as set out in Section C.3: a 0.3 Hz high-pass filter removed drift [**139**], each sweep was referred to the 20 ms before its flash, and sweeps with artefacts were rejected before averaging. For the a-wave and b-wave the average was low-pass filtered at 75 Hz, so that an oscillatory potential could not displace the b-wave peak, a departure from the standard. The oscillatory potentials were band-pass filtered from 75 to 300 Hz, as the standard sets out, at 8 and 25 cd·s/m^2, and the measure is the sum of the first four peaks (OP1 to OP4). The a-wave amplitude was measured from the pre-stimulus baseline and the b-wave amplitude from the a-wave trough.

Dark adaptation in animals 1 and 2 lasted 10 to 23 min, against the standard's minimum of 20 min.

Recordings were grouped by electrode and phase (RM before installation; Jet before switch-on; Jet during cooling; Jet after switch-off; RM after removal).

4.2.8. Ocular Safety Assessments. Both eyes were assessed at the start of each session, before the devices were placed, and again after they were removed. At the start, the staining came after the photographs and tonometry and before the drops and the ERG, which it preceded by 24 to 37 min in animals 1 to 6. After the session the corneal staining was done with the lights off, and tonometry and the photographs followed with the lights on.

Ocular irritation. The ocular surface of each eye was photographed, and bulbar and limbal redness were graded from the photographs in whole steps on the 0-to-4 scales of the Johnson & Johnson Clinical Assessment Guide named in the protocol [**140**], which are adapted from the Efron grading scales [**141**]. Tarsal redness, also listed in the protocol, was not graded.

Intraocular pressure. A handheld rebound tonometer (TonoLab, Icare Finland Oy, Vantaa, Finland) was held perpendicular to the central cornea, with the lids retracted without pressure on the globe, and one to three readings per eye were averaged. The instrument is calibrated for the rodent eye and not validated for the porcine eye, so its readings might carry a systematic offset; they are used only to compare the two eyes of an animal and the same eye before and after a session. At the start of the session the pressure was measured before any topical anesthetic had been applied.

Corneal staining. The eye was flushed with sterile saline and stained with fluorescein, from a wetted sterile strip touched to the conjunctiva away from the central cornea. The excess was flushed away and the surface was photographed in the dark through a cobalt-blue filter. The staining was graded from 0 to 4 for each eye by the author on one photograph per eye and time point, chosen as for redness.

Observation. Between sessions each animal was observed daily for ocular swelling, squinting, and corneal ulceration, and was weighed, scored for body condition, and checked for food and water intake.

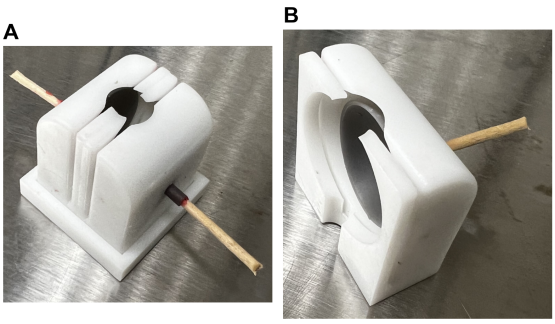


FIGURE 4.5. The 3D-printed guide used to trim the fixed globes for histology, front (A) and side (B). The globe sits in the spherical cavity, centered by a small shell pushed in with the two sticks; the two parallel slots, about 5 mm apart, guide the blade.

Histology. In the terminal session, after the assessments, each animal was euthanized under anesthesia with an intravenous overdose of pentobarbital sodium (Fatal-Plus, 390 mg/mL; 10 mL intravenously), a method acceptable for swine under the AVMA Guidelines for the Euthanasia of Animals [**142**]. Death was confirmed by the absence of respiration, cardiac activity, corneal reflex, and muscle tone. Both eyes were enucleated immediately and fixed, in animal 1 by immersion in formalin and in animal 2 by immersion in formalin with an injection of formalin into the globe. From animal 3 onward modified Davidson's fixative was used, injected into the globe and by immersion, to limit retinal detachment. After fixation each globe was trimmed in a 3D-printed guide designed for this study (Figure 4.5), leaving a central slab containing the optic nerve head and the cooled region. The eyes were sectioned from the cornea to the optic nerve and processed by the UIC histology core for hematoxylin and eosin staining of the cornea, sclera, conjunctiva, and peripheral and posterior retina. The stained sections of all animals are imaged together with a digital microscope (VHX-6000, Keyence, Osaka, Japan) at the Robbins and Jacobs Family Biocompatibility and Implant Pathology Laboratory (Rush University Medical Center, Chicago, IL, USA), and were read by the author. Before the read, the images of each animal were compared masked (Appendix J): each region was graded for epithelial loss, edema,

inflammatory infiltrate, hemorrhage, necrosis, vascular change, and separation, with processing artefacts recorded apart, and the eye with more true change was named.

4.2.9. Data Analysis.

Animals and groups. Animal 1 is exploratory, as the procedure was still being set up (no neuromuscular block, other premedication and calibration, relay failure, assessments reordered, staining not yet in the dark); it is left out of the ERG, intraocular pressure, and ocular surface results and of every test, and appears, marked as exploratory, only in the device operation, ring temperature, systemic course, and histology. The tests were made over animals 2 to 6, the primary animals, and all results are those of animals 2 to 6 unless stated otherwise. Session 3 of animal 3, in which both eyes changed together, pointing to a systemic cause, and after which the animal died (Section 4.3.6), is left out of every measure, so 14 cooling sessions were analyzed. The results are given over both ring temperatures, about 9.5 °C (animals 2 to 4, session 1 of animal 5) and about 4 °C (animal 5 from session 2, animal 6); the two are given apart only where that changes a result.

ERG measures. Cooling prolongs the implicit times and reduces the amplitudes of the ERG [**73, 74, 76**]. By slowing the response it should also widen the b-wave, and more so if the retina cools unevenly, which spreads the timing of its parts. The width was therefore measured on the filtered trace: the time above 80% of the b-wave amplitude, from the a-wave trough, up to the first trough after the peak, so that a later wave is not counted. The primary measures were the b-wave implicit time and amplitude and the summed oscillatory potentials, each at 8 and 25 $\mathrm{cd\,s\,m^{-2}}$.

Evidence of retinal cooling. Each eye was compared with itself, from its Jet recording before cooling to that during cooling, with the device installed: the test eye shows the effect of cooling, the control eye whether the same change occurred through anesthesia, time, or mechanical contact. The two changes were also compared, the more sensitive comparison: the b-wave amplitudes of the two eyes of normal subjects differ by a median of 8 to 11% [**143**], in minipigs no ERG measure differs between the

eyes [**144**], and anesthesia and core temperature act on both alike, whereas one eye recorded on different days must change in amplitude by 25% or more, often 40%, before the change is significant [**145**]. A slower b-wave was taken as evidence of cooling; a change in amplitude alone was not, because amplitude also depends on electrode contact and position. A session was counted as showing cooling when the b-wave of the test eye had both a longer implicit time and a smaller amplitude than before cooling at both primary flash strengths. As an exploratory secondary measure, the same judgment was made on the oscillatory potentials, from a later OP2 peak and a smaller summed amplitude. Each change is given in the unit of the measure; the fall of the summed oscillatory potentials is also given as a percentage of the eye's own earlier value.

Safety. The first safety criterion was the return of the test eye's ERG to its pre-cooling values after rewarming, judged on the Jet recordings before cooling and after switch-off and confirmed on the RM recordings before installation and after removal of the devices, which span the session. A measure was restored when the test eye's change from its own baseline lay within twice the standard deviation of the control eye's change over the same interval, pooled over the primary animals and sessions, for each electrode, flash, and measure. A Jet recording after switch-off counted as a recovery recording only at least 10 min after switch-off; these were made at 11 to 25 min, with the ring 73 to 97% rewarmed. The two made within 2 min (sessions 1 and 2 of animal 2), ring below 13.5 °C, record the end of cooling; they were left out of the comparison and its spread, and those sessions were judged on the RM pair alone. The change from the recording during cooling to the recovery recording shows how far the ERG came back as the ring rewarmed. A change persisting between sessions was sought by comparing the Jet recordings at the start of sessions 2 to 4 with session 1, confirmed with the RM recordings, together with the core temperature, which alters the ERG [**122**]; within a session the control eye shows its effect. When the test eye had not returned, the control eye was checked: if both eyes had changed in the same

direction by more than the control eyes usually changed, the change was attributed to a systemic cause; if the test eye had come part of the way back from its change during cooling while the control eye had not changed, the pattern was read as a recovery not yet complete when recorded.

The second criterion was no harmful increment: intraocular pressure, redness, and corneal staining should rise in the test eye no more than in the control eye, with no increase persisting to the next session, and the histology of the test eye should show no damage absent from the control eye. Safety was not inferred from the absence of a significant difference, which is expected with five animals even when a real effect exists.

Statistical considerations. Each session length was analyzed separately. The change of each eye from before to during cooling, and the test eye's change against the control eye's, were tested with paired t tests over the animals; for each measure and flash, the p value of the test-against-control comparison was Holm-corrected across the three session lengths. Between sessions, the same comparison was made for each later session against session 1, Holm-corrected across sessions 2 to 4. Effect sizes are Hedges' g of the paired test-minus-control differences. For restoration, intraocular pressure, and the comparison between sessions, the 90% confidence interval of the mean test-minus-control difference gives the largest difference the data allow. The redness and staining grades are given as counts by eye and session. The session lengths were always given in the order 30, 60, 120 min in the same animals, so a trend across sessions cannot be separated from repeated anesthesia and cumulative exposure.

4.3. Results

4.3.1. Animals and Sessions. Animals 1, 2, 4, 5, and 6 completed all four sessions. Animal 3 died in recovery from its third cooling session (Section 4.3.6) and had no terminal session. All eighteen cooling sessions were given for their planned

length. Animal 1, exploratory, appears only in the device operation, ring temperature, systemic course, and histology (Section 4.2.9).

4.3.2. Device Operation and Ring Temperature. The ring temperature of the seventeen sessions with a record, corrected for the calibration error (Section 4.2.6), is shown in Figure 4.6. The cooling of session 3 of animal 3 was not recorded because the recording software stopped with an error shortly after it was started, which was not noticed during the session; the session log gives switch-on at 10:42 and switch-off at 12:42.

In all the recorded sessions, the ring, placed on the eye, read 27.6 to 31.0 °C before switch-on. At about 9.5 °C it came within 0.5 °C of its level 0.8 to 2.9 min after switch-on, the controller began to switch the drive 1.0 to 3.1 min after switch-on, and the ring was then held at a mean of 9.1 to 10.0 °C, with a standard deviation within a session of at most 0.14 °C. With the corrected calibration it took 3.9 to 10.3 min to reach its level, 4.0 to 4.1 °C; in the first two (session 1 of animal 6, session 2 of animal 5) the drive stayed on nearly all the time, the device at its limit near a true 4 °C.

With the relay (animals 1 and 2) the drive was interrupted in two of the six sessions. The ring never fell below 3.8 °C, above the 2 °C below which the risk of cold injury to stored organs rises [**116**] (Section 2.2.1). With the MOSFET module (animals 3 to 6, eleven sessions with a record) the drive was not interrupted, and at about 9.5 °C the ring stayed between 8.9 and 10.2 °C from 2 min after the controller began to switch until 1 min before switch-off.

Each animal had the device switched on for 210 to 211 min over its three cooling sessions.

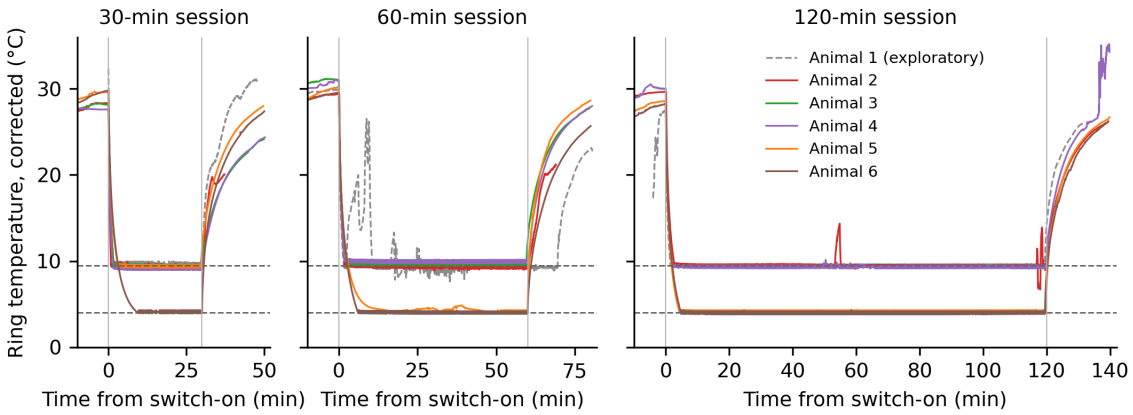


FIGURE 4.6. Temperature of the ring of the active device in each cooling session against time from switch-on, from placement or 10 min before switch-on, whichever is later, to removal or 20 min after switch-off, whichever is earlier; 1-s means of the 10-Hz record, corrected for the calibration error. Dashed lines, the set point in the two calibrations, 9.5 and 4 °C; vertical lines, planned switch-on and switch-off. Animal 1, exploratory, dashed grey line. Session 3 of animal 3 has no record of the cooling. The rise of animal 4 before removal in the 120-min session coincides with the post-cooling ERG, recorded with the device still on the eye.

4.3.3. Electroretinogram. The analysis used 74 recordings, each of both eyes at three flash strengths (444 traces; mean waveforms in Figure 4.7). The trace checks of Section C.3 flagged 9 of the 148 traces at 25 cd·s/m^2, 7 at 8 cd·s/m^2, and 46 at 0.06 cd·s/m^2, where the response is small and the picks are least certain.

Cooling. Of the 15 cooling sessions of animals 2 to 6, 14 were analyzed. Session 3 of animal 3, after which the animal died (Section 4.3.6), was left out: in it the ERG of both eyes moved beyond the usual range of the control eyes in the same direction, with both electrodes, pointing to a systemic cause rather than the cooling. Of the 14, nine were with the ring at about 9.5 °C (four, three, and two after 30, 60, and 120 min) and five at about 4 °C (one, two, and two), so five, five, and four sessions at each session length (Figure 4.8, Tables 4.3 and 4.4).

The b-wave of the test eye slowed in all 14 sessions, by 2.0 to 8.0 ms at 25 cd·s/m^2 and 1.5 to 8.5 ms at 8 cd·s/m^2, while the control eye changed by at most 2.5 ms either way. At 25 cd·s/m^2 the test eye slowed by a mean of The test eye slowed by a

mean of 5.8, 5.5, and 5.2 ms after 30, 60, and 120 min at 25 cd·s/m^2, the control eye by 0.8, 0.2, and −0.5 ms, and by 5.3, 5.5, and 5.1 ms at 8 cd·s/m^2, the control eye by 0.4, −0.5, and −0.4 ms. At 0.06 cd·s/m^2, whose slower rod-driven b-wave peaks later, the change was larger and more variable. Against the control eye the slowing was significant after Holm correction at every session length and at both flashes (p 0.009 to 0.027), with large effects (g 1.5 to 3.3).

The b-wave amplitude of the test eye fell in 10 of the 14 sessions at each flash, by a mean of 67 μV (18%) at 25 cd·s/m^2 and 56 μV (14%) at 8 cd·s/m^2. That of the control eye fell in about half the sessions and rose in the rest, by up to about 130 μV either way, so its mean change was near zero (+12 μV at 25 cd·s/m^2, −2 μV at 8 cd·s/m^2). Against the control eye no session length was significant after Holm correction (p 0.12 to 0.72).

The b-wave of the test eye widened in 13 of the 14 sessions at 25 cd·s/m^2 and in all 14 at 8 cd·s/m^2, and that of the control eye in 12 and 13. The mean changes, 3.2 ms in the test eye against 0.1 ms in the control eye at 25 cd·s/m^2 and 6.5 against 4.5 ms at 8 cd·s/m^2, are dominated by sessions 1 and 2 of animal 6, in which the width of either eye changed by up to 50 ms in either direction; without these two sessions the test eye widened by 4.0 ms against 1.7 ms at 25 cd·s/m^2 and by 2.8 against 1.5 ms at 8 cd·s/m^2.

The oscillatory potentials of the test eye were slower in 13 of the 14 sessions at 25 cd·s/m^2 and 12 at 8 cd·s/m^2, the OP2 peak later by a mean of 1.5 and 1.1 ms, and those of the control eye in 1 and 3, by a mean of −0.5 ms at each flash. Their summed amplitude fell in the test eye in 11 of the 14 sessions at 25 cd·s/m^2 and 13 at 8 cd·s/m^2, and in the control eye in 5 and 6, no more often than it rose. At 25 cd·s/m^2 the fall in the test eye grew with the session length, by 19, 49, and 55%; at 8 cd·s/m^2 it was 39% over the three session lengths. Against the control eye no session length was significant after Holm correction at either flash (p 0.10 to 0.39).

Taken together (Section 4.2.9), a slower and smaller b-wave was found at both flashes in the test eye in 10 of the 14 sessions and in the control eye in 3, and slower and smaller oscillatory potentials, the exploratory measure, in the test eye in 9 and in the control eye in none.

Restoration. Of the 14 sessions, 12 had a recovery recording with the Jet electrode, made 11 to 25 min after switch-off with the ring 73 to 97% rewarmed. In the other two, sessions 1 and 2 of animal 2, the Jet recording after switch-off was made within 2 min, too early to count as a recovery recording, and those two sessions were judged on the RM recordings only.

The b-wave implicit time of the test eye was restored in 11 of the 12 at 25 cd·s/m^2 (limit ±3.7 ms) and in 8 of the 12 at 8 cd·s/m^2 (limit ±2.8 ms). In the others it was still 3.0 to 4.5 ms slower than before cooling: session 1 of animal 3 at both flashes, and session 3 of animal 2, session 2 of animal 3, and session 1 of animal 4 at 8 cd·s/m^2. In each of these the test eye had come part of the way back, by 25 to 46% of its slowing during cooling, while the control eye changed by at most 2 ms: the pattern of a recovery not yet complete in the 11 to 25 min between switch-off and the recording, rather than of a lasting change. The amplitude was restored in all 12 at each flash, the width in all 12 at 25 cd·s/m^2 and 11 at 8 cd·s/m^2, and the summed oscillatory potentials in 11 at 25 cd·s/m^2 (not session 2 of animal 3) and 10 at 8 cd·s/m^2 (not session 3 of animal 2 and session 2 of animal 3). In these three the oscillatory potentials of the test eye were still as low as during cooling, but those of the control eye had also fallen after switch-off, by 107 to 132 μV, so the pattern does not separate an incomplete recovery of the test eye from a change in both eyes. With the RM electrode, over the whole session, the implicit time was restored in all 14 sessions at 25 cd·s/m^2 and 13 at 8 cd·s/m^2, the amplitude in 13 at each flash, the width in 12 at 25 cd·s/m^2 and all 14 at 8 cd·s/m^2, and the oscillatory potentials in all 14. By the 90% confidence intervals, a lasting lengthening of the implicit time by more than 3.6, 3.2, and 5.7 ms after 30, 60, and 120 min at 25 cd·s/m^2, and 3.4, 2.4,

and 4.6 ms at 8 cd·s/m^2, is unlikely with the Jet pair, as is a fall of the amplitude by more than 92 and 231 μV after 60 and 120 min at 25 cd·s/m^2, and 44 and 188 μV at 8; after 30 min the amplitude of the test eye came back higher than that of the control eye (mean +89 μV, 90% confidence interval +34 to +144, at 25 cd·s/m^2; +124 μV, +38 to +209, at 8). With the RM pair, as confirmation, after 60 min the amplitude of the test eye was lower than that of the control eye (mean −123 μV, 90% confidence interval −177 to −69), although each session but one lay within the restoration limit.

In two of the changes not restored both eyes changed together, pointing to a systemic cause: the width at 8 cd·s/m^2 in session 2 of animal 6 and, with the RM electrode, the amplitude in session 2 of animal 3.

Between sessions. The Jet recording at the start of each later session was compared with that of session 1 (Table 4.2). There was no significant difference between the test and the control eye in the b-wave implicit time, amplitude, or width, or in the summed oscillatory potentials, at either flash (smallest p after Holm correction 0.06); the mean width differed from that of session 1 by less than 1 ms. By the 90% confidence intervals, a lasting slowing of the test eye by more than 1.3 ms at the start of session 2 is unlikely at 25 cd·s/m^2, and a slowing of any size at session 3 and at 8 cd·s/m^2, where the upper limits were below zero; a lasting fall of the amplitude by more than 74 and 114 μV at sessions 2 and 3 is unlikely at 25 cd·s/m^2, and by more than 63 and 122 μV at 8. The terminal session rests on four animals, and its intervals are wider. The RM recordings before installation showed changes of the same size, so no slowing of the test eye persisted to the next session. The core temperature was within 1.3 °C of that in session 1.

TABLE 4.2. Change of the Jet ERG at the start of each later session from that of session 1, test minus control eye, mean (90% confidence interval) over animals 2 to 6; p_{Holm}, the smallest over the three sessions of the paired t test. Flash in cd·s/m^2; session 3 of animal 3 is left out; n, animals.

Measure	**Flash**	**Session 2** ($n = 5$)	**Session 3** ($n = 4$)	**Terminal** ($n = 2$)	p_{Holm}
Implicit time (ms)	25	−1.1 (−3.5, +1.3)	−1.8 (−3.0, −0.5)	−0.2 (−5.0, +4.5)	0.13
	8	−1.4 (−2.5, −0.3)	−1.2 (−2.0, −0.5)	+0.0	0.06
Amplitude (μV)	25	−8 (−74, +58)	−33 (−114, +49)	−11 (−517, +494)	1.00
	8	+16 (−63, +95)	−15 (−122, +93)	+2 (−431, +436)	1.00
Summed OPs (μV)	25	+69 (−37, +176)	+54 (+12, +97)	+18 (−224, +259)	0.17
	8	+92 (−3, +187)	+82 (+7, +158)	+60 (−136, +257)	0.25
Width at 80% (ms)	25	+0.8 (−4.1, +5.7)	+0.6 (−1.3, +2.6)	+0.0 (−13.4, +13.5)	1.00
	8	+0.0 (−1.5, +1.5)	+0.1 (−1.5, +1.7)	−0.1 (−2.1, +1.9)	1.00

TABLE 4.3. The b-wave and the summed oscillatory potentials at 25 cd·s/m^2 (the width is a secondary measure): mean change of each eye from the Jet recording before cooling to that during cooling, over both ring temperatures (five, five, and four sessions at 30, 60, and 120 min), with the 95% confidence interval of the test-minus-control difference in parentheses. p_{Holm}, paired t test of the change of the test eye against that of the control eye, corrected across the three session lengths within each measure and flash. g, Hedges' g of the test-minus-control differences, a point estimate (four or five sessions).

Measure	**Min**	**Test eye**	**Control eye**	**Test against control**	p_{Holm}	g
Implicit time (ms)	30	+5.8	+0.8	+5.0 (+1.7, +8.3)	0.027	+1.53
	60	+5.5	+0.2	+5.3 (+1.8, +8.8)	0.027	+1.52
	120	+5.2	−0.5	+5.8 (+2.7, +8.8)	0.027	+2.21
Amplitude (μV)	30	−56	−3	−52 (−100, −4)	0.116	−1.08
	60	−101	+46	−147 (−326, +32)	0.170	−0.82
	120	−41	−13	−28 (−248, +193)	0.718	−0.14
Oscillatory potentials (μV)	30	−40	+46	−86 (−185, +13)	0.139	−0.86
	60	−132	+5	−137 (−258, −17)	0.103	−1.13
	120	−128	+36	−164 (−352, +24)	0.139	−1.01
Width at 80% (ms)	30	+8.6	−5.0	+13.6 (−16.3, +43.6)	0.826	+0.45
	60	−1.8	+4.0	−5.8 (−30.1, +18.5)	0.909	−0.24
	120	+2.6	+1.6	+1.1 (−3.0, +5.2)	0.909	+0.31

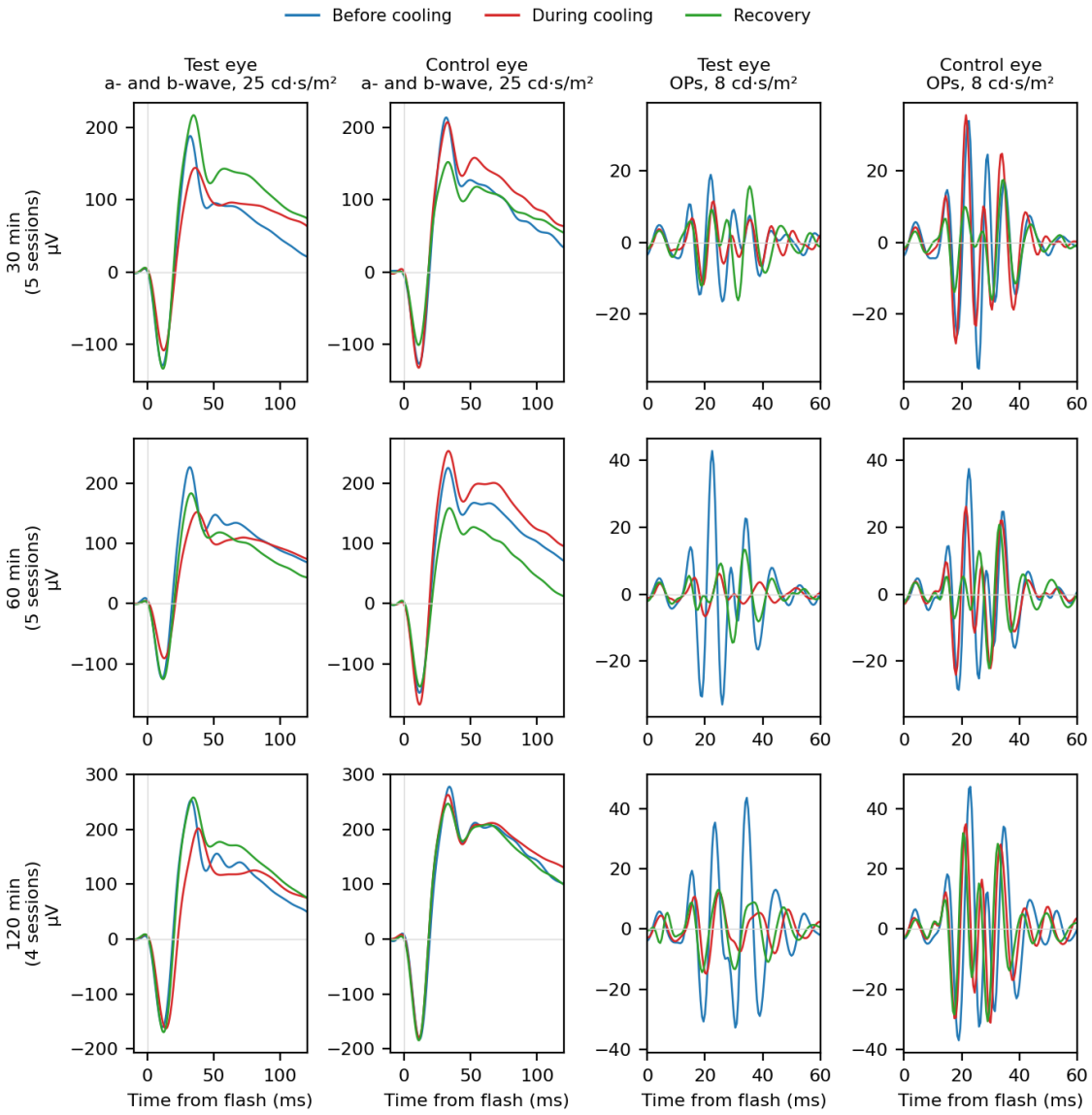


FIGURE 4.7. Mean electroretinograms of animals 2 to 6 after 30, 60, and 120 min of cooling (rows; five, five, and four sessions, both ring temperatures), Jet electrode: before cooling (blue), during cooling (red), and at the recovery recording after switch-off (green). Session 3 of animal 3 is left out. First two columns, a- and b-wave of the test and control eyes at 25 cd·s/m^2; last two, oscillatory potentials at 8 cd·s/m^2 (75–300 Hz). Averaging recordings whose peaks fall at different times lowers and broadens the mean peaks, the oscillatory potentials most.

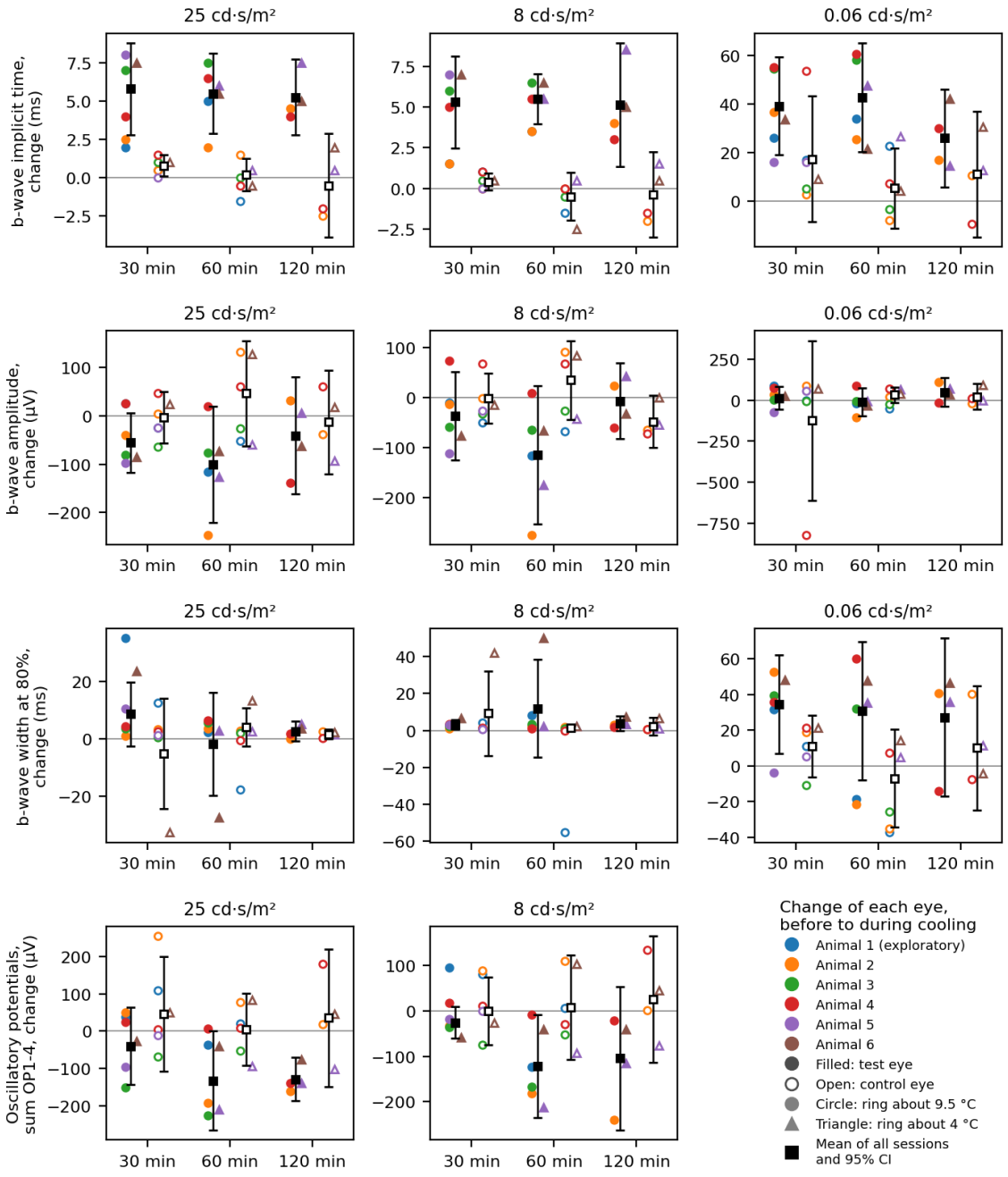


FIGURE 4.8. Change of each eye from its recording before cooling to that during cooling (Jet electrode), test (filled) and control (open), for each animal, by session length: b-wave implicit time, amplitude, and width at 80% of the amplitude, and summed amplitude of oscillatory potentials 1 to 4 (bottom), at the three flash strengths (oscillatory potentials at the two stronger). Circles, ring at about 9.5 °C; triangles, ring at about 4 °C; black squares, mean of all sessions with its 95% confidence interval. Session 3 of animal 3 is left out.

TABLE 4.4. The same measures at 8 cd·s/m^2, as in Table 4.3.

Measure	Min	Test eye	Control eye	Test against control	p_{Holm}	g
Implicit time (ms)	30	+5.3	+0.4	+4.9 (+2.1, +7.7)	0.009	+1.77
	60	+5.5	−0.5	+6.0 (+3.4, +8.6)	0.009	+2.29
	120	+5.1	−0.4	+5.5 (+3.6, +7.4)	0.009	+3.27
Amplitude (μV)	30	−37	−1	−36 (−82, +10)	0.196	−0.77
	60	−115	+35	−150 (−311, +12)	0.186	−0.92
	120	−7	−48	+41 (−57, +140)	0.275	+0.48
Oscillatory potentials (μV)	30	−25	+1	−26 (−100, +49)	0.394	−0.34
	60	−122	+8	−130 (−268, +9)	0.179	−0.93
	120	−104	+26	−131 (−272, +10)	0.179	−1.07
Width at 80% (ms)	30	+3.2	+9.2	−6.0 (−26.5, +14.5)	0.660	−0.29
	60	+11.8	+1.5	+10.4 (−15.6, +36.4)	0.660	+0.40
	120	+3.8	+2.2	+1.6 (+0.4, +2.8)	0.078	+1.50

4.3.4. Intraocular Pressure. From before placement of the devices to after their removal (Figure 4.9), the reading of the control eye rose in all 14 sessions, by a mean of 12.4 mmHg and up to 33.7 mmHg. That of the test eye rose in 7 and fell in 7, by a mean of 3.7 mmHg. Measured from where each eye started, the test eye therefore ended lower than the control eye in 12 of the 14 sessions, by 6.7 to 22.3 mmHg; only in sessions 1 and 2 of animal 2 did it end higher, by 18.0 and 7.5 mmHg. Averaged over the five animals, the test eye rose 8.8 mmHg less than the control eye (95% confidence interval from 18.8 mmHg less to 1.3 mmHg more). By session length (Table 4.5) the 95% confidence interval excluded zero only after 120 min, where the test eye rose 13.6 mmHg less (p_{Holm} 0.075); by the upper limits of the 90% confidence intervals, a rise of the test eye exceeding that of the control eye by more than 7.9 and 0.5 mmHg after 30 and 60 min is unlikely. At the start of the later sessions the reading was within 9 mmHg of that of session 1 in the test eye and within 15 mmHg in the control eye (Figure 4.10).

TABLE 4.5. Change in the intraocular pressure reading from before placement of the devices to after their removal, as in Table 4.3 (relative values: the tonometer is calibrated for the rodent eye).

Measure	Min	Test eye	Control eye	Test against control	p_{Holm}	g
Intraocular pressure (mmHg)	30	+10.2	+14.5	−4.3 (−20.2, +11.6)	0.491	−0.27
	60	+1.1	+10.2	−9.1 (−21.6, +3.4)	0.228	−0.72
	120	−1.3	+12.3	−13.6 (−23.9, −3.3)	0.075	−1.52

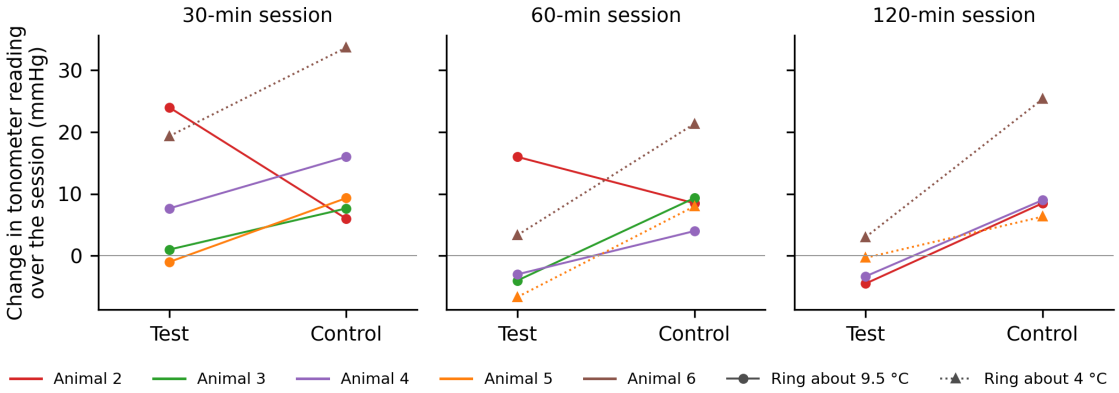


FIGURE 4.9. Change in the tonometer reading of the test and control eyes over each cooling session, from before placement of the devices to after their removal, for animals 2 to 6: ring at about 9.5 °C, solid line; about 4 °C, dotted line. Session 3 of animal 3 is left out.

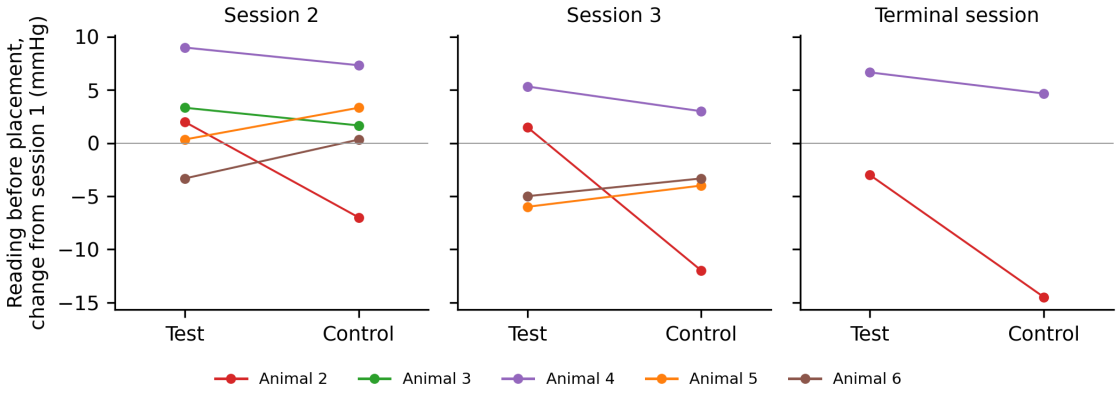


FIGURE 4.10. Tonometer reading of the test and control eyes before placement of the devices at each later session minus that before session 1, for animals 2 to 6. Session 3 of animal 3 is left out.

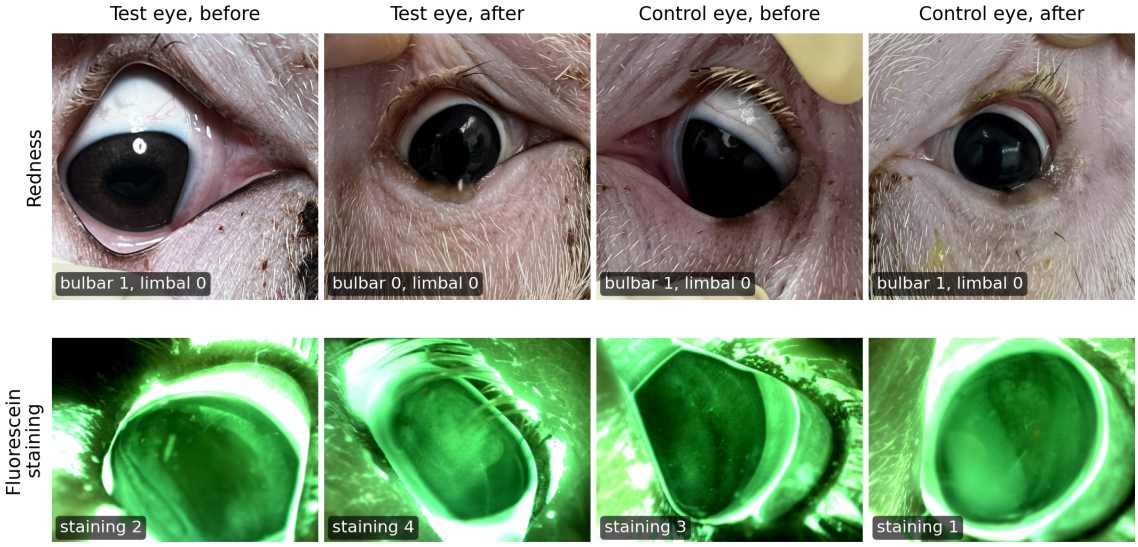


FIGURE 4.11. Representative photographs of the test and control eyes before and after a cooling session (animal 2, 120 min): conjunctival redness (top) and fluorescein staining (bottom), with the grades of record (0 to 4; bulbar and limbal redness, corneal staining).

4.3.5. Ocular Surface. Table 4.6 gives the grades at each session. Before the first session, before either eye had carried a device, in the ten eyes bulbar redness was grade 0 to 2, limbal redness grade 0 or 1, and corneal staining grade 3 in three eyes, all three later cooled (Figure 4.11; all selected photographs in Appendix H). Over the 14 sessions the grade of the test eye changed by at least one step more toward harm than that of the control eye in 2 sessions for bulbar redness, 3 for limbal redness, and 1 for staining. In most of these the grade of the test eye did not rise: both eyes improved or stayed the same, the test eye by less. The redness of the test eye rose only in session 1 of animal 4, limbal, from 0 to 1. Its staining rose in five sessions, in four of them with the control eye rising as much or more; in the fifth, session 3 of animal 2, it rose from 2 to 4 against 3 to 1 in the control eye, and was still grade 4 in the terminal session, against grade 2 in the control eye.

TABLE 4.6. Grades of the ocular surface of the test and control eyes before and after each cooling session and at the terminal session, median (range) over animals 2 to 6 (0 to 4 scales). Session 3 of animal 3 is left out; n, animals.

			Bulbar redness		**Limbal redness**		**Corneal staining**	
Session		n	Test	Control	Test	Control	Test	Control
1 (30 min)	Before	5	1 (1–2)	2 (0–2)	1 (0–1)	1 (0–1)	3 (1–3)	1 (1–2)
	After	5	1 (0–1)	1 (0–2)	1 (0–1)	1 (0–1)	3 (1–3)	3 (1–3)
2 (60 min)	Before	5	1 (1–2)	1 (0–2)	1 (0–3)	1 (0–3)	2 (1–3)	2 (1–2)
	After	5	1 (0–1)	1 (0–2)	1 (0–1)	0 (0–2)	2 (1–3)	3 (2–3)
3 (120 min)	Before	4	1 (0–1)	1 (0–1)	0.5 (0–1)	0.5 (0–1)	2 (2–3)	2.5 (1–3)
	After	4	0 (0–1)	0.5 (0–1)	0 (0–1)	0 (0–1)	2.5 (1–4)	2 (1–2)
Terminal		2	1.5 (1–2)	1.5 (1–2)	0	0.5 (0–1)	2.5 (1–4)	1.5 (1–2)

4.3.6. Systemic Course and Adverse Events. No session was paused or ended for a physiological reason (Figure 4.12), and all values were within the limits of Section 4.2.3. Oxygen saturation stayed at or above 93%, heart rate between 74 and 153 beats per minute, end-tidal carbon dioxide between 35 and 53 mmHg, and core temperature at or above 35.1 °C, its lowest in session 2 of animal 5.

On the day after session 1 of animal 2 the left eyelid, on the control side, was moderately swollen, closing the eye, and the animal scratched at it; the right eyelid was only very mildly swollen, and the right eye appeared normal. The next day both eyes were equally open and the mild inflammation was resolving. After session 2 animal 3 had expiratory wheezing and tracheal irritation, with blood on the endotracheal tube, which may have been too large; it was treated with meloxicam (0.1 mg/kg by mouth) and was much improved the next day. After session 3 it arrested during recovery and was euthanized by the BRL staff with the same agent (Fatal-Plus, 10 mL intravenously). Necropsy found laryngeal necrosis, which the BRL veterinarian and veterinary pathologist attributed most likely to a traumatic intubation in session 2, and cardiac hypertrophy, judged incidental and not the cause of death; neither was attributed to the device (Figure G.1, Appendix G). Both eyes were collected after death.

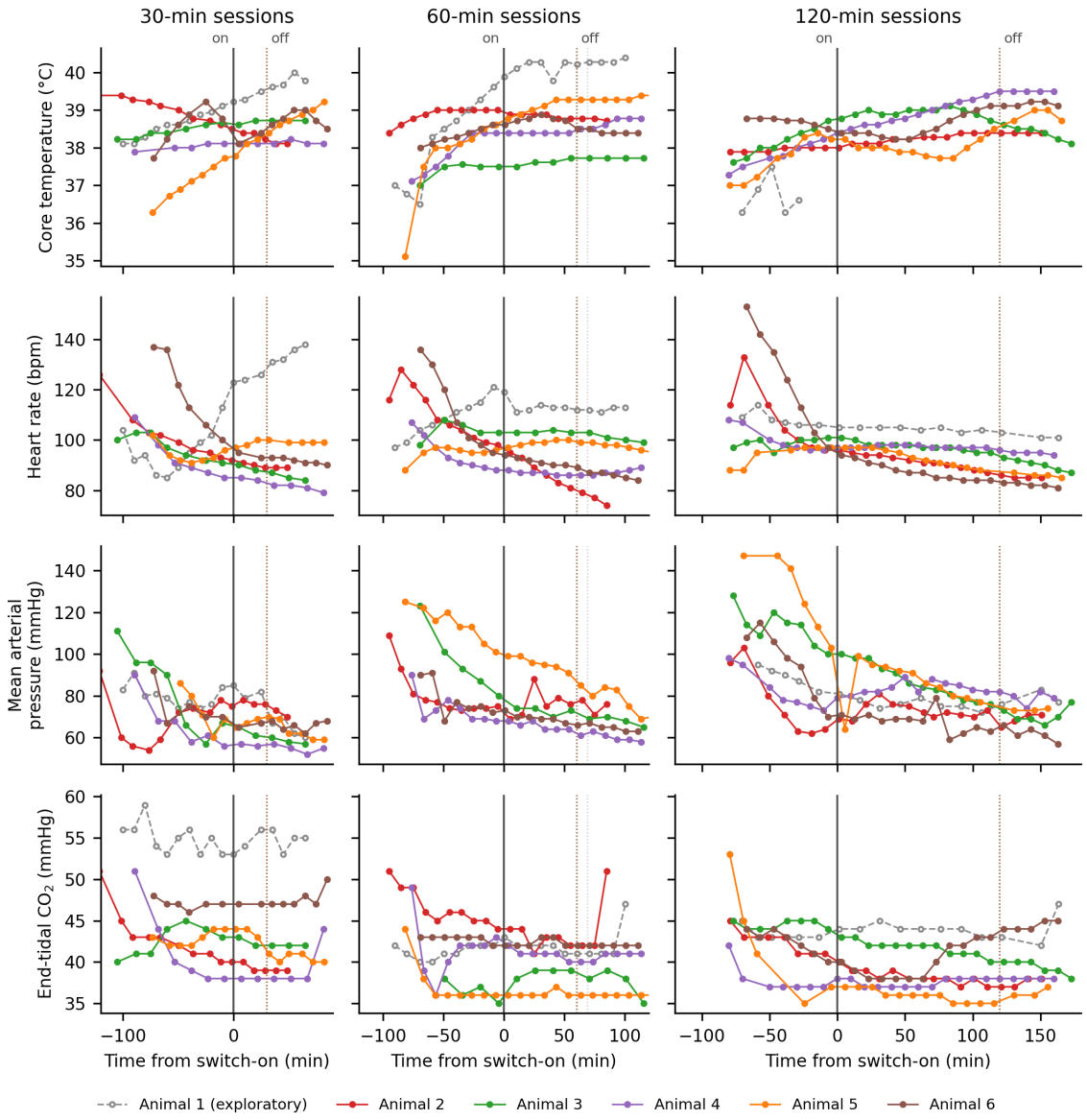


FIGURE 4.12. Core temperature, heart rate, mean arterial pressure, and end-tidal carbon dioxide during each cooling session against time from switch-on, from the anesthesia record (a reading about every 10 min). Solid vertical line, switch-on; dotted lines, switch-off of each session. Animal 1, exploratory, dashed grey line with open circles; its mean arterial pressure was not recorded in session 2, nor its core temperature after switch-on in session 3.

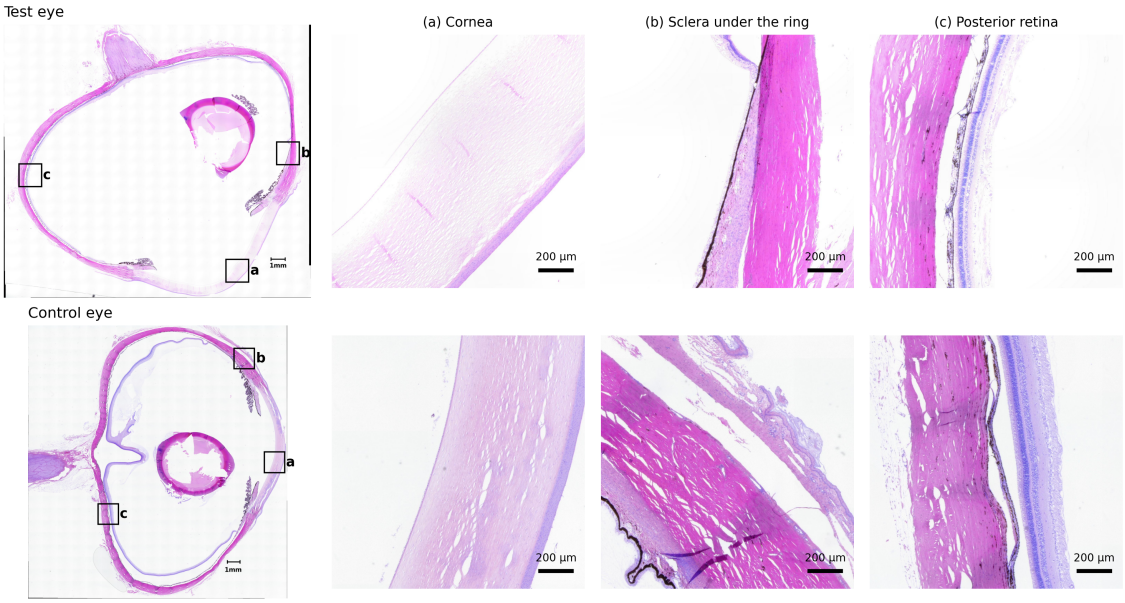


FIGURE 4.13. Hematoxylin and eosin sections of the test (top) and control (bottom) eyes of animal 4, slide 5: the whole section and, enlarged, regions chosen on it: (a) the cornea, (b) the sclera behind the limbus, under the ring, and (c) the posterior retina.

4.3.7. Histology. Both eyes of each animal were enucleated and fixed, those of animal 3 after its death, with a different interval from death to fixation. The sections of the six animals were read (Figure 4.13, animal 4; one slide of each animal in Appendix H). In no animal did the cooled eye show a change absent from the control eye: in the masked comparison the cooled eye was never named as the one with more change. Neither eye of any animal showed inflammation, necrosis, or thrombosis, and the conjunctiva, limbus, and sclera under the ring looked the same in both eyes. The control eyes of animals 3 and 4 had a focal loss of the corneal epithelium, 0.5 to 2 mm wide, at the same place in each section, with no inflammation or healing, so recent: from handling, tonometry, or the electrode near the end of the session. It was not in the cooled eye. In the cooled eye of animal 3 the corneal epithelium was lifted off Bowman's layer as an intact sheet in all three sections, with no damage to the cells beneath, most likely in processing. In both eyes of every animal the retina was detached, its inner layers vacuolated, and the lens cracked, all processing artefacts.

A first set of scans of animal 1, fixed in formalin, showed the retina detached in both eyes, which led to the change of fixative from animal 3 onward; that set was not graded and is not used here.

4.4. Discussion

Apart from the relay interruptions, the device held the ring near its set point for up to 2 h against the heat of the living eye and lids; the relay of the Gen 3 controller was the weak element, and the MOSFET module that replaced it ran without an interruption (Section 4.3.2). Only this Gen 3 device, with the reduced-bulk aluminum ring, was used on an animal (Section 4.2.4); the rest of the Gen 4 design remains to be tested in vivo (Section 5.4).

Three practical problems affected the earlier sessions: the device did not always fit under the eyelid until the lid was drawn over it with tweezers; the ring thermistor sometimes came out of the device, giving poor readings or none, until its insertion point was fixed with silicone; and the channel was first calibrated with a second thermistor in the well that did not touch the walls, so the ring read about 5.5 °C low until a custom piece put the thermistor in contact with the aluminum (Section 4.2.6). With that correction, most of the exposure was at about 9.5 °C, not at the 4 °C intended.

During cooling the b-wave of the cooled eye became slower in every session, at both primary flashes, and wider in all but one, while that of the control eye did not slow (Section 4.3.3, Table 4.3). A longer implicit time is the response expected of a cooler retina [**73, 74, 122, 76**], and the control eye, which carried an identical ring through the same anesthesia, core temperature, and time, shows that the change did not arise from these. The control eye also widened, by less, so part of the widening is shared; the larger widening of the cooled eye fits a slower and possibly uneven response of its retina. The slowing diminished as the ring rewarmed after switch-off, which also points to temperature rather than to a lasting effect of the contact. The

five sessions at about 4 °C slowed within the range of the others and are too few to show a relation to the ring temperature. Ischemia caused by the cold, or a difference in the pressure of the two rings, cannot be excluded as another explanation.

The implicit time, with the control eye that did not slow, not the amplitude, is the better evidence that the slowing of the test eye arises from its temperature. The slowing, about 5 ms, is several times the variability of b-wave timing between sessions on this device, under 1 ms in healthy eyes under light adaptation [**146**], whereas the fall of about 14 to 18% in the b-wave amplitude of the test eye is within the normal difference between the two eyes of a subject, whose 95th percentile is 27 to 36% [**143**]; the amplitude also varied widely between recordings, and no confidence interval of the fall of the summed oscillatory potentials excluded zero (Table 4.3). Anesthesia, too, lowers the amplitude more than it delays the b-wave: in dogs under isoflurane the b-wave amplitude was 54% lower than in the same dogs awake, whereas its implicit time did not differ significantly [**147**], and in anesthetized pigs the amplitude of the multifocal ERG fell over the 150 min after induction [**148**], so a decline of the amplitude in either eye over a long session is expected.

The hypothesis that the ERG returns to its baseline after rewarming was tested only in part: the recovery recording was made before the ring had recovered fully (Section 4.2.9), so whether the difference between the eyes closes completely was not measured. The b-wave implicit time of the test eye returned to within the limit in 11 of the 12 recovery recordings at 25 cd·s/m^2 but in only 8 at 8 cd·s/m^2, where the limit is narrower; in the other four it was still 3.0 to 4.5 ms slower, consistent with a retina not yet back to its temperature.

Whether the cooling left a lasting change in the retina is answered better by the start of the next session than by the recovery recording. At the start of each later session, compared with session 1, before either eye had been cooled, the test eye did not differ from the control eye in the b-wave implicit time, the b-wave amplitude, or the summed oscillatory potentials, at either flash (Section 4.3.3, smallest p_{Holm} 0.06).

Where the means moved, they moved against a lasting effect of the cold: relative to the control eye the b-wave of the test eye was 0 to 1.8 ms faster than in session 1, not slower, and its summed oscillatory potentials were larger by 18 to 92 μV. The slowing still present in four recovery recordings therefore did not persist to the next session, and the cooling did not add up over the sessions of an animal. By its 90% confidence intervals a lasting slowing of more than 1.3 ms is unlikely at the start of sessions 2 and 3 (Table 4.2); the terminal session, on four animals, bounds it less tightly.

The safety endpoints showed no harm attributable to the device within the measurements made. The reading of the control eye, which carried an identical ring that was not cooled, rose over almost every session, so the placement of the device, together with the anesthesia [**149, 150**] and fluids, raised it; that of the cooled eye rose less, or fell, in most sessions (Section 4.3.4), so the cooling reduced or reversed the rise that the device produced on its own. Intraocular pressure under anesthesia moves by several mmHg with induction, intubation, and the agents given, and in anesthetized rats it falls with body temperature when no heat support is given [**151**]; the readings before and after a session were taken at different stages of the anesthetic, but both eyes shared the anesthetic and the body temperature, so these effects cancel in the difference between the eyes.

That local cooling lowers the pressure is established in humans; its mechanism is not. A stream of cold air on one eye lowered the pressure from 14.1 to 12.6 mmHg, with no change in aqueous flow or outflow facility, and the fall was attributed to the episcleral venous pressure, which fell too [**152**]. In a randomized trial, a cold mask worn for 10 min lowered the pressure by 2.0 mmHg and a heated mask raised it [**153**]; the authors attributed the fall to less aqueous production, through vasoconstriction and slower metabolism of the ciliary body, but did not measure it. A coil on the rabbit sclera left the pressure unchanged [**74**]. No study has tested whether cooling offsets a rise produced by a device on the eye. The episcleral veins offer one route: the rise under a scleral lens is attributed to their compression [**22**], the ring rests on the

same region of the sclera, and cold lowers their pressure [**152**]. Neither the episcleral venous pressure nor the aqueous flow was measured here, so this explanation is a hypothesis. Cooling also thickens the cornea, which alters applanation readings [**154**] and may alter a rebound reading; corneal thickness was not measured, so the present readings cannot separate these effects.

Redness and staining were worse in the cooled eye in only a few sessions, of every session length (Section 4.3.5). Bulbar redness fell over a session in both eyes, and in all but two of the 14 sessions that of the cooled eye changed no more toward harm than that of the control eye; limbal redness showed no pattern. Cold is known to reduce conjunctival redness: in a masked trial a cold compress applied for 5 min reduced bulbar hyperemia faster than no treatment [**156**], and cold air on the eye lowered the pressure of the episcleral veins, which drain the same vascular bed [**152**]. A lower bulbar redness in the cooled eye is therefore plausible, but these counts are too few, and the grades too coarse, to show it.

Corneal staining rose over a session in the control eye at least as often as in the cooled eye, so most of it arises from what the two eyes shared: the ring, the corneal ERG electrode, the topical anesthetic, tonometry, and hours of exposure under anesthesia. Corneal ERG electrodes alone cause transient epithelial changes, seen with fluorescein in 62% of normal subjects and associated with the topical anesthetic [**157**]. As both eyes carried all of these, the data cannot say which caused the staining, only that the cooling added nothing that could be measured, so neither redness nor staining points to harm from the cooling. The rises mostly resolved by the next session; the exception, grade 4 persisting to the terminal session in the cooled eye of animal 2, had no counterpart in its histology, in which the corneas of the two eyes looked alike. The staining grades are weak evidence in either direction: three of the ten eyes were graded 3 before any device had touched them, and pooled dye cannot be told from staining (below).

The one death in the study followed a complication of intubation, and the eyelid swelling of animal 2 was on the control side.

The histology showed no change from the cooling: in the masked comparison the cooled eye was never named as the eye with more change, and the conjunctiva, limbus, and sclera under the cooled ring looked like those under the ring that was not cooled (Section 4.3.7). The only losses of corneal epithelium were in control eyes, so they came from what the two eyes shared, not from the cold. The retina was detached and vacuolated in both eyes of every animal by processing, so the sections can show only gross damage to it.

The in-vivo study is based on six animals, and the ERG after 120 min on four (animals 2, 4, 5, and 6: animal 1 is exploratory and session 3 of animal 3 is left out). The main results pool sessions at two ring temperatures, about 9.5 and 4 °C, so they describe cooling over that range rather than at one set point. With the measured variability and five animals the paired comparison has a power above 99% for the implicit time and about 73% for a 30% change in amplitude; for the width, the two outlying sessions of animal 6 widen the spread so much that the power is low (8%). The fixed order of the three session lengths confounds any trend with session length with repeated anesthesia and cumulative exposure. The test eye was always the right eye, fitted and recorded first, so the control eye was slightly longer dark-adapted, before cooling too; in the porcine ERG the b-wave implicit times at strong flashes do not differ significantly between 15 and 25 min of dark adaptation [**158**]. The tonometer gives relative values only and was not used with the devices on the eyes, when a device resting on the sclera can raise the pressure [**155, 22**]. Redness and staining were graded by the author, who knew which eye had been cooled, on one photograph per eye and time point that the author also chose; the grades are whole steps of a 0 to 4 scale, so most sessions tie and one step is the smallest change that can be seen. Dye pooled on the surface cannot be told from staining on some photographs. The histology was read by the author, who knew which eye had been

cooled, and not by a pathologist; it covers sections of a central slab of each globe, from the cornea to the optic nerve head, and there was no slit-lamp, fundus, or corneal endothelial examination. The core temperature of the anesthetized animals should be read against the normal porcine range of 38.5 to 39.5 °C. No temperature was measured inside the eye, and the relation between b-wave timing and retinal temperature has not been established in this preparation, so the size of the delay cannot be converted into a retinal temperature.

CHAPTER 5

General Discussion and Conclusions

5.1. Synthesis Across Aims

The three aims approach one question from three sides: how much of the cooling applied to the anterior sclera reaches the back of the eye. Aim 1 measured it directly, in eyes without a circulation. Aim 2 compared ring designs in the same preparation and used a computational model to estimate what a circulation does to it. Aim 3 looked for its functional trace, a slowing of the electroretinogram (ERG), in the living eye, where no temperature could be measured.

In the isolated eye a ring held at 4 °C produced a steep gradient from front to back (Figure 5.1A). The anterior sclera cooled the most and the proximal optic nerve the least, by a few degrees, and at that site the two species cooled by nearly the same amount, with no difference detected (Table 2.7).

The refinements of Aim 2 left this deep cooling largely unchanged (Chapter 3). In the same donor eyes, the reduced-bulk ring, with about half the volume, and the ceramic ring both gave the proximal optic nerve about the same cooling as the original ring. The reduced-bulk ring cooled it slightly less, by less than the spread between eyes. The other elements of the Gen 4 design were examined only in the model or on the bench.

The isolated eye has no blood flow, and the choroid of the living eye carries heat to the tissue the device is trying to cool. The model of the perfused eye therefore gives less cooling than the bath experiments at the sites away from the ring (Figure 5.1B). This gap is not a measurement of the effect of perfusion: the model has not been compared with a temperature measured in a perfused eye, and most of the gap reflects the difference between the two conditions (Section 3.3.6). Across the model runs, body

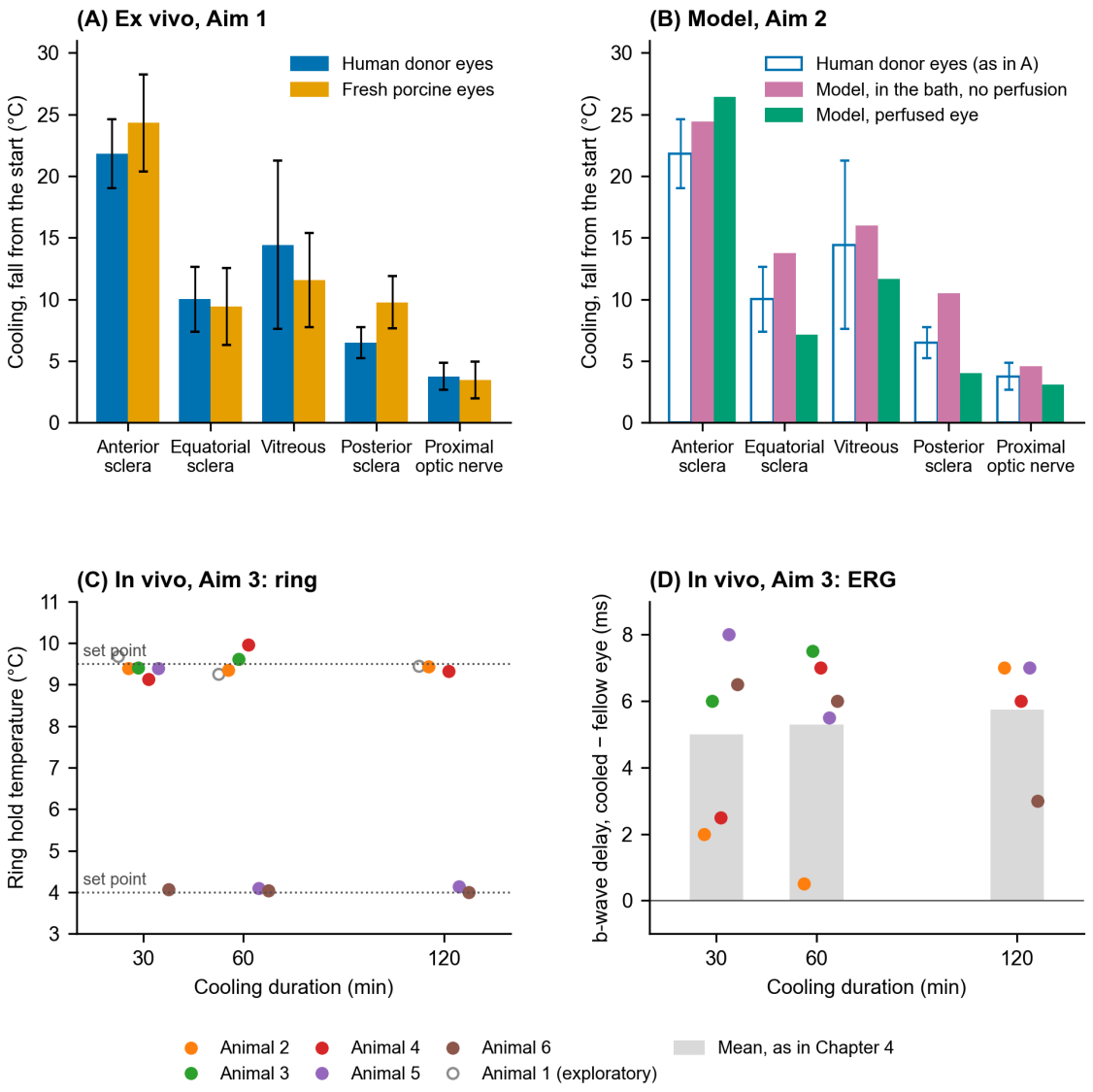


FIGURE 5.1. Cooling at the sites shared by the three aims. (A) Fall in temperature from the start of cooling over the 20-min record in human donor and fresh porcine eyes with the original ring (mean and SD; Tables 2.3 and 2.4). (B) The same fall in the human donor eyes (open bars, as in A), in the model of the enucleated eye in the oil bath without perfusion, with an oil coefficient chosen before the comparison, and in the model of the perfused eye in the orbit (Section 3.3.6); the perfused model is read after 35 min, the measurements over minutes 16 to 20 (Section 3.2.3). (C) Mean ring temperature held in each in-vivo session, corrected, by session length. (D) Delay of the b-wave implicit time of the cooled eye relative to the fellow eye during cooling, at 25 cd·s/m^2; bars, the means over both ring temperatures (Section 4.3.3). Session 3 of animal 3 is left out; animal 1, exploratory, open grey circles (panel C only).

and ring temperature set the temperature at the back of the eye far more than the design of the ring did.

In vivo, the cooling applied at the anterior sclera reached the retina closely enough to change its function, but the ERG does not show by how much the retina cooled. No temperature was measured in vivo, so Chapter 4 addresses the question only through the ERG, whose timing slows as the retina cools [**73, 74, 76**]. The ring was held close to its set point for up to 2 h, and during cooling the b-wave of the cooled eye slowed from its own baseline in every session, and that of the fellow eye did not (Figure 5.1C and D). The relation of the ERG to retinal temperature has not been established in the pig under these conditions, so the delay cannot yet be set against the cooling that the model predicts. No adverse event was attributed to the device, and the histology showed no damage in the cooled eye absent from the control eye.

The isolated eyes fix the size and the pattern of the cooling that the ring produces when no circulation opposes it, and show that it is much the same in human and porcine eyes and with the reduced and ceramic rings. The model places that cooling against a circulation and identifies body temperature and ring temperature as what sets the temperature reached at depth. The in-vivo study shows that the same ring, at the same temperature, on a perfused eye, changes retinal function in the direction expected of cooling, in sessions of up to 2 h that the eye and the animal tolerated as far as was measured. What links the three, a temperature measured at the back of a perfused eye, is still missing.

5.2. Comparison with Prior Work

The approaches reviewed in Section 1.4 fall into two groups. The first cools the inside of the eye directly and reaches the retina or the nerve, but each method requires surgical access [**73, 79, 80**]. The second acts at the surface without surgery [**85, 87, 89**], and for none of these devices has a temperature at the back of the eye been reported. Research preparations in living animals had shown that cold applied

outside the eye lowers posterior temperatures [**73, 74, 64**], and the one study with feedback control delivered its fluid into the anterior chamber [**78**].

The SCEC belongs to the second group by its access and aims at the target of the first. It is placed without surgery, it holds the sclera at a set temperature for tens of minutes to hours instead of seconds, and its effect has been measured at located sites inside the eye, up to the optic nerve. The fresh-porcine part of this work, published as [**1**], established that effect in porcine eyes. This dissertation extends it to human donor eyes and to two further ring designs, and applies the device to the living animal. To the author's knowledge it is the first scleral-contact cooler whose temperature field has been measured in human donor eyes at sites verified by micro-computed tomography. It also differs from the other contact coolers that come closest. The ETX-4143 scleral devices cool for 4 to 10 min from a reservoir frozen before use and are aimed at ocular surface pain [**91, 90**]. Lachapelle's coil cooled the retina but had to be glued to the sclera with the eyelids removed [**74**]. Kataoka's circulating-water contact lens protected the cornea and was not tested at the back of the eye [**84**].

The SCEC trades depth of cooling for the absence of surgery, and the comparison shows this limit of the approach. Direct methods bring cold fluid to the retina or the nerve itself, whereas a ring on the anterior sclera cooled the proximal optic nerve of an isolated eye by a few degrees. In a Pennes model a circulation reduces this further, whereas a multiphase blood model predicts deeper cooling [**54**].

5.3. Clinical Translation

The two clinical settings that motivate the device (Section 1.1) ask different things of it: speed of application in retinal artery occlusion, and tolerance of long exposures in perioperative visual loss. In retinal artery occlusion the injury has already begun when the patient is seen, usually hours after onset [**2, 14, 15, 16**], so cooling could only follow the insult. In perioperative visual loss, of which supine cardiac surgery is

the natural target because direct pressure on the eye is to be avoided in prone surgery [**21**], the period of risk is known in advance, and cooling could begin before it and continue through it. The model places the reach of the device at the retina and the optic nerve head, not at the nerve 1 cm behind the globe (Section 3.4), whereas most ischemic optic neuropathy after spine surgery is posterior [**19**].

The effect is best described as a temperature reduction of a few degrees at the back of the eye, not as the attainment of a target temperature. What the device delivers has to be set against what the laboratory evidence suggests is needed. The measured effect at the proximal optic nerve of the isolated eye was a fall of 3 to 4 °C. Protection of the retina has been reported for temperature reductions of this size (Section 1.3), including the 3 °C step used here as a benchmark [**62**]. Whether a perfused human eye cools by as much is not known. The model suggests somewhat less. With fever the reduction is about the same, but the temperature reached is higher, because it follows body temperature.

Several steps separate the present device from clinical use. The temperature reached at the posterior pole of a perfused eye has to be measured. The exposures tested in vivo, up to 120 min, are shorter than the operations in which perioperative visual loss occurs (a mean anesthetic duration of 9.8 h in the registry cases after spine surgery [**19**]). The tolerance of the anterior sclera, cornea, and lens to longer contact at low temperature is not established. The intraocular pressure with the device on the eye has not been measured. No limit for the cold side has been based on tissue tolerance. The board stops below 0 °C only to prevent freezing, and the controller used in vivo has no cold-side limit at all. The limit of 43 °C used for the warm surfaces is taken from the standard for medical electrical equipment [**129**] and was examined in the model only. Biocompatibility of the materials in contact with the eye, sterility, electrical safety, and the software of the controller have not been addressed.

The in-vivo safety evidence covers three exposures of 30, 60, and 120 min with the ring near 9.5 °C, and near 4 °C in the later sessions, given over about ten days

to an anesthetized pig with an uninjured eye. Within that envelope, in the animals studied, the cooled eye showed no larger rise in intraocular pressure than its fellow once the device was removed. Redness and corneal staining were rated worse in the cooled eye in only a few sessions, without a relation to session length. No adverse event was attributed to the device, and the b-wave timing was restored after switch-off in eleven of the twelve recovery recordings at the stronger flash and eight at the weaker (Section 4.4). The histology, which is the test of injury to the cornea, sclera, and retina, showed none in the cooled eye, but was read by the author, not by a pathologist. The evidence does not extend to longer exposures, to repeated daily use, to awake patients, or to eyes that are already ischemic.

5.4. Limitations

Each chapter gives the limitations of its own work (Sections 2.4, 3.4, and 4.4). Four cross the aims.

No temperature was measured in a perfused eye. The bath is not a model of perfusion, the model has not been validated against such a measurement and depends on perfusion rates taken from the literature, and the ERG is indirect evidence of retinal cooling, which pressure on the globe or cold-induced ischemia can also mimic.

The groups were small: nine human eyes from five donors, five fresh and five preserved porcine eyes, and six animals, the first of them exploratory. The porcine experiments of 2022 were run with the current adjusted by hand and the human experiments under the comparator controller. Conclusions from the eyes rest on excursions, which are better determined than absolute temperatures (Appendix A).

Several analysis settings were set or revised after the data had been seen: the benchmark of 3 °C, the vitreous position limit of 75%, the 3.3-mm spatial screen, the 4-min final window, and the visual exclusions.

SCEC Gen 4 was not assembled as a whole. The in-vivo device was Gen 3 with the reduced-bulk ring (Section 4.2.4). The ceramic ring was tested ex vivo and in the

model, the insulating layer, the corneal lens of v3, and the smaller heat exchangers only in the model, and the integrated board only on the bench, where the fault of its reference voltage remains open.

5.5. Future Work

The first need is a direct measurement of temperature at the back of a perfused eye during cooling, for example with a single fine probe in a terminal experiment. It would test the model, fix the factor by which a circulation reduces the deep cooling, and turn the ERG from the only evidence of retinal cooling into a measure that can be calibrated against temperature.

The model should also be run over the plausible range of choroidal blood flow and then compared with that measurement. The control board needs a revision in which the reference voltage cannot disturb the current limits, a sensor on the warm side, a cold-side limit based on tissue tolerance, and a test of each safety cutout on the hardware. The thermoelectric module can also be developed further, with a higher density of thermoelectric posts and a module printed directly on the ring. The question that motivates the device, whether this degree of cooling protects the retina or the optic nerve, requires a study in an injury model, with cooling begun at a clinically realistic time relative to the insult.

The computational work would be extended in two steps. A complete model of the set-up would couple the eye and orbit to the ring, the thermoelectric module with its Peltier, Joule, and conduction terms, the heat exchanger, and the coolant loop under the control law. It would thus predict tissue temperatures from the drive current instead of from an imposed ring temperature. An optimization on that model would then replace the comparison of a few chosen designs. It would range over the geometry and material of the ring, the module, and the heat exchanger, constrained by the 43 °C limit for warm surfaces, the power of a portable supply, and the size of the device. Both should be run with more than one representation of perfusion,

because Pennes and multiphase models predict opposite effects of the circulation on posterior cooling [**54**].

Several additions to the device itself were conceived in this work and remain to be built or evaluated (Table 3.1). A contact-pressure sensor in the ring would report the force applied to the eye, which bears on the intraocular pressure during treatment. Under scleral lenses the rise in pressure is attributed to compression of the episcleral veins, whose pressure is about 7 to 11 mmHg, and a lens exerts its full compression only after more than 4 h of wear [**22**]. The intraocular pressure would also be measured during application and over exposures of several hours, as it has been during scleral-lens wear [**155**]. The present ring leaves no access for a rebound tonometer. An electrode in the ring would allow the electroretinogram to be recorded through the device, without a separate corneal electrode. Other tools developed in the same laboratory could turn the ERG into a spatial measure of cooling. The contact lens electrode array (CLEAr) records the ERG at many sites on the cornea at once, and the differences between sites reflect where on the retina the response arises [**159, 160, 161**]. The hemispherical stimulus domes of the laboratory stimulate selected regions of the retina. PeriStim presents patterns to the mid and far peripheral retina [**162, 163**], and its LED successor, HoneyDome, delivers full-field and partial-field flashes [**164, 165, 166**]. Used together during cooling, they could show where the retina cools and how far the cooling extends.

Thermistors built into both ceramic plates of the thermoelectric module would measure its cold and its warm side directly, which the 43 °C limit for the warm surfaces and a future cold-side limit based on tissue tolerance both require. A ring that detaches from the module could be replaced or sterilized between patients, and an injection port through the ring would give access to the eye without removing the device. Finally, the silicone potting between the plates of the module, present in every device used here, has not been evaluated on its own for electrical isolation or for protection against moisture.

The reverse polarity used for experimental equilibration could also give the device a therapeutic heating mode, which would need its own validation of temperature control, safety, and efficacy. Existing ocular heating applications, including cyclophotocoagulation, act through different mechanisms [**167**] and do not validate moderate heating by this device.

Thermal analgesia is another possible application. Cooling affects corneal sensory-nerve activity [**168**]. Whether the SCEC relieves pain would have to be tested with a pain endpoint and an appropriate exposure protocol.

Other uses of controlled ocular temperature could follow from the same platform, none of them tested here. Cooling could protect adjacent tissue during laser procedures, as a cooled contact lens did in the rabbit [**84**], or support hypothermic vitreoretinal surgery, in which cold irrigation protected the ischemic rabbit retina [**79**] and the infusion alone already cools the vitreous by about 4 °C in patients [**49**]. Each would need its own temperature target and its own evidence.

Part of this work has been published, and the rest is to be submitted as a series of papers. The fresh-porcine experiments of Aim 1 were published with Y. Nakamura [**1**]. Two manuscripts on the ex-vivo work are in preparation: a methods paper on the ex-vivo system, and the comparison of human donor and porcine eyes. The computational model and the ring designs of Aim 2 will be the subject of a computational manuscript. The in-vivo safety and ERG study of Aim 3 will be submitted for publication, and is also planned for presentation at the ARVO Annual Meeting.

5.6. Conclusions

A thermoelectric ring held at 4 °C on the anterior sclera cooled the proximal optic nerve of isolated human and porcine eyes by a few degrees within the record. The first hypothesis held for the occurrence and timing of this cooling but not for its size. Every eye cooled at that site and approached its final temperature within 15 min, but not every eye reached the 3 °C benchmark. No difference was detected at that

site between the species, or between preserved and fresh porcine eyes. The second hypothesis was supported at the proximal optic nerve, where the reduced-bulk ring and the ceramic ring each cooled it about as much as the original ring. A model of the perfused eye indicates that blood flow reduces the deep cooling, by a factor that remains to be measured, and that body temperature, more than the design of the ring, sets the temperature reached.

In anesthetized pigs the device, fitted with the reduced-bulk ring, held the ring near its set point for up to 2 h. The b-wave of the cooled eye slowed and widened from its own baseline while that of the fellow eye did not slow, a change consistent with retinal cooling, and its summed oscillatory potentials fell. The first part of the third hypothesis was supported, as a longer implicit time and a smaller b-wave amplitude were found together at both primary flash strengths in 10 of the 14 sessions, and slower and smaller oscillatory potentials, an exploratory measure, in 9. After switch-off the b-wave timing returned to its own baseline in 11 of the 12 recovery recordings at 25 cd·s/m^2 and 8 at 8 cd·s/m^2, and the oscillatory potentials in 11 and 10. Since these recordings were made before rewarming was complete, the second part was tested only in part. In the animals studied the intraocular pressure after removal of the device, the ocular surface, and the systemic course showed no harm attributable to the device, and the histology of the cooled eye showed no damage absent from the control eye.

The contributions of this dissertation are the following.

(1) The temperature field produced by a scleral-contact cooler was measured in human donor eyes, at sites inside the eye located by micro-computed tomography.

(2) Human donor and fresh porcine eyes were compared with the same ring, and they cooled in the same spatial pattern and by a similar amount at the proximal optic nerve.

(3) A computational model showed that an alumina ring cools almost as well as an aluminum one, that body and ring temperature matter more at depth than the design of the ring, and that blood flow reduces the deep cooling.

(4) An integrated control board for the device was designed and tested on the bench.

(5) The device was applied to living eyes for the first time, in anesthetized pigs, where it held the ring near its set point, slowed the ERG of the cooled eye, and caused no harm attributable to it in the measures made.

(6) The analysis of the ex-vivo experiments and of the model is documented as code that regenerates every table and figure from the raw data (Appendix C).

Before clinical use, the temperature at the back of a perfused eye must be measured, longer exposures must be shown to be tolerated, and protection of the retina or the optic nerve must be tested in a model of ischemic injury.

Appendices

APPENDIX A

Temperature Measurement Uncertainty

This appendix gives the budget behind Section 2.2.4: an expanded uncertainty ($k = 2$) of ±1.8 °C for a single reading and, for a within-trace excursion ΔT in the human eyes, ±0.40 °C at the proximal optic nerve and ±0.85 °C at the ring. It follows the Guide to the Expression of Uncertainty in Measurement [**202**]: standard uncertainties u_i are combined in quadrature, and Type B contributions bounded by a half-width a are rectangular ($u_i = a/\sqrt{3}$) unless stated otherwise.

A.1. Measurement Chain

Each thermistor (NTC, $R_{25} = 10$ kΩ, $\beta = 3976$ K) sat in a linearized divider and was read through an amplifier by a 16-bit data-acquisition system with a 20 V input range (slope 2.06 °C V^{-1} between the 37 and 20.5 °C plateaus). Human-donor channels were converted with a per-channel least-squares line through plateaus at 4.0, 20.5, and 37.0 °C in a dry-block calibrator, and porcine channels with a fixed linear transfer function trimmed at 20.5 and 37.0 °C (Section 2.2.4). The human lines miss their own plateaus by up to 0.57 °C, more than the 0.34 °C measured on the porcine chain in 2022, so the linearity terms come from them. The tables below are those of the human eyes. For the porcine eyes the slope term doubles, because the two set points are 16.5 rather than 33 °C apart: the expanded uncertainty of an excursion is ±0.33 °C at the proximal optic nerve and ±1.1 °C at the ring, and that of a single reading ±1.8 °C.

The ring channel of the in-vivo study (Section 4.2.6) is not linearized: a straight line misses its plateaus by about 1 °C. Its readings were converted with a model of

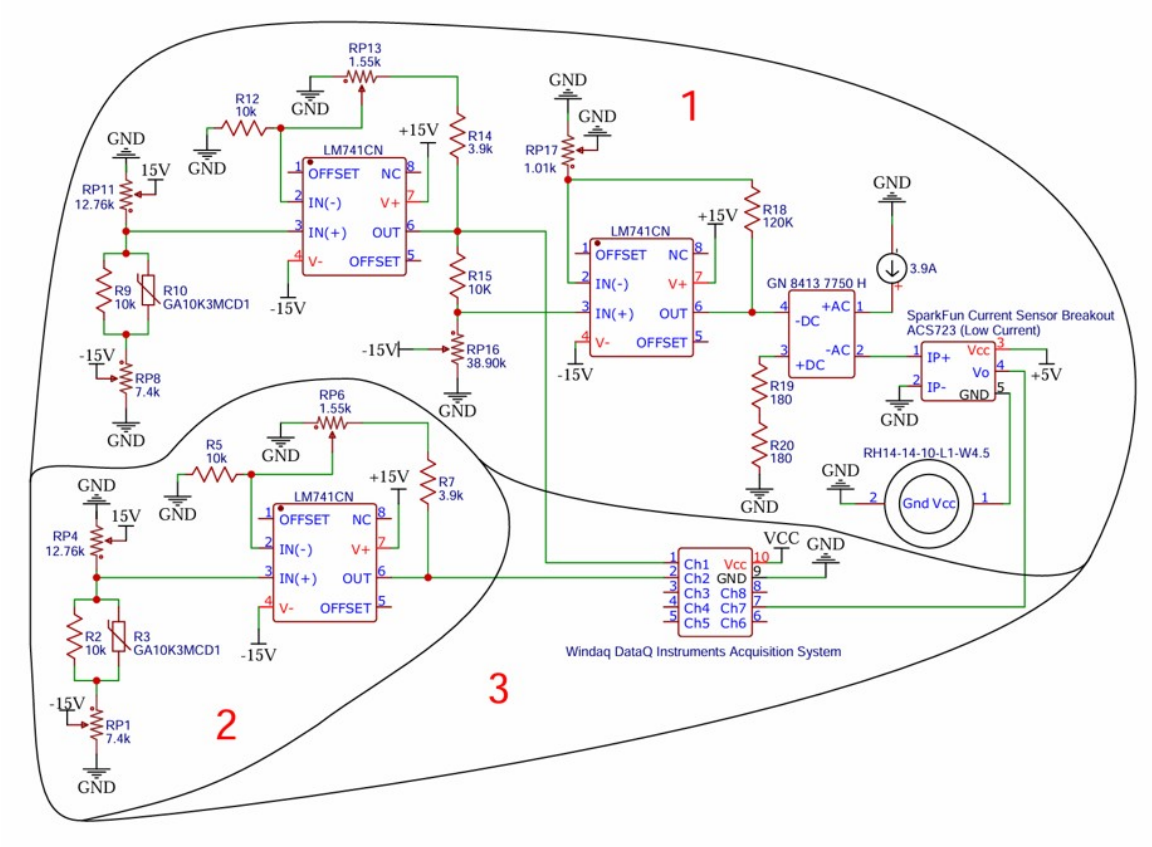


FIGURE A.1. Signal-conditioning and acquisition circuit: area 1, the channel of the SCR feedback thermistor with the automated supply circuit and the current-sensing board; area 2, one of the five identical channels of the implanted sensors; area 3, the data acquisition unit.

the circuit,

$$V = \frac{aR + b}{R + g}, \tag{A.6}$$

fitted to the plateaus recorded in each session, where V is the output voltage of the channel, R the resistance of the thermistor, a and b the gain and offset of the amplifier, and g the effective resistance of the divider in which the thermistor sits. The resistance follows the Steinhart–Hart law of the thermistor type (10K3A),

$$\frac{1}{T} = A + B \ln R + C(\ln R)^3, \tag{A.7}$$

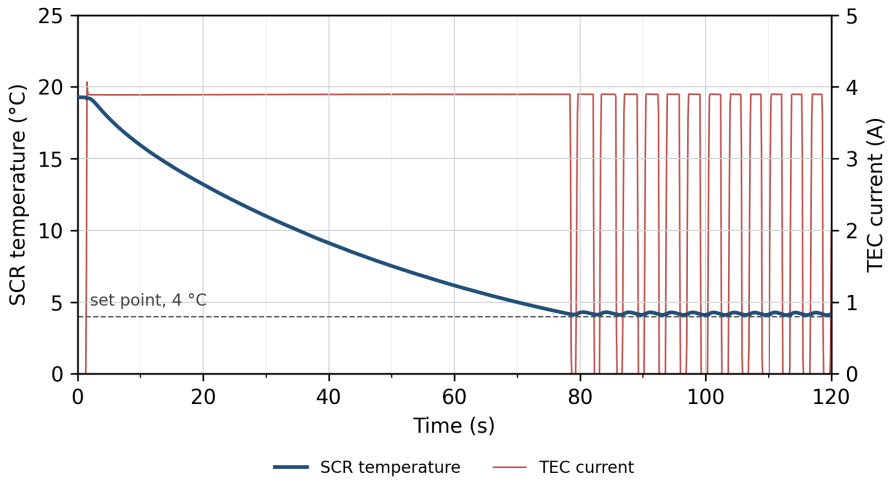


FIGURE A.2. Automatic feedback control of the SCR: the TEC drive current (right axis) is held at its maximum until the SCR temperature from the feedback thermistor (left axis) reaches the 4 °C set point, then the comparator switches the drive on and off to hold it. Bench recording on a preserved porcine eye (2023), not an analyzed run.

where T is the absolute temperature and $A = 1.129241 \times 10^{-3}$, $B = 2.341077 \times 10^{-4}$, and $C = 8.775468 \times 10^{-8}$ are the manufacturer's coefficients for R in ohms. Only a, b, and g were fitted; with five plateaus the residuals were at most 0.16 °C.

A.2. Absolute Uncertainty

Table A.1 lists the contributions to the uncertainty of a single reading. One term, sensor self-heating, accounts for 80% of the variance; without it the expanded uncertainty would be ±0.81 °C.

TABLE A.1. Uncertainty budget for a single absolute temperature reading. u_i is the standard uncertainty; the share is u_i^2/u_c^2.

Source	Basis	u_i (°C)	Share
Self-heating	one-sided bound 1.43 °C, not corrected (see below)	0.826	80%
Linear conversion	largest miss of a human calibration line at its own plateaus, 0.57 °C	0.329	13%
Calibrator accuracy	dry-block specification ±0.25 °C [**126**]	0.144	2%
Calibrator block uniformity	between wells of the aluminum insert, ±0.2 °C (the manufacturer's figure for the 9100S; none is given for the 9102S with a custom insert)	0.115	2%
Sensor interchangeability	thermistor tolerance ±0.2 °C [**125**]	0.115	2%
Dynamic lag	time constant 0.2 s [**125**] at the fastest cooling rate measured, 0.54 °C s^{-1}	0.062	<1%
Drift after calibration	0.05 V recorded after calibration	0.060	<1%
Calibrator stability	±0.05 °C [**126**]	0.029	<1%
Reading noise (Type A)	standard deviation on the settled calibration plateaus	0.021	<1%
Lead resistance	9.4 Ω in series with the sensor, computed from the length and gauge of the leads and cable	0.022	<1%
DAQ accuracy	±5 mV	0.006	<1%
Combined standard uncertainty u_c		0.920	
Expanded uncertainty U ($k = 2$)		1.84	

Self-heating. The current through a thermistor warms it above the tissue it measures. With the sensors in the calibrator, raising the excitation from 3 V to the ±15 V operating supply raised the reading by 4.6 °C on average (3.6 to 6.0 °C by channel).

The calibration was made at the operating supply, and heat leaves the sensor more readily in tissue and oil than in the well, so a reading can only be too low. This underreading was bounded from the study's own data, since at steady state no submerged tissue can be warmer than the oil: in the eleven runs with a recorded oil temperature (36.9 to 37.5 °C) the oil exceeded the starting reading by at least 1.3 °C at the proximal optic nerve and 1.2 °C at the posterior sclera. A bound of 1.43 °C was used for every channel; the vitreous channel, which showed the largest rise in the calibrator, may exceed it. No correction was applied, so the term is rectangular between zero and the bound, $u = a/\sqrt{3}$ with $a = 1.43$ °C. The reported absolute temperatures can therefore be up to about 1.4 °C lower than the true tissue temperature, more likely near that limit than near zero, and not higher. The underreading largely cancels in ΔT; only its change with temperature remains (Table A.2).

A.3. Differential Uncertainty

A cooling excursion is the difference between two readings of one sensor, so the calibrator offset, sensor interchangeability, lead resistance, and most of the self-heating cancel. Three terms grow with the excursion: the slope of the calibration line, the curvature that a straight line ignores, and the change of self-heating between the two temperatures. Table A.2 gives the budget for a small excursion, the proximal optic nerve (34.6 to 30.8 °C), and a large one, the ring (29.7 to 4.1 °C).

The differential value at the proximal optic nerve is the uncertainty of a single excursion at the primary site; differences between groups are judged by their confidence intervals (Section 2.2.6).

TABLE A.2. Uncertainty budget for a within-trace cooling excursion ΔT: standard uncertainty u_i, in °C, for an excursion at the proximal optic nerve and at the ring.

Source	**Basis**	**Optic nerve**	**Ring**
Bath stability	±0.2 °C	0.115	0.115
Drift during the run	as Table A.1	0.060	0.060
Linear conversion, on a difference	human calibration lines, 0.24 and 0.54 °C	0.139	0.311
Reading scatter	standard deviation of single samples about the filtered trace over the final window, median over the included traces	0.054	0.054
Slope of the calibration line	calibrator accuracy and uniformity at two set points, over the excursion	0.030	0.203
Change of self-heating with temperature	bound of 1.43 °C scaled by the change in dissipated power (source resistance 10.8 kΩ, the median estimated from 36 calibration channel-sessions)	0.026	0.140
Thermistor β tolerance	±0.5% [**125**]; the implanted beads are not the calibration set	0.012	0.070
Combined standard uncertainty u_c		0.202	0.427
Expanded uncertainty U ($k = 2$)		0.40	0.85

APPENDIX B

Reconciliation of Published and Reprocessed Fresh-Porcine Values

The fresh-porcine recordings of Nakamura et al. [**1**] were reprocessed for Chapter 2 with the pipeline of Section 2.2.6, which retained all 30 channel traces, so its gates change no value. The initial temperatures agree to 0.1 °C at every site, and the differences come from two definitions. The published final temperature is that of the averaged curve at 1200 s and the reprocessed one the mean over minutes 16–20 of each trace, which gives the uniform −0.1 °C in ΔT. The published t_{90} is read from the averaged curve and the reprocessed one is the mean over the traces, so the largest difference, −27 s at the proximal optic nerve, is mostly one of definition. Tables B.1 and B.2 compare the two sets; no difference in ΔT or final temperature exceeds 0.1 °C, and none changes a conclusion of the published study.

TABLE B.1. Fresh-porcine cooling excursion ΔT, in °C, as published [**1**] (the difference of the published cohort-mean initial and final temperatures) and as reprocessed for Chapter 2; difference, reprocessed minus published. A conclusion is unchanged (last column) when the difference is below the differential uncertainty at the proximal optic nerve (±0.40 °C, Appendix A), as every difference is.

Location	**Published**	**Reprocessed**	**Difference**	**Unchanged?**
Anterior sclera	24.4	24.3	−0.1	Yes
Equatorial sclera	9.5	9.4	−0.1	Yes
Posterior sclera	9.9	9.8	−0.1	Yes
Vitreous	11.7	11.6	−0.1	Yes
Proximal optic nerve	3.6	3.5	−0.1	Yes

TABLE B.2. Fresh-porcine final temperature and time to 90% of the cooling excursion (t_{90}), as published [**1**] and as reprocessed for Chapter 2.

	Final temperature (°C)		t_{90} **(s)**	
Location	**Published**	**Reprocessed**	**Published**	**Reprocessed**
Anterior sclera	10.7	10.8	194	200
Equatorial sclera	25.0	25.1	338	340
Posterior sclera	24.4	24.5	532	519
Vitreous	22.3	22.4	569	566
Proximal optic nerve	30.2	30.3	703	676

APPENDIX C

Analysis Code

This appendix documents the code used to analyze the experiments of Chapter 2 and the computational model of Chapter 3. Each analysis is a Python package that runs from the raw data to every table and figure with one command, holds its parameters in one configuration file, and records the software versions and file checksums with its results.

C.1. Ex-Vivo Analysis Pipeline

The ex-vivo package reproduces the study's analysis notebook (version 4.28) exactly when the settings added after that version are switched off (Figure C.1).

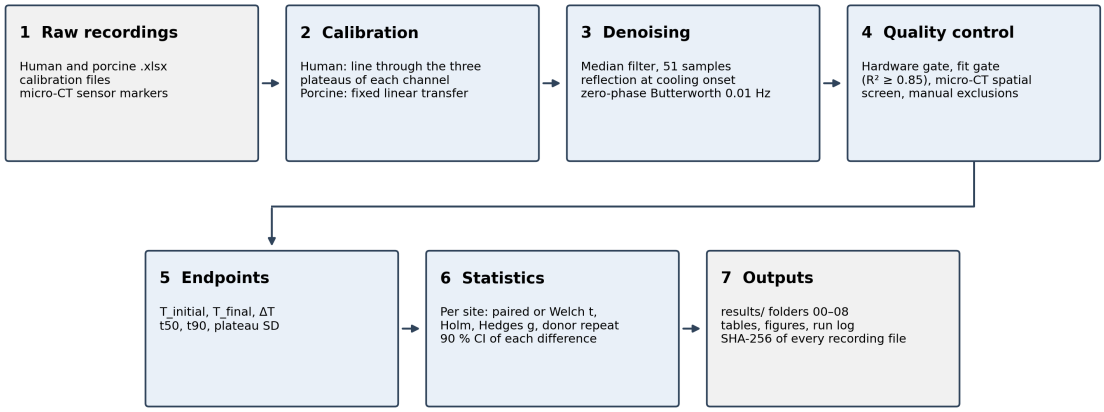


FIGURE C.1. Stages of the ex-vivo analysis, from the raw recordings to the tables and figures of Chapter 2.

C.2. Analysis Parameters

Table C.1 lists the parameters of the method (Section 2.2.6).

Table C.1: Parameters of the ex-vivo analysis.

Stage	Parameter and value
Acquisition	sampling interval 0.1 s
Calibration	human: per-channel least-squares line through the plateaus at 4.0, 20.5, and 37.0 °C, each plateau located by k-means clustering of the voltages (three clusters, seed 42) and read as the median of its last 3 min before the next ramp (last 10 s excluded), or of its longest stretch of 10-s medians within ±0.01 V where those 3 min drifted by more than 5 mV per minute, or of the whole recorded plateau where less than 3 min was recorded; porcine: $T = -2.0656\,V + 20.439$ °C for all channels, with V the recorded channel voltage in volts and T in °C; readings outside 0–43 °C set to missing
Denoising	median filter 51 samples (5.1 s), edge-value padding; point-reflected extension of 300 s before the cooling onset; second-order zero-phase Butterworth low-pass, 0.01 Hz
Channel corrections	human eye 4: equatorial and anterior channels exchanged (the anterior sensor was recorded on the equatorial channel); eye 6: the anterior and posterior workbook columns exchanged, each converted with the other column's calibration
Hardware gate	at least 100 samples; flat line if the raw-voltage SD is below 10^{-4} V; stuck if 90% of samples sit at the median value; railed if 75% of samples lie beyond ±9.9 V; and by hand, the failing anterior-sclera sensor of human eye 9 (original ring)
Fit gate	single-exponential fit $T = A\,e^{-kt} + C$ (bounds $0 \le A \le 50$, $0 \le k \le 1$ s^{-1}, $0 \le C \le 40$ °C), $R^2 \ge 0.85$, at least 20 samples
Spatial screen	sensors identified by their marker labels; the four sensors fixed to tissue (anterior, equatorial, and posterior sclera, proximal optic nerve) of each eye rigidly aligned (rotation and translation) to the cohort mean; limit on the offset from the mean position of one thermistor length (3.3 mm); vitreous sensor excluded if more than 75% of the distance from the center of the globe to its wall on micro-CT; a spatial breach takes precedence over an earlier reason
Manual exclusions	four traces: three in the original-ring data of Chapter 2 (Section 2.2.6) and one, the same human vitreous sensor under the reduced-bulk ring, in the ring comparisons of Chapter 3
Endpoints	final temperature and plateau SD over minutes 16–20 (240 s); t_{50}, t_{90} as the first times the trace has fallen by 50% and 90% of ΔT
Statistics	paired t test when the two groups share at least two eyes, otherwise Welch's t test; Hedges' g; $\alpha = 0.05$; Holm correction over the five tissue sites; 90% confidence interval of each difference (its limits give the largest difference allowed); species comparison repeated on donor means

In the spatial screen, the offset of sensor s in eye e from the mean set of the cohort was

$$r_{e,s} = \left\| \mathbf{R}_e \, \mathbf{x}_{e,s} + \mathbf{t}_e - \bar{\mathbf{x}}_s \right\| , \tag{C.8}$$

where $\mathbf{x}_{e,s}$ is the micro-CT position of sensor s in eye e, $\mathbf{R}_e$ and $\mathbf{t}_e$ the rotation and translation that best map the sensor set of eye e onto the mean set in the least-squares sense, and $\bar{\mathbf{x}}_s$ the mean aligned position of sensor s over the eyes of the cohort. The alignment removes the position and orientation of each eye in the scanner.

C.3. ERG Processing

The width of the b-wave and the number of its peaks (Section 4.2.9) are computed by `erg_cohort.py` from the same filtered trace as the picks. The analysis used the raw record of each flash step, sampled every 0.512 ms, not the instrument's reported waveform. Drift was removed with a 0.3 Hz high-pass filter (second-order Butterworth, forward and backward) [**139**]. Sweeps from 20 ms before to 200 ms after each flash were referred to their 20-ms pre-flash mean and interpolated onto a 0.5-ms grid. A sweep was rejected when the noise of the second before its flash exceeded three times the median over the recording's sweeps, or its b-wave amplitude differed from their median by more than half.

The accepted sweeps were averaged and low-pass filtered at 75 Hz (fourth-order Butterworth, forward and backward). The a-wave was the minimum 2 to 26 ms after the flash, measured from the pre-stimulus baseline, and the b-wave the maximum after that trough, from 8 up to 50 ms (2 to 35 ms, and 12 up to 110 ms, at the weakest flash), measured from the a-wave trough [**139**].

For the oscillatory potentials, the accepted sweeps of each flash step except the first were averaged without the 75 Hz low-pass and band-pass filtered from 75 to 300 Hz (fourth-order Butterworth, forward and backward) [**139**]. They were the positive peaks on the rising limb of the b-wave, from the a-wave trough (not earlier than 10 ms after the flash) to 5 ms after the b-wave peak, each measured from the

trough before it. The measure is the sum of the first four, with the root-mean-square over the same window alongside, at 8 and 25 cd·s/m^2.

Replacing the drift removal by a straight line fitted to the 1 s before each flash changed the b-wave amplitude by a median of 1 μV and the implicit time by a median of 0 ms (at most 77 μV and 1.5 ms) over the 186 recordings at each of the two stronger flashes. Against the waveform the instrument reported, the median correlation was 0.996, 371 of the 372 b-wave amplitudes at the two stronger flashes agreed within 15%, and the oscillatory-potential sums correlated at 0.99 at both flashes, with a median difference of 1%. A recording was flagged when its residual drift before the flash exceeded three times the median of all recordings (67 of the 558 traces of animals 1 to 6), when a pick fell on the edge of its search window (34), or, compared with the instrument's waveform after the same 75 Hz low-pass filter, when the correlation was below 0.95 (45), the b-wave amplitude more than 15% away (23), or the mean level between 150 and 200 ms after the flash beyond $\pm$150 μV, a sign of drift or a slow potential (20). Most flags fell on the dimmest flash, 0.06 cd·s/m^2, whose b-wave is small, and on the exploratory animal; at the two stronger flashes used in the analysis the last three checks flagged 16, 1, and 5 of the 372 traces. Flagged traces were checked against the waveform; none of the checks changes a value. A latency corrected by hand is kept in a separate file with the detector's value, and the phase of each recording was assigned by hand from the session log.

C.4. Globe Outline and Position of the Vitreous Sensor

For imaging, the thermistor wires were cut and the eye on its holder was sealed in a container (Figure C.4), filled with oil for most porcine eyes and for one scan of human eye 2, and with air otherwise. MRI (Agilent, Santa Clara, CA, USA) used a multislice gradient-echo sequence with 0.078 $\times$ 0.078 $\times$ 0.8 mm voxels (repetition time 550 ms, echo time 7.5 ms, flip angle 20°, 36 contiguous slices, 16 averages) and a 72-mm volume coil; the oil is displaced in these images by about 1.1 mm along

the readout direction relative to the water-bearing tissue, which is imaged in place. Micro-CT (North Star Imaging systems, Rogers, MN, USA) used 70 kV and 1200 μA (60 kV and 550 μA for human eyes 5 to 9), reconstructed at isotropic voxels of 32 to 79 μm.

The outline of each globe on micro-CT was found automatically: a smooth closed surface fitted to the tissue edges along rays from the center of the globe in the fresh eyes, and the tissue mask after a 3 mm morphological opening in the folded preserved porcine eyes. Every outline was inspected on fifteen sections and moved by hand onto the wall where it departed from it (18 of the 19 eyes), after which it followed the drawn lines to a median distance of 0.2 mm or less. The center of the globe is the centroid of the automatic outline; the axis is the normal of the holder ring for human eyes 3 to 9 and is found from the nerve stump and the lens for human eyes 1 and 2 and the fresh and three preserved porcine eyes, and from the optic-nerve sensor for preserved eyes 4 and 5. The code is archived with the ex-vivo data (`CT Segmentation/code`) and in `code/ch2-ct-segmentation` of the repository; its results enter the package as `config/vitreous_position_ct.csv`.

The position of the vitreous sensor was the ratio r/R, where r is the distance from the center of the outline to the sensor and R the distance from the center to the outline in the same direction (0 at the center of the globe, 1 at its wall). A vitreous sensor was excluded when r/R exceeded 0.75 (Figures C.2 and C.3), as in human eyes 5 and 8 (0.82 and 0.81).

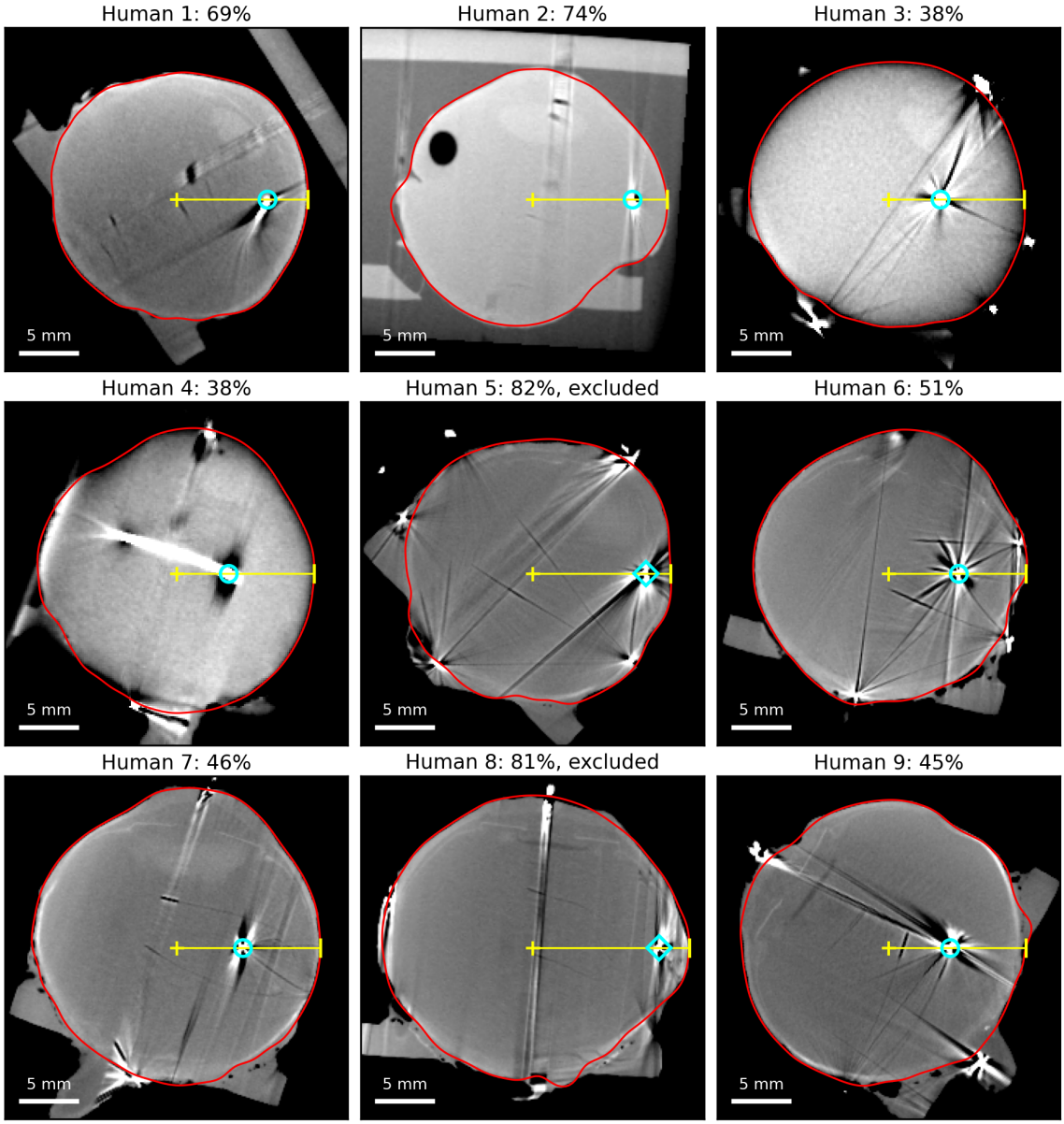


FIGURE C.2. Position of the vitreous sensor in the nine human eyes: micro-CT section through the center of the globe outline (cross) and the sensor (cyan marker), with the outline in red and the line from the center to the wall in yellow. The number is the ratio r/R of Section C.4; a diamond marks a sensor beyond the limit of 0.75, which was excluded.

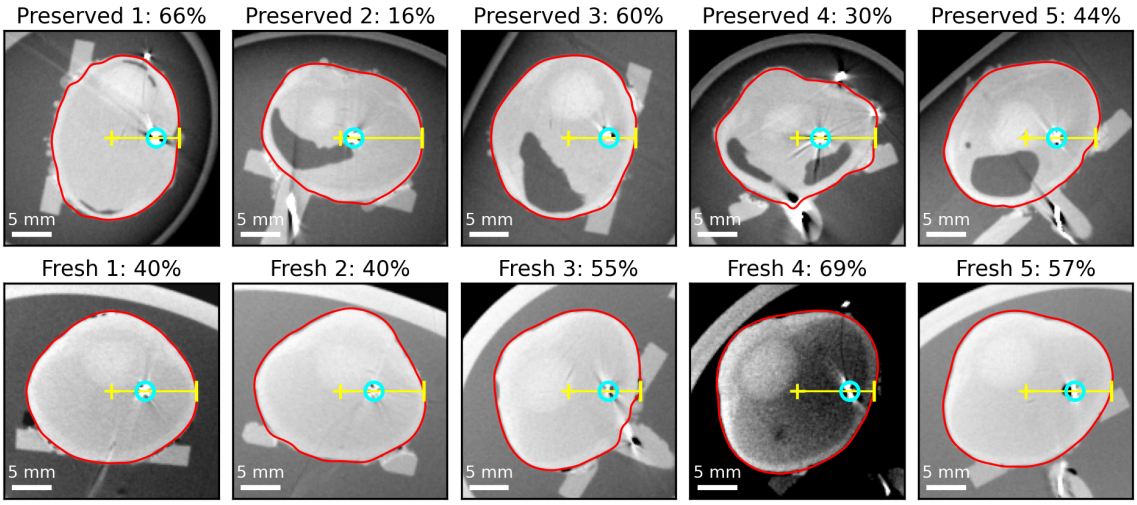


FIGURE C.3. Position of the vitreous sensor in the five preserved (top) and five fresh (bottom) porcine eyes, drawn as in Figure C.2. No porcine vitreous sensor lay beyond the limit of 0.75.

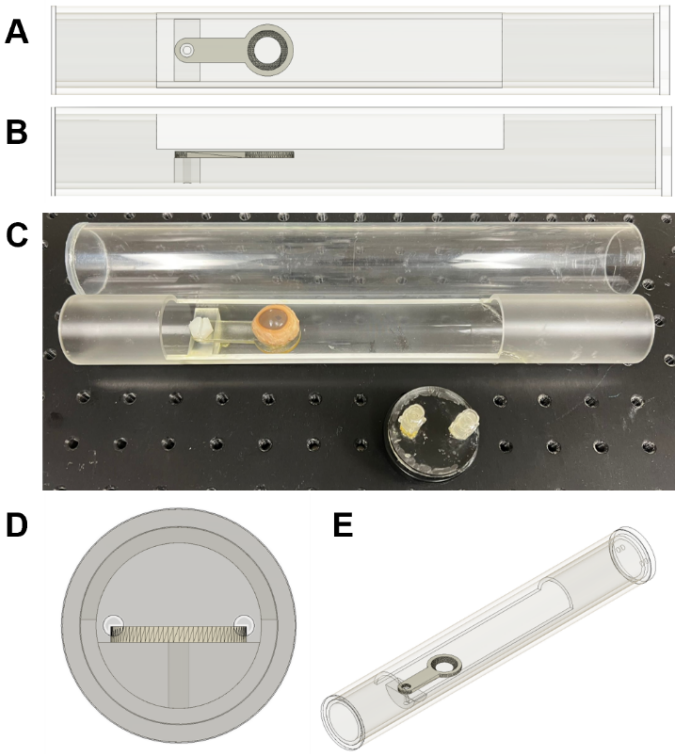


FIGURE C.4. Sealed, oil-filled container used for MRI and micro-CT. Top: top and side views of the CAD model. Middle: the container with an eye on its holder, and the cap with the two valves used to evacuate air. Bottom: front view of the cap and panoramic view of the container (E).

C.5. Reproducing the Analysis

The package was run with Python 3.13, NumPy 2.2.6, pandas 2.3.2, SciPy 1.16.1, statsmodels 0.14.6, scikit-learn 1.9.0, Matplotlib 3.10.6, and openpyxl 3.1.5, pinned in its requirements file because later releases no longer reproduce the stored references exactly. The analysis is run, and its agreement with the notebook checked (later settings off, per-site statistics to a relative tolerance of 10^{-9}), with

```
pip install -r requirements.txt
python run_analysis.py --data <recordings folder> --out results
python tests/test_reproduce.py <recordings folder>
```

`tests/test_default.py` is a regression test of the current settings. Both tests pass on the version used in this dissertation (Section C.7).

C.6. Computational Model Analysis

The COMSOL exports behind Chapter 3 were analyzed with a second package of the same design (Figure C.5), which reads the ex-vivo results and so is run after the ex-vivo package. It reproduces the model-analysis notebook (version 3.8.0) exactly when its corrections are switched off (183 195 checks); eleven corrections, documented in the package, are applied by default, among them the doubling of the heat exported for the face of the module and the division of the heat among eye, eyelids, and air.

The figures of Chapter 3 are drawn by the scripts in `code/figures`, each listed with its inputs and their checksums in the manifest of its folder. The eye-model temperatures and heat values come from the corrected runs, the validation from the replication runs, and the heat-exchanger results from the sweep of `scr-heat-replica`.

Mesh sensitivity. In COMSOL, temperature, velocity, and pressure were interpolated with linear elements, the stationary initial state was solved to a relative tolerance of 10^{-3}, and the transient was integrated with a variable-step backward differentiation formula of order 1 to 2 at a relative tolerance of 0.005. The eye model was solved for each ring on three meshes, from COMSOL's *Coarser*, *Coarse*, and *Normal* presets

with local refinement of the retina, choroid, and tear film (about 14 000, 35 000, and 164 000 to 171 000 vertices without the air and the module layer, as solved; Table 3.2), and the heat exchangers on three meshes in the same way. The grid convergence index [**203**] is stored with the results but not reported, because the refinement was not strictly systematic and the solutions did not converge monotonically at every site.

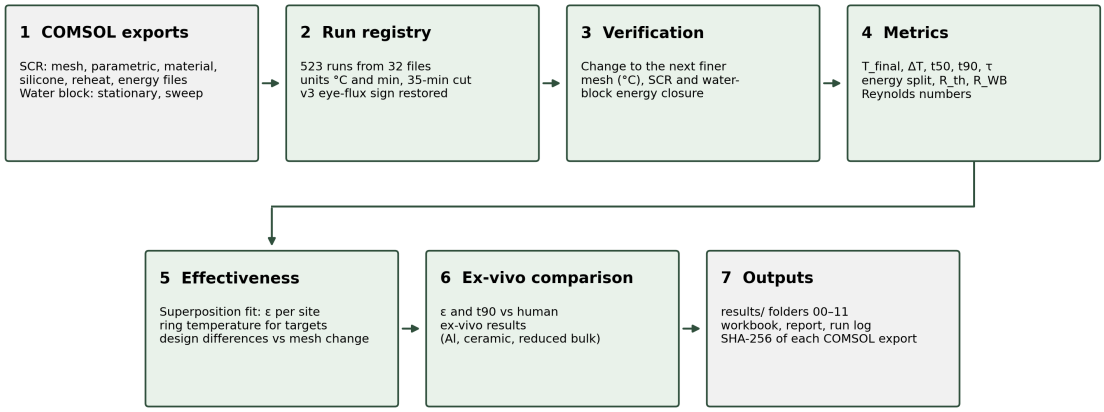


FIGURE C.5. Stages of the computational model analysis, from the COMSOL exports to the results of Chapter 3.

Python finite-element program. The program reads the meshes, the material definitions, and the boundary conditions from the COMSOL model files and solves the heat equation with linear tetrahedral elements and a second-order backward differentiation formula, with fixed steps of 6 s for the first 5 min, 15 s to 15 min, and 30 s afterwards, the first step by backward Euler. In the heat exchangers the coolant velocity was a fixed quadratic profile, without buoyancy. The corrected and extension cases follow Section 3.2.2.1. The flow solver and the reading of the model files are verified only by the agreement with COMSOL.

C.7. Code Availability

The code is kept in a public repository rather than reproduced here: `https://github.com/GigiosGros/eyecooler-thesis`. Folder `code/ch2-exvivo-analysis` holds the ex-vivo analysis of Chapter 2, in the version used here, `code/ch2-bath-vs-perfusion`

the conduction estimate of the oil bath, `code/ch2-ct-segmentation` the segmentation of the micro-CT and MRI scans, `code/ch3-model-analysis` the analysis of the COMSOL exports and `code/ch3-heat-replica` the finite-element program of Chapter 3, and `code/ch4-live-pigs` the processing of the in-vivo recordings (temperature conversion, ERG extraction, and the cohort statistics of Chapter 4). The scripts that draw the figures are in `code/figures`, the figures in `figures`, and this dissertation in `thesis`. The raw recordings, the COMSOL model files, and the inputs and the list of cases of the finite-element program are available from the author on request.

APPENDIX D

Tissue Properties, Boundary Conditions, and Meshes of the Computational Model

The eye measures 24.8 mm from the corneal apex to the choroid at the posterior pole and 25.0 mm across the outer surface of the sclera; the cornea is 11.5 mm across and 0.55 mm thick at its center, the anterior chamber 3.0 mm deep, the lens 9.0 mm across and 4.5 mm thick, and the sclera 0.42 mm thick on average.

Published thermal properties of the eye differ widely and most come from animals [**52**], so each property of the model (Table 3.3) was estimated from the composition of the tissue, averaged over published values, or taken from the IT'IS database [**169**]; the tear film and the ring materials came from the COMSOL library. Regional variation was not represented. The mesh and boundary figures, like the schematics of Appendix F, are at full resolution in the electronic copy.

D.1. Estimates from Tissue Composition

Most ocular tissues were treated as water containing a dispersed solid phase, with the conductivity given by the Hashin–Shtrikman bound with water as the continuous phase [**170**],

$$k = k_w + \frac{1-\phi_w}{\dfrac{1}{k_s - k_w} + \dfrac{\phi_w}{3k_w}}, \tag{D.9}$$

where k is the thermal conductivity of the tissue, ϕ_w its water fraction, $k_w = 0.6\ \mathrm{W\,m^{-1}\,K^{-1}}$ the conductivity of water, and k_s that of the solids: 0.41 (cornea), 0.33 (vitreous), 0.23 (ciliary body), 0.20 (retina), 0.22 (choroid), and $0.43\ \mathrm{W\,m^{-1}\,K^{-1}}$ (sclera). The specific heat c_p and the density ρ followed from the same fractions, $c_p = [\phi_w + 0.4\,(1-\phi_w)] \times 4184\ \mathrm{J\,kg^{-1}\,K^{-1}}$ and $\rho = [\phi_w + (1-\phi_w)\,\rho_s] \times 1000\ \mathrm{kg\,m^{-3}}$,

where 4184 $\mathrm{J\,kg^{-1}\,K^{-1}}$ and 1000 $\mathrm{kg\,m^{-3}}$ are the specific heat and the density of water, 0.4 the specific heat of the solids relative to water, and ρ_s the density of the solids, 1.37, 1.48, 1.35, 1.16, 1.37, and 1.34 $\mathrm{g\,cm^{-3}}$ in the same order (1.34 for the lens).

Checked against published compositions [**171, 28, 172, 173, 174, 175, 176, 177, 178, 179, 169**] and component properties [**180, 181, 182, 183**], the solid-phase values reproduce the conductivities used within 0.01 $\mathrm{W\,m^{-1}\,K^{-1}}$, except for the cornea and the sclera (0.51 and 0.47 against 0.56 and 0.545 used, which lie closer to the measured means [**169**]).

D.2. Perfusion and Metabolic Heat

The perfusion rate of each tissue followed from its measured blood flow, $\omega = Q_{\mathrm{blood}}/(60\,V)$, where ω is the perfusion rate in $\mathrm{s^{-1}}$, Q_{blood} the blood flow in $\mu\mathrm{L\,min^{-1}}$, V the volume of the domain in $\mathrm{mm^3}$, and 60 converts minutes to seconds. The choroid took the mean whole-eye flow measured with microspheres in monkeys, 917.5 $\mu\mathrm{L\,min^{-1}}$ [**40**], over 165 $\mathrm{mm^3}$ (0.0927 $\mathrm{s^{-1}}$; indirect estimates for the human eye are of the same order [**184, 41, 111**]); the retina the mean of six laser Doppler studies in humans, 54.5 $\mu\mathrm{L\,min^{-1}}$ [**29, 30, 31, 33, 32, 185**], over 236 $\mathrm{mm^3}$ (0.0039 $\mathrm{s^{-1}}$); and the ciliary body 133 $\mu\mathrm{L\,min^{-1}}$ [**188**] over 348.25 $\mathrm{mm^3}$. The extraocular muscles, eyelids, and orbital fat used published flows per unit mass [**186, 187**] and the density of the tissue; for the eyelids no measured value was found and the rate is an assumption.

The metabolic heat was the specific metabolic rate of each tissue [**189**] as a volumetric source, $Q_{\mathrm{met}} = \dot{m}\,\rho \times 4184/86\,400$, where Q_{met} is the metabolic heat in $\mathrm{W\,m^{-3}}$, $\dot{m}$ the metabolic rate in $\mathrm{kcal\,kg^{-1}\,day^{-1}}$, ρ the density in $\mathrm{kg\,m^{-3}}$, 4184 the number of joules in a kilocalorie, and 86 400 the number of seconds in a day; for the retina it was derived from the oxygen consumption [**190**], 3443 $\mathrm{W\,m^{-3}}$. The blood density and specific heat, $\rho_b = 1068\ \mathrm{kg\,m^{-3}}$ and $c_b = 3682\ \mathrm{J\,kg^{-1}\,K^{-1}}$, were composition estimates close to the IT'IS values [**169**].

D.3. Values Used

Table D.1 lists the values used, their basis, and the published values compared. The rings of aluminum, aluminum oxide (alpha, polycrystalline), and fused quartz had emissivities of 0.05, 0.95, and 0.90 (the COMSOL models used 0.05 for every ring).

TABLE D.1. Thermal properties of the eye model, their basis, and the published values they were compared with (k in $\mathrm{W\,m^{-1}\,K^{-1}}$, ρ in $\mathrm{kg\,m^{-3}}$, c_p in $\mathrm{J\,kg^{-1}\,K^{-1}}$, ω in $\mathrm{s^{-1}}$).

Domain	**Used (k; ρ; c_p; ω)**	**Basis**	**Published values compared**
Cornea	0.56; 1070; 3707; –	Composition estimate (81% water [**169**])	k 0.35–0.58 [**193, 52, 169, 194**]
Aqueous humor	0.593; 1005; 4159; –	k the mean of four published values [**193, 195, 196, 197**]; ρ from [**196**]; c_p composition estimate	k 0.58 [**193, 169**]
Lens	0.43; 1126; 3255; –	k from its water content [**197**]; ρ, c_p composition estimate (63% water, the nuclear value [**175**])	k 0.40–0.43 [**198, 169**]
Vitreous humor	0.59; 1010; 4134; –	Composition estimate (98% water); viscosity 0.7 Pa s, within the 0.2–1 Pa s range of the liquid phase [**176**]	k 0.58–0.60 [**52, 198, 193, 169**]
Ciliary body	0.49; 1088; 3556; 0.0064	Composition estimate; perfusion from ciliary blood flow [**188**]	k 0.49–0.50 [**169**]
Retina	0.55; 1014; 3958; 0.0039	Composition estimate (91% water [**177**]); perfusion from retinal blood flow [**29, 30, 31, 33, 32, 185**]	k 0.55–0.57 [**199, 169**]
Choroid	0.51; 1074; 3682; 0.0927	Composition estimate (80% water; the IT'IS database, which gives the choroid the properties of blood, lists 80.4% [**169**]); perfusion from choroidal blood flow [**40**]	k 0.52–0.53 [**199, 169**]
Sclera	0.545; 1102; 3431; –	k, ρ, c_p composition estimate (70% water [**171**])	k 0.58–1.0 [**200, 199, 193, 169**]; the value of 1.0 traces to a single measurement and exceeds the conductivity of water
Extraocular muscles	0.49; 1090; 3421; 0.0045	IT'IS database (muscle) [**169**]; perfusion from extraocular muscle blood flow in primates [**186**]	k 0.57 [**53**]
Eyelids	0.34; 1070; 3660; 0.0011	Eyelid dermis [**201**]; perfusion from skin blood flow	k 0.37 (skin) [**169**]
Orbital fat	0.2245; 920; 2500; 0.0004	k the mean of published values; ρ, c_p [**199**]; perfusion from adipose blood flow [**187**]	k 0.20–0.25 (subcutaneous) [**196, 199**]; IT'IS 0.21 [**169**]
Tear film	water	COMSOL library (water); evaporation 12 $\mathrm{W\,m^{-2}}$	12.7–20.5 $\mathrm{g\,m^{-2}\,h^{-1}}$ [**191, 192**]

D.4. Fluid Domains and Boundary Conditions

This section describes the COMSOL models, which the replication followed; the corrected runs differ as stated in Section 3.2.2.1, and their boundary conditions are drawn in Figure D.1.

The aqueous humor, the vitreous humor, and the air were laminar, non-isothermal fluids under gravity (the viscosity of the vitreous is in Table D.1). Aqueous humor entered and left at 1.23×10^{-6} $\mathrm{m\,s^{-1}}$, and the intraocular pressure was 2000 Pa (15 mmHg). The optic nerve had an outer sheath with the properties of the sclera and a core with those of the retina. The outer shell of the orbit was held at body temperature and that of the air domain at ambient temperature; where the two meet on the eyelid, COMSOL and the Python program impose their mean on 227 nodes. The cornea (v1, v2) lost heat by tear evaporation, and surfaces exposed to air radiated to the surroundings with emissivities of 0.98 for tissue and 0.05 for aluminum. The ring was in perfect contact with the eye and the eyelids; a 5-μm film in the Python program raised the anterior sclera by at most 0.14 °C.

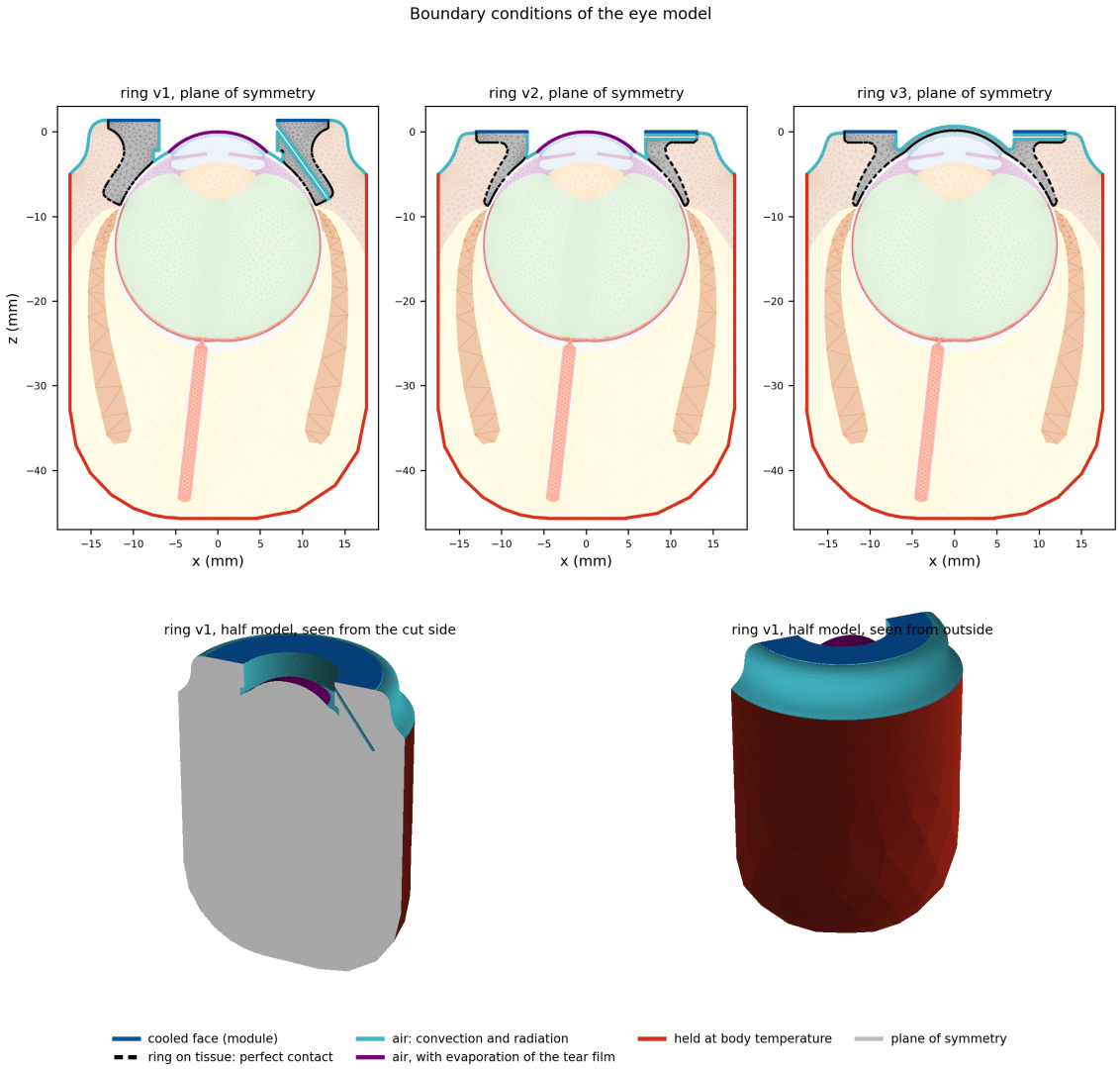


FIGURE D.1. Boundary conditions of the eye model in the corrected runs, in the plane of symmetry for the three rings and in three dimensions for v1. The face of the ring that met the module is cooled; exposed surfaces lose heat to the room by convection and radiation, and the cornea also by evaporation of the tear film (covered by the lens in v3); the outer boundary of the orbit is held at body temperature; the ring is in perfect thermal contact with the tissue.

D.5. Boundary Conditions and Meshes of the Heat Exchanger

Figure D.2 shows the boundary conditions of each heat exchanger (Section 3.2.2.3).

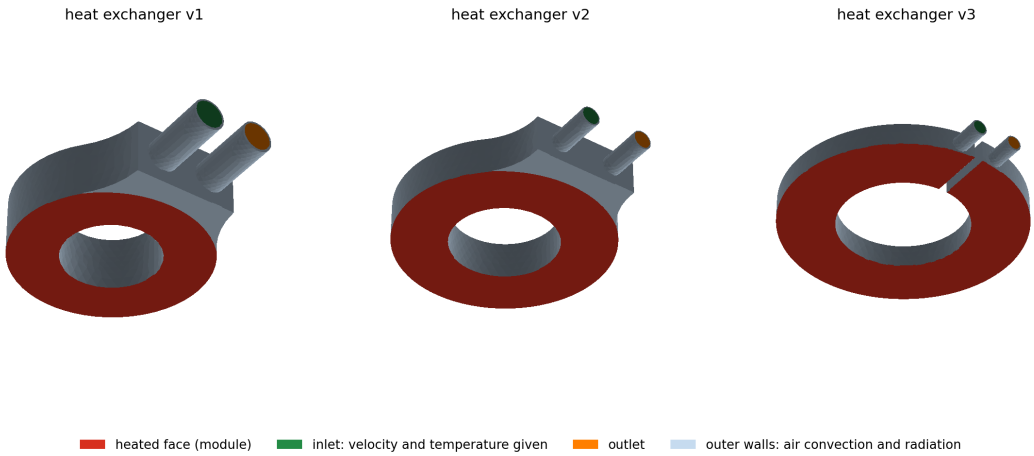


FIGURE D.2. Boundary conditions of the heat exchangers, seen from the module side: the heated face (red); the inlet, with the coolant velocity and temperature given (green); the outlet (orange); and the outer walls, which exchange heat with the ambient air by natural convection and radiation (gray-blue).

The heated face rose above the coolant by 0.24, 0.12, and 0.07 °C per watt in v1 at 0.10, 0.25, and 0.50 L min^{-1}, by 0.15, 0.07, and 0.04 in v2, and by 0.12, 0.05, and 0.03 in v3 (means over the three inlet temperatures, intermediate mesh). In the converged cases these values changed by up to 12% from one mesh to the next, and the order of the designs was the same on every mesh. They are lower bounds: on meshes with resolved wall layers the rise of v1 converges to about 0.32 and 0.18 °C per watt at 0.10 and 0.25 L min^{-1}, and the values here lie 24 to 37% below that limit.

The Reynolds number at the narrowest section of the channels was 170 to 1240 in v1, 360 to 2690 in v2, and 510 to 3800 in v3: laminar in v1 and laminar to transitional in v2 and v3.

D.6. Meshes and Mesh Sensitivity

Figure D.3 shows the meshes of the eye model.

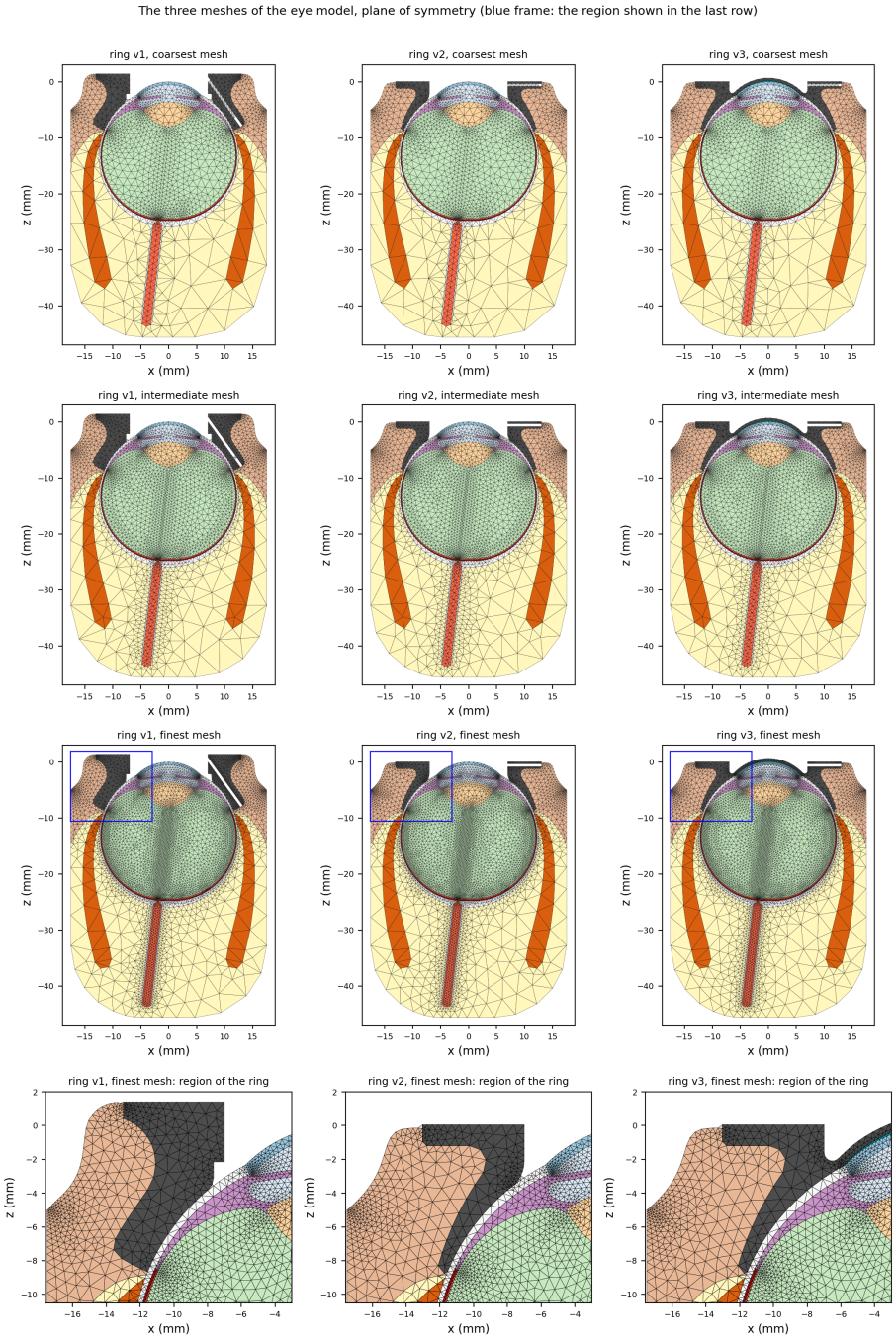


FIGURE D.3. The three COMSOL meshes of the eye model for each ring, in the plane of symmetry, without the air and the module layer as in the corrected runs, from the coarsest (top row) to the finest (third row), on which the results were computed. Bottom row, the region of the ring on the finest mesh enlarged (blue frame in the row above).

In COMSOL the energy balance of the ring closed to within 0.2% after the first 5 min. From the intermediate to the finest mesh the final temperature rose by 0.01 to 0.04 °C at the optic nerve and the posterior sclera, 0.04 to 0.09 °C at the equatorial sclera and the vitreous, and 0.08 to 0.11 °C in the average over the retina (Table D.2); at the anterior sclera the two steps had opposite signs, so no trend can be read there.

This change (bars in Figure D.4) is the threshold below which differences between designs are not interpreted, not the error of the finest mesh, which reads low (proximal optic nerve, vitreous, anterior sclera) by 0.02, 0.06 to 0.08, and 0.36 to 0.62 °C against the same mesh with every element divided into eight, and by 0.03 to 0.04, 0.10 to 0.14, and 0.75 to 1.28 °C against the extrapolated value; anterior-sclera temperatures are therefore lower bounds. Table D.2 and Figure D.4 come from runs of the Python program on these meshes (`mesh_steps.csv` in `figures/aim2/sources/data/python`).

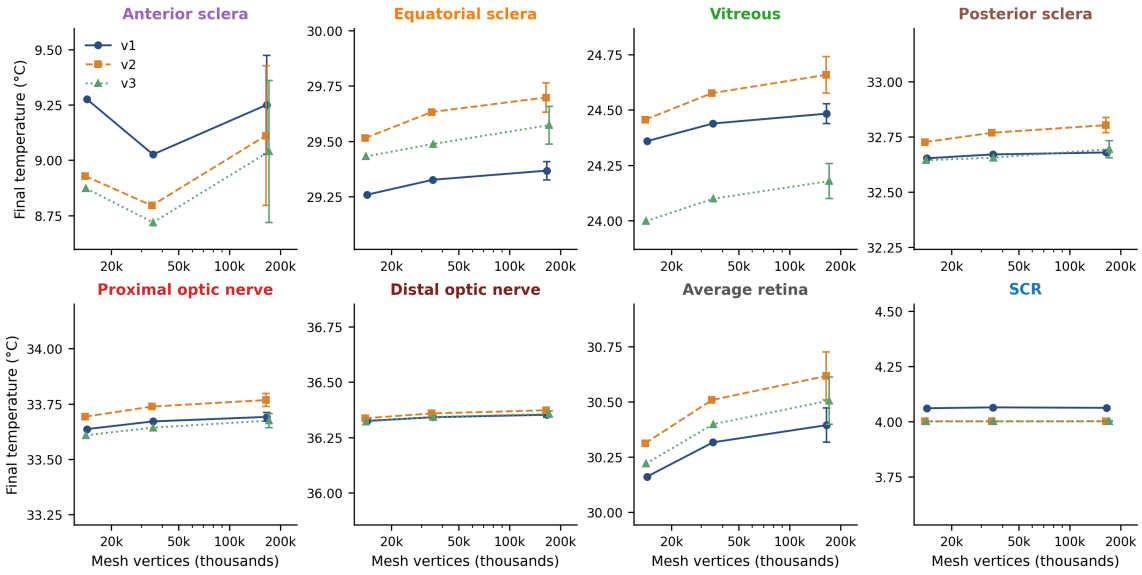


FIGURE D.4. Final temperature at each site of the eye model on the three meshes (vertex counts), for each ring. Each panel spans 1 °C around its own values, so changes can be compared between sites. The bar on the finest mesh spans ± the change from the intermediate to the finest mesh (Table D.2), a step between meshes, not the error of the finest mesh (see text).

From the intermediate to the finest mesh the laminar COMSOL solution of the heat exchanger changed by 0.03 °C at the module face and by 0.02 to 0.03 °C in

TABLE D.2. Change of the final temperature at each site of the eye model from one mesh to the next finer, in °C, at the baseline condition, for each ring: (a) coarsest to intermediate and (b) intermediate to finest (about 14 000, 35 000, and 164 000 to 171 000 vertices, without the air and the module layer).

Site	**v1 (a)**	**v1 (b)**	**v2 (a)**	**v2 (b)**	**v3 (a)**	**v3 (b)**
Anterior sclera	−0.25	+0.22	−0.13	+0.32	−0.15	+0.32
Equatorial sclera	+0.07	+0.04	+0.12	+0.07	+0.06	+0.09
Posterior sclera	+0.02	+0.01	+0.04	+0.03	+0.01	+0.04
Vitreous	+0.08	+0.04	+0.12	+0.08	+0.10	+0.08
Proximal optic nerve	+0.03	+0.02	+0.05	+0.03	+0.03	+0.03
Distal optic nerve	+0.02	+0.01	+0.02	+0.01	+0.02	+0.01
Average retina	+0.16	+0.08	+0.20	+0.11	+0.18	+0.11
SCR	0.00	0.00	0.00	0.00	0.00	0.00

the average temperature of the blocks. The Python program on the same meshes agreed within 0.04 °C at the outlet but only within 0.6 °C at the module face, whose temperature is therefore known to about 1 °C.

APPENDIX E

Selection of the Water-Block Coolant

The heat exchanger of SCEC Gen 3 was run with a 50:50 mixture of propylene glycol and distilled water in the porcine experiments of Chapter 2, and with distilled water alone in the human-donor experiments, in every in-vivo session, and in the water-block model of Chapter 3.

Propylene glycol lowers the freezing point, which was not needed here, since the loop never approached freezing, and every thermal property that governs the loop is less favorable with it. Relative to water, pure propylene glycol has about one third of the thermal conductivity, 60% of the specific heat, and about 55 times the viscosity at 25 °C [**169, 204**], and water–glycol mixtures lie between the two.

The Lynch laboratory at the University of Illinois Chicago allowed the heat flow of each fluid to be measured by differential scanning calorimetry (Figure E.1). Per gram, the 50% mixture absorbed 0.79 and propylene glycol 0.57 times as much heat as distilled water at 10 °C, close to the ratio of the tabulated specific heats, 0.60. Distilled water was therefore used from the human-donor experiments onward.

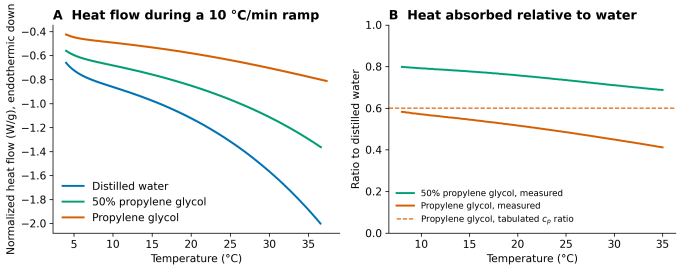


FIGURE E.1. Differential scanning calorimetry of the candidate coolants. (A) Normalized heat flow. (B) Heat flow relative to distilled water (solid) and tabulated ratio for propylene glycol (dashed) [**169, 204**]. Only the comparison between samples is interpreted.

APPENDIX F

The Integrated Control Board

F.1. Hardware and Firmware

An STM32L476RG microcontroller [**205**] sets the drive of an ADN8831 thermoelectric-module controller (Analog Devices) [**206**], which powers the module through an H-bridge of four MOSFETs and senses its current across a 10 mΩ resistor. Trimmers set hard limits on the module current and voltage, independently of the firmware, and the module is off by default. Six thermistor channels read the ring and five ocular sites from a buffered 2.5 V reference; only the ring channel is used for control. The board also measures the module current and voltage, reads two coolant-flow sensors (SLF3S-4000B, Sensirion) [**207**] and an ambient temperature and humidity sensor (AHT20, Aosong) [**208**], has inputs for two coolant-pressure transducers, and drives the pump and a fan (Figure F.1). As designed, the ring signal fed an analog loop in the module controller. On the board as used that connection was cut, and the control loop runs in the microcontroller, which sets the drive through its digital-to-analog converter.

The ring is regulated by a digital proportional–integral–derivative controller at 5 Hz. A safety routine running every 50 ms stops the module and latches the fault when the ring thermistor reads open or outside 0–45 °C, when the 2.5 V reference leaves its band, when the module current exceeds 5 A for 500 ms or, in closed loop, is absent, or when the link to the operator falls silent for more than 2 s over the binary protocol. A hardware watchdog resets the microcontroller if its program stops. Faults on the five measurement channels are reported without stopping cooling. After a fault the pump and fan run at full speed for 30 s (3 s in console runs) to clear the stored heat.

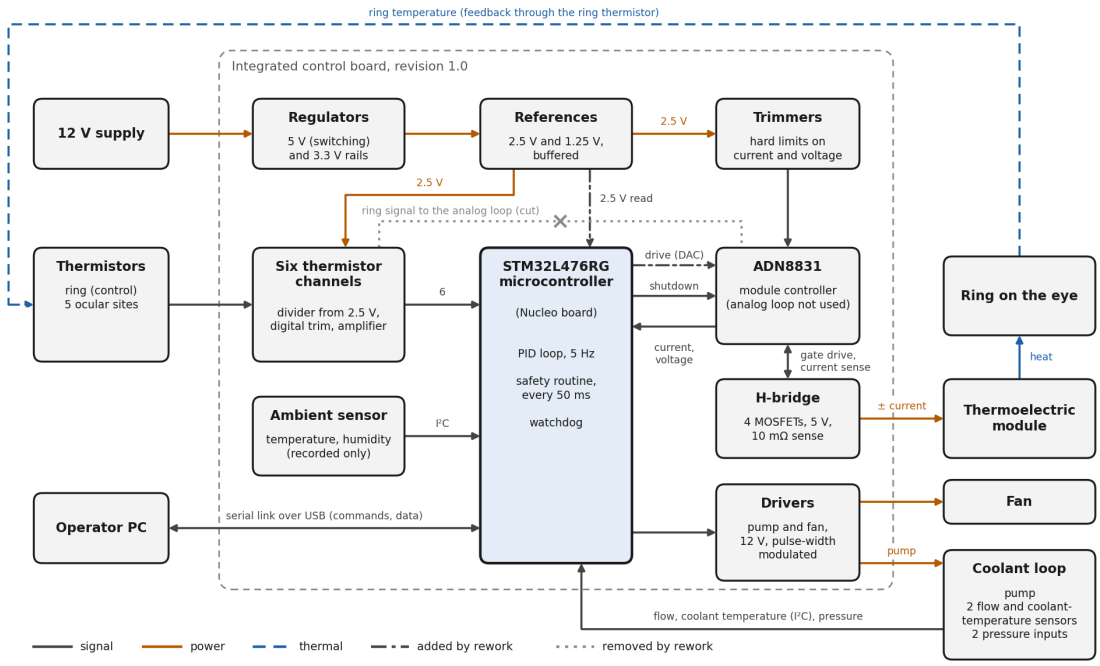


FIGURE F.1. Block diagram of the integrated control board (revision 1.0) as used: solid gray lines, signals; orange, power; dashed blue, the thermal path that closes the loop through the ring thermistor; dash-dot, connections added by rework (the drive set directly by the microcontroller and the reading of the 2.5 V reference); dotted, the connection removed by rework (the ring signal to the analog loop of the module controller).

The module voltage is five times the departure of the drive signal from a 1.25 V center, cooling below it and heating above, with one leg of the H-bridge linear and the other switched at 500 kHz. The board runs from an external 12 V supply, through a buck converter to 5 V and a linear regulator to 3.3 V. Each thermistor forms a divider with a digital potentiometer, followed by a buffer amplifier, and is converted to temperature with a Steinhart–Hart model.

F.2. Bench Verification

Before it drove a ring, the board was verified in stages: the bridge transistors, power-up, the reference, operation into dummy loads to confirm the hard limits, the thermal response, the polarity of the drive, and closed-loop regulation. The last stage was a soak at 4 °C with the device on a preserved porcine eye in the bath, which ended

with the ring thermistor disconnected. In a later session the module was disconnected and the program stalled (the latter with the module off).

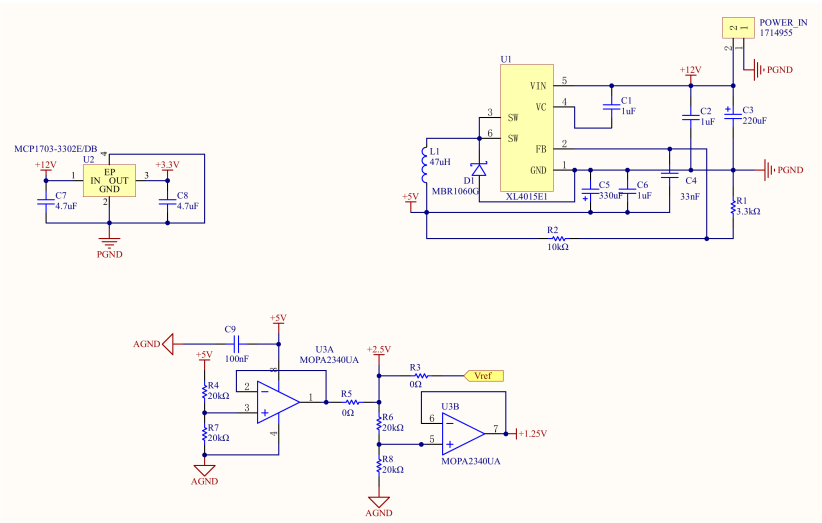
In a second closed-loop run, with chilled coolant and the ring out of the bath, the ring went from 30 to 4.5 °C in 352 s and held 3.98 °C on 1.65 A. The firmware held the drive at 2.9 A, so the pull-down was slower than that of the SCEC Gen 3 ring ex vivo, which completed 90% of its fall in about 2 min.

The other functions of the board worked, with limits. All six thermistor channels read, and in room air five agreed within 3 mV. Both flow sensors read about 0.1 $\mathrm{mL\,min^{-1}}$ with the pump off and 570 to 800 $\mathrm{mL\,min^{-1}}$ with it on, the upper values beyond their range. In later sessions they dropped off their bus during the reference fault of Section 3.3.8 and returned only when restarted, so that a flow reading accompanied 28% of the samples of one session. The board switched the pump and the fan, but neither could be graded: the pump gave no flow below 65% PWM duty, and the fan turned only at full duty. The ambient sensor read 25.3 to 25.9 °C over 12 min and was not compared with a reference. One of the two pressure inputs was connected to a transducer and worked.

F.3. Schematics and Drawings

Figures F.2 and F.3 reproduce the five schematic sheets of the board of Section 3.2.4 (design PCB_Project_LM_260407, revision 1.0) as designed and fabricated; the reworks are not drawn (CS4 moved from PA4 to PH1, PA4 wired to the 2.5 V reference, and the ring signal to IN1N cut). Figure F.4 gives the assembly and fabrication drawings. The board has four copper layers, a total thickness of 1.61 mm, and 741 plated holes. Each sheet is cropped to its drawing, without the frame and title block.

Sheet 1 of 5: Power



Sheet 2 of 5: ADN8831 controller and output stage

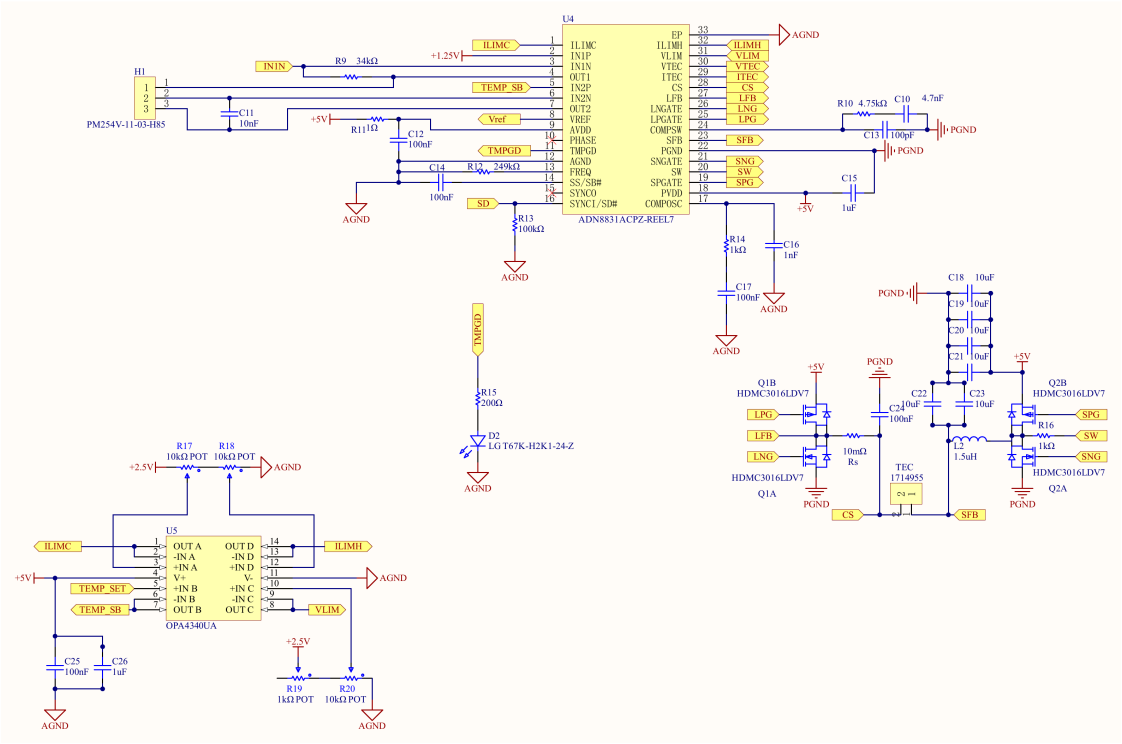
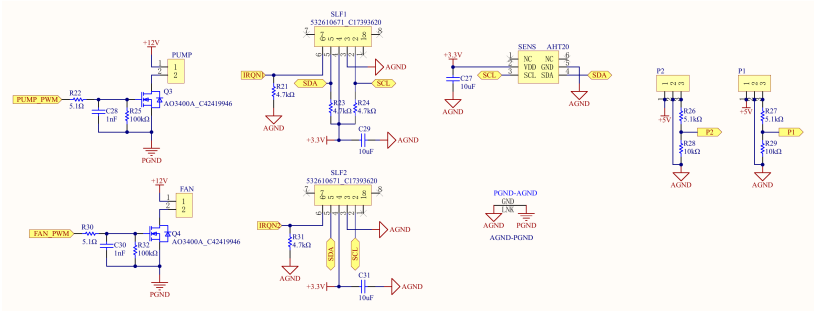
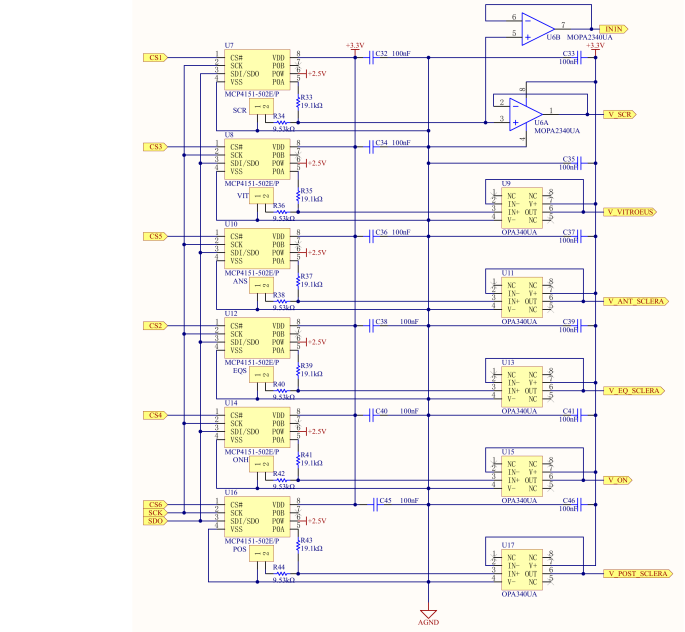


FIGURE F.2. Schematic of the integrated control board, sheets 1 and 2 of 5. Top, power: the 12 V input, the buck converter from 12 to 5 V (XL4015), the linear regulator from 12 to 3.3 V (MCP1703), and the buffered 2.5 V and 1.25 V references. Bottom, the thermoelectric-module controller (ADN8831), its output stage and H-bridge of four MOSFETs with the current-sense resistor and filter inductor, and the trimmers that set the current and voltage limits.

Sheet 3 of 5: Peripherals



Sheet 4 of 5: Thermistor channels



Sheet 5 of 5: Microcontroller

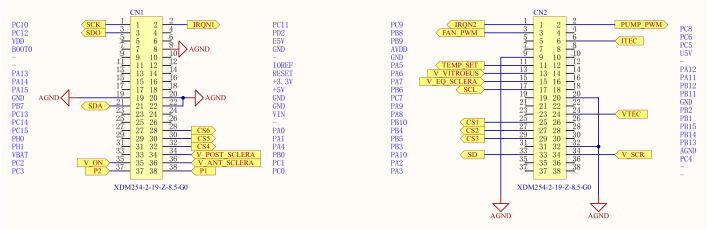


FIGURE F.3. Schematic of the integrated control board, sheets 3 to 5 of 5. Top, peripherals: low-side drivers of the pump and fan, the two coolant-flow sensors, the ambient temperature and humidity sensor, and the inputs of the two pressure transducers. Middle, thermistor channels: for each of the six thermistors, a digital potentiometer forming the divider from the 2.5 V reference and a buffer amplifier feeding the converter of the microcontroller. Bottom, microcontroller interface: the connections to the microcontroller development board through its expansion headers.

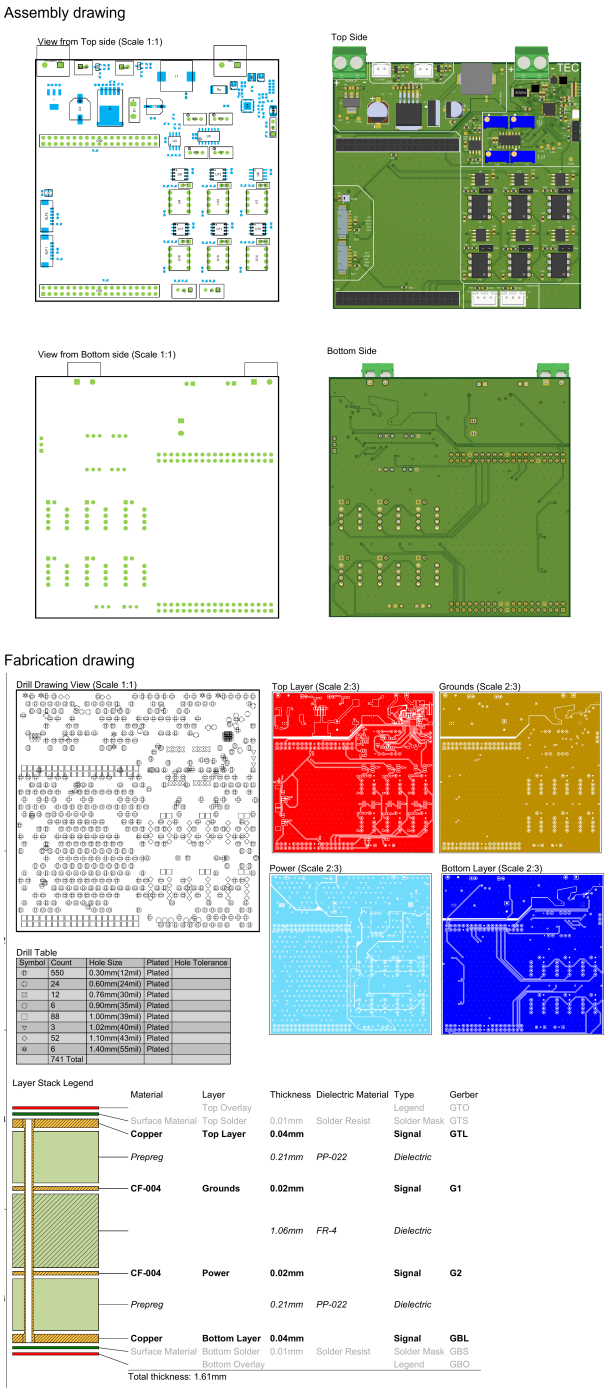


FIGURE F.4. Assembly (top) and fabrication (bottom) drawings of the integrated control board. Assembly: component placement on the top and bottom sides at full scale (left) and rendered views of the populated board (right). Fabrication: drill drawing and drill table, the four copper layers (top, ground, power, bottom), and the layer stack, 1.61 mm in total.

APPENDIX G

Anesthesia Records

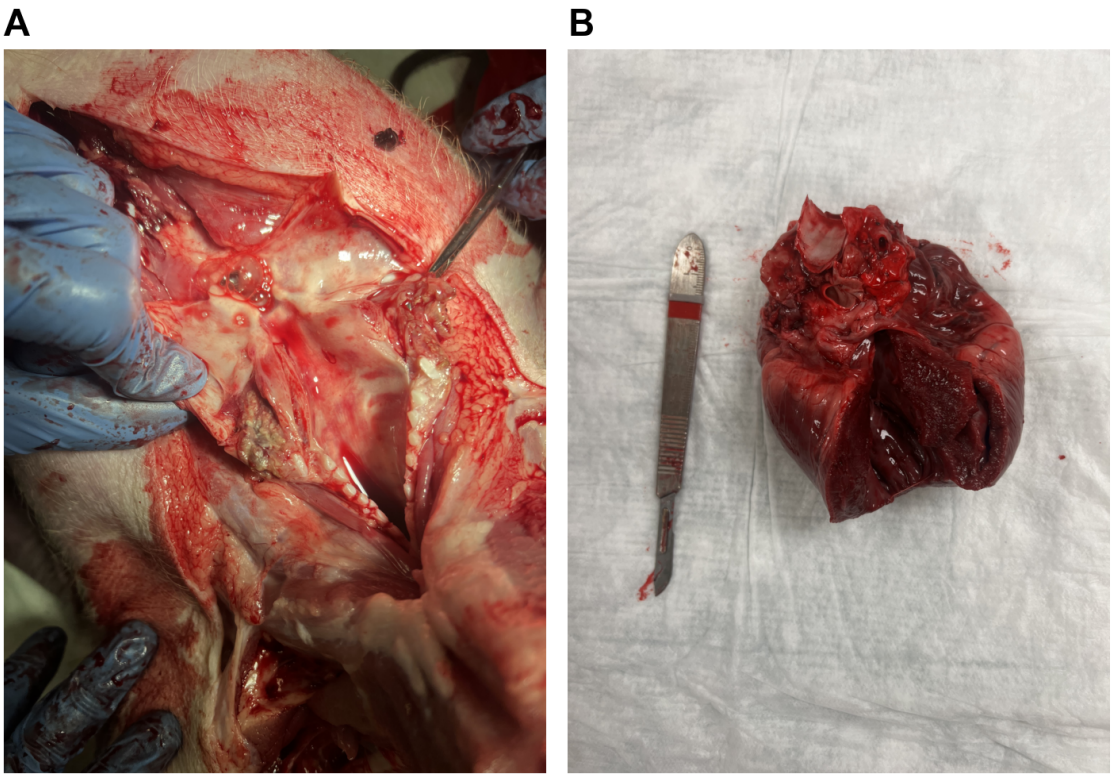


FIGURE G.1. Necropsy of animal 3, which died in recovery from session 3 (Section 4.3.6): (A) the opened larynx and trachea, with the laryngeal necrosis, most likely from a traumatic intubation in session 2; (B) the opened heart, with a scalpel for scale, with the cardiac hypertrophy, judged incidental. Neither was attributed to the device.

Table G.1 gives the drugs and doses of the BRL anesthesia record of each session (Section 4.2.3). After rocuronium the animal was weaned from the ventilator until spontaneous breathing resumed, and normal respiration and oxygen saturation were confirmed before it was returned to its housing.

TABLE G.1. Drugs given in the sessions, from the BRL anesthesia records. A, animal; S, session; IM, intramuscular; IV, intravenous.

Drug	Dose	Route	Given in
Ketamine and medetomidine (20 mg/mL)	Protocol 10 and 0.08 mg/kg; recorded 5 and 0.2 mg/kg in A2; up to 13.6 and 0.32 mg/kg in A1 S3	IM	All sessions except A1 S1 and A1 S4
Tiletamine–zolazepam, xylazine, butorphanol (100, 40, 6 mg/mL)	0.8 mL, 4.4 mg/kg of the combined drugs	IM	A1 S1
Tiletamine–zolazepam with xylazine (300 mg/mL combined)	1.4 mL (14.7 mg/kg), 0.7 mL (5.8 mg/kg), and 0.5 mL (4.6 mg/kg)	IM	S4 of A1 and A4; A6 S2
Isoflurane	1–2%	Inhaled	All
Rocuronium (10 mg/mL)	20 mg, then 10 mg about every 30 min; every 20 min in A2 S4, up to 60 min in A2 S2	IV	From A2 onward
Sugammadex	2 mg/kg (0.7 mL in A2 S1, S2 and A4 S1); none recorded in the other sessions	–	Session end
Flunixin	Protocol 2.2 mg/kg; 3.7 mg/kg recorded in A2 S1	IM, IV	All
Propofol	1.5 and about 1.8 mg/kg, when premedication did not allow intubation	IV	A1 S1, S3
Hydromorphone	Two doses in A1 S1; in A6 S1 ticked, amount not recorded	–	A1 S1, A6 S1
Lidocaine and ketamine infusion	Adjunct to isoflurane	IV	A2 S3, S4; all sessions of A3 to A6

APPENDIX H

Selected Ocular Surface Photographs and Histological Sections

This appendix gives, for every session, the photographs selected for grading: redness (top) and fluorescein staining (bottom), of the test and control eyes, before and after the session, with the grades given to each, on the Efron scale from 0 to 4 (Section 4.2.8). All grades are the author's. Animal 1 is exploratory. The redness photographs are cropped to a centered square, and the staining photographs lose only the banner of the instrument.

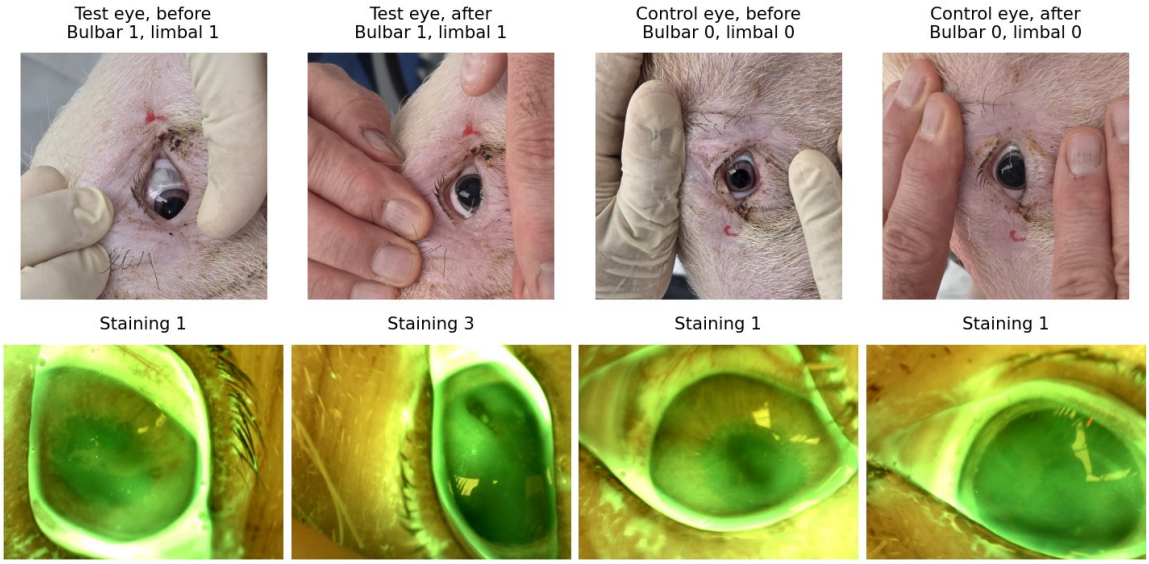


FIGURE H.1. Selected photographs of animal 1, session 1 (30-min session), with the grades of record.

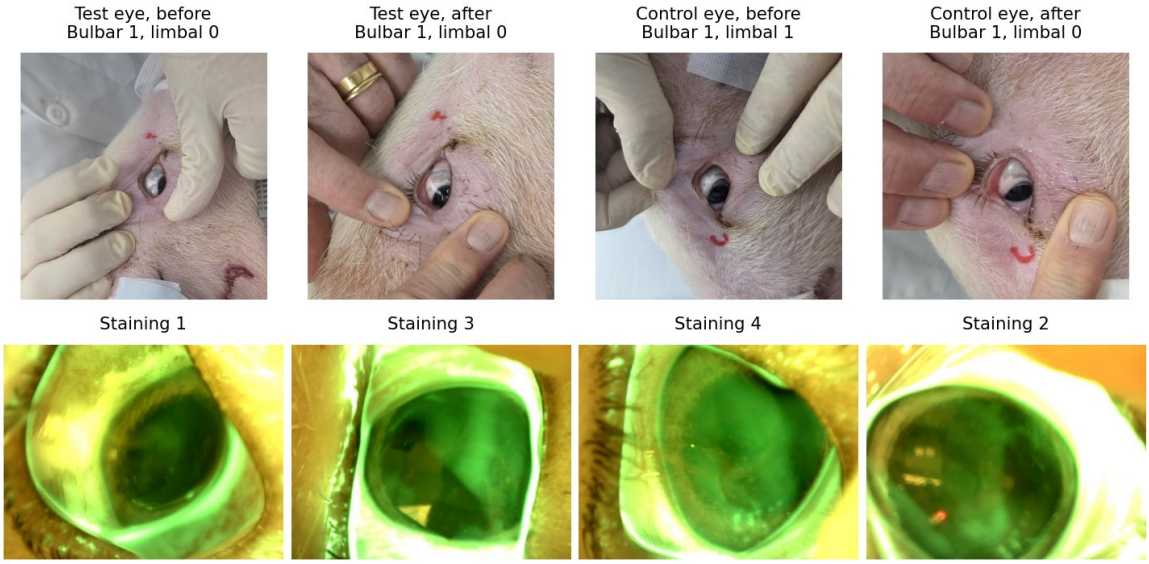


FIGURE H.2. Selected photographs of animal 1, session 2 (60-min session), with the grades of record.

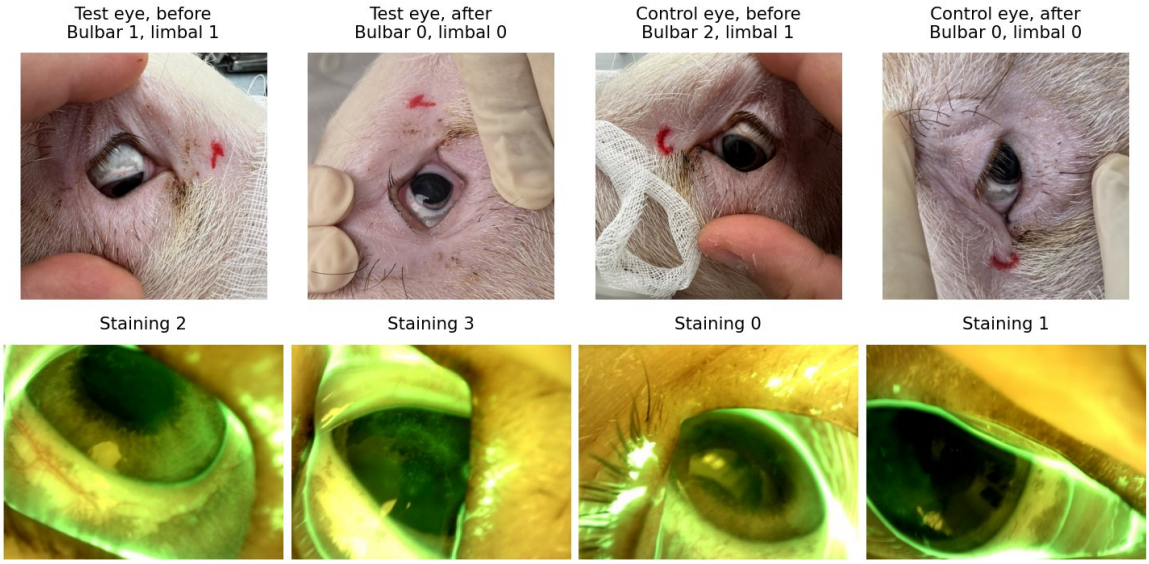


FIGURE H.3. Selected photographs of animal 1, session 3 (120-min session), with the grades of record.

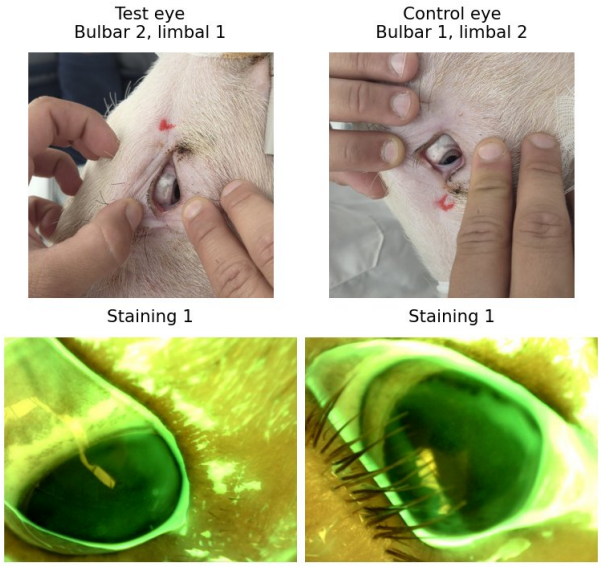


FIGURE H.4. Selected photographs of animal 1, session 4 (end-of-life session), with the grades of record.

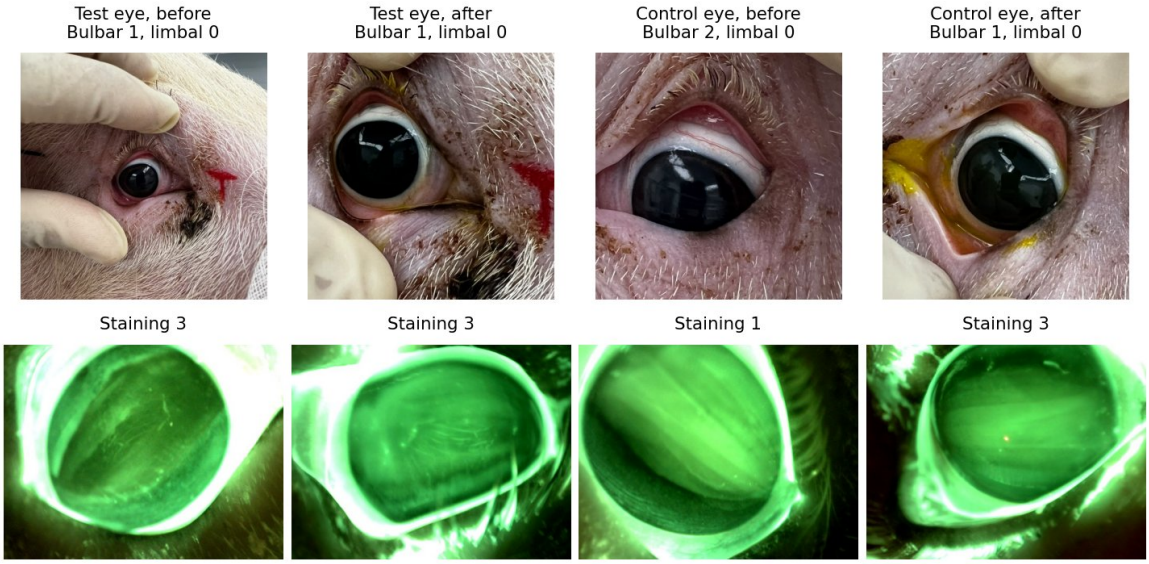


FIGURE H.5. Selected photographs of animal 2, session 1 (30-min session), with the grades of record.

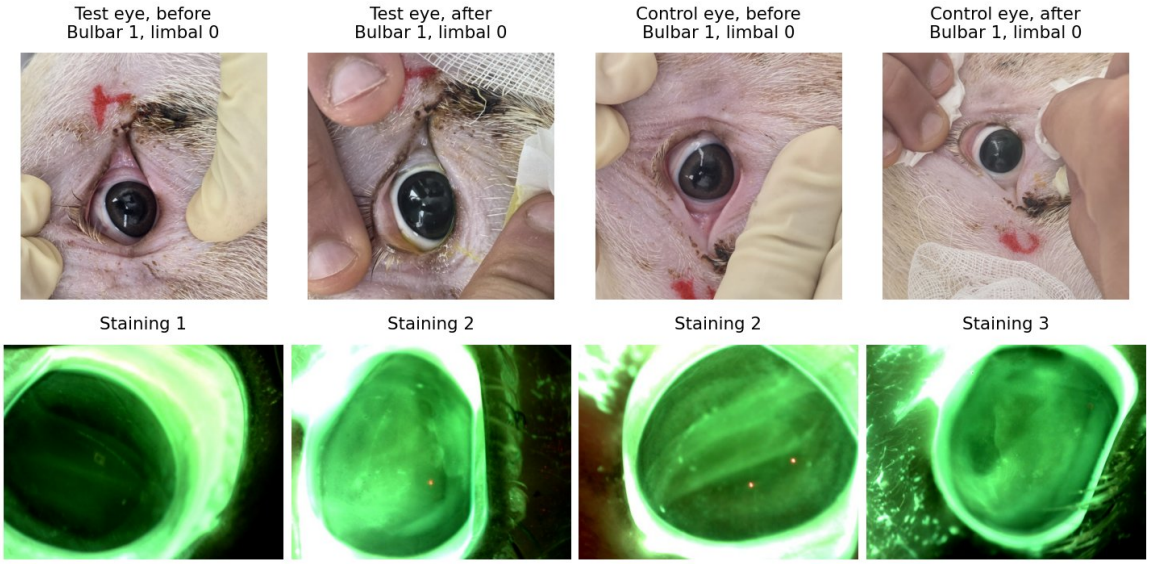


FIGURE H.6. Selected photographs of animal 2, session 2 (60-min session), with the grades of record.

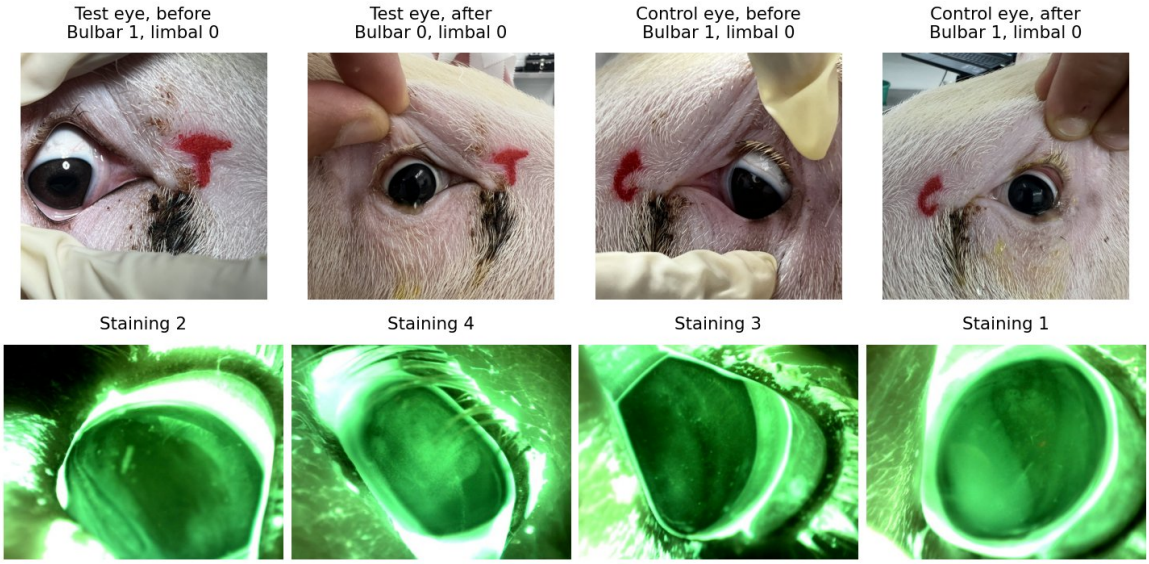


FIGURE H.7. Selected photographs of animal 2, session 3 (120-min session), with the grades of record.

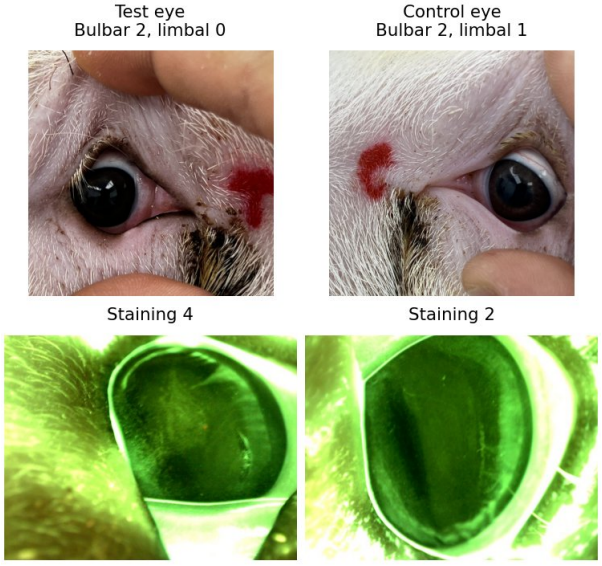


FIGURE H.8. Selected photographs of animal 2, session 4 (end-of-life session), with the grades of record.

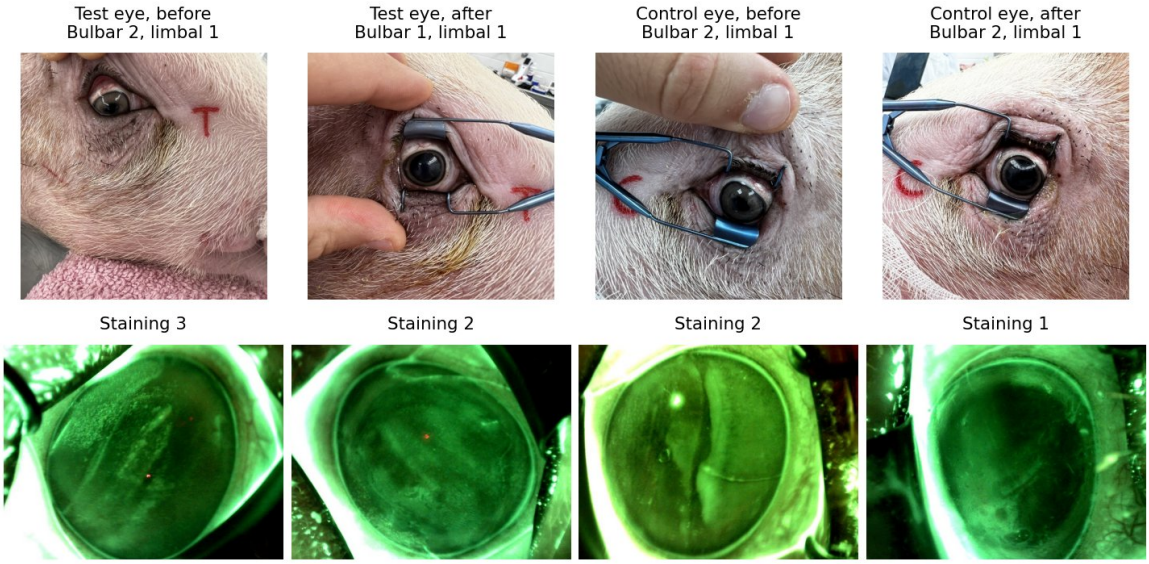


FIGURE H.9. Selected photographs of animal 3, session 1 (30-min session), with the grades of record.

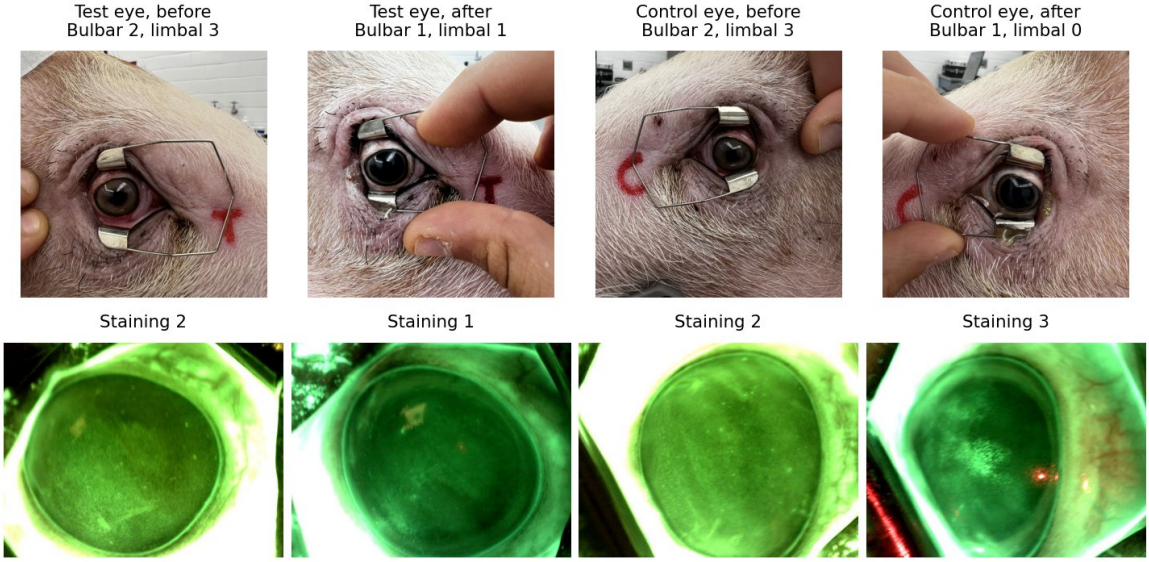


FIGURE H.10. Selected photographs of animal 3, session 2 (60-min session), with the grades of record.

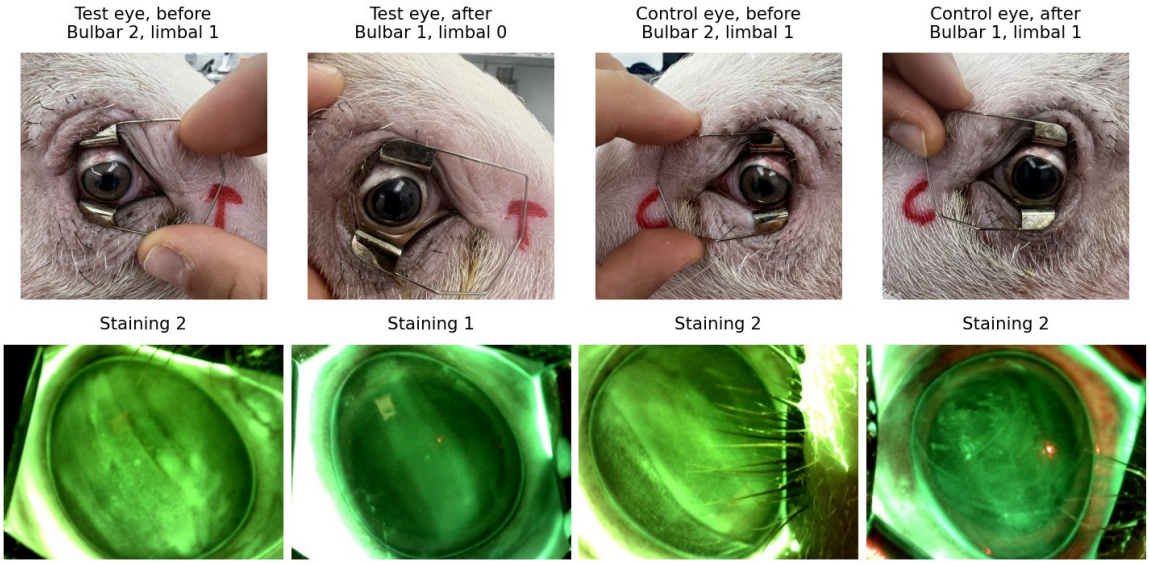


FIGURE H.11. Selected photographs of animal 3, session 3 (120-min session), with the grades of record.

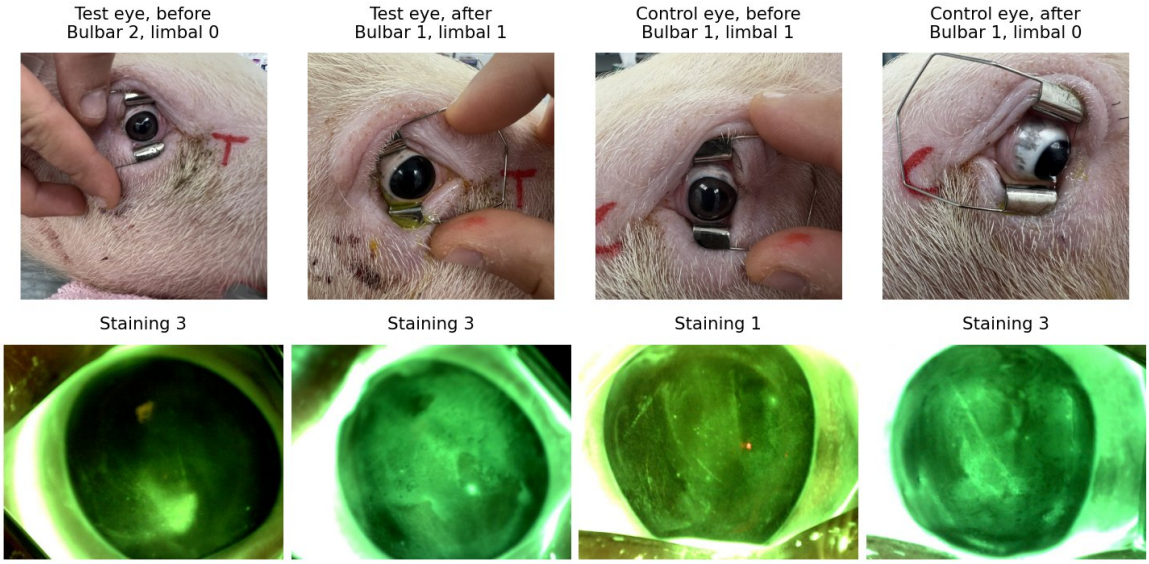


FIGURE H.12. Selected photographs of animal 4, session 1 (30-min session), with the grades of record.

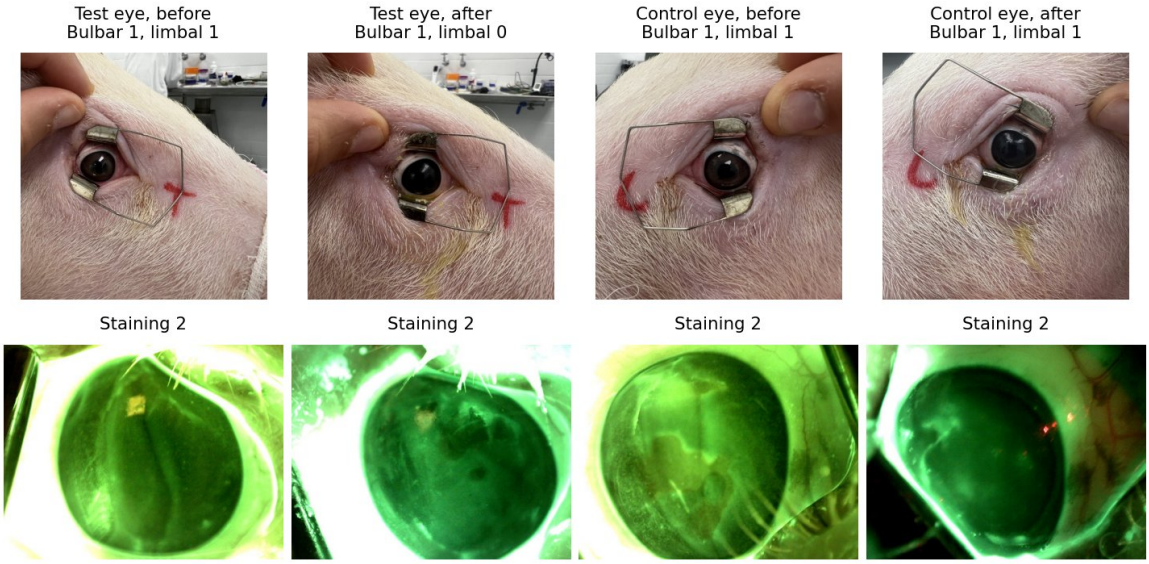


FIGURE H.13. Selected photographs of animal 4, session 2 (60-min session), with the grades of record.

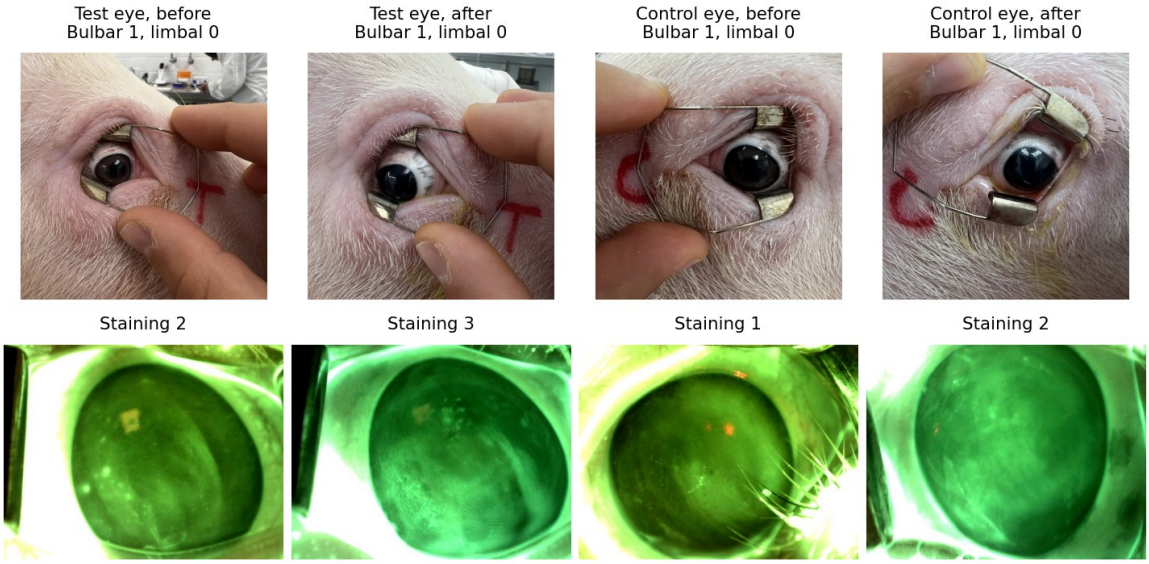


FIGURE H.14. Selected photographs of animal 4, session 3 (120-min session), with the grades of record.

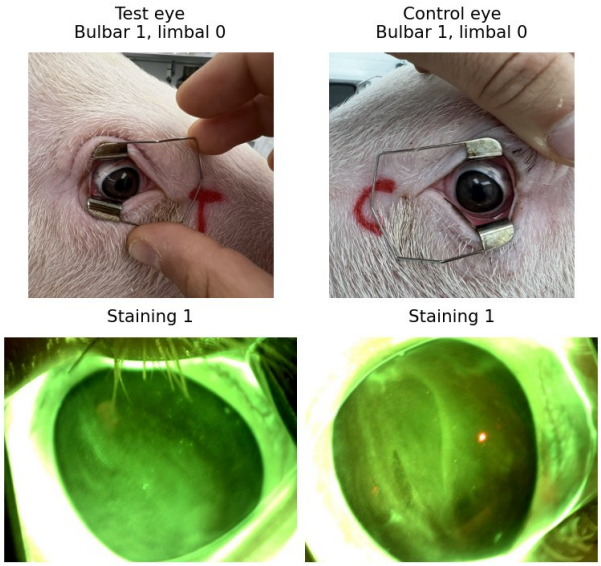


FIGURE H.15. Selected photographs of animal 4, session 4 (end-of-life session), with the grades of record.

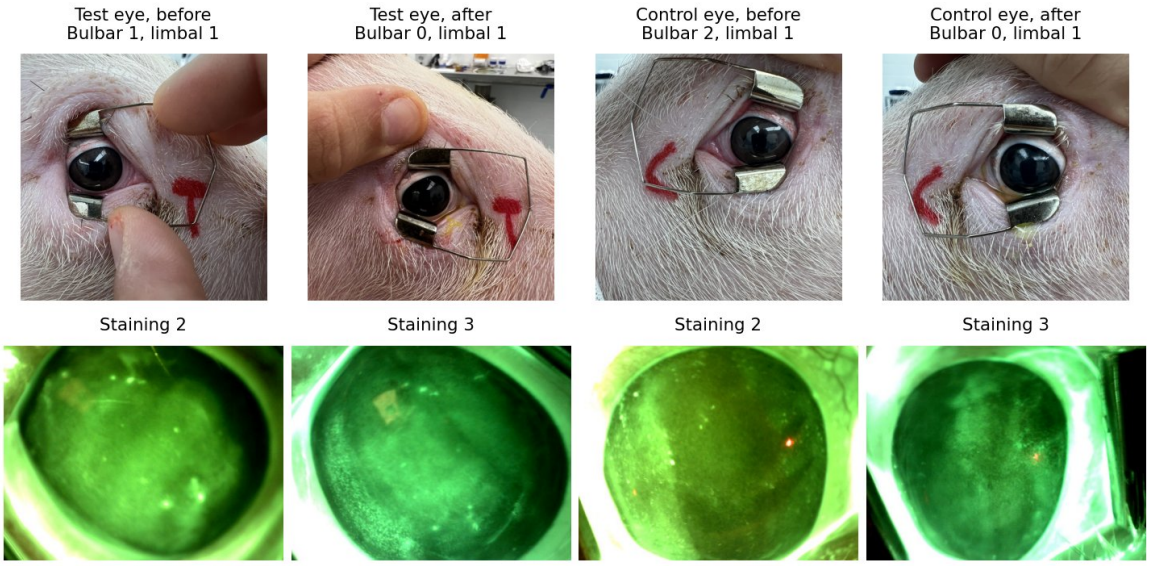


FIGURE H.16. Selected photographs of animal 5, session 1 (30-min session), with the grades of record.

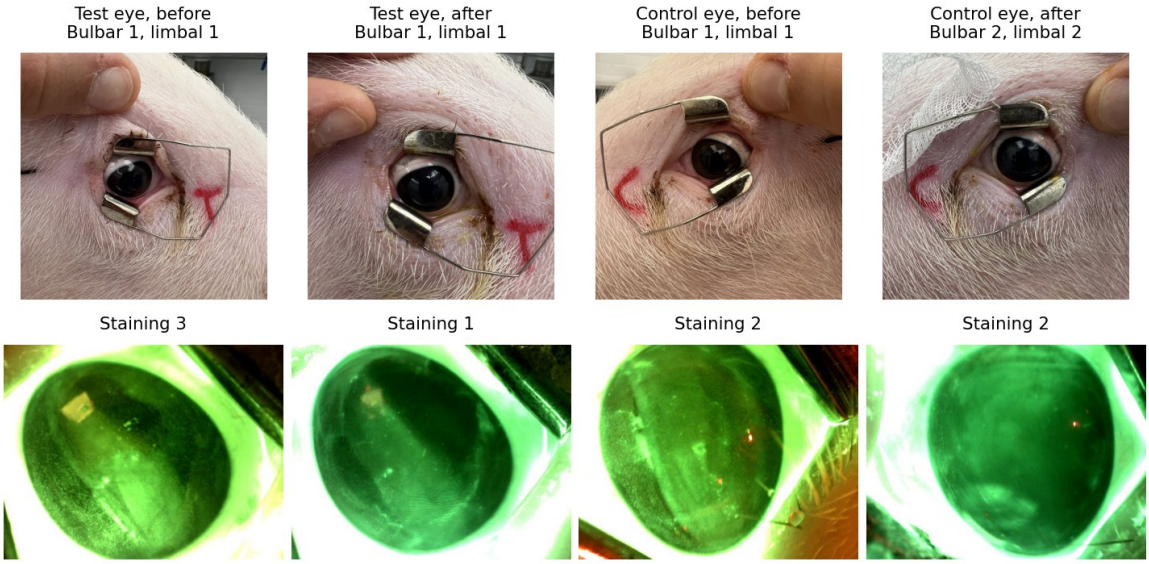


FIGURE H.17. Selected photographs of animal 5, session 2 (60-min session), with the grades of record.

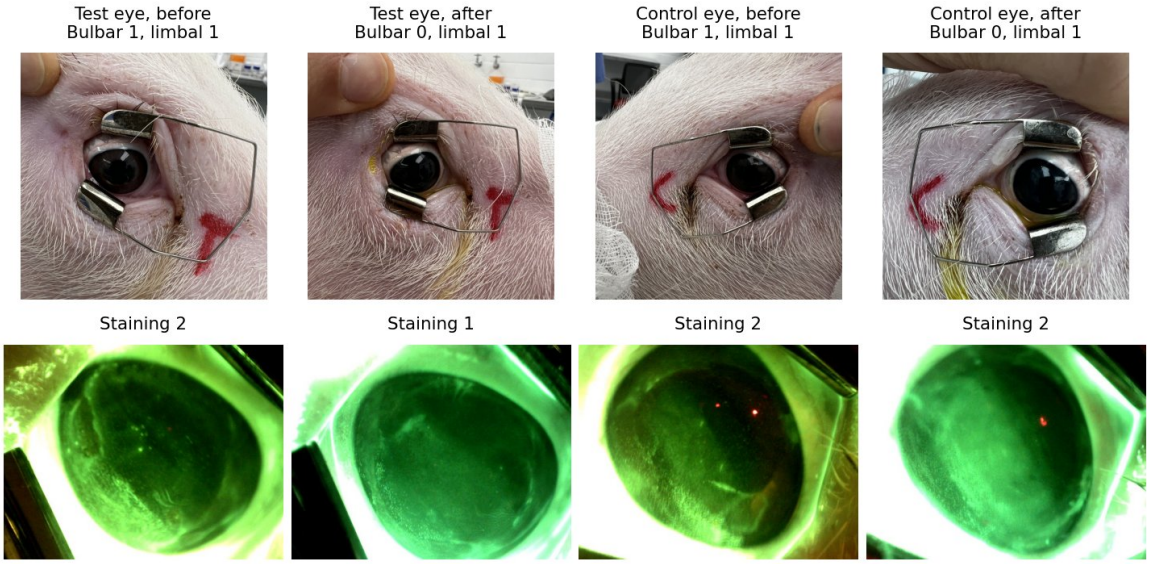


FIGURE H.18. Selected photographs of animal 5, session 3 (120-min session), with the grades of record.

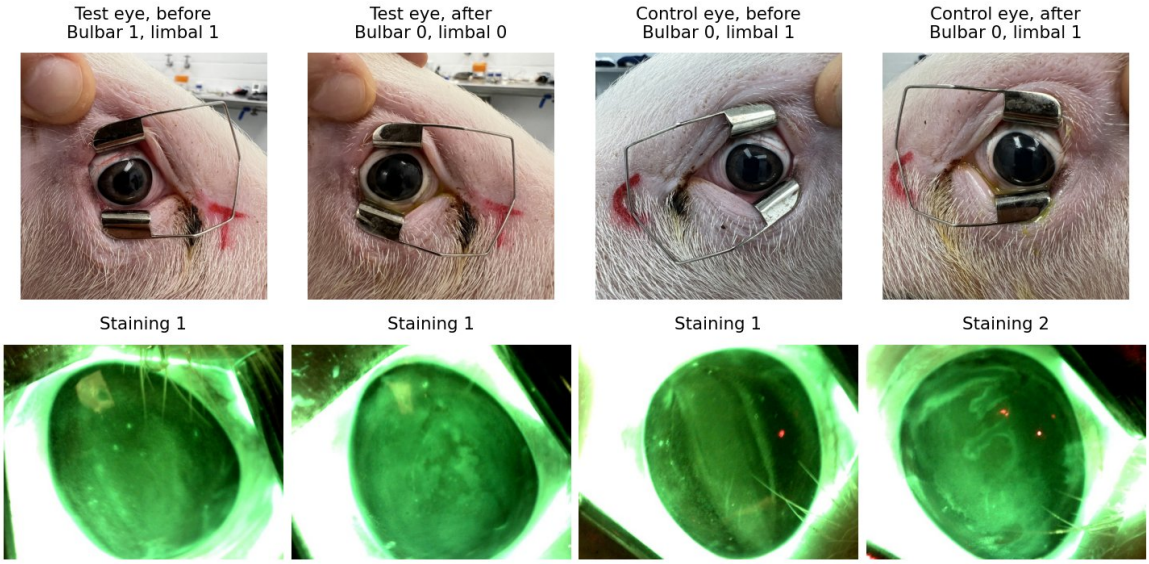


FIGURE H.19. Selected photographs of animal 6, session 1 (30-min session), with the grades of record.

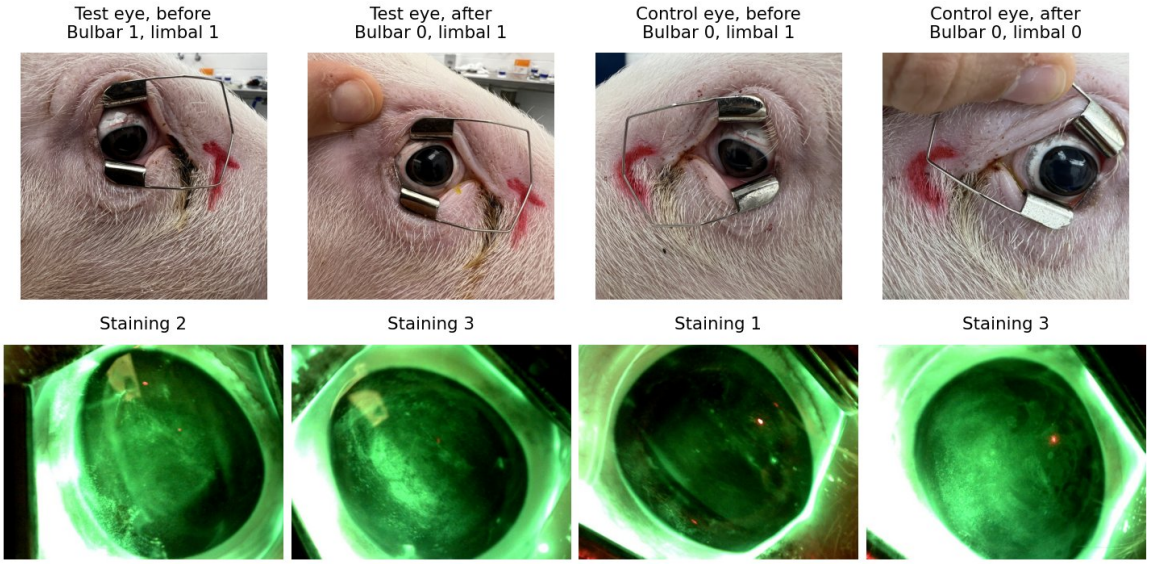


FIGURE H.20. Selected photographs of animal 6, session 2 (60-min session), with the grades of record.

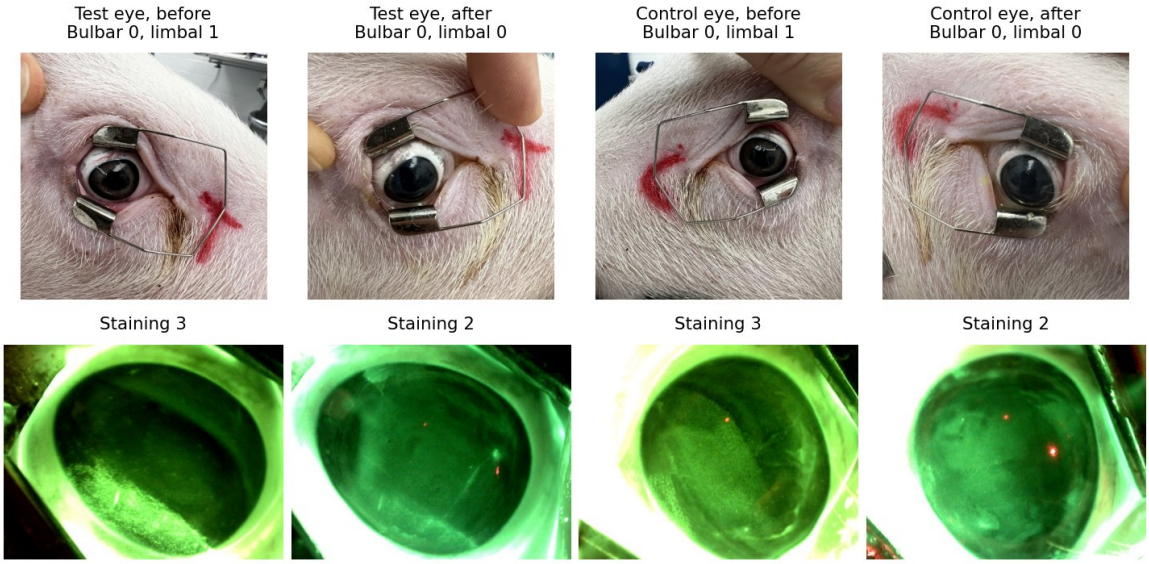


FIGURE H.21. Selected photographs of animal 6, session 3 (120-min session), with the grades of record.

Histological Sections

Slide 5 of the hematoxylin and eosin sections of the test and control eyes of each animal, as scanned (Section 4.2.8); for animal 1, which has no slide 5, slide 10. Animal 3 died after its third session and its eyes were collected after death.

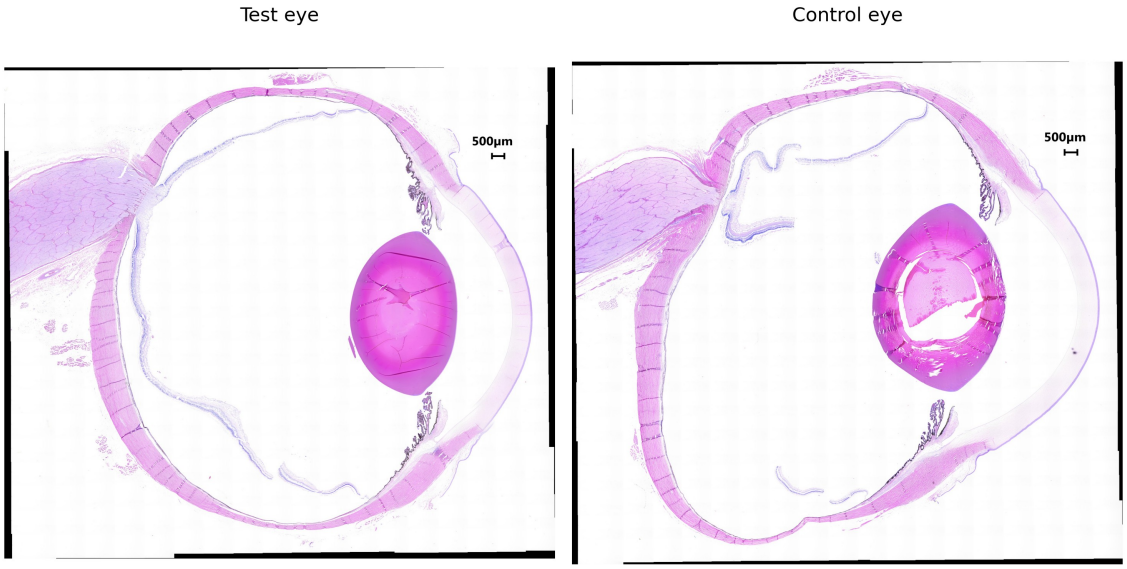


FIGURE H.22. Hematoxylin and eosin sections of animal 1 (exploratory), slide 10: test eye (left) and control eye (right).

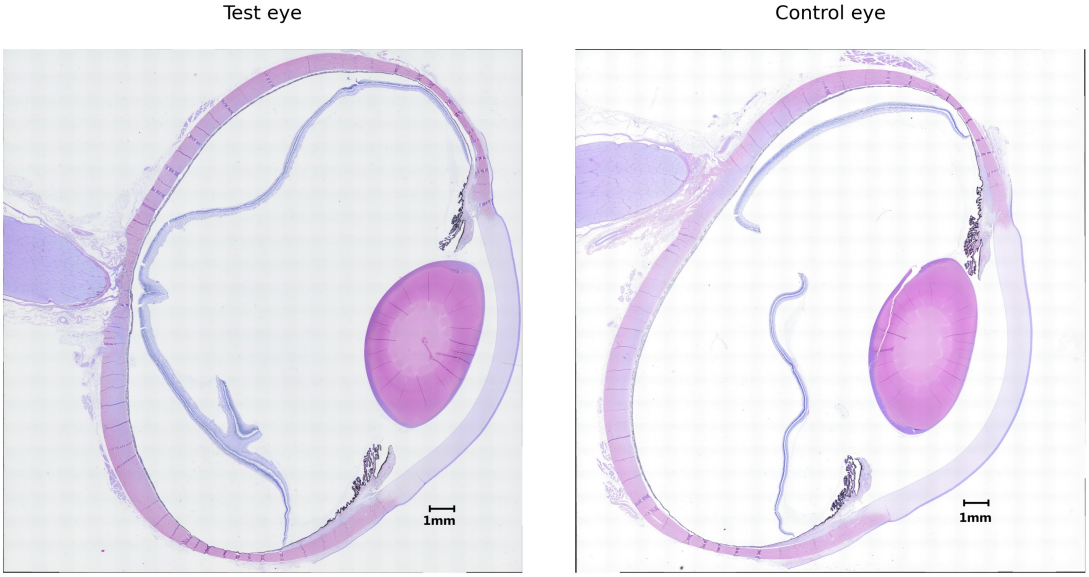


FIGURE H.23. Hematoxylin and eosin sections of animal 2, slide 5: test eye (left) and control eye (right).

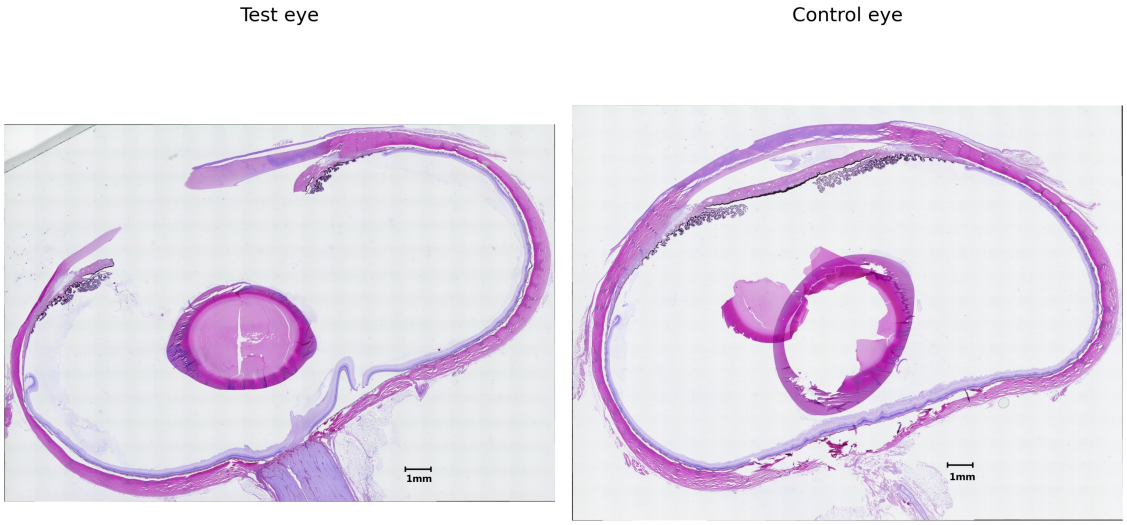


FIGURE H.24. Hematoxylin and eosin sections of animal 3, slide 5: test eye (left) and control eye (right).

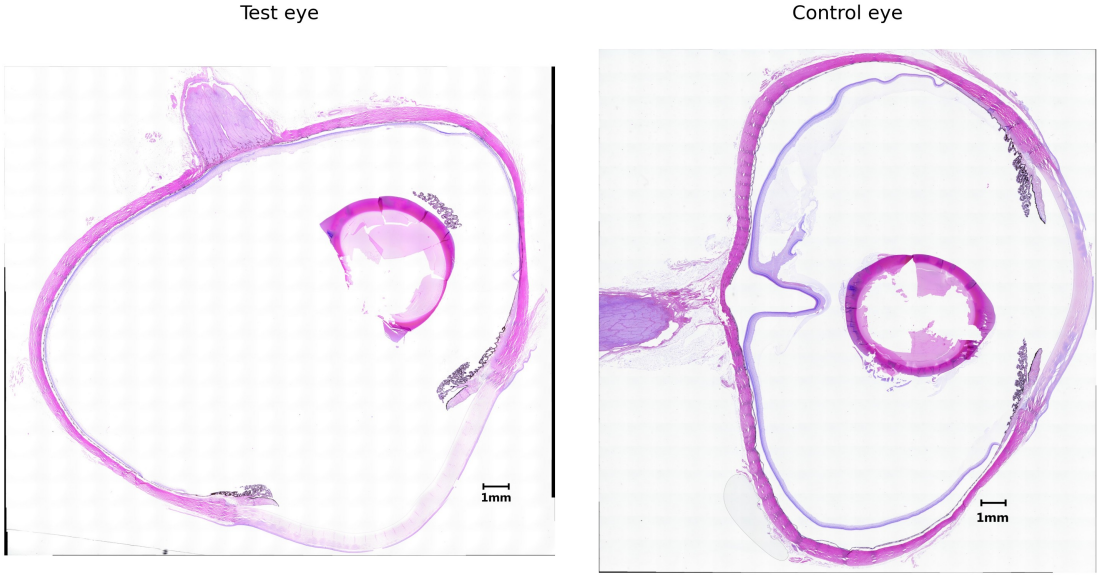


FIGURE H.25. Hematoxylin and eosin sections of animal 4, slide 5: test eye (left) and control eye (right).

APPENDIX I

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APPENDIX J

Use of Generative Artificial Intelligence

A generative artificial intelligence tool (Claude, Anthropic, accessed via Claude Code) was used during the preparation of this dissertation. Its use, as described below, was approved by all five members of the dissertation committee by e-mail between October 23 and November 4, 2026.

Writing. The tool assisted with literature search and organization, verification of cited sources against their original publications, drafting and revision of prose, and formatting of the LaTeX source. All AI-assisted text was reviewed, checked against the primary literature, and edited by the author.

Code. Under the author's direction, the tool helped write, revise, test, and run the analysis code of Chapters 2 and 3 and the scripts that generate their figures and tables, and helped transfer the results into the text. It wrote the Python finite-element program of Chapter 3 and its test scripts, and it reviewed the code and results of that program, which was accepted only after it had reproduced the solutions of the commercial software and exact solutions (Section 3.3.1). Every analytical decision was made by the author.

Histology. Before the author read the histological sections of Chapter 4, the tool made a masked comparison of the two eyes of each animal, coded and partly mirrored, in five independent passes per animal (Section 4.2.8). The author then read the sections, knowing which eye had been cooled, and agreed with it; the findings reported are the author's.

The tool was not used to design experiments or to collect data. The author takes full responsibility for the accuracy and content of this dissertation.

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- Engineering Intern, R&D, RetMap Inc., Chicago, Illinois, Summers 2023, 2024, and 2025.
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TEACHING

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PUBLICATIONS

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- L. Mecacci, Y. Nakamura, K. Fong, N. Shirdade, and J. R. Hetling, "An ex vivo system to study clinically feasible induction of therapeutic hypothermia in porcine eyes," *Journal of Neuroscience Methods* (manuscript pending submission).
- L. Mecacci, Y. Nakamura, and J. R. Hetling, "Ocular hypothermia delivery system: in vitro validation using a non-perfused pig eye model," ARVO Annual Meeting, 2023 (poster).
- L. Mecacci, Y. Nakamura, and J. R. Hetling, "Ocular hypothermia, in-vitro validation of a novel therapeutic apparatus using non perfused porcine eye

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- L. Mecacci and J. R. Hetling, "Ocular hypothermia delivery system: in vitro validation using non-perfused human donor eyes," ARVO Annual Meeting, 2026 (poster).

PATENTS AND GRANTS

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